# Cluster Editing above a $P_3$ packing: W[1]-hardness in the excess and the treewidth of the intersection graph

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**Abstract**

Cluster Editing asks for the fewest edge insertions and deletions that turn a graph into a disjoint union of cliques. Li, Pilipczuk and Sorge parameterize it by the excess $\ell$ of the solution size over a modification-disjoint packing $H$ of induced $P_3$'s, and leave open for which packings the problem is fixed-parameter tractable. We use the treewidth $w$ of the intersection graph $I(H)$ of the packing as the structural measure. Our main result is a negative answer to the question whether $\ell + w$ suffices: the problem is W[1]-hard parameterized by $\ell + w$, and also by $\ell + \mathrm{pw}(I(H))$ and by $\ell + c$, where $c$ is the maximum number of packed $P_3$'s through a vertex, even for pair-maximal packings. Hence there is no algorithm in time $f(\ell, w)n^{O(1)}$ unless FPT = W[1], and under ETH none in time $f(\ell, w)n^{o(\sqrt{\ell})}$. The reduction is from Grid Tiling with $\leq$; it uses a selection chain of bounded width whose position is released by cutting one of many "tunnels", and it is computer-assisted at four finite steps, which we list (the chain lemma for chains of length $m \geq 10$ rests on a finite computation); its counting core and several structural lemmas are also machine-checked in Lean 4, which does not cover the graph-level statements of the reduction. The matching upper bounds come from a CSP view: for $\ell = 0$ the problem is a constraint satisfaction problem with domain size 3 whose scopes are cliques of $I(H)$; for general $\ell$, guessing at most $\ell$ edits gives domain size 5. This yields algorithms in time $3^{w+1}n^{O(1)}$ and $5^{w+1}n^{2\ell+O(1)}$, and $n^{\ell}f(\ell, w)5^{w+\ell+1}n^{O(1)}$ after a locality argument. Under ETH there is no $2^{o(w)}n^{O(1)}$ algorithm even for $\ell = 0$, and neither $w$ nor $\ell$ alone suffices. We isolate a configuration, a *soft triangle*, as the source of difficulty: without soft triangles the zero-excess solutions are closed under every majority operation, so $\ell = 0$ is polynomial for every $c$ and $w$; the hardness reduction shows that soft triangles suffice. At $\ell = 0$ and $c \leq 2$ a certificate of at most four remote pairs always exists, even with soft triangles. Further results (structure of optimal solutions, components of $I(H)$, locality of inserted pairs, repacking) are stated in a short section with proofs in the appendices. Still open are hardness for bounded $c$ (our reduction has $c = O(\ell)$), the case $c = 2$, bounded certificates for $\ell \geq 1$, and the exact exponent of $n$ under ETH.

## 1 Introduction

Cluster Editing asks for the fewest edge insertions and deletions that turn a graph $G$ into a disjoint union of cliques. Every induced $P_3$ must be hit by an edit, so a packing $H$ of induced $P_3$'s that pairwise share no vertex pair (a *modification-disjoint* packing) gives the lower bound $|H|$ on the solution size. Li, Pilipczuk and Sorge [15] parameterize by the *excess* $\ell$ over this bound (problem CEaMP): is there a solution of size at most $|H| + \ell$? They show NP-hardness already for $\ell = 0$, with every vertex in at most 23 packed $P_3$'s, and an $n^{2\ell+O(1)}$ algorithm when every vertex lies in

at most two packed $P_3$'s; whether the latter case is fixed-parameter tractable is open [15]. For vertex-disjoint packings the problem is fixed-parameter tractable in $\ell$ [17].

The question of this paper is which measurable *interaction structure* of $H$ makes the local repair choices (one edit per packed $P_3$, in three ways) combine into a global solution in $f(\ell, w)\, n^{O(1)}$ time. We take $w = \mathrm{tw}(I(H))$, where the intersection graph $I(H)$ has the packed $P_3$'s as nodes and joins two of them when they share a vertex. The answer is negative: the problem is W[1]-hard parameterized by $\ell + w$ (Theorem 5.1), already for $\ell + \mathrm{pw}(I(H))$ and for $\ell + c$, where $c \le w + 1$ is the maximum number of packed $P_3$'s through a vertex. The matching upper bounds are XP in $\ell$ with single-exponential dependence on $w$.

### 1.1 Results

Our main theorem collects the results of Sections 3–6.

**Theorem 1.1** (Main theorem)**.** *Let $(G, H, \ell)$ be an instance of CEaMP, $w = \mathrm{tw}(I(H))$, and assume a tree decomposition of $I(H)$ of width $w$ is given.*

*(a)* W[1]-hardness. *CEaMP is W[1]-hard parameterized by $\ell+w$, by $\ell+\mathrm{pw}(I(H))$ and by $\ell+c$, even for pair-maximal packings; the hard instances have $\ell = k^2$, $c \le 4p+2$ and $\mathrm{pw}(I(H)) \le 24kp^2$ with $p = O(k^2)$. Hence there is no algorithm in time $f(\ell, w)n^{O(1)}$ unless* FPT = W[1]*, and, unless ETH fails, none in time $f(\ell, w)N^{o(\sqrt{\ell})}$ on instances with $N$ vertices. So the XP algorithms in (c) are optimal up to the exponent of $n$. The proof is computer-assisted at four finite steps (Section 5.9).*

*(b)* CSP representation. *$(G, H, 0)$ is a YES instance iff a constraint satisfaction problem with one variable of domain $\{1, 2, 3\}$ per packed $P_3$ is satisfiable; every constraint has at most three variables, and its scope is a clique of $I(H)$. For $\ell > 0$, for each of the at most $n^{2\ell}$ sets $U$ of at most $\ell$ hard pairs there is such a problem with domain size 5 (the five edit sets of a packed $P_3$ that leave it a cluster graph, Lemma 3.3) and unary costs, and $(G, H, \ell)$ is a YES instance iff for some $U$ its minimum cost is at most $|H| + \ell - |U|$.*

*(c)* Algorithms. *CEaMP is solvable in time $3^{w+1}n^{O(1)}$ for $\ell = 0$ and in time $5^{w+1}n^{2\ell+O(1)}$ for general $\ell$. The latter is XP in $\ell$ with single-exponential dependence on $w$; by (a) it is not a fixed-parameter algorithm and cannot be made one unless* FPT = W[1].

*(d)* ETH lower bound in the exponent of 2. *Unless ETH fails, there is no $2^{o(w)}n^{O(1)}$ algorithm for $\ell = 0$, even if every vertex lies in at most 23 packed $P_3$'s.*

*(e)* Neither parameter suffices alone. *CEaMP is NP-hard when $w = 0$ (with $\ell$ part of the input) and, by [15], when $\ell = 0$.*

*(f)* Soft triangles. *Call a vertex triple a* soft triangle *if its three pairs are packed by three distinct members. If $(G, H)$ has no soft triangle (in particular if $I(H)$ is triangle-free, e.g. a forest), then the set of zero-excess state vectors is closed under every majority operation; consequently $(G, H, 0)$ is decidable in polynomial time for every $c$ and $w$, and every lock core has at most two anchors.*

*(g)* Certificates at $\ell = 0$. *If $c \le 2$ (soft triangles allowed) and no zero-excess solution avoids remote edits, there is a certificate of at most 4 remote pairs (proved, with a finite classification by computer); with soft triangles minimal certificates are nevertheless unbounded at $c = 2$.*

Part (a) is Theorem 5.1 and Corollary 5.2 (Section 5; the reduction is given there, the proofs of the long technical lemmas are in Appendix A, and four finite steps of the proof are computer-assisted, as listed in Section 5.9); parts (b) and (c) are Theorems 3.5 and 3.6 (and Theorem I.2); (d) and (e) are Theorem 4.1 and Proposition 4.3; (f) is Theorem 6.4 with Corollaries 6.5 and 6.7, and (g) is Theorem 6.11 with Theorem B.16 (Section 6). The CSP view and the dynamic program are standard once the interaction structure is identified (Observation 3.2); (d) is a consequence of the reduction of [15]. The hardness reduction is from Grid Tiling with $\leq$; it combines a bounded-width selection chain, blown-up vertices that cannot be split, guests that forbid positions, and hard edges (tunnels) between far vertices whose cut releases two chains at once.

**Organisation.** Section 2 fixes notation, Section 3 gives the CSP representation and the algorithms, Section 4 the lower bounds for $\ell = 0$ and for $w$ alone, and Section 5 the hardness reduction (the longest section). Section 6 explains why hardness needs soft triangles and proves a bounded-certificate statement at $\ell = 0$, $c \leq 2$. Section 7 lists, with statements only, the structural lemmas, the component theorem, the $n^{\ell}$ algorithm, the locality of inserted pairs and the repacking results; Section 8 lists open problems; Section 9 states precisely what is computer-assisted and what is machine-checked in Lean 4. The appendices contain the proofs of the long lemmas of Section 5 (Appendix A) and the full text of the results of Sections 6 and 7; Appendix R is a table of the scope of every result.

### 1.2 Related work

van Bevern, Froese and Komusiewicz [17] parameterize $F$-free editing above packings; for Cluster Editing they obtain fixed-parameter algorithms for vertex-disjoint $P_3$ packings and leave open the modification-disjoint case, which Li, Pilipczuk and Sorge [15] settle negatively for $\ell = 0$ and study for $1/c$-integral packings. Modification-disjoint $P_3$ packings are also used as a lower-bound tool for cluster vertex splitting [8]; for broader context see the survey [5]. To our knowledge, the treewidth of the intersection graph of the packing has not been used as a parameter before. Komusiewicz and Uhlmann [13] show that Cluster Editing is NP-hard on graphs of maximum degree 6 and has no $2^{o(k)}n^{O(1)}$ algorithm under ETH. The critical-clique (twin) property used in Appendix E goes back to [10, 7]; the half-degree property of Lemma G.2 is the same single-vertex-move argument. Cubic expanders of linear treewidth are cited from Grohe and Marx [9].

## 2 Preliminaries

**Graphs and solutions.** Graphs are simple and undirected, $n = |V(G)|$. A *pair* is a 2-subset of $V(G)$; for a set $S$ of pairs, $G \triangle S$ flips the adjacency of each pair in $S$. A graph is a *cluster graph* iff it has no induced $P_3$, i.e. no vertex triple with exactly two edges. A *solution* is a set $S$ of pairs with $G \triangle S$ a cluster graph; $\mathrm{OPT}(G)$ is the minimum size of a solution. Solutions correspond to partitions of $V$ into *clusters* (the cliques of $G \triangle S$); the *cost* of a partition is the number of pairs $ab$ with $[ab \in E] \neq [a, b \text{ in one cluster}]$, and an *optimum* is a partition of minimum cost. We write $\alpha(\cdot)$ for the independence number and $\nu(\cdot)$ for the matching number.

**Packings, soft and hard pairs.** A *packing* $H = \{P_1, \ldots, P_p\}$ is a set of induced $P_3$'s $P_i = (a_i, b_i, c_i)$ with centre $b_i$ ($a_ib_i, b_ic_i \in E$, $a_ic_i \notin E$); its elements are the *members* or *packed $P_3$'s*. Write $\mathrm{pairs}(P_i)$ for the three pairs inside $V(P_i)$. $H$ is *modification-disjoint* if the sets $\mathrm{pairs}(P_i)$ are pairwise disjoint; this is equivalent to: any two members share at most one vertex. All packings

below are modification-disjoint. A pair is *soft* if it lies in $\bigcup_i \text{pairs}(P_i)$ (its *owner* is then unique) and *hard* otherwise; a *hard edge* is a hard pair in $E$ and a *hard non-edge* a hard pair outside $E$. For a vertex $v$, $Q(v)$ is the set of soft partners of $v$, $N_h(v)$ the set of hard neighbours and $d_h(v) = |N_h(v)|$. $G_{\text{soft}}$ is the *soft graph* on $V$ whose edges are the soft pairs.

**Problem CEaMP.** Given $G$, a modification-disjoint packing $H$ and $\ell \geq 0$, decide whether there is a solution $S$ with $|S| \leq |H| + \ell$ (the convention of [15]; $\ell$ is the *excess*). We call $(G, H, \ell)$ a *YES instance* if $\text{OPT}(G) \leq |H| + \ell$.

**The intersection graph.** The *intersection graph* $I(H)$ has node set $H$ and an edge $PP'$ iff $V(P) \cap V(P') \neq \emptyset$. The members through one vertex form a clique of $I(H)$; hence (Helly property of tree decompositions) they lie in a common bag. Put $c = c(H) = \max_v |\{P \ni v\}|$ and $w = \text{tw}(I(H))$ (with $w = 0$ if $I(H)$ has no edge), so $c \leq w + 1$. Tree decompositions of width $w$ are assumed *given* in the algorithmic statements; one of width $w$ is computed in time $2^{O(w^3)}n$ [3, 6], and one of width at most $2w + 1$ in time $2^{O(w)}n$ [14], which would change the bases below (e.g. $7^{w+1}$ to $7^{2w+2}$).

**Lemma 2.1** (Degrees)**.** *For every vertex $v$, $|Q(v)| = 2\,|\{P \in H : v \in V(P)\}|$. Hence $D(H) := \max_v |Q(v)| = 2c \leq 2(w + 1)$.*

*Proof.* Each member $P \ni v$ contributes the two soft pairs of $P$ at $v$, i.e. two partners. Two members through $v$ cannot have a common partner $x \neq v$, since they would then share the two vertices $v, x$, hence a pair. So the partner sets are disjoint and $|Q(v)| = 2|\{P \ni v\}|$. □

Below, $D$ denotes a *number* with $D \geq \max_v |Q(v)|$ (for instance $D = D(H) = 2c$), and $D(H)$ the function value.

**Lemma 2.2** (Solutions decompose)**.** *Let $S$ be any set of pairs such that $G \triangle S$ is a cluster graph, $x_i = S \cap \text{pairs}(P_i)$ and $U = S \setminus \bigcup_i \text{pairs}(P_i)$ (the* hard edits*). Then every $x_i \neq \emptyset$, $|S| = \sum_i |x_i| + |U|$, and $|U| \leq |S| - |H|$.*

*Proof.* If $x_i = \emptyset$ the triple $V(P_i)$ keeps its adjacencies in $G \triangle S$ and is an induced $P_3$. The pair sets are disjoint, so $|S| = \sum |x_i| + |U| \geq |H| + |U|$. □

Consequently the *excess* of a solution is $\sum_i (|x_i| - 1) + |U|$, and every solution of size at most $|H| + \ell$ has $|U| \leq \ell$.

## 2.1 Maximalisation and pair-maximal packings

A family $\Phi$ of induced $P_3$'s is *hard over $H$* if the sets $\text{pairs}(P)$, $P \in \Phi$, are pairwise disjoint and disjoint from the soft pairs of $H$; then $H \cup \Phi$ is a packing with $|H| + |\Phi|$ members. $H$ is *pair-maximal* if no induced $P_3$ of $G$ has all three pairs hard, i.e. if $H$ is maximal among modification-disjoint packings. This is strictly stronger than maximality among vertex-disjoint packings (Proposition Q.2).

**Lemma 2.3** (Hard families)**.** *Let $(G, H, \ell)$ be a YES instance and $\Phi$ a family hard over $H$. Then $|\Phi| \leq \ell$. In general* $\text{OPT} \geq |H| + |\Phi|$.

*Proof.* $H \cup \Phi$ is a modification-disjoint packing of $|H| + |\Phi|$ induced $P_3$'s. Lemma 2.2, applied to $H \cup \Phi$ and an optimum solution, gives $\text{OPT} \geq |H| + |\Phi|$. In a YES instance $\text{OPT} \leq |H| + \ell$. □

**Lemma 2.4** (Maximalisation)**.** *Let $H$ be any packing and $M$ a maximal family hard over $H$ (inclusion-maximal; found greedily in polynomial time). Put $H' = H \cup M$, $\ell' = \ell - |M|$. Then:*

*(a)* $\mathrm{OPT} \geq |H'|$. *In particular* $|M| > \ell$ *implies that* $(G, H, \ell)$ *is a NO instance.*

*(b)* $\mathrm{OPT} \leq |H| + \ell$ *iff* ($|M| \leq \ell$ *and* $\mathrm{OPT} \leq |H'| + \ell'$). *So* $(G, H, \ell)$ *and* $(G, H', \ell')$ *are equivalent whenever* $\ell' \geq 0$.

*(c)* $H'$ *is a pair-maximal packing ("hard" now means hard with respect to* $H'$*; the pairs of* $M$ *have become soft).*

*(d)* $\mathrm{tw}(I(H')) \leq \mathrm{tw}(I(H)) + |M|$, *hence* $w' \leq w + |M| \leq w + \ell$.

*(e)* $c' \leq \min\{c + |M|,\ w' + 1\}$ *and* $D(H') = 2c' \leq D(H) + 2|M|$.

*A tree decomposition of* $I(H')$ *of width at most* $w + |M|$ *is obtained in polynomial time from one of* $I(H)$.

*Proof.* (a) is Lemma 2.3 (second sentence: $\mathrm{OPT} \geq |H| + |M|$). (b) If $\mathrm{OPT} \leq |H| + \ell$ then $|M| \leq \ell$ by (a), and $|H'| + \ell' = |H| + \ell$; the converse is the same identity.

(c) The members of $M$ have pairwise disjoint pair sets, disjoint from those of $H$, so $H'$ is a packing. Suppose an induced $P_3$ $Q$ has all its pairs hard with respect to $H'$. Then $M \cup \{Q\}$ is a family hard over $H$ ($Q \notin M$ because the pairs of $Q$ are not in pairs($M$)), contradicting maximality of $M$.

(d) Let $(\mathcal{T}, \{\beta_t\})$ be a tree decomposition of $I(H)$ of width $w$ (if $H = \emptyset$ take a single empty bag). $I(H')[H] = I(H)$. Put $\beta'_t = \beta_t \cup M$. Every node of $H$ still occupies a connected subtree. Every node of $M$ occupies all of $\mathcal{T}$. An edge of $I(H')$ with both ends in $H$ is covered as before; an edge with an end in $M$ is covered by any bag containing its other end (if both ends are in $M$ every bag covers it). Bags have at most $w + 1 + |M|$ nodes.

(e) At most $|M|$ members of $M$ pass through a vertex, so $c' \leq c + |M|$; $c' \leq w' + 1$ is the inequality $c \leq w + 1$ for $H'$. By Lemma 2.1, $D(H') = 2c' \leq D(H) + 2|M|$. □

**Example 2.5** (Tightness of Lemma 2.4(d))**.** Let $G$ be the star with centre $v$ and leaves $p, q, a_1, b_1, \ldots, a_m, b_m$, $H = \{(p, v, q)\}$ and $M = \{(a_i, v, b_i) : 1 \leq i \leq m\}$. $M$ is a maximal family hard over $H$: every induced $P_3$ of a star is $(x, v, y)$ with leaves $x, y$, and every pair $xv$ is already soft. $I(H)$ is one node, $\mathrm{tw} = 0$; $I(H \cup M) = K_{m+1}$, $\mathrm{tw} = m = 0 + |M|$.

**Convention: pair-maximal packings and primed parameters.** Appendix G–L and the appendices state their results for a *pair-maximal* $H$; where a result needs a YES instance or an optimal solution, it says so. For an arbitrary input packing $H$ let $M$ be as in Lemma 2.4 and read the statements for the *primed instance* $(G, H', \ell')$ with parameters $w' \leq w + |M|$, $c' \leq c + |M|$, $D' \leq D + 2|M|$, $\ell' = \ell - |M|$. The family $M$ is not unique (it depends on the greedy order), so quantities defined from $H'$ (such as $\omega$ in Appendix I) are parameters of the primed instance, not functions of the input $(G, H, \ell)$. Only Theorems 3.5, 3.6, H.1, H.10 and I.2 are statements about the input $H$ in terms of $w$, $\ell$ and $n$; the table in Appendix R lists the scope of every result.

**Notation.** $P_i = (a_i, b_i, c_i)$ with centre $b_i$; $c$ is the maximum number of members through a vertex (unsubscripted), and $\kappa_i \geq 1$ the number of edited pairs among pairs($P_i$). $U = U^- \cup U^+$ is the set of hard edits, $U^-$ the deleted hard edges and $U^+$ the inserted hard non-edges, $u = |U|$. $\ell' = \ell - |M|$ is the primed excess and $\bar{\ell}$ the excess left after component accounting (Appendix H). $\mathcal{C}_j$ are the components of $I(H)$ and $W_j$ the vertex sets they span. $R = V \setminus V(H)$ (Appendix E). $B = 2D + 3\ell$, $Z = B + D + 1$ and BIG are defined in Appendix G. Hard components $K$ and the cliques $\mathcal{K}_v$ of the constructions in Appendix I are distinct objects.

## 3 The interaction CSP and its dynamic programs

**Observation 3.1.** *Let $S$ be a solution with $|S| \le |H|$. Then $S$ has exactly one pair in each* pairs$(P_i)$ *and no other pair, and $G \triangle S$ is a cluster graph iff no vertex triple is an induced $P_3$ of $G \triangle S$. Consequently $(G, H, 0)$ is a YES instance iff the following CSP is satisfiable: variables $x_i \in \{1, 2, 3\}$ (which pair of $P_i$ is flipped), and for every vertex triple $t$ the constraint "$t$ is not an induced $P_3$ after flipping", whose truth depends on the owners' variables of the soft pairs of $t$ only (hard pairs keep their status in $G$).*

*Proof.* By Lemma 2.2, $|x_i| \ge 1$ for all $i$ and $|S| = \sum |x_i| + |U| \le |H|$ force $|x_i| = 1$ and $U = \emptyset$. The rest is the definition of a cluster graph. □

The CSP has $\Theta(n^3)$ constraints, not $O(|H|)$.

**Observation 3.2.** *The set of owners of the soft pairs of any triple is a clique of $I(H)$ (at most three nodes). Hence the primal graph of the CSP is a subgraph of $I(H)$ (not every edge of $I(H)$ is witnessed by a triple), and a tree decomposition of $I(H)$ is one of the primal graph.*

*Proof.* Two pairs of one triple share a vertex $v$; their owners both contain $v$, so they coincide or are adjacent in $I(H)$. □

**Lemma 3.3** (Five states)**.** *Let $P = (a, b, c)$ be a member, with edges $ab, bc$ and non-edge $ac$. Of the seven non-empty subsets $x \subseteq$ pairs$(P)$, exactly five leave no induced $P_3$ on $\{a, b, c\}$ after editing: $\{ab\}$, $\{bc\}$, $\{ac\}$ (cost 1, states 1, 2, 3), $\{ab, bc\}$ (cost 2, state 4) and $\{ab, bc, ac\}$ (cost 3, state 5). The sets $\{ab, ac\}$ and $\{bc, ac\}$ leave an induced $P_3$.*

*Proof.* After editing $\{ab, ac\}$ the edges on $\{a, b, c\}$ are $bc, ac$, after $\{bc, ac\}$ they are $ab, ac$: two edges, an induced $P_3$. The other five sets leave the edge sets $\{bc\}$, $\{ab\}$, $\{ab, bc, ac\}$, $\emptyset$ and $\{ac\}$, none of which has exactly two edges. □

Hence in every solution each $x_i$ is one of these five sets (its triple $V(P_i)$ must be a cluster graph), and the variables of the dynamic programs below may range over five values; this changes no solution and no cost.

The natural parameter of the CSP is therefore the treewidth of its primal graph, which is at most $w$; the ETH bound of Section 4 is stated for $I(H)$.

**Lemma 3.4** (Dynamic programming over $I(H)$)**.** *Let there be one variable $x_i$ per member, with a domain $\mathcal{D}_i$ of size at most $d$ and a unary cost $\gamma_i : \mathcal{D}_i \to \mathbb{N}$, and let $\mathcal{C}$ be a set of constraints, each given by a test of its satisfaction on $\prod \mathcal{D}_i$ in polynomial time, whose scope is a clique of $I(H)$. Given a tree decomposition of $I(H)$ of width $w$, a satisfying assignment of minimum total cost is found in time $d^{w+1}(|H| + |\mathcal{C}|)^{O(1)}$.*

*Proof.* Let $(\mathcal{T}, \{\beta_s\})$ be the tree decomposition. By the Helly property each clique lies in a bag; assign each constraint to one such bag. For a node $s$ and an assignment $\alpha$ of the variables of $\beta_s$, let $f_s(\alpha)$ be the minimum cost over the variables in the subtree of $s$, among assignments extending $\alpha$ and satisfying all constraints assigned in the subtree. The standard recurrences (introduce, forget, join; see [6]) compute all $f_s$; each table has at most $d^{w+1}$ entries. □

**Theorem 3.5.** *Given $(G, H)$ and a tree decomposition of $I(H)$ of width $w$, CEaMP with $\ell = 0$ is solvable in $3^{w+1} n^{O(1)}$ time.*

*Proof.* Observations 3.1 and 3.2 give a CSP with domains of size 3, all unary costs 1 and $O(n^3)$ constraints with clique scopes; apply Lemma 3.4. □

**Theorem 3.6** (CSP algorithm)**.** *Given $(G, H, \ell)$ and a tree decomposition of $I(H)$ of width $w$, CEaMP is solvable in $5^{w+1}n^{2\ell+O(1)}$ time.*

*Proof.* Let $\mathcal{D}_i$ be the 5 subsets of pairs($P_i$) of Lemma 3.3. By Lemma 2.2, a solution of size $\leq |H|+\ell$ has the form $S = U \cup \bigcup x_i$ with $x_i \in \mathcal{D}_i$ and $U$ a set of at most $\ell$ hard pairs; there are at most $\sum_{j\leq\ell} \binom{n^2/2}{j} \leq n^{2\ell}$ candidates for $U$.

Fix $U$ and put $G_U = G\triangle U$. Since $U$ avoids soft pairs, every $P_i$ is still an induced $P_3$ of $G_U$ with the same pairs, so $H$ is still a modification-disjoint packing. For a triple $t$, the status of each hard pair of $t$ in $G_U\triangle x$ is fixed, and each soft pair is determined by $G_U$ and the variable of its owner. Hence $t$ gives a constraint $C_t$ ("$t$ is not an induced $P_3$ in $G_U\triangle x$") whose scope is the owner set of $t$, a clique of $I(H)$ (the proof of Observation 3.2 does not use $G$ itself). Triples with empty scope are tested directly, in $O(n^3)$ time. Seek $x \in \prod \mathcal{D}_i$ satisfying all $C_t$ and minimizing $\sum_i |x_i|$; by Lemma 3.4 this takes $5^{w+1}n^{O(1)}$ time. Accept iff for some $U$ the minimum $m_U$ satisfies $m_U + |U| \leq |H| + \ell$. Soundness: any feasible $(U, x)$ gives the solution $U \cup \bigcup x_i$. Completeness: Lemmas 2.2 and 3.3. □

Theorem 3.5 is the case $\ell = 0$ of Theorem 3.6 with the smaller base 3. Together with Observations 3.1 and 3.2 and Lemma 3.4 these are parts (b) and (c) of Theorem 1.1. The exponent of $n$ can be halved: by Corollary G.7 it suffices to guess one end of each hard edit, which gives time $(n+1)^\ell Z_*^{2\ell} 5^{w+\ell+1} n^{O(1)}$ with $Z_* \leq 6w + 9\ell + 7$ for every packing (Theorem I.2, Section 7); this is still XP in $\ell$. The algorithm of Theorem 3.6 is not fixed-parameter tractable: $\ell$ is part of the input and appears in the exponent of $n$, and Proposition 4.3 shows that $w$ alone does not suffice. By Theorem 5.1 this cannot be repaired by a fixed-parameter algorithm in $\ell + w$ unless FPT = W[1].

*Remark* 3.7 (Domain $\{1, 2, 3\}$ does not suffice for $\ell > 0$). In the encoding above, a variable ranges over the five states of Lemma 3.3. For $\ell > 0$ it cannot be restricted to the three single-pair values: an edited packed $P_3$ can receive two or three edits in every optimum (the states 4 and 5). For two: $n = 5$, $E = \{02, 04, 12, 23, 24\}$, $H = \{(1, 2, 3)\}$, OPT = 2, unique optimum $\{12, 23\}$. For three: $n = 6$, $E = \{01, 02, 03, 04, 12, 14, 15, 24, 25, 35, 45\}$, $H = \{(0, 3, 5)\}$, OPT = 3, unique optimum $\{03, 05, 35\}$ (both verified by enumerating all set partitions). We do not claim that 5 is the smallest possible domain size in other encodings. For $c = 2$, $I(H)$ is subcubic but may have unbounded treewidth, so Theorem 3.6 is a different tradeoff from the $n^{2\ell}$ algorithm of [15], not a recovery of it.

# 4 Limits of the parameterization

## 4.1 An ETH lower bound in the exponent

**Theorem 4.1.** *Assume ETH. There is no algorithm that decides CEaMP with $\ell = 0$ in time $2^{o(\mathrm{tw}(I(H)))}n^{O(1)}$. This holds even when every vertex lies in at most 23 packed $P_3$'s (so $I(H)$ has maximum degree at most 66), and whether or not a tree decomposition is supplied.*

The theorem is a corollary of the reduction of Li, Pilipczuk and Sorge.

*Proof.* By the construction behind [15, Thm. 1] there is a polynomial reduction from 3-SAT to CEaMP with $\ell = 0$ in which every vertex lies in at most 23 packed $P_3$'s and the graph has $|V| = O(N+m)$ vertices for a formula with $N$ variables and $m$ clauses (each variable occurrence and each clause is replaced by a gadget of constant size). By the Sparsification Lemma [11] (see also [6]) we may assume $m = O(N)$ at the cost of $2^{o(N)}$ extra time, so $|V| = O(N)$.

Every member has three vertices and every vertex lies in at most 23 members, hence $3|H| = \sum_v |\{P \ni v\}| \leq 23|V|$, i.e. $|H| \leq 23|V|/3 = O(N)$. A single bag containing all members is a tree

decomposition of $I(H)$, so $\mathrm{tw}(I(H)) \le |H| - 1 = O(N)$, and such a decomposition can be supplied. An algorithm with running time $2^{o(\mathrm{tw}(I(H)))}n^{O(1)}$, which may compute its own decomposition, would therefore decide 3-SAT in time $2^{o(N)}$, contradicting ETH. The degree bound: a member meets at most $3 \cdot 22 = 66$ others. □

*Remark* 4.2. (1) The bound is on the exponent only: it does not show that the base 3 (or 5) is tight, and $2^{o(w)}$ excludes $3^{o(w)}$ and $5^{o(w)}$ alike. A SETH-type bound $(3-\varepsilon)^w$ is open. (2) On the instances of the theorem every vertex lies in at most 23 members, so $\mathrm{tw}(I(H)) \le |H| - 1 \le 23n/3$ (for general packings $|H|$ can exceed $n$); hence on these instances a $2^{o(\mathrm{tw}(I(H)))}n^{O(1)}$ algorithm is in particular a $2^{o(n)}$ algorithm, the constant 23/3 being absorbed in the $o(\cdot)$, so the theorem is no stronger than the statement that CEaMP at $\ell = 0$ has no $2^{o(n)}$ algorithm under ETH; it says that the exponent of Theorem 3.5 cannot be improved to $o(w)$ on instances with $|H| = \Theta(n)$.

### 4.2 Other parameters do not suffice

**Proposition 4.3.** *CEaMP is NP-hard when $I(H)$ has no edge ($w = 0$, vertex-disjoint packing), with $\ell$ part of the input. Hence $w$ alone is not enough, and (by Theorem 1 of [15]) neither is $\ell = 0$ alone.*

*Proof.* Cluster Editing is NP-hard (e.g. already on graphs of maximum degree 6 [13]). Given $(G, k)$ compute greedily a maximal set $H$ of vertex-disjoint induced $P_3$'s (if none, $G$ is a cluster graph). Vertex-disjoint implies modification-disjoint and $I(H)$ is edgeless. Since $\mathrm{OPT} \ge |H|$, $(G, k)$ is a YES instance iff $(G, H, k - |H|)$ is (a NO instance if $k < |H|$). □

Two further facts are in Appendix Q: bounded $\mathrm{tw}(G)$ does not make $\ell + w$ small (Proposition Q.1, a star), and pair-maximality cannot be replaced by vertex-maximality (Propositions Q.2 and Q.3), so edits touching vertices outside $V(H)$ cannot be excluded.

## 5 W[1]-hardness

Theorems 3.5 and 3.6 solve CEaMP in time $5^{w+1}n^{2\ell+O(1)}$, and Theorem I.2 halves the exponent of $n$; all of these are XP in $\ell$. This section shows that the exponent of $n$ cannot be removed: the main question of this paper has a negative answer. The reduction is described completely and its key proofs are given here; the proofs of the long technical lemmas (the chain lemma, the profiles of the hosts, the parameter counts, the pathwidth bound) are in Appendix A. Four finite steps of the proof use a computer (Section 5.9).

**Theorem 5.1** (W[1]-hardness)**.** *There is a polynomial-time reduction that maps an instance $(k, n, (S_{ij}))$ of Grid Tiling with $\le$ to an instance $(G, H, \ell)$ of CEaMP with the following properties. Let $p$ be the least even integer with $p \ge k^2 + 5$.*

*(i)* *$\ell = k^2$; $H$ is a pair-maximal modification-disjoint packing; $|V(G)| = O(k^2n^2p)$.*

*(ii)* *$c(H) \le 4p + 2$, and $\mathrm{tw}(I(H)) \le \mathrm{pw}(I(H)) \le 24\,k\,p^2$, independently of $n$; $I(H)$ has at most $2k$ components.*

*(iii)* *$(G, H, \ell)$ is a YES instance iff the Grid Tiling instance has a solution; if it has one, then $\mathrm{OPT}(G) = |H| + k^2$.*

*Consequently CEaMP is W[1]-hard parameterized by $\ell + \mathrm{tw}(I(H))$, by $\ell + \mathrm{pw}(I(H))$ and by $\ell + c$, even for pair-maximal packings. Unless* FPT = W[1] *there is no algorithm with running time $f(\ell, w)\, n^{O(1)}$, and unless ETH fails there is no algorithm with running time $f(\ell, w)\, N^{o(\sqrt{\ell})}$, where $N = |V(G)|$.*

## 5.1 Consequences and remaining gaps

**Corollary 5.2** (The ETH bound that follows)**.** *Assume ETH. There are no computable $f$ and no algorithm that decides CEaMP in time $f(\ell, \mathrm{tw}(I(H)))\, N^{o(\sqrt{\ell})}$ on instances with $N$ vertices.*

*Proof.* Suppose such an algorithm runs in time $f(\ell, w)N^{h(\ell)}$ with $h(\ell) = o(\sqrt{\ell})$. Apply the reduction of Theorem 5.1 to a Grid Tiling instance with parameter $k$: then $\ell = k^2$, $w \le 24k(k^2+6)^2 =: g(k)$ and $N \le \mathrm{poly}(k)\, n^2$. The running time is at most $f(k^2, g(k))\, (\mathrm{poly}(k)n^2)^{h(k^2)}$, and $h(k^2) = o(k)$, so it is at most $f'(k)\, n^{o(k)}$ for a computable $f'$ (the factor $\mathrm{poly}(k)^{h(k^2)} = 2^{o(k \log k)}$ is absorbed in $f'$). This solves Grid Tiling with $\le$ in time $f'(k)n^{o(k)}$, contradicting ETH [6]. □

Nothing stronger follows from this reduction: since $\ell = k^2$, the lower bound is $n^{o(\sqrt{\ell})}$, while the upper bounds of Theorems 3.6 and I.2 have exponent $2\ell$ and $\ell$. The gap between $\sqrt{\ell}$ and $\ell$ is open (Open Problem 3). Since $w$ is a polynomial in $k$ on the produced instances, no statement in terms of $\ell + w$ jointly is claimed beyond Corollary 5.2.

*Matching the XP algorithm.* Theorem 5.1 shows that the exponent of $n$ in Theorems 3.6 and I.2 cannot be made independent of $\ell$, even though $w$ and $c$ are bounded by functions of $\ell$ alone on the hard instances. Up to the exact exponent of $n$ ($\Omega(\sqrt{\ell})$ under ETH, against $\ell$ resp. $2\ell$ proved), the complexity of the pair $(\ell, w)$ is therefore settled.

*What is still open.* (1) The reduction has $c = 4p + 2$ with $p \ge k^2 + 5$, so $c$ grows with $\ell$; whether CEaMP is W[1]-hard (or fixed-parameter tractable in $\ell$) when $c$ is bounded by a constant, in particular for $c = 2$, is open (Open Problem 1). The reason is local to the proof: $p$ is the size of the cliques into which every chain vertex is blown up, and the argument that no clique is split (Lemma 5.9) needs $p > \ell$. (2) The exact exponent under ETH (Open Problem 3). (3) Packings with $w \le 1$ with tree components of unbounded size, and soft-triangle-free packings (Open Problems 5 and 2); the instances here have many soft triangles and large $w$ (the width is $\Theta(p^2)$ per chain).

## 5.2 Overview of the construction

The reduction follows the plan of the tunnel attempt of Appendix N, and realises the abstract problem of Theorem 6.9 with soft triangles; the obstruction of Theorem 6.10 is circumvented because the instances are not soft-triangle-free. We use the language of zero-excess (ZE) solutions: a member $(a, b, c)$ with centre $b$ is in state $L$, $R$ or $M$ if its single edit deletes $ab$, deletes $bc$, or inserts $ac$ (states $1, 2, 3$ of Appendix N).

**A selection chain.** A *thin chain* $\mathrm{OH}_m$ ($m \ge 7$) is an explicit tight instance on the vertices $s_0, \ldots, s_{2m}, e_0, \ldots, e_m$ with spine members $P_j = (s_{2j}, s_{2j+1}, s_{2j+2})$, further interior members $G_j, F_j$ that carry the vertices $e_j$, two caps at the ends and four hard edges (Appendix A). Its ZE partitions are exactly $m$ *positions* $a \in \{0, \ldots, m-1\}$, in which the spine members are in states $R^a M L^{m-1-a}$ (Lemma 5.3). This is a tight instance of bounded $c$ and bounded width in which a "position" is selected, so that a cut at any one of arbitrarily many places repairs it (Remark 5.15).

**Thick chains.** Every thin vertex $v$ becomes a *super-vertex* $K_v = \{(v,i) : i \in \mathbb{Z}_p\}$, glued by hard edges, and every thin member $(a,b,c)$ is replaced by the $p^2$ *Latin members* $((a,i),(b,j),(c,i+j))$. These cover every pair between $K_a, K_b$, between $K_b, K_c$ and between $K_a, K_c$ exactly once. All pairs of such a triple of super-vertices are soft; so if $i+j \neq k$, the triple $(a,i),(b,j),(c,k)$ is a soft triangle. The purpose is that a ZE partition cannot split a super-vertex (the glue is hard) and that, for an arbitrary solution of excess at most $\ell$, splitting a super-vertex costs more than $\ell$ hard edits when $p \geq \ell + 5$ (Lemma 5.9). Hence the thick chain behaves like the thin chain, while a vertex outside the chain can be attached to a super-vertex at a cost that is independent of the position.

**Guests, locks and comparators.** A *guest* is a single new vertex $y$ with a set $W_y$ of super-vertices to which it is soft (through $p/2$ *pad members* $(y,(v,i),(v,k))$ for a perfect matching of $K_v$); all other pairs of $y$ are hard non-edges. A guest can join the cluster of a base vertex $h$ only if the entire cluster of $h$ lies in $W_y$; so the set $W_y$ forbids exactly those positions of the chain at which the cluster of $h$ contains a vertex outside $W_y$ (Lemma 5.4). Each chain is divided into $U = 1 + n(n+1)$ *groups* of two consecutive positions. A *permanent guest* sits in the cluster of a host vertex of each group and, for groups that carry no tile, forbids both positions of the group. For a tile $(a,b) \in S_{ij}$ the chain $A_{ij}$ has a *lock guest* and an *anchor* at the slot group $(a,b)$, and the chain $B_{ij}$ at the slot group $(b,a)$; the two anchors of one tile are joined by a hard edge, a *tunnel*. An uncut tunnel forces the lock guest into the host cluster, hence forbids the group; cutting it (one hard edit, cost one) frees the group. So a chain whose tunnels are all uncut has no position, and a chain with exactly one cut tunnel is in a position of the group of that tunnel; as the same hard edit releases the chains $A_{ij}$ and $B_{ij}$, *it chooses the tile $s_{ij}$ in both orders $(a,b)$ and $(b,a)$ at the price of one edit.* Finally, for consecutive chains of a row (of columns) and a separator group $\alpha < n$, a *comparator guest* $y$ with a member $(z,y,z')$ ($z,z'$ the permanent guests of the two chains) must join exactly one of the two chains, and can do so at the first chain only if its position is before the separator and at the second only if it is after; this is the clause $(\beta \leq \alpha) \vee (\beta' > \alpha)$ on the blocks $\beta, \beta'$ of the two chosen tiles, and the $n-1$ clauses together express $\beta \leq \beta'$. This is Proposition 5.7.

**Soundness by counting.** The completeness direction is Proposition 5.7. For soundness, an arbitrary solution $S$ of excess at most $\ell = k^2$ is analysed without assuming its form. By Lemma 5.9 no super-vertex is split. Partition the pairs into regions $\Pi_c$, one for each of the $2k^2$ chains; every region needs at least one edit beyond one per member (Lemma 5.10), because with all tunnels uncut the chain has no position. A pair of two regions is a *cross-anchor pair* (a pair between the anchor sets of the two chains of one cell); only these pairs can serve two regions, and they are hard. If $q$ of them are edited, the excess is at least $\max(q, 2k^2 - q) \geq k^2$ (Theorem 5.11); if it equals $k^2$, then $S$ is a set of exactly one tunnel cut per cell plus one edit per member, and Proposition 5.7 gives a tiling. Thus the instance has $\mathrm{OPT} \leq |H| + k^2$ iff the Grid Tiling instance is solvable. In this argument the restricted model of Appendix N is not assumed; it is derived.

### 5.3 Conventions and the thin chain

Soft and hard pairs, members, states and zero-excess (ZE) solutions are as in Section 2 and Appendix N. A *ZE partition* of an instance $(G,H)$ is a partition of $V(G)$ whose cost is $|H|$; by Lemma 2.2 and Lemma N.1 it is a partition in which no hard pair is edited (hard edges lie inside clusters, hard non-edges join different clusters) and every member has exactly one edited pair. For a member $(a,b,c)$ with centre $b$ the state is $L$ if the single edit deletes $ab$ ($b \sim c$, $a$ apart), $R$ if it deletes $bc$, and $M$ if it inserts $ac$; these are the states $1,2,3$ of Appendix N. Write $x \sim y$ for "$x,y$ are in one cluster". The *excess* of a solution $S$ is $\mathrm{ex}(S) = |S| - |H|$. Grid Tiling instances are assumed

to have all $S_{ij} \neq \emptyset$ and $n \geq 2$ (an instance with an empty $S_{ij}$ is mapped to a fixed NO instance; $n = 1$ is trivial).

**Definition.** Fix $m \geq 7$. The *thin chain* $\mathrm{OH}_m$ has the vertices $s_0, \ldots, s_{2m}$ and $e_0, \ldots, e_m$ and the following members, listed as $(a, b, c)$ with centre $b$:

- spine members $P_j = (s_{2j}, s_{2j+1}, s_{2j+2})$, $0 \leq j \leq m-1$;
- $G_j = (e_j, s_{2j+1}, e_{j+2})$ for $2 \leq j \leq m-4$ and $F_j = (e_j, s_{2j+2}, e_{j+1})$ for $2 \leq j \leq m-3$;
- left cap: $(e_0, s_2, e_1)$, $(s_1, e_1, s_3)$, $(e_0, s_4, e_2)$;
- right cap: $(s_{2m-5}, e_{m-3}, s_{2m-1})$, $(e_{m-2}, s_{2m-2}, e_m)$, $(s_{2m-4}, e_{m-1}, e_m)$, $(s_{2m-3}, e_{m-1}, s_{2m})$, $(e_{m-2}, e_{m-1}, s_{2m-1})$, $(s_{2m-3}, e_m, s_{2m-1})$.

The edges are the pairs $ab, bc$ of the members and the four *hard edges* $s_0e_1$, $e_0s_3$, $s_{2m-3}e_{m-2}$, $s_{2m}e_m$; every other pair of vertices is a hard non-edge. The members have pairwise disjoint pair sets and are induced $P_3$'s (checked by computer, and by the pair-disjointness argument of Lemma 5.14 for the thick version). The *cell* of a vertex is $\lfloor t/2 \rfloor$ for $s_t$ and $t$ for $e_t$; every member has its vertices in at most three consecutive cells, and every soft pair joins vertices in cells at distance at most 2. A *position* is an integer $a \in \{0, \ldots, m-1\}$; a *group* is a set $\{2u, 2u+1\}$, $0 \leq u \leq U-1$, where $m = 2U$.

**The partitions $P_a$.** Put $\varepsilon(0) = e_1$, $\varepsilon(1) = e_0$, $\varepsilon(j) = e_j$ for $2 \leq j \leq m-2$ and $\varepsilon(m-1) = e_m$. For $0 \leq a \leq m-1$ let $P_a$ be the partition of $V(\mathrm{OH}_m)$ whose clusters $C_0, \ldots, C_{m-1}$ are

$$C_j = \begin{cases} \{s_{2j}, s_{2j+1}\} & j < a \\ \{s_{2j}, s_{2j+1}, s_{2j+2}\} & j = a \ \cup \ \{\varepsilon(j)\} \quad (\text{for } j \leq m-2), \\ \{s_{2j+1}, s_{2j+2}\} & j > a \end{cases}$$

together with the following changes at the right end. If $a \leq m-2$ then

$$C_{m-1} = \{s_{2m-1}, s_{2m}, e_{m-1}, e_m\};$$

if $a = m-1$ then $C_{m-2}$ contains in addition $e_{m-1}$ and $C_{m-1} = \{s_{2m-2}, s_{2m-1}, s_{2m}, e_m\}$. In $P_a$ the spine members $P_0, \ldots, P_{m-1}$ are in the states $R^a M L^{m-1-a}$.

**Lemma 5.3** (Chain lemma)**.** *For every $m \geq 7$ the ZE partitions of $\mathrm{OH}_m$ are exactly $P_0, \ldots, P_{m-1}$; they are pairwise distinct.*

*Proof (computer-assisted).* Both parts are finite statements about a family indexed by $m$ and are proved by computer; the argument that makes the finite check cover all $m \geq 7$ is as follows, and is written out in Appendix A. Every constraint of $\mathrm{OH}_m$ (a member, or a hard edge) involves vertices of at most three consecutive cells, so a condition at a cell depends only on the relation of $a$ to the cell, and on the distance to the left or the right end up to a constant; interior windows are isomorphic, left-end windows do not depend on $m$, and right-end windows depend only on the distance to the right end. **(CA1)** (part (a)): for every $7 \leq m \leq 60$ and every $a$, $P_a$ was checked by computer to have cost exactly $|H|$ with one edit per member and no hard pair edited; all window types occur for $m \leq 20$. **(CA2)** (part (b)): the ZE partitions are counted by a transfer-matrix dynamic programme over the cells, with six states per interior cell and an explicit integer $6 \times 6$ matrix $M$, the same at every interior cell, with $\chi_M(x) = x^4(x-1)^2$; hence $N_m$ is an affine function of $m$ for large $m$, and

a direct computation gives $N_m = m$ for $7 \le m \le 40$. Together with the $m$ distinct partitions $P_a$ of part (a) these are all. Thus the chain lemma is established by a finite computation combined with a translation-invariance argument rather than by a closed-form proof: the claim that the transfer matrix is the same at every interior cell of every $m$ rests on the translation invariance of the member list. The chain lemma for $m \ge 10$ is therefore computer-assisted; the Lean formalisation covers $m = 7, 8, 9$ only (Section 9). □

**Profiles.** Let $m = 2U$. The *host* of the group $u$ is $h_u = s_{4u+2}$; it is the last vertex of $P_{2u}$ and the first vertex of $P_{2u+1}$. Write $\varepsilon_u^- = \varepsilon(2u)$ and $\varepsilon_u^+ = \varepsilon(2u+1)$; $\varepsilon_u^- \ne \varepsilon_u^+$ for all $u$.

**Lemma 5.4** (Profile of a host)**.** *For the group $u$ the positions fall into four classes $A_0 = \{a \le 2u-1\}$, $A_1 = \{2u\}$, $A_2 = \{2u+1\}$, $A_3 = \{a \ge 2u+2\}$, and for $a \in A_i$ the cluster of $h_u$ in $P_a$ is $\{h_u\} \cup \Pi_i$ with*

$$\Pi_0 = \{s_{4u+1}, \varepsilon_u^-\}, \quad \Pi_1 = \{s_{4u}, s_{4u+1}, \varepsilon_u^-\}, \quad \Pi_2 = \{s_{4u+3}, s_{4u+4}, \varepsilon_u^+\}, \quad \Pi_3 = \{s_{4u+3}, \varepsilon_u^+\}.$$

*($A_0 = \emptyset$ for $u = 0$ and $A_3 = \emptyset$ for $u = U-1$.) In particular $\Pi_0 \subset \Pi_1$, $\Pi_3 \subset \Pi_2$, $s_{4u} \in \Pi_1 \setminus \Pi_0$, $s_{4u+4} \in \Pi_2 \setminus \Pi_3$, and $\Pi_0, \Pi_1$ are disjoint from $\Pi_2, \Pi_3$.*

*Proof* in Appendix A: the cluster of $h_u$ is read off the definition of $P_a$, with a check of the exceptional $\varepsilon$-labels at the two caps by computer **(CA3)**.

**Allowed sets and forbidden positions.** For a set $W$ of thin vertices of one chain with $h_u \in W$ let $F_u(W) = \{a :$ the cluster of $h_u$ in $P_a \not\subseteq W\}$ be the set of positions *forbidden* by $W$. Define, for the group $u$,

$$\mathrm{Pad}_u = \{h_u\} \cup \Pi_0 \cup \Pi_3, \qquad X_u = \{h_u\} \cup \Pi_0, \qquad Y_u = \{h_u\} \cup \Pi_3, \qquad \mathrm{All}_u = \{h_u\} \cup \Pi_0 \cup \Pi_1 \cup \Pi_2 \cup \Pi_3.$$

**Corollary 5.5.** *$F_u(\mathrm{Pad}_u) = \{2u, 2u+1\}$, $F_u(X_u) = \{a \ge 2u\}$, $F_u(Y_u) = \{a \le 2u+1\}$ and $F_u(\mathrm{All}_u) = \emptyset$.*

*Proof.* By Lemma 5.4: for $\mathrm{Pad}_u$ the clusters at $A_0$ and $A_3$ are inside $\mathrm{Pad}_u$, while at $A_1$ the vertex $s_{4u} \notin \mathrm{Pad}_u$ and at $A_2$ the vertex $s_{4u+4} \notin \mathrm{Pad}_u$. For $X_u$: $A_0$ is allowed, $A_1$ is forbidden by $s_{4u}$, and $A_2, A_3$ by the non-empty sets $\Pi_2, \Pi_3$ disjoint from $\Pi_0$ (if $A_3 = \emptyset$ there is nothing to forbid). $Y_u$ is symmetric: $A_3$ is allowed, $A_2$ is forbidden by $s_{4u+4}$, and $A_1, A_0$ by $\Pi_1, \Pi_0$, which are disjoint from $\Pi_3$. $\mathrm{All}_u$ contains every cluster. □

## 5.4 Thick chains and guests

Let $p \ge 6$ be even. A fixed *one-factorisation* $\mathcal{F}_1, \dots, \mathcal{F}_{p-1}$ of $K_p$ on the vertex set $\mathbb{Z}_p$ (round robin) is used throughout.

**Thick chain.** The thick version of a thin chain with tag $c$ has the vertices $(v, i)$, $v$ a thin vertex, $i \in \mathbb{Z}_p$; $K_v = \{(v, i) : i \in \mathbb{Z}_p\}$ is the *super-vertex* of $v$. For each thin member $(a, b, c')$ there are the $p^2$ *Latin members* $((a, i), (b, j), (c', i+j))$ with centre $(b, j)$, $i, j \in \mathbb{Z}_p$. For each thin hard pair $uv$ all $p^2$ pairs between $K_u$ and $K_v$ are hard, edges or non-edges as $uv$. Inside each $K_v$ all pairs are hard edges (the *glue*) except the pairs used as pad or host pairs below. Every pair between $K_a$ and $K_b$ ($ab$ a pair of a thin member) is covered by exactly one Latin member: the pair $(a, i), (b, j)$ by the member with these two vertices, the pair $(b, j), (c', k)$ by $i = k - j$, and the pair $(a, i), (c', k)$ by $j = k - i$. So the Latin members have pairwise disjoint pair sets, and if $i + j \ne k$ the triple $(a, i), (b, j), (c', k)$ is a *soft triangle*.

**Guests.** A *guest* is a single vertex $y$ together with a set $W_y$ of thin vertices (of one or two chains) such that all pairs between $y$ and the super-vertices $K_v$, $v \in W_y$, are soft and all other pairs of $y$ are hard non-edges, apart from the single-vertex members specified below. Realisation: for every $v \in W_y$ a perfect matching $\mathcal{M}(y,v)$ of $K_p$ is chosen, a different class $\mathcal{F}_r$ for different guests at the same $v$; for every $\{i,k\} \in \mathcal{M}(y,v)$ $(i < k)$ there is the *pad member* $(y,(v,i),(v,k))$ with centre $(v,i)$ (edges $y(v,i)$ and $(v,i)(v,k)$, non-edge $y(v,k)$). A *permanent guest* $z$ with a *host* $h \in W_z$ has instead, for the matching $\mathcal{M}(z,h)$, the *host members* $((h,i),z,(h,k))$ with centre $z$ (edges $z(h,i)$, $z(h,k)$, non-edge $(h,i)(h,k)$), and pad members for all $v \in W_z \setminus \{h\}$. The pairs of $K_v$ in a matching used at $v$ are soft; the glue is the complete graph on $K_v$ minus these pairs. All pairs between different super-vertices that are not soft and not thin hard edges are hard non-edges, and so are all pairs between guests (except those in the members $(x,y,z)$ and $(z,y,z')$ below).

**Thin-guest model.** A *thin configuration* is a partition $\pi$ of the set of thin vertices, guests, anchors and stubs of the instance such that: (T1) a pair that is a hard edge in the thin picture (a thin hard edge, an uncut tunnel, a pair $(z,h)$ of a permanent guest and its host) lies inside a cluster; every other pair of two elements is split, except the *free pairs* $(y,v)$, $v \in W_y$, and the pairs of the members; (T2) every thin chain member and every single-vertex member $(x,y,z)$, $(z,y,z')$ has exactly one edited pair.

**Lemma 5.6** (Blow-up)**.** *Assume that the glue graph of every $K_v$ is connected (Lemma 5.8). Then the ZE partitions of a thick instance are exactly the lifts (replace each thin vertex $v$ by $K_v$) of the thin configurations. In particular in a ZE partition (a) every $K_v$ lies in one cluster; (b) a guest $y$ is together with $K_v$ only if $v \in W_y$; (c) a permanent guest is together with its host $K_h$; (d) the thick members of a thin member are all in the state of the thin member.*

*Proof.* Let $\sigma$ be a ZE partition. No hard pair is edited, so the connected glue puts each $K_v$ into one cluster, which is (a). A pair between $K_u$ and $K_v$ that is a thin hard pair is hard of the same type for all $p^2$ pairs, so $K_u \sim K_v$ iff the thin pair is an edge. For a thin member $(a,b,c')$ the $p^2$ Latin members see the same togetherness pattern of the three super-vertices, so they are in the same state, which is a state of the thin member (one edit); this is (d). A pad member $(y,(v,i),(v,k))$ with $(v,i) \sim (v,k)$ has the togetherness patterns $(1,1,1)$ (if $y \sim K_v$) with one edit, inserting $y(v,k)$, and $(0,1,0)$ (if $y \not\sim K_v$) with one edit, deleting $y(v,i)$; so a pad member imposes no condition. If $v \notin W_y$ the pairs $y(v,i)$ are hard non-edges, so $y \not\sim K_v$, which is (b). A host member $((h,i),z,(h,k))$ with $(h,i) \sim (h,k)$ has one edit iff $z \sim (h,i)$ (pattern $(1,1,1)$: insert $(h,i)(h,k)$); if $z$ is apart it has two edits. This is (c). Conversely, the lift of a thin configuration has no edited hard pair (hard pairs between super-vertices and glue edges are respected by (T1), pairs of guests with $K_v$, $v \notin W_y$, are split) and one edit in every thick member, by the same case analysis. Note that in a lift the host pair $(h,i)(h,k)$ is a soft non-edge inside a cluster and is edited, while a pad pair is a soft edge inside a cluster and is not. □

## 5.5 The reduction

Let $(k,n,(S_{ij}))$ be given, $n \geq 2$, and let $p$ be the least even integer $\geq k^2+5$.

**Chains and groups.** For every cell $(i,j) \in [k]^2$ there are two chains $A_{ij}$ and $B_{ij}$, each a thick copy of $\mathrm{OH}_m$ with $m = 2U$, $U = 1+n(n+1)$ (so $m \geq 14$). For $\alpha \in [n]$ the *block* $\alpha$ has the *slot groups* $u = (\alpha-1)(n+1)+s$, $s \in [n]$, and is followed by the *separator group* $u_\alpha = \alpha(n+1)$; group 0 is the left end pad, and the last separator group $u_n = U-1$ is the right end pad.

**Tiles and tunnels.** A tile $(a,b) \in S_{ij}$ has its *slot* in $A_{ij}$ at the group $(a-1)(n+1)+b$ (block $a$) and in $B_{ij}$ at the group $(b-1)(n+1)+a$ (block $b$). The groups that are slots of tiles are *tile groups*.

**Guests.** For every chain $c$ and every group $u$ there is a permanent guest $z_{c,u}$ with host $h_u$; if $u$ is a tile group, $W_z = \mathrm{All}_u$, and in addition there are a *lock guest* $y_{c,u}$ with $W_y = \mathrm{Pad}_u$, a vertex $x_{c,u}$ (the *anchor*) and the single-vertex member $(x_{c,u}, y_{c,u}, z_{c,u})$ with centre $y_{c,u}$. If $u$ is not a tile group, $W_z = \mathrm{Pad}_u$ (a *permanent lock*). All other pairs of $x, y, z$ are hard non-edges.

**Tunnels.** For every cell and every tile $e \in S_{ij}$ the two anchors of $e$ (in $A_{ij}$ and in $B_{ij}$) are joined by a hard edge, the *tunnel* of $e$. The anchors have no other hard edge.

**Comparators.** For $\alpha \in [n-1]$, for every row $i$ and $j < k$, let $c_1 = A_{ij}$, $c_2 = A_{i,j+1}$, and for every column $j$ and $i < k$ let $c_1 = B_{ij}$, $c_2 = B_{i+1,j}$. In each case there is a *comparator guest* $y$ with $W_y = X_{u_\alpha}(c_1) \cup Y_{u_\alpha}(c_2)$ (sets of thin vertices of the two chains) and the single-vertex member $(z_{c_1,u_\alpha}, y, z_{c_2,u_\alpha})$ with centre $y$.

**The instance.** $G$ is defined by the members above (edges $ab, bc$ of every member), the hard edges (thin hard edges, glue, tunnels) and no other edges. $H$ is the set of all members: Latin members, host members, pad members, lock members $(x,y,z)$ and comparator members. The excess is $\ell = k^2$. The instance is denoted $I_{\mathrm{tile}}$; for a set $C$ of tunnels, $I_C$ is $I_{\mathrm{tile}}$ with the pairs of $C$ made non-edges (they stay hard). A *stub instance* is the same with some anchors' tunnels ending at stubs (vertices of degree one).

Lemma 5.8 below shows that $H$ is a modification-disjoint packing of induced $P_3$'s, that the glue graphs are connected and that matchings suffice, so Lemma 5.6 applies to $I_{\mathrm{tile}}$, to every $I_C$, and to the stub instances used below.

### 5.6 Tunnel-release semantics

For a chain $c$ and a cell let $\beta(c) \in [n]$ denote the *block* of the tile whose tunnel is cut: $\beta(A_{ij}) = a$ and $\beta(B_{ij}) = b$ if the tile is $(a,b)$.

**Proposition 5.7** (Tunnel release)**.** *Let $t_{ij} = (a_{ij}, b_{ij}) \in S_{ij}$ be a tile of every cell, $C$ the set of their tunnels, and $\sigma$ a ZE partition of $I_C$. Then in $\sigma$ every chain $c$ is in a position $a_c$ that lies in the group of its cut tunnel, and $a_{ij} \le a_{i,j+1}$, $b_{ij} \le b_{i+1,j}$ for all cells. Conversely, if $(t_{ij})$ satisfies these inequalities, $I_C$ has a ZE partition.*

*Proof.* ($\Rightarrow$) By Lemma 5.6 take the thin configuration. Its restriction to a chain $c$ is a ZE partition of $\mathrm{OH}_m$ (all non-soft pairs inside the chain are hard), so by Lemma 5.3 it is $P_{a_c}$ for one position $a_c$.

*Permanent guests.* By Lemma 5.6(c), $z = z_{c,u} \sim K_{h_u}$, so $z$ is in the cluster of $h_u$ in $P_{a_c}$; for each $v$ in that cluster the pair $zK_v$ is not a hard non-edge, so $v \in W_z$ by (b). Hence $a_c \notin F_u(W_z)$. If $u$ is not a tile group, Corollary 5.5 gives $a_c \notin \{2u, 2u+1\}$.

*Locks.* Let $u$ be a tile group whose tunnel $xx'$ ($x = x_{c,u}$) is not cut. It is a hard edge of $I_C$ with $N_h(x) = \{x'\}$ and $Q(x) \cap Q(x') = \emptyset$ (the soft partners are guests of different chains), so by Lemma N.2 the cluster of $x$ is $\{x, x'\}$; in particular $x \not\sim y, z$ for the members $(x,y,z)$. In the member $(x,y,z)$ with centre $y$ this leaves the single edit "delete $xy$" only if $y \sim z$ (if $y \not\sim z$ also $yz$ is deleted: two edits). So $y \sim z$ and $y$ is in the cluster of $h_u$; as above $a_c \notin F_u(\mathrm{Pad}_u) = \{2u, 2u+1\}$.

Exactly one tunnel of the chain $c$ is cut (the tunnel of $t_{ij}$ serves both $A_{ij}$ and $B_{ij}$). All groups except the group of that tunnel exclude their two positions, and the groups partition $\{0, \dots, m-1\}$;

hence $a_c$ lies in the group of the cut tunnel. With the slot indices above, this gives for the group $u_c$ of $c$: $u_c = (a_{ij}-1)(n+1)+b_{ij}$ for $A_{ij}$ and $(b_{ij}-1)(n+1)+a_{ij}$ for $B_{ij}$. Since $s \in [n]$, the group $u_c$ lies in the block $\beta(c)$: $u_c < u_\alpha$ iff $\beta(c) \le \alpha$, and $u_c > u_\alpha$ iff $\beta(c) \ge \alpha+1$; so $a_c \le 2u_\alpha - 1$ iff $\beta(c) \le \alpha$ and $a_c \ge 2u_\alpha + 2$ iff $\beta(c) \ge \alpha+1$.

*Comparators.* Let $y$ be the comparator guest of $(c_1, c_2, \alpha)$, $u = u_\alpha$. In the member $(z_1, y, z_2)$ ($z_i = z_{c_i,u}$) the state $M$ would give $z_1 \sim z_2$, impossible because $z_1, z_2$ lie in clusters of different chains (a pair of thick vertices of different chains is a hard non-edge). So the state is $L$ ($y \sim z_1$) or $R$ ($y \sim z_2$). In the first case the cluster of $h_u(c_1)$ is inside $W_y$, i.e. inside $X_u(c_1)$ (the part of $W_y$ in $c_1$), so $a_{c_1} \notin F_u(X_u)$, i.e. $a_{c_1} \le 2u-1$, i.e. $\beta(c_1) \le \alpha$. In the second case $a_{c_2} \notin F_u(Y_u)$, i.e. $a_{c_2} \ge 2u+2$, i.e. $\beta(c_2) \ge \alpha+1$. So for every $\alpha \in [n-1]$: $\beta(c_1) \le \alpha$ or $\beta(c_2) > \alpha$, which is equivalent to $\beta(c_1) \le \beta(c_2)$ (if $\beta(c_1) > \beta(c_2)$ take $\alpha = \beta(c_2) \le n-1$). For rows, $\beta(A_{ij}) = a_{ij}$, for columns $\beta(B_{ij}) = b_{ij}$, which are the inequalities of Grid Tiling.

($\Leftarrow$) Put every chain $c$ at the position $a_c = 2u_c$, $u_c$ the group of its cut tunnel. Build the thin configuration: the base clusters are those of $P_{a_c}$. *Permanent guests:* $z_{c,u}$ joins the cluster of $h_u$; this is allowed: for a tile group the set is $\mathrm{All}_u$, for the others $a_c \notin \{2u, 2u+1\}$ and Corollary 5.5 applies. *Locks:* for a tile group $u \ne u_c$ (tunnel not cut) the lock guest $y_{c,u}$ joins the cluster of $z_{c,u}$ (allowed, $a_c \notin \{2u, 2u+1\}$), and the anchors $x, x'$ at the two ends of that tunnel form the cluster $\{x, x'\}$; the member $(x, y, z)$ has the single edit "delete $xy$" (state $L$). For $u = u_c$ the tunnel is cut, $y_{c,u}$ is not joined to anything but its anchor, and the anchor forms the cluster $\{x, y\}$ (state $R$: delete $yz$). The pair $xx'$ is a cut tunnel, a hard non-edge, and the two anchors are in different clusters. *Comparators:* the guest $y$ of $(c_1, c_2, \alpha)$ joins $z_1$ if $\beta(c_1) \le \alpha$ (allowed: then $a_{c_1} \le 2u_\alpha - 1$, so the cluster of $h_{u_\alpha}$ is inside $X_{u_\alpha}$) and joins $z_2$ otherwise (then $\beta(c_2) > \alpha$ and $a_{c_2} \ge 2u_\alpha + 2$, inside $Y_{u_\alpha}$). Every guest is in at most one cluster with a base vertex and two guests are never together, except in the pairs $y, z$ of members: guests join base clusters only through a permanent guest; two hosts $h_u, h_{u'}$ are never in one cluster (their positions in the spine differ by at least four); a tile group carries one lock guest and no comparator, and a separator group carries up to two comparator guests on the same $z$ (one as first chain with condition $\beta \le \alpha$, one as second chain with $\beta \ge \alpha+1$), which cannot both join. All thin chain members have one edit since $P_{a_c}$ is ZE, the members $(x, y, z)$ and $(z_1, y, z_2)$ have one edit as described, and every remaining pair is as required in (T1). By Lemma 5.6 this lifts to a ZE partition of $I_C$. □

### 5.7 Soundness and completeness

**Lemma 5.8** (Parameters and validity)**.** *(a) $H$ is a modification-disjoint packing of induced $P_3$'s of $G$. (b) Let $u(v)$ be the number of matchings used at $K_v$ (pad matchings and the host matching). Then $u(v) \le 3$ for all $v$ and $u(v) \le 2$ if $k \le 2$; at a vertex lying in four thin members $u(v) \le 2$, and at a vertex in at most three thin members $u(v) \le 3$. (c) The glue graph of every $K_v$ is connected, and $p-1 \ge 3$ matchings are available. (d) $c(H) \le 4p+2$ and $|V(G)| = O(k^2n^2p)$, $|H| = O(k^2n^2p^2)$.*

*Proof* in Appendix A; it is a hand count from Lemma 5.4, whose result $u_{\max} \le 3$, $c \le 4p+2$ is also confirmed by computer **(CA4)**.

**Lemma 5.9** (No super-vertex is split)**.** *Let $S$ be a solution of $I_{\mathrm{tile}}$ with $\mathrm{ex}(S) \le k^2$. Then every $K_v$ lies in one cluster of $G \triangle S$. Consequently no hard glue pair and no pad pair (a soft edge inside $K_v$; in particular no pad pair of a comparator pad member) is edited. It is* not *true that no pair inside a super-vertex is edited: the host pairs $(h,i)(h,k)$ are soft non-edges inside $K_h$ and are edited (Lemma 5.6).*

*Proof.* Suppose $K_v$ meets at least two clusters; let $X \subset K_v$ be its smallest part, $x = |X| \in [1, p/2]$, $Y = K_v \setminus X$. Every pair between $X$ and $Y$ that is an edge of $G$ is deleted. A vertex of $X$ has

at most $u(v) \le 3$ soft pairs inside $K_v$, so at least $p - x - u(v)$ hard edges to $Y$; altogether at least $\varphi(x) = x(p - x - u(v))$ hard edits. Since $\varphi$ is concave, $\varphi(x) \ge \min\{\varphi(1), \varphi(p/2)\}$. We have $\varphi(1) = p - 1 - u(v) \ge p - 4 \ge k^2 + 1$. For $k \ge 3$: $\varphi(p/2) \ge \frac{p}{2}(\frac{p}{2} - 3) \ge \frac{(k^2+5)(k^2-1)}{4} > k^2$. For $k \le 2$ we have $u(v) \le 2$ and $p \in \{6, 10\}$: $\varphi(p/2) \ge 3 > 1$ resp. $\ge 15 > 4$. So there are more than $k^2$ hard edits, but by Lemma 2.2 the number of hard edits is at most $\mathrm{ex}(S) \le k^2$, a contradiction. Since $K_v$ lies in one cluster, every glue pair and every pad pair (all of them edges inside $K_v$) is an edge inside a cluster, hence not edited. □

(The host pairs $(h,i)(h,k)$ are soft *non*-edges inside $K_h$; they are edited whenever the permanent guest $z$ joins its host, which happens in every ZE partition (state $M$, the single edit of the host member). What the proofs below use is only: each $K_v$ is inside one cluster, so no hard glue pair and no comparator pad pair is edited.)

**Regions.** For a chain $c$ let $R_c$ be the set of its thick vertices and of its guests $z_{c,u}, y_{c,u}$ and anchors $x_{c,u}$ (not the comparator guests). Let $T_c$ be the set of tunnel partners of anchors of $c$ (they are anchors of the partner chain, $A_{ij} \leftrightarrow B_{ij}$). The sets $R_c$ are pairwise disjoint, and $T_c \subseteq R_{c'}$ for the partner chain $c'$. Let $H_c$ be the set of members inside $R_c$ (Latin, host, pad members of its permanent and lock guests, lock members), $H_{\mathrm{cmp}}$ the set of members that contain a comparator guest (their pad members and the comparator members); $H$ is the disjoint union of the $H_c$ and $H_{\mathrm{cmp}}$. Let $\Pi_c$ be the set of pairs inside $R_c \cup T_c$ except the pairs with both ends in $T_c$. A *cross-anchor pair* is a pair $\{x, x''\}$ of an anchor $x$ of $c$ carrying a tunnel and an anchor $x''$ of the partner chain carrying a tunnel; cross-anchor pairs are hard (tunnels and hard non-edges). A pair lies in two sets $\Pi_c, \Pi_d$ iff it is a cross-anchor pair (and then $d$ is the partner of $c$); every other pair lies in at most one $\Pi$. Pairs of $H_{\mathrm{cmp}}$ members other than the pad pairs involve a comparator guest or join $z_{c_1}$ to $z_{c_2}$ and lie in no $\Pi_c$.

**Lemma 5.10** (Region bound). *Let $S$ be a solution with $\mathrm{ex}(S) \le k^2$. For a chain $c$ and a member $M$ let $x_M = S \cap \mathrm{pairs}(M)$; let $h_c$ be the number of edits in $\Pi_c$ on pairs that belong to no member of $H_c$, let $E_c = \sum_{M \in H_c}(|x_M| - 1) \ge 0$ be the* member excess *(members of $H_c$ with two or three edits), and put $r_c = |S \cap \Pi_c| - |H_c|$. Then $r_c = h_c + E_c \ge \max\{1, h_c\}$.*

*Proof.* By Lemma 5.9 the pad pairs of comparator pad members (which lie in $\Pi_c$ but belong to members of $H_{\mathrm{cmp}}$, not of $H_c$) are unedited, so the edits in $\Pi_c$ are on pairs of members of $H_c$ or on pairs of no member at all; this gives $|S \cap \Pi_c| = \sum_{M \in H_c} |x_M| + h_c$, i.e. $r_c = h_c + E_c$. Each member has at least one edit, so $E_c \ge 0$ and $r_c \ge h_c$; if $h_c \ge 1$ we are done. Suppose $h_c = 0$ and $E_c = 0$, i.e. $r_c = 0$: then every member of $H_c$ has exactly one edit and $S$ has no other edit in $\Pi_c$. In particular no tunnel at $c$ is cut. Two stubs $x_1', x_2' \in T_c$ (partners of anchors $x_1, x_2$ of $c$) are not together: otherwise $x_1 \sim x_1' \sim x_2' \sim x_2$ (tunnels unedited), but $x_1 x_2$ is an unedited hard non-edge. So the partition restricted to $R_c \cup T_c$ has no edit on pairs with both ends in $T_c$ either, and is a ZE partition of the instance $G_c = G[R_c \cup T_c]$ with the packing $H_c$ (the comparator pad pairs inside $R_c$ are plain hard edges of $G_c$, and the vertices of $T_c$ are stubs: each has as only edge its tunnel). $G_c$ is a stub instance, so Lemma 5.6 applies. The proof of Proposition 5.7 ($\Rightarrow$) for the single chain $c$, with all tunnels present, uses only Lemma 5.3, Corollary 5.5, the permanent guests and the locks, and no comparators; it shows that the position $a_c$ lies outside every group (every group is excluded either by a permanent lock or by an uncut tunnel), a contradiction since the groups cover all positions. Hence $r_c \ge 1$ whenever $h_c = 0$, and $r_c \ge \max\{1, h_c\}$ in all cases. □

**Theorem 5.11** (Soundness). *Every solution $S$ of $I_{\mathrm{tile}}$ has $\mathrm{ex}(S) \ge k^2$. If $\mathrm{ex}(S) = k^2$, then $S$ consists of one cut tunnel in every cell and one edit in every member, and nothing else; hence $S \setminus C$ is a ZE partition of $I_C$ with $C$ the set of cut tunnels, and Grid Tiling has a solution.*

*Proof.* Let $\mathrm{ex}(S) \le k^2$ (otherwise there is nothing to show). Let $q$ be the number of edited cross-anchor pairs and $S_{\mathrm{rest}}$ the set of edits outside $\bigcup_c \Pi_c$. Since only cross-anchor pairs lie in two sets, $|S| = \sum_c |S \cap \Pi_c| - q + |S_{\mathrm{rest}}|$. Every member of $H_{\mathrm{cmp}}$ has an edit; its pad pair is unedited because no comparator pad pair is edited (Lemma 5.9), and its other pairs are in no $\Pi_c$; these members are pair-disjoint, hence $|S_{\mathrm{rest}}| \ge |H_{\mathrm{cmp}}|$. With $r_c = |S \cap \Pi_c| - |H_c|$ (Lemma 5.10) and $|H| = \sum_c |H_c| + |H_{\mathrm{cmp}}|$,

$$\mathrm{ex}(S) \;=\; \sum_c r_c - q + \big(|S_{\mathrm{rest}}| - |H_{\mathrm{cmp}}|\big), \qquad r_c = h_c + E_c \ge \max\{1, h_c\}, \qquad |S_{\mathrm{rest}}| - |H_{\mathrm{cmp}}| \ge 0.$$

All three slacks $E_c$, $r_c - \max\{1, h_c\}$ and $|S_{\mathrm{rest}}| - |H_{\mathrm{cmp}}|$ are non-negative, so $\mathrm{ex}(S) \ge \sum_c \max\{1, h_c\} - q$. Let $Z = \{c : h_c = 0\}$ and let $h_{\mathrm{in}}$ be the number of hard edits lying in exactly one $\Pi_c$. A cross-anchor edit counts in two chains, so $\sum_c h_c = h_{\mathrm{in}} + 2q$, and $\sum_c \max\{1, h_c\} = |Z| + \sum_{c \notin Z} h_c = |Z| + h_{\mathrm{in}} + 2q$. Hence $\mathrm{ex}(S) \ge |Z| + h_{\mathrm{in}} + q$. Moreover, for $c \notin Z$ we have $r_c = h_c + E_c$, so keeping the member excess of these chains gives $\mathrm{ex}(S) \ge |Z| + h_{\mathrm{in}} + q + \sum_{c \notin Z} E_c$ (the sum runs over $c \notin Z$ only: for $c \in Z$ the term $\max\{1, h_c\} = 1$ already accounts for $r_c \ge 1$ and $E_c$ must not be added again). The number of chains not in $Z$ is at most $\sum_c h_c$, so $|Z| \ge 2k^2 - h_{\mathrm{in}} - 2q$, and therefore $\mathrm{ex}(S) \ge 2k^2 - q$. On the other hand every edited cross-anchor pair is a hard edit, so $q \le |U| \le \mathrm{ex}(S)$ (Lemma 2.2). Thus $\mathrm{ex}(S) \ge \max\{q, 2k^2 - q\} \ge k^2$.

Let $\mathrm{ex}(S) = k^2$. Then $q = k^2$ (as $\mathrm{ex} \ge q$ and $\mathrm{ex} \ge 2k^2 - q$ give $q = k^2$), and every inequality above is tight: from $\mathrm{ex}(S) \ge |Z| + h_{\mathrm{in}} + q + \sum_{c \notin Z} E_c$ we get $Z = \emptyset$, $h_{\mathrm{in}} = 0$ and $\sum_{c \notin Z} E_c = 0$, hence $\sum_c E_c = 0$ because $Z = \emptyset$, and from the first display $|S_{\mathrm{rest}}| = |H_{\mathrm{cmp}}|$. Every chain has $h_c \ge 1$ and $\sum_c h_c = 2q = 2k^2$, so $h_c = 1$ for all $2k^2$ chains: the only non-member edit at a chain is one cross-anchor pair. All hard edits are these $k^2$ pairs (as $|U| \le \mathrm{ex} = k^2 = q$). Since $E_c = 0$, all members of every $H_c$ have exactly one edit, and since $|S_{\mathrm{rest}}| = |H_{\mathrm{cmp}}|$ with one edit at least per member, $S_{\mathrm{rest}}$ has exactly one edit per member of $H_{\mathrm{cmp}}$. In a cell $(i, j)$ exactly one cross-anchor pair is edited (each chain has $h_c = 1$ and sees only cross pairs of its own cell). It is a tunnel: suppose the edited pair is a non-edge $x_A x_B$ of anchors of $A_{ij}$ and $B_{ij}$ whose tunnels are $x_A x_B^*$ and $x_A^* x_B$ (with $x_B^* \ne x_B$). It is inserted, so $x_A \sim x_B$; the tunnel $x_A x_B^*$ is an unedited edge, so $x_A \sim x_B^*$; hence two distinct anchors of $B_{ij}$ are together, but their pair is an unedited hard non-edge ($h_{B_{ij}} = 1$ is used up by the cross pair), a contradiction. So the edited cross pair is a tunnel of the cell, deleted, and $S$ is the set $C$ of $k^2$ cut tunnels (one per cell) together with one edit per member. By Lemma 5.9 the partition is a partition with $K_v$ intact, and it is a ZE partition of $I_C$. Proposition 5.7 ($\Rightarrow$) gives a tiling: the tiles $t_{ij}$ of the cut tunnels satisfy $a_{ij} \le a_{i,j+1}$ and $b_{ij} \le b_{i+1,j}$. □

**Theorem 5.12** (Completeness)**.** *If the Grid Tiling instance has a solution $(s_{ij})$, then* $\mathrm{OPT}(I_{\mathrm{tile}}) = |H| + k^2$.

*Proof.* Let $C$ be the tunnels of the tiles $s_{ij}$. By Proposition 5.7 ($\Leftarrow$) the instance $I_C$ has a ZE partition, whose edit set (relative to $G$) is the set of its $|H|$ member edits together with the $k^2$ deleted tunnels. So $\mathrm{OPT} \le |H| + k^2$, and $\mathrm{OPT} \ge |H| + k^2$ by Theorem 5.11. □

### 5.8 Pathwidth, pair-maximality and proof of Theorem 5.1

**Lemma 5.13** (Pathwidth)**.** *Every component of $I(H)$ is contained in the set of members of the $k$ chains of one row $\{A_{i,1}, \dots, A_{i,k}\}$ or one column $\{B_{1,j}, \dots, B_{k,j}\}$ together with their guests and comparator guests; so $I(H)$ has at most $2k$ components, and* $\mathrm{pw}(I(H)) < 24kp^2$*, independently of $n$.*

*Proof* in Appendix A (a level sweep: members at distance two in level cannot share a vertex, and a level has $O(kp^2)$ members).

**Lemma 5.14** (Pair-maximality)**.** *$H$ is pair-maximal.*

*Proof* in Appendix A (every non-adjacent pair inside a component of the hard-edge graph is soft).

*Proof of Theorem 5.1.* The instance $(G, H, k^2)$ is constructed in time polynomial in $k$ and $n$ (Lemma 5.8(d)). (i) holds by definition, Lemma 5.8(a),(d) and Lemma 5.14; (ii) is Lemmas 5.8(d) and 5.13 ($\mathrm{tw} \le \mathrm{pw}$); (iii) is Theorems 5.11 and 5.12 (a solution of size at most $|H| + k^2$ has excess at most $k^2$, hence exactly $k^2$, hence a tiling exists by Theorem 5.11). The parameters $\ell + \mathrm{tw}(I(H))$, $\ell + \mathrm{pw}(I(H))$ and $\ell + c$ are bounded by $k^2 + 24k(k^2+6)^2$, which is a function of $k$ only, and the reduction is polynomial: it is a parameterized reduction from Grid Tiling with $\le$ parameterized by $k$, which is W[1]-hard. An algorithm of running time $f(\ell, w)n^{O(1)}$ would decide Grid Tiling in time $f'(k)\, n^{O(1)}$, so FPT = W[1]. The ETH statement is Corollary 5.2. □

*Remark* 5.15 (The chain object). Consider one chain with $N$ tile groups, each with a lock guest and an anchor whose tunnel ends at a stub, all other groups being permanent locks. The proof of Proposition 5.7 ($\Rightarrow$) shows that with all tunnels present there is no ZE partition, and the proof of ($\Leftarrow$) shows that cutting any single tunnel gives one. In the language of Appendix N the set of all $N$ tunnels is therefore a core, of unbounded size $N$, while $c \le 4p+2$ and the pathwidth do not depend on $N$ (the bound of Lemma 5.13). This is the family of tight instances with bounded $c$ and width and cores of unbounded size that the discussion of Appendix N asks for; by Corollary 6.7 it must have soft triangles, and it does (the Latin members). The argument is the proof above applied to a single chain. The hypotheses $p$ even, $p \ge 8$ suffice for this statement (the tunnel semantics do not use the no-split lemma); for the full problem $p > \ell$ is needed.

### 5.9 Computer assistance

The proof of Theorem 5.1 is *computer-assisted*. Precisely, the following finite statements are established by computation and are used as lemmas; all the other steps (Lemmas 5.6, 5.9, 5.10, Proposition 5.7, Theorems 5.11 and 5.12, the pathwidth bound, the pair-maximality) are proved by hand from them.

(CA1) *Lemma 5.3(a).* The explicit partitions $P_a$ of the thin chain are ZE partitions for all $m \ge 7$ and $a$. Each constraint involves at most three consecutive cells; windows at the left end do not depend on $m$, windows at the right end depend only on the distance to the right end, and interior windows are isomorphic, so a check for $7 \le m \le 60$ and every $a$ (no failure) covers all $m$.

(CA2) *Lemma 5.3(b).* The chain has no other ZE partition. The number $N_m$ of ZE partitions is computed by a transfer-matrix dynamic programme over the cells (six states per interior cell; an explicit integer matrix of size $6 \times 6$, identical for all interior cells, with characteristic polynomial $x^4(x-1)^2$); this makes $N_m$ an affine function of $m$ for large $m$, and a computation gives $N_m = m$ for $7 \le m \le 40$. The statement "the transfer matrix is the same at every interior cell of every $m$" rests on translation invariance of the member list and was checked at all cells $7, \ldots, 19$ of $m = 26$.

(CA3) *Lemma 5.4.* The partner sets of the host vertices (the clusters of the hosts at the four classes of positions). The statement is read off the formula of the partitions $P_a$; it was in addition checked by computer for all even $m \in [8, 60]$ and all groups (1,836 cases), because the two cap groups have exceptional $e$-labels.

(CA4) *Counts.* The values $c \le 4p+2$ and $u_{\max} \le 3$ (Lemma 5.8) are obtained by a count that is easily checked by hand from the sets of Lemma 5.4; the count was also carried out by computer for $k \le 4$.

Further computational experiments (independent implementations of the chain lemma, tests of the tunnel-release semantics on random Grid Tiling instances, exact optima of full thick instances for $k \le 3$) support the construction but are not used in the proof; they are summarised in Appendix S.

### 5.10 Where soft triangles are used

Without soft triangles every instance satisfies the conclusions of Theorem 6.4 (Section 6): lock cores have at most two anchors, and tunnel releases cannot realise the abstract hardness mechanism (Theorem 6.10). A single chain with its anchors has a core of unbounded size (Remark 5.15), so by Lemma N.6 its set of ZE state vectors is closed under no majority operation. The Latin members are the soft triangles: the reduction shows that soft triangles suffice for hardness.

### 5.11 Consequences for the rest of the paper

Theorem 5.1 does not contradict any result below: Theorem H.1 needs components of bounded size, but $I(H)$ here has components of size growing with $n$; Theorem I.3 is fixed-parameter tractable in $(\ell, D, \omega)$ where $D = 2c = O(k^2)$ is bounded by a function of $k$, so $\omega$ is unbounded on these instances (the tunnels are hard pairs between far vertices and make the graph $\Gamma'$ of Appendix I have unbounded treewidth; compare Proposition I.4); Theorem 6.4 is about soft-triangle-free packings. Two consequences follow from results proved earlier. By Proposition O.19, the problem of locating a repacking (Open Problem 8) has no fixed-parameter algorithm unless FPT = W[1]; and a reduction lemma in the sense of Open Problem 7 cannot exist unless FPT = W[1], since by Theorem I.3 it would give fixed-parameter tractability in $\ell + w$.

## 6 Soft triangles

For zero-excess solutions the interaction between packed $P_3$'s is as simple as a constraint system can be unless the packing contains one specific configuration, a *soft triangle*. Without it, zero excess is polynomial for every $c$ and $w$ (Corollary 6.5) and every lock core has at most two anchors (Corollary 6.7); a far-lift model of excess is fixed-parameter tractable and on forests the full problem is fixed-parameter tractable in $\ell$ conditionally on one bounded-certificate hypothesis (Appendix B). The instances of Section 5 are full of soft triangles, so soft triangles are what hardness needs along every route that we can analyse, and they suffice. Every statement below is marked *proved* (a proof is given, in an appendix when long), *by computer* (a finite case check that is part of the proof) or *experimental* (computer experiments that are not part of any proof).

**Conventions.** States are as in Appendix N: a member $(a, b, c)$ with centre $b$ is in state $1, 2, 3$ if its single edited pair is $ab, bc, ac$. Put $t_{ab} = (0, 1, 1)$, $t_{bc} = (1, 0, 1)$, $t_{ac} = (0, 0, 1)$ for the states $(1, 2, 3)$: $t_e(s) = 1$ iff the pair $e$ of the member is inside one cluster after the edit. $\Sigma = \Sigma(G, H) \subseteq \{1, 2, 3\}^H$ is the set of zero-excess (ZE) state vectors; by Lemma 2.2, $s \in \Sigma$ iff no vertex triple is an induced $P_3$ of $G$ after the edits $s$. A *majority operation* is $m : \{1, 2, 3\}^3 \to \{1, 2, 3\}$ with $m(a, a, b) = m(a, b, a) = m(b, a, a) = a$; applied to vectors it acts coordinatewise.

## 6.1 Zero excess without soft triangles is easy

**Definition 6.1** (Soft triangle)**.** A vertex triple $\{x, y, z\}$ is a *soft triangle* of $(G, H)$ if its three pairs are soft and are owned by three pairwise distinct members. $(G, H)$ is *soft-triangle-free* if there is none. (This depends on $H$ only, not on the edges of $G$.)

**Lemma 6.2** (Soft triangles and $I(H)$)**.** *(a) The owners of a soft triangle form a triangle of $I(H)$. (b) If $c \leq 2$, every triangle of $I(H)$ consists of the owners of a soft triangle. (c) If $I(H)$ is triangle-free, in particular if $I(H)$ is a forest ($w \leq 1$), there is no soft triangle. For $c \geq 3$ a triangle of $I(H)$ need not come from a soft triangle (three members through one vertex).*

*Proof.* (a) Two pairs of a triple share a vertex, so their owners intersect. (b) Let $A, B, C$ be pairwise adjacent. If $A \cap B$, $B \cap C$ and $A \cap C$ shared a vertex, it would lie in three members, contradicting $c \leq 2$; so the three intersection vertices $x \in A \cap B$, $y \in B \cap C$, $z \in A \cap C$ are distinct, $\{x, y\} \subseteq V(B)$, $\{y, z\} \subseteq V(C)$, $\{x, z\} \subseteq V(A)$ are soft pairs with three distinct owners. (c) follows from (a). □

**Lemma 6.3** (Constraint structure)**.** *Let $H$ be any packing and classify a vertex triple $\tau$ by the owners of its soft pairs.*

*(i) No soft pair: a* static *condition on $G$ (if $\tau$ is an induced $P_3$ of $G$, then $\Sigma = \emptyset$).*

*(ii) One soft pair, owner $A$: a* unary *constraint on $s_A$.*

*(iii) Two soft pairs, from distinct members $A, B$: these share a vertex $q \in V(A) \cap V(B)$, $\tau = \{p, q, r\}$ with $p \in V(A) \setminus \{q\}$, $r \in V(B) \setminus \{q\}$, and $pr$ is hard; a* binary *constraint on $(s_A, s_B)$, namely $t_{pq}(s_A) + t_{qr}(s_B) + [pr \in E] \neq 2$.*

*(iv) All soft pairs from one member: $\tau = V(A)$, no restriction on $\Sigma$ beyond the states.*

*(v) Three soft pairs from three distinct members: a soft triangle, a* ternary *constraint.*

*Hence if $(G, H)$ is soft-triangle-free, $\Sigma$ is empty or the set of vectors satisfying unary lists $\Lambda_A \subseteq \{1, 2, 3\}$ and, for every pair $A, B$ of adjacent members sharing $q$, the binary relation $R_{AB} \subseteq \{1, 2, 3\}^2$ that is the conjunction of the four triples $\{p, q, r\}$.*

*Proof.* Two soft pairs of one triple share a vertex. If their owners coincide, the triple is $V(A)$, which is case (iv). If they differ, the shared vertex lies in both members, so it is the unique common vertex $q$; the third pair $pr$ is not soft, since an owner $A$ or $B$ would then contain two vertices of the other member (forcing $|V(A) \cap V(B)| \geq 2$) and a third owner gives (v). The constraint of (iii) is "not exactly two edges". The final sentence is the classification, as $R_{AB}$ depends only on $A, B$ and the bits $[pr \in E]$ (the cross pairs). □

**Theorem 6.4** (Majority closure)**.** *If $(G, H)$ is soft-triangle-free, then $\Sigma(G, H)$ is closed under* every *majority operation $m$ applied coordinatewise, with the same $m$ at all coordinates: for $u, v, w \in \Sigma$, $m(u, v, w) \in \Sigma$. No pair-maximality is needed, and $c$ and $w$ are arbitrary.*

*Proved; the classification of the binary relations $R_{AB}$ is by computer* (Appendix C.1). The relation $R_{AB}$ depends on the roles of $q$ in $A$ and in $B$ and on the four cross bits: 144 configurations, giving exactly 40 distinct relations, all of the shapes empty, singleton, 2-element set, a full row or column through $\{1, 2\}$, a 3-element subset of $\{1, 2\}^2$, or a "cross" (row ∪ column through $\{1, 2\}$); each shape is closed under every majority by a short argument. The classification was confirmed by an independent construction from real graphs, and all 729 majority operations were checked against all relations. The point of the theorem is that single triples are not closed under any uniform majority; only the conjunction of the four triples at a shared vertex is.

**Corollary 6.5** (Zero excess without soft triangles is polynomial)**.** *For soft-triangle-free $(G, H)$, whether $(G, H, 0)$ is a YES instance is decidable in polynomial time. No bound on $c$ or $w$ and no tree decomposition is needed.*

*Proof.* By Lemma 6.3, $(G, H, 0)$ is YES iff there is no static violation and the binary CSP with $|H|$ variables of domain $\{1, 2, 3\}$, unary lists $\Lambda_A$ and relations $R_{AB}$ is satisfiable. By Theorem 6.4 every constraint is closed under a majority operation. By the theorem of Jeavons, Cohen and Cooper [12], a set of constraints closed under a near-unanimity operation (here a majority, of arity 3) is decided by enforcing strong consistency of a fixed level, which for binary constraints and a majority is path consistency; this takes time polynomial in $|H|$. Static triples are tested in $O(n^3)$. □

*Remark* 6.6 (No conflict with the hardness of $\ell = 0$). Corollary 6.5 does not contradict the NP-hardness of $\ell = 0$ for $c \leq 23$ [15]: the variable gadget of [15, Sect. 4.1.1] (three cliques of five vertices over $\mathbb{F}_5$, one member for each arithmetic progression, 25 members, $c = 5$) packs every pair between consecutive cliques and a computer count finds 100 soft triangles in it (any triple with one vertex in each clique that is not an arithmetic progression is one). The hardness proof therefore lives on ternary constraints. Likewise Example N.11 ($c = 2$) has four soft triangles and is not closed under the operation $m_3$, and the flip gadget of Example N.10 has soft triangles. At $c \leq 2$ a soft triangle is a triangle of $I(H)$ (Lemma 6.2), hence $w \geq 2$, so for $w \leq 1$ the hardness of $\ell = 0$ cannot occur (this is also clear from Theorem 3.5).

**Corollary 6.7** (Lock cores have at most two anchors)**.** *If $(G, H)$ is soft-triangle-free, every core (minimal non-face of anchors, Appendix N) has at most two anchors. In particular this holds for every $H$ with $c \leq 2$ and $I(H)$ triangle-free, whatever* $\mathrm{tw}(I(H))$.

*Proof.* Theorem 6.4 gives closure under a majority operation, and Lemma N.6 applies. □

This settles, for these instances, the question whether the hypothesis of Lemma N.6 holds: it holds exactly when there is no soft triangle in the sense that matters. The statement that the specific operation $m_3$ closes $\Sigma$ for $c \leq 2$ is false with triangles (Example N.11) and true without soft triangles.

## 6.2 What soft triangles allow

W[1]-hardness through hard edges between far vertices needs an operation of cost one that repairs an obstruction at one of many interior positions. We separate the abstract problem from its realisation: the abstract problem is W[1]-hard (Theorem 6.9); tunnel releases cannot realise it without soft triangles (Theorem 6.10); and the reduction of Section 5 realises it with soft triangles.

**Definition 6.8** (LPPH)**.** *Linked-pair path hitting*: a forest, a family of *obstruction paths* in it, a set of *links* (pairs of nodes) and a number $L$. An operation is a single node or a link; it hits every path containing one of its nodes. Can $L$ operations hit all paths?

**Theorem 6.9** (Abstract hardness)**.** *LPPH is W[1]-hard parameterized by $L$, already for a forest of $2k$ paths and a set of links that is a perfect matching (every node in at most one link), with $L = k^2$.*

*Proved* (Appendix C.8), by a reduction from Grid Tiling with $\leq$ [6]; also confirmed by exhaustive search on small instances. It is the abstract form of the mechanism "a far two-sided edit shares cost between two places"; an operation can hit an interior node of an obstruction, which is the feature the reduction uses.

**Theorem 6.10** (Tunnel releases cannot realise Theorem 6.9 without soft triangles)**.** *Let $(G, H)$ be soft-triangle-free and consider the tunnel-release model (Appendix N).*

*(i) Let the* conflict graph $\mathcal{C}$ *have as vertices the anchors that carry a tunnel or a stub, an edge $xy$ if $\{x, y\}$ is not a face, and a loop at $x$ if $\{x\}$ is not a face. Releasing a set $R$ of these anchors (cutting their tunnels) gives a feasible instance iff $R$ is a vertex cover of $\mathcal{C}$ (loops included); a cut at $x$ repairs exactly the conflicts at $x$, and every conflict has at most two ends.*

*(ii) Let $X$ be a set of anchors and $R_0$ a set of released anchors disjoint from $X$ such that the instance with release set $R_0$ is infeasible but $R_0 \cup \{x\}$ is feasible for every $x \in X$. Then $|X| \leq 2$. In particular an obstruction that must be repaired by cutting a tunnel at* any *one of $N \geq 3$ anchors (as Theorem 6.9 needs, with $N = |S_{ij}|$ positions) cannot occur.*

*Proof.* (i) A set of locked anchors is feasible iff it is a face, iff (Corollary 6.7, using that a non-face contains a non-face of size at most two) it contains no loop and no edge of $\mathcal{C}$, i.e. iff the released set covers $\mathcal{C}$. (ii) $R_0$ is not a vertex cover, so some edge or loop $e$ of $\mathcal{C}$ is uncovered; for each $x \in X$ the set $R_0 \cup \{x\}$ covers $e$, so $x$ is an end of $e$, and $e$ has at most two ends. □

*Remarks (heuristic; not part of Theorem 6.10).* Breaks (double-edited members) and other operations (inserted far pairs, cuts inside hard components of size at least three) act only through unary changes at their ends (Lemma B.3); we have no proof that no combination of operations simulates interior hits without soft triangles.

### 6.3 Certificates at excess zero

A *certificate* for a budget $|H| + \ell$ is a set $T$ of *remote* hard pairs (pairs not adjacency-local to the packing, Appendix B.3) such that every solution of size at most $|H| + \ell$ contains a pair of $T$; the natural branching algorithm (Theorem B.13) branches on the pairs of a minimal certificate, and a leaf test is fixed-parameter tractable (Theorem B.14). Its running time is governed by the size of certificates.

- Without soft triangles every minimal certificate at $\ell = 0$ has at most 12 pairs, for every $c$ and $w$ (Theorem B.15; proved).
- With soft triangles minimal certificates are unbounded already at $c = 2$, $w = 2$, $\ell = 0$ (Theorem B.16; proved, the block structure by computer): for every $N$ there is $G_N$ with a pair-maximal packing of $3N$ members, $c = 2$, $\mathrm{tw}(I(H_N)) = 2$, $N$ soft triangles and a minimal certificate of $2N + 4$ remote pairs. This is not a hardness result: in $G_N$ a small certificate exists.

The following statement shows that the second item is a feature of minimal certificates, not of $\ell = 0$ at $c \leq 2$.

**Theorem 6.11** (Bounded certificates at $\ell = 0$, $c \leq 2$)**.** (Proved; the classification of finitely many relation types is by computer.) *Let $c \leq 2$ (soft triangles allowed; pair-maximality is not needed). If there is no ZE solution without remote edits, then there is a certificate for $\ell = 0$ with at most* 4 *remote pairs (*3 *if it is a static violation or an empty lock). The constant* 4 *is attained by an explicit instance (verified by computer). Such a certificate can be found in time $f(w)n^{O(1)}$.*

Proof outline (Appendix D). A triple that contains a remote pair has at most one soft pair; the zero-excess lists it can impose on a member are the table of Observation B.6, and every singleton

entry is produced by one triple, i.e. by at most two remote pairs. Hence (Theorem D.4, any $c$) a certificate of size at most $\max\{3, 2\kappa_1\}$ exists, where $\kappa_1$ is the *unit-lock Helly number* of the set $\Sigma_0$ of ZE vectors for the triples without remote pair. For $c \leq 2$ the set $\Sigma_0$ is an intersection of lifted relations of four types (link, $K_3$, diamond, $K_4$), enumerated by computer: the first three are closed under one majority operation, so Lemma B.5 gives $\kappa_1 \leq 2$; for $K_4$ (a whole component of $I(H)$ on four members) $\kappa_1 \leq 2$ is computed directly.

*What is not proved.* The version for $\ell \geq 1$ is open (Open Problem 2): Appendix D proves a sufficient criterion (Lemma D.6: $\ell + 1$ zero-excess cores with pairwise disjoint regions give a certificate of at most $4(\ell + 1)$ pairs) and the pattern-wise statement (Lemma D.5), and the natural proof by weighted gluing is impossible already for soft-triangle-free link relations over the five states (SAT-verified). *Experimental evidence only:* exact minimum certificates found by local search on forests $I(H)$ with $n \leq 20$ were $4, 8, 10, 11, 15$ for $\ell = 0, \ldots, 4$, never above $4(\ell + 1)$; for $c \geq 3$ with soft triangles even $\ell = 0$ is open (by Theorem D.4 it is equivalent up to a factor 2 in the upper bound to boundedness of the unit-lock Helly number of $\Sigma_0$, and Lemma B.18 shows that a uniform near-unanimity polymorphism does not exist for the $c \geq 3$ language).

### 6.4 Summary

1. *Soft-triangle-free instances reduce to one hypothesis.* Zero excess is polynomial, lock cores have at most two anchors, the far-lift and tunnel-release models are fixed-parameter tractable (Theorem B.7, Corollary B.1), the leaf test is fixed-parameter tractable and minimal certificates at $\ell = 0$ are bounded. For forests the full problem is fixed-parameter tractable in $\ell$ conditionally on the bounded-certificate hypothesis BC (Theorem B.11); BC for $\ell \geq 1$ is open.

2. *Hardness needs soft triangles along the routes we can analyse, and soft triangles suffice* (Theorems 6.10 and 5.1). Theorem B.16 alone is not a hardness result, and the exclusion of other operations in the soft-triangle-free case is heuristic.

3. *With soft triangles at $c \leq 2$ ($w \geq 2$):* the bounded-certificate statement holds at $\ell = 0$ (Theorem 6.11); that the lock structure is 2-decomposable (Remark B.2) is supported by experiments only. The reduction of Section 5 has $c \geq 26$; hardness for bounded $c$ and for $c = 2$ is open (Open Problem 1).

## 7 Further structure

The results below describe optimal solutions, the components of $I(H)$, an $n^\ell$ algorithm, locality of inserted pairs, and repacking. They are all proved, in Appendices E–P, and none is needed for the proofs in Sections 3–5; the proofs are not repeated here. Unless said otherwise, $H$ is pair-maximal (for an arbitrary packing read the primed instance of Lemma 2.4), $S$ is an optimal solution of size at most $|H| + \ell$, $U$ is its set of $u \leq \ell$ hard edits, $D \geq \max_v |Q(v)|$, and $Z = O(w + \ell)$. Appendix R tabulates for every result whether it holds for the input packing or for a pair-maximal extension.

**Structural lemmas.**

- *Twins* (Lemma E.1). If $u \neq v$ are adjacent with $N[u] = N[v]$, they lie in one cluster of every optimum, with no hypothesis on $H$.
- *Half-degree and diameter* (Lemma G.2). In an optimal solution every vertex $v$ of a cluster $C$ has at least $(|C| - 1)/2$ neighbours in $C$, and two non-adjacent vertices of $C$ have a common neighbour in $C$.

- *MAX* (Lemma 2.4, Section 2). Adding a maximal hard family to $H$ makes it pair-maximal and raises $w$ and $c$ by at most the size of the family, which is at most $\ell$ in a YES instance.
- *HUB*$^+$ (Lemma F.1). For any packing, in a YES instance, for every vertex $u$ the graph of hard non-edges on the hard neighbourhood $N_h(u)$ has matching number at most $\ell$, a vertex cover of size at most $2\ell$, and $\alpha(G[N_h(u)]) \le 2\ell + c + 1$.
- *SM, small merges* (Lemma F.4). Let the *pieces* of a solution be the components of the graph of its kept hard edges. If a cluster $C$ is the union of at least two pieces, then $|C| \le \max\{D + u + 1, 2D, 2u\}$.
- *No edit inside the uncovered part* (Theorem E.5). Every cluster of an optimum meets $V \setminus V(H)$ in the empty set or in exactly one whole class of true twins, so no edit has both ends outside $V(H)$; edits with one end outside $V(H)$ cannot be excluded (Proposition Q.3).
- *Normal form and candidates* (Lemma G.6, Corollary G.7). There are at most $2\ell$ clusters containing an end of a hard edit, each connected of size at most $2D + \ell + 1$ and of bounded hard degree; one end of a hard edit determines at most $Z^2$ candidates for the other, which lie at distance at most 2.

**Components of $I(H)$ (fixed-parameter tractability in $(\ell, s)$).** Let $s$ be the maximum number of vertices spanned by a connected component of $I(H)$. CEaMP is solvable in time $1.62^{(\binom{s}{2}+1)\ell} n^{O(1)} + n3^s$ and has a kernel with at most $(s^2 - s + 2)\ell$ vertices (Theorem H.1); in particular a perfect matching $I(H)$ is fixed-parameter tractable in $\ell$ (Corollary H.2). This is the reduction rule of van Bevern, Froese and Komusiewicz [17] applied to components; the soundness of the rule, the algorithm of Böcker [2] and the kernel of Chen and Meng [4] are cited, the rest is proved. For components of arbitrary size, after reducing tight reducible components at most $2\ell$ components remain (Theorem H.10); what is left is a few arbitrarily large components (Open Problem 5). Since $s$ is not bounded by $w$ this is not a fixed-parameter algorithm in $\ell + w$.

**An $n^\ell$ algorithm and a stronger parameter.** By Corollary G.7 it suffices to guess one end of each hard edit: CEaMP is solvable in time $(n+1)^\ell Z_*^{2\ell} 5^{w+\ell+1} n^{O(1)}$, $Z_* \le 6w + 9\ell + 7$, for every packing (Theorem I.2), which halves the exponent of Theorem 3.6. Following clusters instead of guessing gives a fixed-parameter algorithm in $(\ell, D, \omega)$, where $\omega$ is the treewidth of the graph of edges and soft pairs outside the large hard components (Theorem I.3); $\omega$ is not bounded by any function of $(\ell, w)$ (Proposition I.4, from cubic expanders [9]), and is unbounded on the hard instances of Section 5.

**Locality of inserted pairs (T1/T2).** Every inserted hard pair $ab$ of an optimum has a witness $z$ (a common neighbour of $a$ and $b$ in $G$ in their cluster), at least one of whose steps $az, zb$ is soft, and is of type T1 (both steps soft, owned by two distinct members through $z$; local in every tree decomposition of $I(H)$, Proposition L.5) or of type T2 (one step a kept hard edge) (Theorem L.2). Merges of two pieces joined by no soft pair are of type T1 only (Corollary L.3); merges joined by a soft pair can be of type T2 (instances with $n = 7$ and $n = 8$, Proposition L.7). For every $m \ge 3$ there is an instance with $w = 1$, $c = 2$, $\ell = 1$ in which every optimum deletes a hard edge whose ends lie in packed $P_3$'s at distance $m - 1$ in $I(H)$ (Proposition L.8, the "FAR family").

**Repacking.** The parameters belong to the packing: exchanging one packed $P_3$ for an induced $P_3$ keeps $|H|$ and raises $w$ by at most one, and a pair-maximal repair costs at most $\ell$ more (Lemma O.1). Every hard edit of an optimum lies in an induced $P_3$ and is bridged (local in every decomposition) or absorbable by one exchange (Lemma O.5); no hard edit has two uncovered ends (Lemma O.7).

- *The FAR artifact.* The FAR family has intrinsic excess one, but after one exchange $I(H^*)$ is a cycle and its only hard edit is at an uncovered vertex; the spine family has excess zero and $\mathrm{tw}(I(H^*)) \leq 2$ after one exchange and one augmentation (Propositions O.10 and O.11).
- *PL, a one-sided algorithm.* In time $5^{w+1}(Z^2+1)^{3\ell(w+1)}n^{O(1)}$ one computes an upper bound on OPT that is exact whenever some optimum has only hard edits that are co-bagged or at local hubs (Theorem O.13); the promise cannot be tested.
- *Theorem N.* No polynomial procedure outputs a repacking with excess and width bounded in $\ell$ alone unless P = NP (and, under ETH, none with $\mathrm{tw} \leq |H|^{1-\varepsilon}$) (Theorem O.15); bounded-radius augmentation cannot find the good packing (Proposition O.17); the naive normal form "hard edits have an uncovered end" is false (Proposition O.16), and a corrected normal form is open (Conjecture O.18).
- *Locating the repacking* in fixed-parameter time is impossible unless FPT = W[1], since it would give fixed-parameter tractability in $\ell + w$ (Proposition O.19 and Corollary O.20, with Theorem 5.1).

# 8 Open problems

The main question, whether CEaMP is fixed-parameter tractable in $\ell + w$, is answered negatively (Theorem 5.1); the XP algorithms of Theorems 3.6 and I.2 are optimal up to the exact exponent of $n$. Positive results that remain are fixed-parameter tractability in $\ell$ for $w = 0$ [17], in $(\ell, s)$ for bounded components (Theorem H.1), in $(\ell, D, \omega)$ (Theorem I.3), and for soft-triangle-free packings the reduction of the question to one hypothesis (Section 6). What is open is the fine structure.

**Open Problem 1** (Hardness for bounded $c$; $c = 2$)**.** Is CEaMP W[1]-hard parameterized by $\ell$ when $c$ is bounded by a constant? In particular, is CEaMP with $c = 2$ fixed-parameter tractable in $\ell$, or W[1]-hard? The reduction of Theorem 5.1 has $c = 4p + 2 \geq 26$ with $p \geq k^2 + 5 = \ell + 5$, because a blown-up clique of size $p$ cannot be split by an edit set of excess $\ell$ only if $p > \ell$ (Lemma 5.9); the zero-excess semantics (Proposition 5.7) hold for every even $p \geq 8$. What is missing is a gadget that holds a vertex of the chain together at bounded $c$ against $\ell$ edits. The complexity of $\ell = 0$ for $3 \leq c \leq 22$ is open in [15], whose authors say a W[1]-hardness result for $c = 2$ "would also not be surprising". For soft-triangle-free packings $\ell = 0$ is polynomial (Corollary 6.5), so a hardness proof for $\ell = 0$ at $c = 2$ needs triangles in $I(H)$, hence $w \geq 2$.

*Bounded certificates at $\ell = 0$, $c \leq 2$.* By Theorem 6.11 (proved; finite classification by computer), at $\ell = 0$ and $c \leq 2$ a bounded certificate always exists: if no zero-excess solution avoids remote edits, there is a certificate of at most 4 remote pairs, found in time $f(w)n^{O(1)}$, and this holds with soft triangles. So the unbounded minimal certificates of Theorem B.16 are not an obstruction to branching at $\ell = 0$, $c \leq 2$; this does not decide the complexity at $c = 2$ and $\ell \geq 1$.

**Open Problem 2** (Bounded certificates BC for $\ell \geq 1$)**.** Is there $K(\ell, D)$ such that a certificate of at most $K$ remote pairs always exists (for forests $I(H)$, soft-triangle-free packings, and $c \leq 2$)? Proved: $\ell = 0$, $c \leq 2$ (Theorem 6.11) and $\ell = 0$ for every $c$ in terms of the unit-lock Helly number (Theorem D.4); for $\ell \geq 1$ only the sufficient criterion of Lemma D.6, the pattern-wise statement

of Lemma D.5, and the obstruction to the natural weighted-gluing proof. *Computational evidence only:* a local search on forests found minimum certificates of sizes $4, 8, 10, 11, 15$ for $\ell = 0, \dots, 4$ (below $4(\ell+1)$), and periodic chains with $c \le 3$ at $\ell = 0$ showed no growth. For $c \ge 3$ with soft triangles even $\ell = 0$ is open. For forests, BC and an oracle assumption imply fixed-parameter tractability in $\ell$ (Theorem B.11).

**Open Problem 3** (ETH-tight exponent)**.** Corollary 5.2 excludes $f(\ell, w)N^{o(\sqrt{\ell})}$ (from $\ell = k^2$), while the algorithms run in time $N^{2\ell+O(1)}$ resp. $N^{\ell+O(1)}$ times $f(\ell, w)$. Is there an algorithm with exponent $O(\sqrt{\ell})$, or is $N^{o(\ell)}$ excluded? A reduction with $\ell = O(k)$ would be needed for the latter.

**Open Problem 4** (Base of the exponent)**.** For $\ell = 0$, does a $(3-\varepsilon)^w n^{O(1)}$ algorithm exist, or is there a SETH-type lower bound? Theorem 4.1 gives only the exponent.

**Open Problem 5** ($w = 1$ with unbounded tree components)**.** If $I(H)$ is a forest with components of unbounded size, is CEaMP fixed-parameter tractable in $\ell$? For forests there is no soft triangle: $\ell = 0$ is polynomial, the far-lift model is fixed-parameter tractable (Theorem B.7), and the full problem is fixed-parameter tractable conditionally on BC and the oracle (O) (Theorem B.11); the question is thus reduced to Open Problem 2.

**Open Problem 6** (Soft triangles at $c \le 2$)**.** For $c \le 2$ with triangles in $I(H)$, is $\Sigma$ always 2-decomposable (equivalently, do lock cores of the tunnel model have at most two anchors)? Theorem 6.11 proves the corresponding Helly property for the unit locks that matter for certificates; for the formulation of Remark B.2 only evidence is available; Example N.11 shows that closure under the specific operation $m_3$ fails.

**Open Problem 7** (Pointers of type T2 and deleted hard edges; reduction lemma; repacking)**.** Can the far ends of inserted pairs of type T2 and of deleted hard edges between far members be handled in a dynamic program of width $g(\ell, w)$ on a restricted class (soft-triangle-free, bounded $c$)? For the full problem the answer is no unless FPT = W[1], since such a program, a polynomial-time reduction lemma (given $(G, H, \ell)$ with $H$ pair-maximal, output NO correctly or an equivalent pair-maximal instance $(G^*, H^*, \ell^*)$ with $\ell^* \le \ell$, $w(H^*) \le f_1(\ell, w)$ and $\mathrm{tw}(\Gamma'(G^*, H^*)) \le f_2(\ell, w)$), or a fixed-parameter algorithm *locating* a repacking of bounded width and excess on YES instances (LOC, Open Problem 8) would each give fixed-parameter tractability in $\ell + w$ (Proposition O.19, Theorem I.3); by Theorem 5.1 none exists unless FPT = W[1]. Whether a merge pointer between two pieces joined by a soft pair can be forced at unbounded distance is open, as is the corrected normal form (Conjecture O.18).

# 9 Computer assistance and formal verification

*What is computer-assisted.* The proof of Theorem 5.1 uses a computer at four finite steps (CA1)–(CA4) (Section 5.9): that every $P_a$ is a ZE partition of the thin chain for $7 \le m \le 60$, the transfer-matrix count $N_m = m$ for $7 \le m \le 40$ (so the chain lemma for $m \ge 10$ is computer-assisted and rests on a translation-invariance argument), the host profiles, and the parameter counts. Theorem 6.4 uses a computer classification of 144 configurations of binary relations, and Theorems B.16 and 6.11 use finite classifications; explicit small instances in the paper were checked by enumerating all set partitions. Further computations (independent implementations, random tests, exact optima of full thick instances for $k \le 3$) are supporting experiments, are not used in any proof, and are summarised in Appendix S.

*Lean 4.* Part of the argument was formalised in Lean 4 with Mathlib (the repository is given in Appendix S). All files are free of `sorry` and of `native_decide`; every theorem depends at most

on the axioms `propext`, `Classical.choice` and `Quot.sound`; the finite computations use `decide +kernel`. *Machine-checked:* the CSP representation of Section 3 (clusterings versus cluster graphs, OPT $\geq |H|$, Observations 3.1 and 3.2 without the bound "at most three owners" and without the treewidth statements, and the five-state lemma); the twin lemma, half-degree and diameter at most 2 of an optimal cluster and the separation of vertices at distance at least 3, the lemma on the maximal family $M$ (MAX), the lemma on the hub (HUB, in the bounds $\nu(\Lambda_u) \leq \ell$, vertex cover of size at most $2\ell$ and $|Y| \leq 2\ell + 1$), and the small-merge lemma in abstract form with its hypotheses taken as assumptions; the block encoding of Grid Tiling and the equivalence of the comparator conditions with the Grid Tiling constraints (the last step of Proposition 5.7, $\Rightarrow$); the counting core of Theorem 5.11 as an abstract inequality system; the chain lemma for $m = 7, 8, 9$ only; the algebra of the transfer matrix; and the parameter arithmetic of Theorem 5.1 including the no-split inequality. *Not machine-checked:* Theorem 5.11 at graph level; the blow-up lemma; host profiles (Lemma 5.4); tunnel release at graph level; the region bound (Lemma 5.10); pathwidth and pair-maximality of the reduction instance; the chain lemma for $m \geq 10$; the link between the graph $\mathrm{OH}_m$ and the transfer matrix (so the steps (CA1)–(CA4) remain computer-assisted, with Lean covering only $m \leq 9$ and the matrix algebra); the dynamic programmes and running-time theorems; the ETH statements; and the reduction as a whole (Theorem 5.1). The hypotheses of the counting core are taken for granted, so the Lean result certifies the arithmetic of Theorem 5.11, not the claim that the instance $I_{\mathrm{tile}}$ satisfies them. A detailed list is in Appendix S.

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# Appendix map

The appendices contain, in this order: the proofs of the long lemmas of the reduction (Appendix A); the full text of the soft-triangle analysis, its proofs, and the proofs of the bounded-certificate statements (Appendices B, C, D); the proved results summarised in Section 7, namely twins, vertices outside the packing and a shrink rule (E), counting lemmas (F), the structure of optimal solutions (G), components of $I(H)$ (H), the $n^{\ell}$ algorithm (I), the dynamic program for $\omega$ (J), conflict weights (K), locality (L), a family on which reduction on unions of members fails (M), tunnel gadgets (N) and repacking (O, P); small examples (Q); the scope table (R); and the computer verifications, the Lean development and the reproducibility information (S).

# A Proofs for the reduction

This appendix contains the proofs of the lemmas of Section 5 that are not given there. The computer-assisted steps (CA1)–(CA4) are marked as in Section 5.9.

## A.1 Proof of the chain lemma

*Proof of Lemma 5.3. (a) Each $P_a$ is a ZE partition.* This is a conjunction of local conditions: every member, and every hard pair that is an edge, involves vertices of at most three consecutive cells, and the pairs of vertices that must be apart are only constrained through clusters, which have vertices in at most three consecutive cells. A condition at a cell therefore depends only on the relation of the position $a$ to this cell (smaller by at least 3, or $a - j \in \{2, 1, 0, -1, -2\}$, or larger), on the distance to the left end up to a constant, and on the distance to the right end up to a constant. Windows at the left end do not depend on $m$, windows at the right end depend only on the distance to the right end, and interior windows are isomorphic. All window types occur for $m \le 20$. **(CA1)**: for every $7 \le m \le 60$ and every $a$ the partition $P_a$ was checked by computer to have cost exactly $|H|$ with one edit per member and no hard pair edited (no failure). By the window argument this covers all $m \ge 7$.

*(b) There are no other ZE partitions.* **(CA2)**. Process the cells $0, 1, \dots, m$ from left to right. After cell $c$, a *partial ZE partition* is a partition of the vertices of the cells $\le c$ in which every hard pair and every member with all its vertices in these cells is respected. A vertex of cell $c'$ can be together with a vertex of a cell $c'' \ge c' + 3$ only if the pair is soft or a hard edge, which never happens (soft pairs join cells at distance at most 2 and the hard edges join vertices in cells at distance at most 1). So only the partition induced on the vertices of the last two cells matters for the future, together with the information which blocks already contain an older vertex (such a block is *closed*). The state after cell $c$ is the set partition of the vertices of the cells $c-1, c$ with its set of closed blocks, with blocks renamed canonically and vertices described by (type, cell offset). At every interior cell there are exactly six reachable states, and the transition matrix between consecutive interior cells is the same matrix

$$M = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \chi_M(x) = x^4(x-1)^2,$$

with a Jordan block of size two at the eigenvalue 1 (computed with exact integer arithmetic; the equality of the matrices was verified at the cells $7, \dots, 19$ of $m = 26$, and the transition rule at a cell involves only members and hard pairs within three consecutive cells, while the member list is translation invariant in the interior, so the matrix is the same at every interior cell of every $m$). The number $N_m$ of ZE partitions is $N_m = u^\top M^{m-c_0} v_m$ where $u$ is the vector after the left cap, which does not depend on $m$, and $v_m$ is determined by the right cap and depends only on the distance to the right end, hence is the same for all $m$ after the shift by the exponent. Since $x^4$ divides $\chi_M$, the generalised eigenspace of the eigenvalue 0 is annihilated by $M^4$, and $M^t$ is for $t \geq 4$ an affine function of $t$ (the Jordan block at 1). Hence $N_m$ is an affine function of $m$ for $m \geq m_1$. A direct computation gives $N_m = m$ for $7 \leq m \leq 40$, so $N_m = m$ for all $m \geq 7$. Together with (a), which provides $m$ distinct ZE partitions (their spine words $R^a M L^{m-1-a}$ differ), these are all of them. □

*Remark* A.1. As a consistency check that is not used in the proof, an independent SAT enumeration of all ZE partitions of the thin chain returned exactly the $m$ partitions $P_a$ for $m = 7, \dots, 14$, for $m = 18, 26, 42$, and for a sample of larger $m$.

**Translation invariance, precisely.** The member list of $\mathrm{OH}_m$ consists of a *left cap* (the members on the first few cells), a *right cap* (the members on the last few cells) and the interior families $P_j, G_j, F_j$. The map $s_t \mapsto s_{t+2}$, $e_t \mapsto e_{t+1}$ shifts every cell by one and is an isomorphism between the sub-instances induced on the cells $\{j, j+1, j+2\}$ and $\{j+1, j+2, j+3\}$ whenever both windows avoid the caps. Consequently: (i) the constraints at the cells of the left cap involve only the cap members and the first interior members, the same for every $m$, so *left cap windows are independent of* $m$; (ii) the constraints near the right end involve the right cap and the last interior members, the same for all $m$ after the shift of the index by $m$, so *right cap windows depend only on the distance to the right end*; (iii) every other window is interior and isomorphic to every other interior window. In (CA1) every window type of (i)–(iii) occurs for $m \leq 20$, so a check of all $m \in [7, 60]$ is a check of every window type, and the set of window types does not grow for larger $m$. In (CA2) the vector $u$ after the left cap is independent of $m$, the interior transfer matrix is the same $M$ at every interior cell (by the isomorphism above), and the right-cap vector $v$ depends only on the distance to the right end, so $N_m = u^\top M^{m-c_0} v$ with $u, v, c_0$ fixed. The finite computation and the affine form of $M^t$ then determine $N_m$ for all $m \geq 7$. This is a finite certificate combined with the invariance argument above, not a closed-form proof.

## A.2 Proof of the profile lemma

*Proof of Lemma 5.4.* Read off the definition of $P_a$: the vertex $s_{4u+2}$ is $s_{2j+2}$ for $j = 2u$ and $s_{2j}$ for $j = 2u+1$. If $a < 2u$ the cluster $C_{2u}$ has the form $\{s_{4u+1}, s_{4u+2}\} \cup \{\varepsilon(2u)\}$ and $C_{2u+1} \not\ni s_{4u+2}$; if $a = 2u$ the cluster is $C_{2u} = \{s_{4u}, s_{4u+1}, s_{4u+2}, \varepsilon(2u)\}$; if $a = 2u+1$ it is $C_{2u+1} = \{s_{4u+2}, s_{4u+3}, s_{4u+4}, \varepsilon(2u+1)\}$; if $a \geq 2u+2$ it is $C_{2u+1} = \{s_{4u+2}, s_{4u+3}, \varepsilon(2u+1)\}$. At the caps, the label $\varepsilon$ takes the exceptional values stated before Lemma 5.3. **(CA3)**: the statement was in addition verified by computer for all even $m$ with $8 \leq m \leq 60$ and all $u$ (1,836 cases), including the groups $u = 0$ and $u = U - 1$. □

## A.3 Proof of the parameter lemma

*Proof of Lemma 5.8.* (a) Every pair between guests and super-vertices, and every pair inside a super-vertex that is used as a pad or host pair, lies in exactly one member because different guests use different classes of the one-factorisation at the same $v$ and a pair of a matching lies in exactly

one pad (or host) member; Latin members are pair-disjoint as shown above and have no pair inside a super-vertex; the single-vertex members $(x, y, z)$, $(z, y, z')$ use pairs between guests and anchors only. Every member $(a, b, c)$ has the edges $ab, bc$ and the non-edge $ac$ by definition, and these adjacencies are consistent because a soft pair belongs to one member.

(b) **(CA4)** A hand count from Lemma 5.4: the sets $W$ contain $h_u$, $\Pi_0, \Pi_3$ (pad), $\Pi_0$ (X), $\Pi_3$ (Y) or all $\Pi_i$ (All) of a single group $u$, so a super-vertex is used only by guests of the group $u$, except $s_{4u+4} = s_{4(u+1)}$, which occurs in $\Pi_2$ of the group $u$ and $\Pi_1$ of the group $u+1$. A group carries: a tile group the permanent guest $z$ and a lock guest $y$ (both $W \ni h_u$); a separator group (with $W_z = \mathrm{Pad}_u$) the permanent guest and at most two comparator guests (one with the chain as first chain, using $X_u$, one as second chain, using $Y_u$); any other group only $z$. At the host $h_u$: the permanent guest $z$ (host matching) plus either the lock guest $y$ of a tile group (a tile group carries no comparator), giving $u = 2$, or, at a separator group, the up to two comparator guests (one with $h_u$ in $X_u$ of the first chain, one in $Y_u$ of the second), giving $u \le 3$; a group is a tile group, a separator group or neither, never two of these. At $s_{4u+1}, \varepsilon_u^-$: $z$ plus $y$ or the comparator guest of the first chain: at most 2; at $s_{4u+3}, \varepsilon_u^+$: $z$ plus $y$ or the comparator of the second chain: at most 2; at $s_{4u}, s_{4u+4}$: only in All, i.e. only the permanent guests of tile groups, at most one from each of two groups: at most 2. If $k = 1$ there are no comparators, and if $k = 2$ each chain has at most one neighbour, so the host has at most 2. So a host carries 2 (host matching and lock-guest pad) or 3 (host matching and two comparator pads at a separator) matchings, and every non-host super-vertex at most 2. The value 3 occurs, at a separator host, only for $k \ge 3$; the count of $u_{\max}$ by computer (thin-guest model, all tiles present, $1 \le k \le 5$, $2 \le n \le 4$) gave 2 for $k \le 2$ and 3 for $k \ge 3$, never more. The thin member counts: a vertex lying in four thin members is an $e$-vertex, and the hosts $s_{4u+2}$ lie in three; this, and the above counts, were confirmed by computer for $k \le 4$ and $n \le 3$ (maximum of $u(v)$ and of the number of thin members through every vertex, with the exceptional values at the caps).

(c) A vertex of $K_v$ has glue degree at least $p - 1 - u(v)$. For $p \ge 8$ and $u(v) \le 3$ this is at least $p - 4 \ge (p-1)/2$, hence the glue graph is connected (diameter at most 2). For $p = 6$ we have $k = 1$, so $u(v) \le 2$ and the degree is at least $3 \ge (p-1)/2$. At most $3 \le p - 1$ matchings are used at any vertex.

(d) A thick vertex $(v, i)$ lies in $p$ Latin members of each thin member through $v$, and in $u(v)$ guest members. By (b) this is at most $4p+2$ at a vertex in four thin members and at most $3p+3 \le 4p+2$ otherwise. The sizes: $2k^2$ chains with $O(n^2)$ cells each and $O(p)$ thick vertices per thin vertex; each thin member gives $p^2$ Latin members; guests contribute $O(p)$ members each. □

## A.4 Proofs of the pathwidth and pair-maximality lemmas

*Proof of Lemma 5.13.* Members of different chains share a vertex only through a comparator guest or through its pad members (they share vertices with Latin members of both chains); tunnels are hard pairs and are not members. This gives the components. Define a *level* $\lambda(M)$ of a member: for a Latin member of a thin member, the smallest cell of the thin member (all three cells lie in $[\lambda, \lambda+2]$); for every member belonging to a guest of the group $u$ (host, pad, lock, comparator members), $\lambda = 2u$. A guest of group $u$ has pad and host vertices in cells $[2u, 2u+2]$ (the vertices $s_{4u}, \dots, s_{4u+4}, \varepsilon_u^{\pm}$), so for every thick vertex $(v, i)$ all members through it have levels in $[\mathrm{cell}(v) - 2, \mathrm{cell}(v)]$, and all members through a guest have the same level. Hence two members that share a vertex have levels differing by at most 2. For a component $Q$ put $B_j = \{M \in Q : \ j \le \lambda(M) \le j+2\}$; every member lies in the bags $j \in [\lambda(M)-2, \lambda(M)]$, an interval, and two members sharing a vertex have a common bag; so $(B_j)_j$ is a path decomposition. At one level a chain has at most 6 thin members (at most 6 at the right cap, at most 3 elsewhere), hence at most $6p^2$ Latin members. At an even level $2u$, the

guest members of one chain are at most $3.5p$ (permanent guest) $+2.5p+1$ (lock) in a tile group and $2.5p$ in a separator group, and the comparator guests of a row or column at a separator group contribute at most $k-1$ guests with at most $3p+1$ members each. So a level of a component has at most $6kp^2+k(6p+1)$ members, and $|B_j| \le 3 \cdot 6kp^2 + 2k(6p+1) \le 24kp^2$ for $p \ge 6$. The bound does not depend on $n$ because the number of members at one level is a function of $p$ and $k$ only: the cell structure is the same in every chain, guests and comparators of a group sit at the same level $2u$ in every chain, and the level range of a member is at most 2. □

*Remark* A.2. As a consistency check that is not used in the proof, the path decomposition with bags $\{M : \lambda'(M) \le t \le \lambda''(M)\}$ (where $[\lambda'(M), \lambda''(M)]$ is the interval spanned by the coordinates of the vertices of $M$) was verified by computer for small parameters: for every vertex the intervals of the members through it have a common point, every member spans at most 2 cells, and for $k=1$, $p=8$ and $n=2,\dots,5$ the largest bag has between 1296 and 1304 members, independent of $n$, as the proof states.

*Proof of Lemma 5.14.* An induced $P_3$ with three hard pairs consists of two hard edges sharing a vertex and a hard non-edge. The hard edges form the graph $\Gamma_h$ whose components are: (a) for a thin vertex without a thin hard edge, the glue of $K_v$; (b) for each of the four thin hard edges $uv$ of a chain, $K_u \cup K_v$ with the glue inside both and all edges between them; (c) the tunnels $xx'$ (the anchors carry no other hard edge, guests carry none). In the components (a) and (b) the pairs that are not adjacent are exactly the pad and host pairs inside a super-vertex, which are soft; a component (c) is a single edge. So in every component every non-adjacent pair is soft, and an induced $P_3$ with two hard edges has its three vertices in one component of $\Gamma_h$, hence its third pair is soft or an edge. Hence no induced $P_3$ has all pairs hard. (The statement was also confirmed by computer for small parameters.) □

# B Soft triangles: the remaining statements

This appendix collects the statements of the soft-triangle analysis that are not in Section 6: Corollary B.1, the experimental Remark B.2, the far-lift model, the five-state exact model and the abstract models, and certificates for far edits. Conventions are those of Section 6; the proofs are in Appendix C.

**Corollary B.1** (Tunnel releases are fixed-parameter tractable without soft triangles)**.** *In the tunnel-release model of Appendix N, for a soft-triangle-free packing the minimum number of cuts is computable in time $2^{\ell}n^{O(1)}$, with no dependence on $c$ or $w$.*

*Proof.* A set $S$ of tunnels is lockable iff the anchors at the ends of its tunnels in $V(H)$ form a face of $G$ with all tunnels removed (Lemma N.3). By Corollary 6.7 a non-face contains a non-face of at most two anchors, hence of at most two tunnels, so every core has at most two tunnels. Proposition N.9(iv) with $d=2$ applies, with the $\ell=0$ test of Corollary 6.5. □

*Remark* B.2 (With triangles at $c \le 2$: experimental evidence only). For $c \le 2$ with triangles in $I(H)$ the set $\Sigma$ can fail to be closed under a given majority (Example N.11). Whether it is always 2-*decomposable* (equal to the set of vectors whose binary projections lie in the binary projections of $\Sigma$), which is what Lemma N.6 actually needs, is open. Experimentally, all tight triangle blocks and tight $K_4$ gadgets, random tight instances on the triangular prism and random instances with $c \le 2$ gave 2-decomposable $\Sigma$ and no core with three anchors, although some of the sets $\Sigma$ of the blocks are not closed under all majorities; among random instances with soft triangles the non-2-decomposable

ones all had $c = 3$. Theorem 6.11 proves the corresponding Helly property for the unit locks that govern certificates at $\ell = 0$.

## B.1 Far hard pairs: what they can and cannot do

Call a hard pair $\{p, r\}$ a *cross pair* if $p \in V(A) \setminus \{q\}$ and $r \in V(B) \setminus \{q\}$ for two members $A \neq B$ with $V(A) \cap V(B) = \{q\}$ (these are the cross bits of Lemma 6.3); every other hard pair is *far*. A far pair never lies in a binary triple.

**Lemma B.3** (U-only identity and locality of hard edits)**.** *Call a solution* U-only *if every member receives exactly one edit, so that its other edits form a set $U$ of hard pairs. For every packing, the least size of a U-only solution is $|H| + \min\{|U| : \Sigma(G \triangle U, H) \neq \emptyset\}$. For soft-triangle-free $(G, H)$, $\Sigma(G \triangle U, H)$ differs from $\Sigma(G, H)$ only through (a) the static triples (these gate non-emptiness: an all-hard induced $P_3$ created or destroyed by $U$ empties or un-empties $\Sigma$ globally), (b) the lists $\Lambda_A$ of the members $A$ that contain an end of a pair of $U$, and (c) the relations $R_{AB}$ of adjacent members $A, B$ for which $U$ contains a cross pair of $(A, B)$. In particular a far pair changes only static triples and the lists at the members through its two ends.*

*Proof.* $U$ avoids soft pairs, so every member is still an induced $P_3$ of $G \triangle U$, with the same pairs, and a vector is the single-edit part of a solution of $G$ with hard edits $U$ iff it lies in $\Sigma(G \triangle U, H)$. A hard pair $e$ enters only the triples containing it; by Lemma 6.3 these are static, unary on a member through an end of $e$, or (if $e$ is a cross pair of $(A, B)$) in a binary triple of $R_{AB}$ or in a list of $A$ or $B$. □

The model is a sub-model: one edit per member is not always optimal, as in Remark 3.7; for instance $n = 5$, $E = \{02, 03, 12, 23, 24\}$, $H = \{(1, 2, 4)\}$ has $\mathrm{OPT} = |H| + 1$ but least U-only value $|H| + 2$ (by enumeration of all partitions).

**Lemma B.4** (Strictness)**.** (by computer; finite check) *Let $A, B$ be adjacent members of a packing, $R$ the relation $R_{AB}$, and $\Lambda_A, \Lambda_B$ the lists that the cross bits of $(A, B)$ impose on $A$ and on $B$. For $X \subseteq \{1, 2, 3\}$ put $\mathrm{img}(X) = \{s_A : \exists s_B \in X \cap \Lambda_B,\ (s_A, s_B) \in R\} \cap \Lambda_A$. If $X \subseteq Y \subseteq \{1, 2, 3\}$ and $\mathrm{img}(X) \neq \mathrm{img}(Y)$, then $|\mathrm{img}(X)| \leq 1$.*

Across an edge of $I(H)$, a restriction of the neighbour's list that really changes the image leaves at most one state; locks propagate as unit propagation along edges of $I(H)$. This is a one-edge statement, verified by computer for all 144 configurations and all pairs $X \subseteq Y$. It is used only through Lemma B.5.

**Lemma B.5** (Helly number two for unit lists)**.** *Let $\Sigma \subseteq D^V$ be closed under some majority operation (for any finite domain $D$) and let $J$ be a family of unit constraints $x_{v_j} = s_j$. If $\Sigma$ and $J$ are jointly infeasible, then $\Sigma$ together with some at most two members of $J$ is infeasible.*

*Proof.* Take a minimal infeasible $J$ with $|J| = k \geq 3$ and, for each $j$, $t_j \in \Sigma$ satisfying $J \setminus \{j\}$. For three indices $a, b, c$ put $u = m(t_a, t_b, t_c) \in \Sigma$. For $j \notin \{a, b, c\}$ all three tuples have value $s_j$ at $v_j$; at $v_a$ the tuples $t_b, t_c$ both have value $s_a$, so $u$ has $s_a$ there (and symmetrically for $b, c$). Hence $u$ satisfies all of $J$, a contradiction. (If two constraints concern one coordinate with different values, the same argument applies.) □

**Observation B.6** (Lists of an outside vertex)**.** *Let $A = (a, b, c)$ and let $t \notin V(A)$ be such that the three pairs $\{x, t\}$, $x \in V(A)$, are hard; put $S = N(t) \cap V(A)$. The three triples $\{a, b, t\}, \{b, c, t\}, \{a, c, t\}$ allow exactly the following states of $A$:*

| $S$ | $\emptyset$ | $\{a\}$ | $\{b\}$ | $\{c\}$ | $\{a,b\}$ | $\{a,c\}$ | $\{b,c\}$ | $\{a,b,c\}$ |
|---|---|---|---|---|---|---|---|---|
| *allowed in* $\{1,2,3\}$ | $\{1,2,3\}$ | $\{1\}$ | $\emptyset$ | $\{2\}$ | $\{2\}$ | $\emptyset$ | $\{1\}$ | $\{3\}$ |
| *allowed in* $\{1,\dots,5\}$ | $\{1,\dots,5\}$ | $\{1,4\}$ | $\{4,5\}$ | $\{2,4\}$ | $\{2\}$ | $\{5\}$ | $\{1\}$ | $\{3\}$ |

*(Direct check; the second row is for the five states of Lemma 3.3: $S$ must be empty or a single part of the cluster partition of $\{a,b,c\}$ after the edit.) So a zero-excess list of an outside vertex is trivial, empty, or a* unit *list; there are no two-element lists. This is why forked locks never occur for zero excess.*

**Theorem B.7** (Far-lift zero-excess model)**.** *Let $(G,H)$ be soft-triangle-free, with $c$ and $w$ arbitrary. It is decidable in time $6^\ell n^{O(1)}$ whether there is a set $W$ of at most $\ell$ far pairs with $\Sigma(G\triangle W,H)\neq\emptyset$ (the* far-lift model*). A YES answer gives a solution of CEaMP of size $|H|+|W|$.*

*Proved* (Appendix C.2). Algorithm: (1) if $G'=G\triangle W$ has an all-hard induced $P_3$, branch on its far pairs (none: NO); (2) otherwise accept if $\Sigma(G')\neq\emptyset$ (Corollary 6.5); (3) delete all *liftable* unit lists (those of Observation B.6 for which at least one of the three pairs is far); if the rest is already infeasible, answer NO; (4) otherwise find a minimal infeasible family of liftable lists, which has at most two members by Lemma B.5 (the rest of $\Sigma$ is closed under all majorities by Theorem 6.4), and branch on the at most 3 far pairs of each: at most 6 branches, depth at most $\ell$. A far pair changes only liftable lists (Lemma B.3), so some far pair of the family must be edited. *Experimental check*: the algorithm agreed with exhaustive search on random small instances; the largest minimal family had two lists.

The model is restricted (no double edits, no edits of cross pairs): it is an upper bound on the excess, exact when some optimum uses neither. It contains the tunnel-release model of Corollary B.1 and removes the need for tunnels: *arbitrary* far hard edits, including those that close static violations, are handled. It also shows what the cores of far edits look like: a pair of unit lists, a vertex-cover-like structure whose branching factor depends neither on $n$, nor $c$, nor $w$.

## B.2 Five states, an exact model, and the abstract models

By Lemma 3.3 the variables of the dynamic programs of Section 3 range over five states: besides $1,2,3$, the state 4 (edits $ab,bc$; extra cost 1) and 5 (edits $ab,bc,ac$; extra cost 2). For a graph $G'$ with the same packing let $\mathrm{Ex}(G')$ be the least value of $|F|+\sum_i \mathrm{extra}(x_i)$ over sets $F$ of cross pairs and state vectors $x\in\{1,\dots,5\}^H$ such that $G'\triangle F\triangle$(soft edits of $x$) is a cluster graph ($\infty$ if none); extra $=0,0,0,1,2$ for the states $1,\dots,5$.

**Lemma B.8** (Exactness)**.** *For every packing,* $\mathrm{OPT}(G)=|H|+\min_W\big(|W|+\mathrm{Ex}(G\triangle W)\big)$*, where $W$ ranges over sets of far pairs.*

*Proof.* Split a solution as in Lemma 2.2 into its soft edits (one non-empty edit set per member, which by Lemma 3.3 is a state; its cost is $1+$ extra), its edits of cross pairs $F$, and its edits of far pairs $W$. Conversely every $(W,F,x)$ is a solution of that size. □

Hence the *far* edits are the only edits whose position is not controlled by the dynamic program, as in Appendix I.

**Breaks.** Let RM2 be the following abstract model on a forest: a set of *conflicts* (pairs of nodes, with the path joining them) and two kinds of operations, a *break* at a node (cost 1; it hits every conflict whose path contains the node) and a *tunnel*, a link in a graph of maximum degree $d$ (cost 1; it hits every conflict that has an end at one of its two nodes). We want at most $L$ operations hitting all conflicts.

**Proposition B.9** (RM2 is fixed-parameter tractable)**.** *RM2 is solvable in time* $(2d+1)^L n^{O(1)}$ *by a search tree.*

*Proof.* Root every tree and let $Q$ be an unhit conflict whose top node $v$ is deepest. If an optimal solution hits $Q$ by a break at a node $w \in Q$, replace it by a break at $v$: every other unhit conflict $R$ through $w$ has its top at depth at most $\mathrm{depth}(v)$ and contains $w$, hence contains the whole path from $w$ to its top, which passes through $v$. If $Q$ is hit by a tunnel, the tunnel has an end at an end of $Q$ ($2d$ choices). So the search tree branches $2d+1$ ways to depth $L$. □

(The search tree agreed with exhaustive search on random small instances.) RM2 is a natural abstraction of CEaMP on forests; it is not a faithful one.

**Proposition B.10** (RM2 is not faithful)**.** *Let* $n = 11$ *(vertices* $0, \ldots, 10$*), let* $G$ *have the twelve edges*

$$01,\ 02,\ 04,\ 05,\ 06,\ 13,\ 28,\ \{2,10\},\ 34,\ 37,\ 38,\ 78$$

*and let* $H = \{M_0, M_1, M_2, M_3\}$ *with* $M_0 = (2,0,1)$, $M_1 = (4,3,1)$, $M_2 = (5,0,6)$, $M_3 = (2,8,7)$. *This instance is valid, pair-maximal,* $I(H)$ *is a star with centre* $M_0$, $c = 2$ *and there is no soft triangle. Then* $\mathrm{OPT} = |H| + 3$ *(enumeration of all set partitions). The members* $M_1$ *and* $M_2$ *cannot be in any of the states* $1, 2, 3$ *in a solution without hard edits (an outside vertex gives them the list* $\{4,5\}$ *of the second row of Observation B.6), while* $M_0, M_3$ *can; each of* $M_1, M_2$ *is repaired alone by one break (state* $4$*), so the break model predicts excess* $2$*, but the true excess is* $3$*: two breaks next to a common member put the cluster-views* $(0,0)$ *at two vertices of* $M_0$*, which then cannot be in a state of* $\{1,2,3\}$ *either. The cheapest vectors without hard edits are* $(1,5,4,1)$ *and* $(4,4,4,1)$*, both of cost* $7$*.*

The proposition (verified by exhaustive enumeration) says that a break is not a node-hit: it locks its neighbours. Consequently the abstract model of "hitting conflict paths by breaks and tunnels" cannot be used to decide CEaMP on forests, and the difficulty of the interior of a chain, where breaks are placed, stays inside the dynamic program (as the Core-Bound analysis below also shows).

**The full problem on forests, conditionally.** Let $I(H)$ be a forest ($w \le 1$, so $c \le 2$, $D = 2c \le 4$, no soft triangle). The *sites* of $(G, H)$ are the members and the edges of $I(H)$; a triple that contains a far pair has as sites the owners of its soft pairs and the edges at which its cross pairs lie. For a set $C$ of sites let $\mathrm{Ex}_C(G')$ be $\mathrm{Ex}(G')$ computed after deleting every triple that contains a far pair and has no site in $C$; $\mathrm{Ex}_C$ is monotone in $C$. Let $V(C)$ be the set of vertices of the members in $C$ and of the members incident to an edge in $C$.

**Hypothesis BC$(t)$ (bounded certificates; open).** There is $K(t, D)$ such that for every $G'$ (with the same forest $H$, and no static violation) with $\mathrm{Ex}(G') > t$, every inclusion-minimal set $C$ of sites with $\mathrm{Ex}_C(G') > t$ has $|C| \le K(t, D)$.

**Assumption (O).** $\mathrm{Ex}_C(G')$ can be computed in time $f(w)n^{O(1)}$, by a dynamic program over a tree decomposition whose bags are enlarged by the members at distance one. (Every triple with a soft pair or a cross pair has its vertices in one such bag; this is assumed and not proved here.)

**Theorem B.11** (Forests, conditional)**.** *Let* $I(H)$ *be a forest. Algorithm* $\mathcal{A}$ *of Appendix C.3 decides CEaMP; it is correct unconditionally (given the oracle). Its running time is* $(6KZ^2)^\ell \cdot T_{\mathrm{O}} \cdot n^{O(1)}$*, where* $Z = 3D + 3\ell + 1$*,* $T_{\mathrm{O}}$ *is the time of the oracle (O) and* $K$ *is the largest minimal set* $C$ *met. Hence, if BC*$(t)$ *holds for* $t < \ell$ *and (O) holds, CEaMP on forests is fixed-parameter tractable in* $\ell$*.*

*Proved implication*; BC and (O) are not proved. The hypothesis BC affects the *running time only*, not correctness. BC(0) holds with $K = 2$ (Lemma B.5). The case $\ell \leq 1$ is already polynomial by Theorem 3.6 and is not progress. Experimental evidence for BC: in computer searches minimal certificates had at most $t+2$ sites. A proof would need a weighted Helly statement for the five-state problem; breaks interact through locks (Proposition B.10), so an exchange argument in the style of Proposition B.9 fails.

### B.3 Certificates for far edits: a branching scheme and its unbounded branching

We now use a different split of the hard pairs, tuned to a branching algorithm. For a vertex $v$ let $\mathcal{M}(v)$ be the set of members through $v$. A pair $\{u, v\}$ is *local* if $u, v \in V(H)$ and some member through $u$ equals or shares a vertex with some member through $v$, and *remote* otherwise; soft pairs are local, every pair with an end outside $V(H)$ is remote. Let $\mathcal{N}$ be the set of remote hard pairs. For $T \subseteq \mathcal{N}$ let $F'(T)$ be the least number of edits in an assignment of a status to every soft pair (each member gets a non-empty subset of its pairs) and to every local hard pair, such that every vertex triple *all of whose remote pairs lie in* $T$ is not an induced $P_3$ (remote pairs of $T$ keep their status in $G$; triples with a remote pair outside $T$ are dropped).

**Lemma B.12** (Certificates)**.** *(a) $F'$ is monotone: $T \subseteq T'$ implies $F'(T) \leq F'(T')$. (b) If $S$ is a solution with $S \cap T = \emptyset$, then $F'(T) \leq |S|$. (c) $F'(\mathcal{N})$ is the least size of a solution with no remote edit. Hence if $F'(T) > |H| + \ell$ every solution of size at most $|H| + \ell$ contains a pair of $T$ (such a $T$ is a* certificate*).*

**Theorem B.13** (Certificate branching)**.** *Algorithm: if $F'(\mathcal{N}) \leq |H| + \ell$, accept; if $\ell = 0$, reject; otherwise delete pairs from $\mathcal{N}$ one by one while $F' > |H| + \ell$ stays true, and for every pair $e$ of the resulting minimal certificate $T$ recurse on $(G \triangle e, \ell - 1)$. The algorithm is correct for every packing; depth at most $\ell$ and branching factor $d = \max |T|$ over the certificates met, so $d^\ell(|\mathcal{N}| + 2)$ oracle calls.*

*Proof.* (a) fewer constraints; (b) $S$ restricted to soft and local pairs is an assignment, and every triple with its remote pairs in $T$ has them unedited; (c) with $T = \mathcal{N}$ nothing is dropped. Soundness: a solution of $G \triangle e$ of size $\leq |H| + \ell - 1$ gives $S \triangle \{e\}$, and $H$ is still a packing since $e$ is hard. Completeness: if no solution of size $\leq |H| + \ell$ avoids all remote pairs then $F'(\mathcal{N}) > |H| + \ell$, a minimal certificate $T$ exists, and every solution contains some $e \in T$. □

So the question is how large minimal certificates are, and how $F'$ is computed. For the second, note that with this definition of *local* the difficulty that a triple with two local pairs and one remote pair need not lie in a bag of a decomposition of $I(H)$ does not arise.

**Theorem B.14** (The leaf test is fixed-parameter tractable)**.** *Given a tree decomposition of $I(H)$ of width $w$, $F'(T)$ for any $T \subseteq \mathcal{N}$, in particular $F'(\mathcal{N})$, is computable in time $f(w)n^{O(1)}$.*

*Proved* (Appendix C.4; radius 3 in $I(H)$ is attained and suffices). Remark: this is true for the *adjacency-defined* notion of local used here; for locality defined through the bags of a decomposition of $I(H)$ the corresponding statement is not proved. Together, Theorems B.13 and B.14 give an algorithm in time $d^\ell f(w)n^{O(1)}$; the question is the size of $d$.

**Theorem B.15** (Core bound at $\ell = 0$ without soft triangles)**.** *Let $H$ have no soft triangle and $\ell = 0$. Every minimal $T \subseteq \mathcal{N}$ with $F'(T) > |H|$ has $|T| \leq 12$ ($\leq 3$ if it is a static violation), for every $c$ and $w$.*

*Proved* (Appendix C.5; the proof uses the far pairs as *guards*, unit or two-element lists on one member, together with Theorem 6.4 and a Helly lemma for 2-element lists on a 3-element domain). The bound 12 is not claimed tight. At $\ell = 0$ the algorithm of Theorem B.13 never branches, so Theorem B.15 is a statement about the structure of $\ell = 0$ certificates; its algorithmic use needs an $\ell \geq 1$ version, which is open (an Erdős–Pósa-type argument has not been completed). *Experimentally*, in random soft-triangle-free instances the largest minimal inconsistent family of guards had two members, and minimal certificates found by deletion had at most 4 pairs.

With soft triangles the corresponding statement is false, already at $c = 2$ and $w = 2$.

**Theorem B.16** (Unbounded minimal certificates)**.** *For every $N \geq 1$ there is a graph $G_N$ with a pair-maximal packing $H_N$ of $3N$ members ($c = 2$, $\mathrm{tw}(I(H_N)) = 2$, $N$ soft triangles, $6N+3$ vertices) and a set $T_N$ of $2N + 4$ remote pairs with $F'(T_N) \geq |H_N| + 1$ and $F'(T_N \setminus \{e\}) = |H_N|$ for every $e \in T_N$. So $T_N$ is a minimal certificate for $\ell = 0$ of unbounded size, and the minimal-certificate branching of Theorem B.13 has unbounded branching factor.*

*Proved* (Appendix C.6; the block structure of the set of ZE vectors of $G_N$ is by computer, for the first blocks and by isomorphism of interior blocks). Two comments. (1) The proof gives $F'(T_N) \geq |H_N| + 1$; the equality $F'(T_N) = |H_N| + 1$ was verified by computer ($N \leq 10$) and is not needed. (2) At $\ell = 0$ the set $T_N$ is never branched on. For $\ell = 1$: the union of two disjoint copies, $T = T_N \cup T'_N$, is a minimal certificate of size $2(2N + 4)$ because $F'$ is additive over disjoint unions (every triple meeting two copies has a remote pair outside $T$ and is dropped); and for *connected* instances, a chain $G_{2N}$ with one further pendant $w_*$ adjacent to the glue vertex $b_N$ and the remote pairs $w_*b_N, w_*r_N, w_*p_{N+1}$ added to $T_{2N}$ gives $|T| = 4N + 7$ with $F'(T) - |H| = 2 > 1$ and $F'(T \setminus \{e\}) - |H| = 1$ for every $e$ (verified with exact solvers for $N = 1, \ldots, 4$). So the branching factor of Theorem B.13 with an arbitrary minimal certificate is unbounded already at $\ell = 1$.

The mechanism is a chain of soft triangles that carries one bit through the glue vertices; a 2-element guard $\{1, 2\}$ ("the pair $ac$ is apart", obtained from a pendant vertex adjacent to one end of the pair) switches off the escape state 3 of its block. This is exactly the *multi-valued unary guard* that cannot occur without a soft triangle (Theorem B.15).

*Remark* B.17 (Not a hardness result; minimum certificates). In $G_N$ the real optimum is $|H| + N$: every pendant edge forces its anchor to be a singleton, impossible for a centre, so all $N$ stubs are cut; the persistent bit exists only in the relaxation. Moreover there is always a small certificate there: if a far hard edge $wx$ ends at the centre $x$ of a member $(y, x, y')$ and $wy, wy'$ are far hard non-edges, then $\{wx, wy, wy'\}$ is a certificate at $\ell = 0$ (both pairs at $x$ would have to be apart, two edits in one member). Every $G_N$ has such a pendant at each centre of $A_j$. So Theorem B.16 refutes "every minimal certificate is bounded", not "some certificate is bounded"; this is a property of the instance design, and computing a minimum certificate is itself not known to be fixed-parameter tractable. Whether a certificate of bounded size can always be chosen is open.

**Lemma B.18** (No conservative near-unanimity polymorphism with soft triangles, $c \geq 3$)**.** *Let $R_k = \{111, 113, 122, 131, 212, 221, 222, 232, 311, 331\}$ (states written as three-digit words in $\{1, 2, 3\}$). Any constraint language that contains $R_k$ and all unary relations has no conservative near-unanimity polymorphism of any arity $r \geq 3$.*

*Proved* (Appendix C.7). The relation $R_k$ is the conjunction of all triple constraints whose scope lies in $\{A, B, C\}$ for three pairwise adjacent members $A, B, C$ on six vertices, in a packing with $c = 3$ that has four further members, each owning one of the four pairs of these six vertices that are neither in $A, B, C$ nor hard non-edges (an explicit instance, found and checked by computer). It requires $c \geq 3$; for $c = 2$, where Theorem B.16 lives, the lemma says nothing. What explains

Theorem B.16 at $c = 2$ is that a near-unanimity operation of arity $r$ bounds the cores of unit lists but not of 2-element lists, since it only controls coordinates where the other $r - 1$ values coincide. The lemma shows that the route through a uniform near-unanimity polymorphism is closed for the full language with soft triangles; it is not a hardness proof.

# C Proofs for Section 6

Notation and the labels "proved", "by computer" and "experimental" are as in Section 6.

## C.1 Proof of Theorem 6.4

By Lemma 6.3, $\Sigma$ is empty or the intersection of unary lists and of binary relations $R_{AB}$; it suffices to show that each of these is closed under every majority operation $m$. A static violation makes $\Sigma$ empty, which is closed.

*Lists.* Let $S \subseteq \{1, 2, 3\}$ and $u, v, w \in S$. If two of them coincide then $m(u, v, w)$ is their common value, which lies in $S$. If all three are distinct then $S = \{1, 2, 3\}$.

*Binary relations (classification by computer).* $R_{AB}$ is determined by the roles of the shared vertex $q$ in $A$ and in $B$ (three each) and the four cross bits, 144 configurations. Computing the relation from the formula of Lemma 6.3(iii) gives exactly 40 distinct relations, with the following shapes, where $p, q$ range over $\{1, 2\}$ (the states that edit a pair at the shared vertex):

| shape | number |
|---|---|
| $\emptyset$ | 1 |
| singletons | 9 |
| two-element sets | 18 |
| rows $\{p\} \times \{1, 2, 3\}$, columns $\{1, 2, 3\} \times \{q\}$ (4) and the 3-element subsets of $\{1, 2\}^2$ (4) | 8 |
| crosses $\big(\{p\} \times \{1, 2, 3\}\big) \cup \big(\{1, 2, 3\} \times \{q\}\big)$ | 4 |

The same list results from an independent construction: all 576 graphs on five vertices consisting of two members sharing exactly one vertex (all $6 \times 6$ orientations, 16 patterns of the four cross pairs), with each triple $\{p, q, r\}$ tested for being an induced $P_3$ after editing, without the formula.

*Closure of each shape.* Let $m$ be a majority and $t_1, t_2, t_3 \in R$, $u = m(t_1, t_2, t_3)$. (1) $|R| \leq 2$: at every coordinate at most two values occur among $t_1, t_2, t_3$, so $u$ is the coordinatewise majority; if, say, $t_1$ occurs at least twice among the three tuples, then $u = t_1 \in R$. (2) A row $\{p\} \times \{1, 2, 3\}$: the first coordinate is constantly $p$, so $u_1 = p$ and $u \in R$; columns are symmetric. (3) A 3-element subset of $\{1, 2\}^2$: all values lie in $\{1, 2\}$, so $m$ acts as Boolean majority, and every binary relation on $\{1, 2\}$ is closed under it (it is definable by clauses with at most two literals). (4) A cross $R = (\{p\} \times \{1, 2, 3\}) \cup (\{1, 2, 3\} \times \{q\})$: suppose $u_1 \neq p$; we show $u_2 = q$. If two of the three first coordinates coincide, $u_1$ is their value, which differs from $p$, so these two tuples are not in the row and are of the form $(x, q)$; two second coordinates equal $q$, so $u_2 = q$. If the three first coordinates are distinct, at most one of them equals $p$, so at least two tuples are of the form $(x, q)$ and again $u_2 = q$. The case $u_2 \neq q$ is symmetric. So $u_1 = p$ or $u_2 = q$, i.e. $u \in R$. □

*Orientation.* The state names $1, 2$ depend on the orientation $(a, b, c)$ versus $(c, b, a)$ of a member (they are exchanged). The set of all majority operations is invariant under relabelling of the states by any permutation, so closure under *all* majorities does not depend on this convention (closure under one fixed operation would).

## C.2 Proof of Theorem B.7

Call a pair of vertices a *cross pair* or *far* as in Appendix B.1. For a graph $G' = G \triangle W$ and a member $A = (a, b, c)$ and vertex $t \notin V(A)$ whose three pairs $\{x, t\}$, $x \in V(A)$, are hard, the *list of* $(A, t)$ is the set of states of Observation B.6. It is *liftable* if at least one of the three pairs is far. A far pair lies in no binary triple (the third pair of a binary triple is a cross pair) and in no triple with two soft pairs. A triple $\{x, y, t\}$ with one soft pair $xy$ of $A = (x, y, z)$ (up to naming) and a far pair $xt$ also has $yt$ and $zt$ hard: if $yt$ (or $zt$) were soft, with owner $B$, then $B$ shares the vertex $y$ (or $z$) with $A$ and $xt$ would be a cross pair. So the triple belongs to the list of $(A, t)$, which is liftable. By Lemma B.3 a far pair therefore changes only static triples and liftable lists. Let $\Sigma_{\text{perm}}(G')$ be the set of vectors satisfying all constraints of $G'$ except the liftable lists; it is closed under every majority operation by Theorem 6.4 (only unary lists and the relations $R_{AB}$ remain), and $\Sigma_{\text{perm}}(G \triangle W) = \Sigma_{\text{perm}}(G)$ for every set $W$ of far pairs.

*Algorithm* solve$(G', b)$, deciding whether some set $W$ of at most $b$ far pairs has $\Sigma(G' \triangle W) \neq \emptyset$. (1) If $G'$ has an induced $P_3$ with three hard pairs, branch on its far pairs, $b \to b - 1$ (if there is none, or $b = 0$, answer NO). (2) Else if $\Sigma(G') \neq \emptyset$, answer YES; if $b = 0$, NO. (3) If $\Sigma_{\text{perm}}(G') = \emptyset$, NO. (4) Else choose a minimal family $\mathcal{F}$ of liftable lists with $\Sigma_{\text{perm}}(G') \cap \mathcal{F} = \emptyset$ (greedy deletion). By Lemma B.5, $|\mathcal{F}| \leq 2$ (a list is a unit list, or empty; empty is a family of one). For each list of $\mathcal{F}$ and each of its far pairs $e$ (at most 3) call solve$(G' \triangle e, b - 1)$.

*Correctness.* Soundness is clear. Completeness by induction on $b$: let $W$ be a smallest set of at most $b$ far pairs with $\Sigma(G' \triangle W) \neq \emptyset$. If $W = \emptyset$ step (2) answers YES (static violations are absent by (1) and then $\Sigma \neq \emptyset$ is tested). Otherwise $b \geq 1$. A static all-hard induced $P_3$ of $G'$ has to be hit by a far pair of $W$ (only far pairs are edited), so (1) branches correctly. In (4), $W$ changes no relation and no permanent list, so $\Sigma(G' \triangle W) \subseteq \Sigma_{\text{perm}}(G') \cap \{\text{unchanged liftable lists}\}$; as $\Sigma(G' \triangle W) \neq \emptyset$ and $\Sigma_{\text{perm}}(G') \cap \mathcal{F} = \emptyset$, $W$ changes some list of $\mathcal{F}$, i.e. contains a far pair $\{x, t\}$ with $x \in V(A)$ of it. Recursing on $G' \triangle e$ with $W \setminus \{e\}$ completes the induction (the pair $e$ stays far in $G' \triangle e$, and a far pair is a far pair of $H$ whichever graph is considered).

*Time.* Each node takes polynomial time (Corollary 6.5), at most 6 children, depth at most $\ell$. □

## C.3 Algorithm $\mathcal{A}$ and the proof of Theorem B.11

Let $I(H)$ be a forest. Write $\text{Ex}_C(G')$ as in Appendix B.2. For a graph $G'$ (obtained from the input by flipping hard pairs) without an all-hard induced $P_3$, every triple whose three pairs are far is not an induced $P_3$, so it can be deleted without changing Ex; hence $\text{Ex}_C(G') = \text{Ex}(G')$ if $C$ is the set of all sites.

*Algorithm* $\mathcal{A}(G', r)$ (excess budget $r$; accept iff $\text{OPT}(G') \leq |H| + r$). (1) If $G'$ has an induced $P_3$ with three hard pairs: if $r = 0$ reject; else call $\mathcal{A}(G' \triangle e, r - 1)$ for each of its three pairs $e$ (cross pairs included; the recursion is on graphs, so this stays sound) and accept iff one call accepts. (2) Now $H$ is pair-maximal in $G'$. If $\text{Ex}(G') \leq r$, accept; if $r = 0$, reject. (3) Compute an inclusion-minimal set $C$ of sites with $\text{Ex}_C(G') > r - 1$ by greedy deletion (at most $2|H|$ oracle calls). (4) With $B = 2D + 3r$ let $L_B$ be the set of Appendix G, computed from $G', H$, and $P(x)$ the set of $y \in \text{Ball}_{G'[L_B]}(x, 2)$ with $xy$ far. For every $x \in V(C)$ and $y \in P(x)$ call $\mathcal{A}(G' \triangle xy, r - 1)$; accept iff one accepts.

*Soundness.* Acceptance in (2) gives a solution of size $\leq |H| + r$ by Lemma B.8 (take $W = \emptyset$); in (1) and (4) a solution of $G' \triangle e$ of size $\leq |H| + r - 1$ gives one of $G'$ of size $\leq |H| + r$.

*Completeness.* Suppose $\text{OPT}(G') \leq |H| + r$, there is no static violation and $\text{Ex}(G') > r$ (otherwise the algorithm accepts). In step (1), a solution contains one of the three pairs of the static $P_3$, and removing it gives a solution of $G' \triangle e$ of size $\leq |H| + r - 1$. In (4): by Lemma B.8 choose a set

$W$ of far pairs minimising $|W| + \mathrm{Ex}(G' \triangle W) \le r$, with minimising $F, x$; $(W, F, x)$ is an optimum of $G'$. As $\mathrm{Ex}(G') > r$, $W \neq \emptyset$ and $\mathrm{Ex}(G' \triangle W) \le r - 1$. Let $Z(W)$ be the set of members through an end of a pair of $W$ together with all edges of $I(H)$ at these members. Every triple that contains a pair of $W$ has all its sites in $Z(W)$ (an owner of a soft pair of it is a member through an end of the pair; a cross pair $\{x, z\}$ of it lies at an edge incident to a member through $x$ or $z$). Let $C'$ be the set of sites outside $Z(W)$. The triples kept in $\mathrm{Ex}_{C'}(G')$ contain no pair of $W$, so they have the same status in $G'$ and in $G' \triangle W$, and $(F, x)$ is feasible for $\mathrm{Ex}_{C'}(G')$ with the same cost: $\mathrm{Ex}_{C'}(G') \le \mathrm{Ex}(G' \triangle W) \le r - 1$. The set $C$ of step (3) has $\mathrm{Ex}_C(G') > r - 1$, so by monotonicity $C \not\subseteq C'$: $C$ contains a site of $Z(W)$, a member through an end $x$ of some $e = xy \in W$, or an edge incident to such a member, and in both cases $x \in V(C)$. Since $(W, F, x)$ is an optimum of $G'$ of size $\le |H| + r$ and $H$ is pair-maximal in $G'$, Corollary G.7 gives $y \in \mathrm{Ball}(x, 2)$ of $G'[L_B]$, so $y \in P(x)$. The recursion on $G' \triangle e$ with $W \setminus \{e\}$ satisfies $|W \setminus \{e\}| + \mathrm{Ex} \le r - 1$.

*Time.* The depth is at most $r \le \ell$; a node has at most $\max\{3, |V(C)|Z^2\} \le 6|C|Z^2$ children, $Z = 3D + 3r + 1$; each node makes at most $2|H| + 1$ oracle calls. Correctness did not use BC; only $|C| \le K$ does. □

## C.4 Proof of Theorem B.14

Variables of the CSP: the state of each member (seven non-empty edit sets, cost = size) and a bit for each local hard pair (cost 1 if edited); remote pairs are constants (in $T$) or absent. A vertex triple $\tau$ whose remote pairs all lie in $T$ gives the constraint "$\tau$ is not an induced $P_3$"; its scope consists of the owners of its soft pairs and the bits of its local hard pairs. For a triple $\tau$ let $P(\tau)$ be its non-remote pairs and $W(\tau)$ the union of $\mathcal{M}(u)$ over the ends $u$ of pairs of $P(\tau)$ (a set of members).

**Lemma C.1** (Triple scopes)**.** *If $P(\tau) \neq \emptyset$, then $W(\tau)$ lies in a ball of radius 3 of $I(H)$ around one member.*

*Proof.* One pair $uv$: there are $A \in \mathcal{M}(u)$ and $B \in \mathcal{M}(v)$ equal or adjacent; every member of $\mathcal{M}(u)$ is within distance 1 of $A$ (it shares $u$ with $A$), every member of $\mathcal{M}(v)$ within 1 of $B$, so $W(\tau)$ lies within distance 2 of $A$. Two pairs $uv, vw$ (they share $v$): there are $A \in \mathcal{M}(u)$, $B, B' \in \mathcal{M}(v)$, $C \in \mathcal{M}(w)$ with $d(A, B) \le 1$, $d(B', C) \le 1$, and $d(B, B') \le 1$ as both contain $v$. Around $B$: $\mathcal{M}(v)$ is within 1, $\mathcal{M}(u)$ within 2, $C$ within 2 and $\mathcal{M}(w)$ within 3. Three pairs: two of them share a vertex and already involve all three vertices of $\tau$, so the argument for two pairs applies. □

Radius 3 is attained: take six members forming a path $A'$–$A$–$B$–$B'$–$C$–$C'$ in $I(H)$ with $u \in A \cap A'$, $v \in B \cap B'$, $w \in C \cap C'$, so that $uv$ and $vw$ are local and $uw$ is remote; the minimal radius covering $W(\tau)$ is 3.

*Proof of Theorem B.14.* Let $(\mathcal{T}, \{\beta_t\})$ be the given decomposition and $B_t$ the set of members within distance 3 of $\beta_t$ in $I(H)$; $|B_t| \le (w+1)(1 + \Delta + \Delta^2 + \Delta^3)$ with $\Delta \le 3(c-1) \le 3w$. For a member $m$ the set $\{t : m \in B_t\}$ is the union of the subtrees $\{t : y \in \beta_t\}$ over the members $y$ within distance 3 of $m$; these subtrees are chained along the connected ball, so the union is a subtree. A bit for a local pair $uv$ is placed in the bags $t$ with $\mathcal{M}(u) \cup \mathcal{M}(v) \subseteq B_t$, an intersection of subtrees, non-empty: take $t$ with $A \in \beta_t$ for $A \in \mathcal{M}(u)$ adjacent to some $B \in \mathcal{M}(v)$; then $\mathcal{M}(u)$ and $B$ are within 1, $\mathcal{M}(v)$ within 2. By Lemma C.1, all variables of a triple are in $B_t$ for any $t$ whose bag contains the centre member of the lemma. Hence this is a tree decomposition of the primal graph, with at most $|B_t|$ members and $\binom{3|B_t|}{2}$ bits per bag, and min-sum dynamic programming over it (at most $7^{|B_t|} 2^{\binom{3|B_t|}{2}}$ states per bag) solves the CSP in time $f(w) n^{O(1)}$. Dropping triples with a remote pair outside $T$ is the definition of $F'(T)$. □

(As a consistency check, the leaf test and the algorithm of Theorem B.13 were compared with independent exact solvers on small random instances, with no disagreement; on random tree-like instances radius 3 sufficed.)

## C.5 Proof of Theorem B.15

Let $\ell = 0$. Then $F'(T) > |H|$ iff there is no zero-excess assignment (states in $\{1,2,3\}$, no local hard edit) satisfying all triples whose remote pairs lie in $T$.

**Lemma C.2** (Guards). *Let $\tau$ be a triple with a remote pair, $\ell = 0$. (i) If all three pairs are remote, or the non-remote pairs of $\tau$ are hard, then $\tau$ is a* static test*: violated iff $\tau$ is an induced $P_3$ of $G$; it is active iff its (at most three) remote pairs lie in $T$. (ii) If exactly one pair $xz$ of $\tau$ is non-remote and it is soft with owner $M$, with remote pairs $xy, zy$, the triple is a* guard*, a list on $M$ active iff $xy, zy \in T$: $t_{xz}(s_M) = 1$ if both are edges, $= 0$ if exactly one is an edge, no condition if none. For the three pairs of $M$ the lists are $\{2,3\},\{1,3\},\{3\}$ (for $t = 1$) and $\{1\},\{2\},\{1,2\}$ (for $t = 0$). (iii) If two pairs $uv, vw$ are non-remote they are not both soft (their owners would share $v$ and $uw$ would be a cross pair, hence local); one soft and one local hard gives a guard on the soft owner, active iff the remote pair $uw \in T$, of size 1 or 2; two local hard pairs give a static test.*

*Proof.* Read off the definitions: a local hard pair is unedited at $\ell = 0$ and a soft pair of a member in state $s$ is together iff $t_e(s) = 1$; the triple is not a $P_3$ iff the number of edges is not 2. □

Let $\Sigma_0$ be the set of zero-excess vectors that satisfy all triples with no remote pair (and are empty if one of these triples is a static violation). If $H$ has no soft triangle, $\Sigma_0$ is an intersection of unary lists and of the relations $R_{AB}$ (all four cross pairs are local), so it is closed under every majority operation (proof of Theorem 6.4).

**Lemma C.3** (Helly lemma for lists of size at most two). *Let $|D| = 3$ and $\Sigma \subseteq D^V$ closed under all majority operations. Let $L_i \subseteq D$, $|L_i| \le 2$, for $i$ in a finite index set $I$, such that $\Sigma \cap \prod_i L_i = \emptyset$ but for every $i$ there is $x^i \in \Sigma$ with $x^i_l \in L_l$ for all $l \neq i$. Then $|I| \le 3$.*

*Proof.* Suppose $|I| \ge 4$; then $x^i_i \notin L_i$. For $i,j,k \in I$ distinct and a majority operation $m$, $u = m(x^i, x^j, x^k) \in \Sigma$. For $l \notin \{i,j,k\}$ the three values lie in $L_l$, $|L_l| \le 2$, so two coincide and $u_l \in L_l$. At $i$: if $x^j_i = x^k_i$ then $u_i = x^j_i \in L_i$. Otherwise $L_i = \{x^j_i, x^k_i\}$ and $(x^i_i, x^j_i, x^k_i)$ is a permutation $\pi_i$ of $D$; $u_i \in L_i$ iff $m(\pi_i)$ differs from the *first* entry of $\pi_i$. Likewise at $j$ ($m(\pi_j) \neq$ second entry) and at $k$ (third entry). The values of $m$ on the six permutations are free, so a suitable $m$ exists unless $\pi_i = \pi_j = \pi_k$ (then $m(\pi)$ would have to avoid all three entries), i.e. unless $L_i, L_j, L_k$ are 2-lists whose missing values are pairwise distinct. Among four coordinates two missing values coincide in every triple that contains them, so some triple has a suitable $m$; then $u \in \Sigma \cap \prod_i L_i$, a contradiction. □

*Proof of Theorem B.15.* Let $T$ be a minimal certificate. If one of its active static tests is violated, the violated triple has at most three remote pairs, all in $T$ by minimality, so $|T| \le 3$. Otherwise the active guards of $T$ have empty intersection with $\Sigma_0$. Let $\mathcal{G}$ be a minimal family of active guards with $\Sigma_0 \cap \bigcap \mathcal{G} = \emptyset$ and, for each member $i$, let $L_i$ be the intersection of the guards of $\mathcal{G}$ at $i$ ($|L_i| \le 2$). Take a minimal set $I$ of coordinates with $\Sigma_0 \cap \prod_{i \in I} L_i = \emptyset$; then for each $i \in I$ some $x^i$ exists as in Lemma C.3, so $|I| \le 3$, and by minimality of $\mathcal{G}$ all guards of $\mathcal{G}$ are at coordinates of $I$. At one coordinate, the partial intersections of its guards (non-redundant by minimality) strictly shrink: $\le 2 \to \le 1 \to 0$. If some $L_i = \emptyset$, then $\mathcal{G}$ consists of the at most three guards at $i$; otherwise there are at most two guards at each coordinate. Hence $|\mathcal{G}| \le 6$. Each guard has at most two remote pairs, and the remote pairs of $\mathcal{G}$ form a certificate contained in $T$, so equal to $T$: $|T| \le 12$. □

## C.6 The family $G_N$ and the proof of Theorem B.16

*Construction.* For $j = 1, \dots, N$ take vertices $p_j, q_j, r_j, c_j, b_j$; add a vertex $a_1$ and put $a_j := b_{j-1}$ for $j \geq 2$. Members (centre in the middle)

$$A_j = (q_j, p_j, a_j), \qquad B_j = (q_j, r_j, b_j), \qquad C_j = (p_j, c_j, r_j).$$

The edges of $G_N$ are the pairs $q_jp_j, p_ja_j, q_jr_j, r_jb_j, p_jc_j, c_jr_j$ of the members and the hard edges $c_jq_j$ $(1 \leq j \leq N)$; all other pairs among these vertices are hard non-edges. Pendant vertices (in no member): $w_j$ adjacent to $p_j$ only $(1 \leq j \leq N)$, $w_{N+1}$ adjacent to $a_1$ only, $w_{N+2}$ adjacent to $b_N$ only; all their pairs are remote. Then $|V| = 6N + 3$ and $H_N$ has $3N$ members. Every vertex lies in at most two members ($p_j$ in $A_j, C_j$; $q_j$ in $A_j, B_j$; $r_j$ in $B_j, C_j$; $b_j$ in $B_j, A_{j+1}$), $I(H_N)$ is a chain of $N$ triangles joined by single edges (tw $= 2$), each triple $\{p_j, q_j, r_j\}$ is a soft triangle, and the hard edges form a matching, so no induced $P_3$ has three hard pairs: $H_N$ is pair-maximal (all by direct inspection; also checked by computer for $N \leq 3$).

$$T_N = \{w_jp_j,\, w_jr_j : 1 \leq j \leq N\} \cup \{w_{N+1}a_1,\, w_{N+1}p_1,\, w_{N+2}b_N,\, w_{N+2}r_N\}, \qquad |T_N| = 2N + 4.$$

**Lemma C.4** (Structure). *Write $s_j = (A_j, B_j, C_j)$ for the states of block $j$. Then $\Sigma_0(G_N)$ (all zero-excess vectors satisfying every triple that has no remote pair) is the set of vectors with $s_j \in \{121, 212, 223\}$ for all $j$ and no $j < N$ with $s_j = 212$ and $s_{j+1} = 121$.*

*Proof (by computer, for the first blocks, and by isomorphism of the interior blocks).* A triple without remote pair lies in $W_j \cup W_{j+1}$ for some $j$, where $W_j$ is the vertex set of block $j$ (with $a_j, b_j$), by the definition of local. Inside a block the triples force the three states $121, 212, 223$ (the relation of a triangle block, by enumeration). The only triples with vertices of two blocks that carry soft pairs have both soft pairs at the glue vertex $b_j = a_{j+1}$ (the pairs $r_jb_j$ of $B_j$ and $p_{j+1}a_{j+1}$ of $A_{j+1}$) and the hard non-edge $r_jp_{j+1}$ as third pair: it is a $P_3$ iff both soft pairs stay edges, i.e. $B_j$ and $A_{j+1}$ are both in $\{1, 3\}$, which within the allowed block states means $B_j = 1$ ($s_j = 212$) and $A_{j+1} = 1$ ($s_{j+1} = 121$). The other mixed triples have one soft pair and two non-edges. Enumeration over all $3^{3N}$ vectors for $N \leq 4$ gives $|\Sigma_0| = 3, 8, 21, 55$, equal to the transfer-matrix count, and contains every interior configuration (a block with both neighbours); interior blocks are isomorphic. □

*The active guards of $T_N$.* A triple is active iff its remote pairs lie in $T_N$. A triple containing $w_j$ $(j \leq N)$ has two remote pairs at $w_j$, so it is $\{w_j, p_j, r_j\}$; this is the guard of Lemma C.2(ii) with $xz = p_jr_j$, the pair $ac$ of $C_j$, $w_jp_j$ an edge and $w_jr_j$ a non-edge: $t_{ac}(C_j) = 0$, i.e. $C_j \in \{1, 2\}$. Likewise $\{w_{N+1}, a_1, p_1\}$ gives $t_{bc}(A_1) = 0$, i.e. $A_1 = 2$, and $\{w_{N+2}, b_N, r_N\}$ gives $B_N = 2$. No other triple is active, and each guard has exactly two remote pairs.

*Proof of Theorem B.16. Lower bound.* $F'(T_N) \geq |H| + 1$: otherwise there would be a zero-excess vector in $\Sigma_0$ with $C_j \neq 3$ for all $j$, $A_1 = 2$ and $B_N = 2$. By Lemma C.4 and $C_1 \neq 3$, block 1 is 121 or 212; $A_1 = 2$ excludes 121, so $s_1 = 212$, $B_1 = 1$. The link forbids $s_2 = 121$, and $C_2 \neq 3$ excludes 223, so $s_2 = 212$; inductively $s_N = 212$, $B_N = 1$, contradicting $B_N = 2$. *Minimality.* Removing a pair $e \in T_N$ deactivates exactly one guard; for each guard a zero-excess vector exists: without the $A_1$-guard take all blocks 121 (the link needs $s_j = 212$, never); without the $B_N$-guard all blocks 212 ($A_{j+1} = 2$); without the $C_k$-guard, blocks $j < k$ equal to 212, block $k$ equal to 223 and blocks $j > k$ equal to 121 (the link applies only after a block 212, and the next block is then 212 or 223). These satisfy all remaining guards, and $F'(T_N \setminus \{e\}) = |H|$. Also $F'(\emptyset) = |H|$ by $\Sigma_0 \neq \emptyset$. □

(The two further statements, $F'(T_N) = |H| + 1$ for $N \leq 10$ and the connected family at $\ell = 1$, are supported by computation only, as stated in Appendix B.3. The value $\mathrm{OPT}(G_N) = |H| + N$ was computed exactly for $N \leq 5$.)

### C.7 Proof of Lemma B.18

Identify $\{1,2\}$ with $\{0,1\}$ and put $R_5 = \{(x,y,z) \in \{0,1\}^3 : x+y+z \neq 1\}$; then $R_k \cap \{1,2\}^3 = R_5$ (the words $111, 122, 212, 221, 222$). Let $f$ be a conservative near-unanimity operation of arity $r \geq 3$ preserving $R_k$ and all unary relations. Conservativity gives $f(\{1,2\}^r) \subseteq \{1,2\}$, so $f$ preserves $R_5$ on $\{0,1\}$. Let $\mathcal{N}_0 = \{B \subseteq [r] : f(\mathbf{1}_B) = 0\}$ ($\mathbf{1}_B$ the indicator vector). For $B, C \in \mathcal{N}_0$ apply $f$ to the three rows $\mathbf{1}_{B\cup C}, \mathbf{1}_B, \mathbf{1}_C$; every column is $(1,1,1), (1,1,0), (1,0,1)$ or $(0,0,0)$, hence in $R_5$, so $(f(\mathbf{1}_{B\cup C}), 0, 0) \in R_5$, i.e. $f(\mathbf{1}_{B\cup C}) = 0$ and $B \cup C \in \mathcal{N}_0$. Near-unanimity of arity $r \geq 3$ gives $f(\mathbf{1}_{\{i\}}) = 0$ (all but one argument equal 0), so every singleton is in $\mathcal{N}_0$ and by union-closure $[r] \in \mathcal{N}_0$, i.e. $f(1, \ldots, 1) = 0$, contradicting idempotence ($f(x, \ldots, x) = x$). □

The relation $R_k$ itself is realised as follows (by computer): six vertices $0, \ldots, 5$, members $A = (2,1,0)$, $B = (4,3,1)$, $C = (5,0,3)$ (centres $1, 3, 0$), hard non-edges $25, 45$, and four further members that own the pairs $24, 23, 40, 51$ (using fresh vertices; each of the vertices $0, \ldots, 4$ lies in three members, so $c = 3$). The conjunction of the triples of the six vertices that contain none of these four pairs is $R_k$, with the states written $1, 2, 3$ for $ab, bc, ac$.

### C.8 Proof of Theorem 6.9

*Grid Tiling with* $\leq$ [6]: given sets $S_{ij} \subseteq [n]^2$ ($1 \leq i, j \leq k$), choose $s_{ij} = (a_{ij}, b_{ij}) \in S_{ij}$ with $a_{ij} \leq a_{i,j+1}$ and $b_{ij} \leq b_{i+1,j}$; it is W[1]-hard in $k$ (and has no $f(k)n^{o(k)}$ algorithm under ETH). (This is the version of the cited chapter after exchanging the two coordinates.)

*Construction.* For each cell $(i,j)$ create two *arms*: $A_{ij}$ with one node per tile of $S_{ij}$ ordered by $(a,b)$, and $B_{ij}$ with one node per tile ordered by $(b,a)$. Row $i$ is the path obtained by concatenating $A_{i,k}, A_{i,k-1}, \ldots, A_{i,1}$ (each in increasing order), column $j$ the path $B_{k,j}, \ldots, B_{1,j}$. The forest consists of these $2k$ paths. Obstruction paths: (1) every arm; (2) for every row $i$, $j < k$ and $\alpha \in [1, n-1]$ the set $\rho$ of nodes of $A_{i,j+1}$ with $a \geq \alpha + 1$ together with the nodes of $A_{ij}$ with $a \leq \alpha$ (a subpath of the row, as $A_{i,j+1}$ immediately precedes $A_{ij}$); (3) the same for columns with the $b$-coordinate and the arms $B$. Links: the node of the tile $e$ in $A_{ij}$ with the node of $e$ in $B_{ij}$ (a perfect matching). $L = k^2$.

*Correctness.* ($\Rightarrow$) For a tiling use, for every cell, the link of its chosen tile: all arms are hit; a path $\rho$ of type (2) is hit iff $a_{ij} \leq \alpha$ or $a_{i,j+1} \geq \alpha + 1$, and $a_{ij} \leq a_{i,j+1}$ gives this for every $\alpha$; columns likewise. ($\Leftarrow$) The $2k^2$ arms are pairwise disjoint and an operation meets at most two of them (a link only the two arms of its cell), so with $L = k^2$ every operation is a link, one per cell, hence a tile in every $S_{ij}$. If $a_{ij} > a_{i,j+1}$, the path $\rho$ for $\alpha = a_{i,j+1}$ contains neither chosen node (the chosen nodes of other cells are on other arms), a contradiction; columns likewise. An empty $S_{ij}$ gives an unhittable empty arm (NO, correctly), and an empty $\rho$ means $a_{ij} > a_{i,j+1}$ in every choice (NO). The size is polynomial and $L = k^2$, so an $f(L)n^{O(1)}$ algorithm would give an $f(k^2)n^{O(1)}$ algorithm for Grid Tiling with $\leq$. □

(The reduction was also confirmed by two independent solvers on small instances with $k = 2, 3$ and $n = 2, 3, 4$.)

## D Bounded certificates at excess zero

This appendix proves Theorem 6.11 and the related statements of Section 6.3. Throughout, *local* and *remote* pairs, $\mathcal{N}$ and $F'(T)$ are as in Appendix B.3, a *certificate for* $\ell = 0$ is a set $T \subseteq \mathcal{N}$ with $F'(T) > |H|$, and the states $1, 2, 3$ of a member $(a, b, c)$ are also written $L, R, M$ (edit $ab$, $bc$, $ac$). At $\ell = 0$, $F'(T) \leq |H|$ iff there is a state vector $s \in \{L, R, M\}^H$ (one edit per member, no edit of a local hard pair) such that no vertex triple all of whose remote pairs lie in $T$ is an induced $P_3$ after

the edits; all pairs other than the soft pairs are then constants. Let $\sigma(\tau)$ be the number of soft pairs of a triple $\tau$.

**Lemma D.1** (Triples with a remote pair)**.** *If a triple $\tau$ has a remote pair then $\sigma(\tau) \le 1$. If $\sigma(\tau) = 1$ then $\tau = (z; x, y)$, the pair $xy$ is owned by a member $A$, $z \notin V(A)$, and $zx, zy$ are local hard or remote. If $\sigma(\tau) = 0$ all three pairs are hard.*

*Proof.* Two soft pairs of distinct members $A, B$ share a vertex $q \in V(A) \cap V(B)$, and the third pair of the triple joins a vertex of $A$ to a vertex of $B$, so it is local. Two soft pairs of one member force the third pair of the triple to be soft too. □

Hence every constraint is a *static* test ($\sigma = 0$), a *unary list* on a member ($\sigma = 1$), or a binary (adjacent members) or ternary (soft triangle) relation without remote pair. Let $\Sigma_0 \subseteq \{L, R, M\}^H$ be the set of vectors satisfying all triples that contain no remote pair; then the vectors feasible for $T$ are those of $\Sigma_0$ that satisfy the lists activated by $T$.

**Lemma D.2** (Lock table)**.** *Let $z \notin V(A)$ with all three pairs from $z$ to $A = (a, b, c)$ hard (constants), and $S = N(z) \cap V(A)$. The three triples $(z; x, y)$, $xy \in \mathrm{pairs}(A)$, allow exactly the states of $\{L, R, M\}$ shown in the first row of Observation B.6. Every singleton entry is produced by one single triple, hence by at most two remote pairs: "ab apart" gives $\{L\}$, "bc apart" gives $\{R\}$, "ac together" gives $\{M\}$. The empty entries ($S = \{b\}$ or $S = \{a, c\}$) need two triples (three remote pairs); a single triple with a non-singleton list is a 2-element guard.*

*Proof.* Direct check of the table of Observation B.6: for $S = \{a\}$ the triple $(z; a, b)$ gives $\{L\}$, for $S = \{c\}$ or $S = \{a, b\}$ the triple $(z; b, c)$ gives $\{R\}$, for $S = \{b, c\}$ the triple $(z; a, b)$ gives $\{L\}$, and for $S = \{a, b, c\}$ the triple $(z; a, c)$ gives $\{M\}$. □

**Lemma D.3** (Strengthening)**.** *Let $T$ be a certificate for $\ell = 0$ and $T^{\mathrm{cl}}$ its closure under: if a remote pair $zx$ is in $T^{\mathrm{cl}}$ and $x$ lies in a member $A$ with $z \notin V(A)$, add every remote pair between $z$ and $V(A)$. Then $T^{\mathrm{cl}}$ is a certificate, and every list active under $T^{\mathrm{cl}}$ is the full list of Lemma D.2 of some pair $(z, A)$: a singleton, empty, or trivial.*

*Proof.* $T^{\mathrm{cl}} \supseteq T$ and $F'$ is monotone. By Lemma D.1 each active list comes from a triple $(z; x, y)$ with $xy \in \mathrm{pairs}(A)$; if one of its remote pairs $zx, zy$ is in $T^{\mathrm{cl}}$, the closure puts all remote pairs between $z$ and $V(A)$ in $T^{\mathrm{cl}}$, so all three triples of $(z, A)$ are active and their intersection is the full list. □

**Theorem D.4** (Reduction to the unit-lock Helly number, any $c$)**.** *Let $\kappa_1$ be the maximum size of an inclusion-minimal family of unit locks $x_A = s$ (two locks on one coordinate allowed) that is inconsistent with $\Sigma_0$. If there is a certificate for $\ell = 0$, there is one with at most 3 pairs (an empty lock or a static violation) or at most $2\kappa_1$ pairs.*

*Proof.* By Lemma D.3, $T^{\mathrm{cl}}$ is a certificate whose active lists are full locks. If one is empty, its at most 3 remote pairs already form a certificate (the triples of that lock need only the pairs of $(z, A)$); if $\Sigma_0 = \emptyset$ the empty set is a certificate, and a static violation needs at most 3 pairs. Otherwise the full locks form a family of unit locks inconsistent with $\Sigma_0$; take a minimal inconsistent subfamily, of size at most $\kappa_1$, and replace each lock by the single triple that produces it (Lemma D.2), at most two remote pairs. The lists active under the resulting set contain these locks, so the set is a certificate. □

In the soft-triangle-free case $\Sigma_0$ is closed under every majority operation (Theorem 6.4), so $\kappa_1 \le 2$ by Lemma B.5, and some certificate has at most 4 pairs for every $c$ and $w$ (Theorem B.15 bounds *every* minimal certificate by 12).

## D.1 The case $c \leq 2$ with soft triangles

Let $c \leq 2$. Members share at most one vertex and every vertex is in at most two members, so $I(H)$ has maximum degree 3, and its triangle-connected components are $K_3$ (one soft triangle), the diamond ($K_4$ minus an edge) and $K_4$ (the octahedron case, a whole component of $I(H)$). By Lemma D.1 every constraint of $\Sigma_0$ is a unary list, a relation on one triangle-connected component (its members' vertices, all pairs between them constants or soft), or a link relation $R_{AB}$ for an edge of $I(H)$ in no triangle (an "outside" pair that is soft for a member not in the component would make that member part of the component, except for the pair of the two private vertices of a diamond, which is allowed for in the enumeration). So $\Sigma_0$ is a product over the components of $I(H)$ of intersections of lifted relations of the four types *link, $K_3$, diamond, $K_4$* and lists.

*Classification (by computer, exhaustive over centre choices and all hard bits).* Let $m_M$ be the conservative majority $m_M(x, y, z) = x$ if two arguments are equal and $M$ otherwise. Up to the relabelling $L \leftrightarrow R$ of a member, the numbers of distinct relations over $\{L, R, M\}^k$ and the closure under $m_M$ are:

| type | configurations | distinct relations | closed under $m_M$ |
|---|---|---|---|
| link (2 members) | $9 \cdot 16$ | 32 | all |
| $K_3$ | $27 \cdot 64$ | 92 | all |
| diamond | $81 \cdot 2^8 \cdot 3$ | 332 | all |
| $K_4$ | $81 \cdot 8$ | 254 | 248 (6 fail) |

For $K_4$ the operation $m_M$ fails on six relations, but a $K_4$ component has only four coordinates and its unit-lock Helly number is 2 anyway: $\kappa_1 \in \{1, 2\}$ was computed for all 254 relations and for all 840 distinct sub-relations $R \cap$ (product of arbitrary non-empty lists).

*Proof of Theorem 6.11.* $\Sigma_0$ is a product over the connected components of $I(H)$ and lists act inside components, so an inconsistent family of unit locks lives in one component. For a $K_4$ component $\kappa_1 \leq 2$ by the computation above. For any other component, $\Sigma_0$ restricted to it is an intersection of link, $K_3$ and diamond relations and lists, each closed under $m_M$ (the table; lists are preserved by every conservative operation, and $m_M$ is conservative). By Lemma B.5 (Helly number two for unit lists under a majority) $\kappa_1 \leq 2$. Apply Theorem D.4: a certificate of at most $\max\{3, 4\} = 4$ pairs exists. To find one, enumerate at most two locks (at most $n^{O(1)}$ choices) and test each candidate $T$ with the leaf test of Theorem B.14, which takes time $f(w)n^{O(1)}$. The constant 4 is attained: an explicit instance with $n = 13$, $c = 2$, four members and no soft triangle (found by computer search) has a minimum certificate of exactly four remote pairs, and every set of three of its 54 remote pairs is feasible (verified by computer). □

*Experimental check of the classification* (not used in the proof): on a large sample of random instances with $c \leq 2$, $\Sigma_0$ was closed under $m_M$ except for some octahedron instances, and exact minimum certificate sizes never exceeded 4. The family $G_N$ of Theorem B.16 has minimum certificate 3 for $N \leq 4$ (a pendant at a centre), and by Theorem 6.11 this is forced: no choice of pendants makes every certificate large at $\ell = 0$ when $c \leq 2$.

## D.2 Excess $\ell \geq 1$: what is proved

For budget $|H| + \ell$ the objects that cost are *defect objects*: a member edited at least twice (extra cost at least 1) and an edited local hard pair (cost 1); the cost is $|H|$ plus the sum of extras plus the number of flips, at least $|H| +$ #defect objects. A *pattern* is a set of at most $\ell$ defect objects with values; $\Sigma_0^\pi$ is the zero-excess set of the instance with the pattern applied (flips applied, defect members replaced by their pair statuses, which become hard constants).

**Lemma D.5** (Pattern-wise Helly; by computer)**.** *For $c \le 2$ and every pattern $\pi$, $\kappa_1(\Sigma_0^\pi) \le 2$. So every single pattern is killed by at most two locks (at most four remote pairs).*

*Proof.* Flips change hard bits of enumerated configurations; freezing a member turns its soft pairs into hard constants, i.e. replaces a $K_3$, diamond or $K_4$ relation by a slice that is a relation of the same enumerated families (intersections with lists); lists are arbitrary. The classification above covers all of them. □

For a zero-excess core $C$ (two locks forming a minimal inconsistent sub-CSP on a member set $M(C)$) put

$D(C) = M(C) \cup \{\text{local hard pairs occurring in a triple with } \sigma \ge 1 \text{ whose soft owners lie in } M(C)\}$.

**Lemma D.6** (Disjoint regions)**.** *If $T$ contains $\ell+1$ zero-excess cores with pairwise disjoint $D(C_i)$, then $F'(T) > |H| + \ell$. Hence if there are $\ell+1$ such cores among all locks, a certificate with at most $4(\ell+1)$ pairs exists.*

*Proof.* An assignment feasible for $T$ with no defect object in $D(C_i)$ restricts to $M(C_i)$ as a state vector satisfying all constraints used by the core and its locks, a contradiction. So the assignment has a defect object in each $D(C_i)$; these are distinct objects, so its cost is at least $|H| + \ell + 1$. □

So a minimum certificate can exceed $4(\ell+1)$ pairs only if the maximum number of cores with disjoint regions is at most $\ell$.

*What is not proved.* Boundedness of certificates for $\ell \ge 1$ (Open Problem 2) and for $c \ge 3$ with soft triangles. The natural proof would glue three optimal assignments, each violating one lock, whose defect sets form a sunflower, by a coordinatewise majority; this step is impossible already for the simplest soft-triangle-free link relations over the five states (SAT-verified: no operation on the sunflower patterns closes the relations, even if its output may be defective; with non-defective outputs the system is unsatisfiable for a single link relation and for 53 of the 785 distinct five-state $K_3$ relations). A proof would need a bound on families of locks with independent single-lock witnesses; that is the exact remaining gap. *Experimental evidence only* (local search and random search, not a proof): exact minimum certificates on forests $I(H)$ with $n \le 20$ were $4, 8, 10, 11, 15$ for $\ell = 0, \dots, 4$, all below $4(\ell+1)$; random periodic chains with $c \le 3$ at $\ell = 0$ showed no growth in the number of repetitions. For $c \ge 3$, Lemma B.18 shows that the scope language has no conservative near-unanimity polymorphism, so by the Baker–Pixley theorem some primitive-positive definition has unbounded unit-lock cores; whether one can be realised with bounded treewidth as a packing is open.

# E Twins, vertices outside the packing, and a shrink rule

By Lemma 2.2 the whole difficulty of $\ell > 0$ is the set $U$ of at most $\ell$ hard edits. This appendix proves what can be said about the vertices outside $V(H)$ for pair-maximal packings $H$. The results are not used in the main text.

**Conventions.** Partitions, clusters, costs and optima are as in Section 2; $\Pi(v)$ is the cluster of $v$. Unless said otherwise, $H$ is pair-maximal and $R = V \setminus V(H)$; every pair with an end in $R$ is hard.

## E.1 Twins

**Lemma E.1** (Twins, strict form)**.** *Let $u \neq v$ be adjacent with $N[u] = N[v]$. For every partition $\Pi$ with $\Pi(u) \neq \Pi(v)$, one of the two partitions obtained by moving $u$ into $\Pi(v)$, respectively $v$ into $\Pi(u)$, has cost smaller than $\Pi$ by at least 1; more precisely the two cost changes sum to exactly $-2$. Hence $u, v$ lie in one cluster in* every *optimum, with no hypothesis on $H$.*

*Proof.* For a vertex $y$ and a set $Y \not\ni y$ let $f(y, Y) = |Y \setminus N(y)| - |Y \cap N(y)|$. Put $C_u = \Pi(u) \setminus \{u\}$ and $C_v = \Pi(v) \setminus \{v\}$ (possibly empty); $C_u \not\ni v$, $C_v \not\ni u$, and neither contains $u$ or $v$. Moving $u$ out of $\Pi(u)$ changes the status of the pairs $uz$, $z \in C_u$: before, $uz$ is an edit iff it is a non-edge; after, iff it is an edge; the change is $-f(u, C_u)$. Moving $u$ into $\Pi(v) = C_v \cup \{v\}$ changes, for $z \in C_v \cup \{v\}$, the status from "edit iff edge" to "edit iff non-edge"; the change is $f(u, C_v \cup \{v\}) = f(u, C_v) - 1$ because $uv \in E$. All other pairs keep their status. So

$$\delta_1 = -f(u, C_u) + f(u, C_v) - 1, \qquad \delta_2 = -f(v, C_v) + f(v, C_u) - 1.$$

Since $N(u) \setminus \{v\} = N(v) \setminus \{u\}$ we have $f(u, Y) = f(v, Y)$ for all $Y \subseteq V \setminus \{u, v\}$, hence $\delta_1 + \delta_2 = -2$. Thus $\min(\delta_1, \delta_2) \leq -1$. □

This is the critical-clique property used in kernelizations of Cluster Editing [7, 10] (it is also recalled in [17] when motivating Reduction Rule 6.2); the argument above is self-contained and shows that it is strict.

## E.2 Viability of a hard edit

**Lemma E.2** (Viability of a hard edit)**.** *Let $S$ be any solution, $U$ its hard part and $uv \in U$. Let $\mathrm{Bd}(uv)$ be the set of $z \notin Q(u) \cup Q(v) \cup \{u, v\}$ such that $\{u, v, z\}$ is an induced $P_3$ of $G \triangle \{uv\}$. Then for every $z \in \mathrm{Bd}(uv)$ one of $uz, vz$ lies in $U$, and $|\mathrm{Bd}(uv)| \leq |U| - 1$.*

*Proof.* All three pairs of $\{u, v, z\}$ are hard ($z \notin Q(u) \cup Q(v)$). If neither $uz$ nor $vz$ is in $S$, the triple has in $G \triangle S$ the same pattern as in $G \triangle \{uv\}$, an induced $P_3$. Distinct $z$ give distinct pairs, all different from $uv$. □

This gives a kernel-style condition (deleting $uv$ needs fewer than $\ell$ common neighbours outside $Q(u) \cup Q(v)$) but no bound on the number of candidate pairs: all pendant edges are viable.

## E.3 The vertices outside $V(H)$

**Lemma E.3** (Structure of $R$)**.** *Let $H$ be pair-maximal. For $r, s \in R$: if $rs \in E$ then $N[r] = N[s]$; if $rs \notin E$ then $N(r) \cap N(s) = \emptyset$. Consequently:*

1. *every component $X$ of $G[R]$ (a* class*) is a clique all of whose vertices have the same closed neighbourhood $N[X]$ (they are true twins);*
2. *$A_X := N(X) \setminus X$ is contained in $V(H)$;*
3. *the sets $A_X$, over the classes $X$, are pairwise disjoint.*

*Proof.* Every pair with an end in $R$ is hard. Let $rs \in E$ and $z \neq r, s$ be adjacent to exactly one of $r, s$. Then $\{r, s, z\}$ has exactly two edges, i.e. is an induced $P_3$ with all three pairs hard, contradicting pair-maximality; so $N[r] = N[s]$. If $rs \notin E$ and $z$ is a common neighbour, then $r\,z\,s$ is an induced $P_3$ whose pairs $rz, zs, rs$ all have an end in $R$, again a contradiction.

(1) Adjacent vertices of $R$ have equal closed neighbourhoods; along a connected component all closed neighbourhoods coincide, and then any two vertices $r, t$ of the component satisfy $r \in N[r] = N[t]$, so they are adjacent. (2) A neighbour of $x \in X$ lying in $R$ is in the component of $x$, hence in $X$. (3) If $a \in A_X \cap A_Y$ for classes $X \neq Y$, then $a$ is a common neighbour of some $x \in X$ and $y \in Y$, which are non-adjacent (different components of $G[R]$); this contradicts the second clause. □

For $x \in X$ and $a \in V(H)$ we thus have $xa \in E$ iff $a \in A_X$; in particular vertices of one class are twins towards $V(H)$ as well. (Lemma E.3 fails for vertex-maximal packings, see Proposition Q.2, whose instance also shows that Theorem E.5 fails without pair-maximality.)

**Lemma E.4** (Exchange formula)**.** *Let $\Pi$ be any partition, $X$ a class with $X \subseteq C$ for a cluster $C$, $x = |X|$ and $a = |A_X \cap C|$. Let $\Pi'$ be obtained by moving $X$ out of $C$ into a cluster of its own. Then* $\mathrm{cost}(\Pi') - \mathrm{cost}(\Pi) = x\,(x + 2a - |C|)$.

*Proof.* Only the pairs between $X$ and $C \setminus X$ change status. By Lemma E.3, a vertex of $X$ has exactly $a$ neighbours in $C \setminus X$, namely $A_X \cap C$. Before the move such a pair is an edit iff it is a non-edge: $x(|C| - x - a)$ edits. After the move it is an edit iff it is an edge: $xa$ edits. The difference is $xa - x(|C| - x - a) = x(x + 2a - |C|)$. □

**Theorem E.5** (C1: no edit inside $R$)**.** *Let $H$ be pair-maximal. In* every *optimum $\Pi$, every cluster meets $R$ in the empty set or in exactly one whole class. Hence no edit of $\Pi$ has both endpoints in $R$, and for every class $X$ the cluster $\Pi(X)$ equals $X \cup T_X$ with $T_X := \Pi(X) \cap V(H)$.*

*Proof.* By Lemma E.3(1) and Lemma E.1, every class lies inside one cluster (a class of size 1 trivially). Suppose a cluster $C$ contains two distinct classes $X_1, X_2$. Put $x_i = |X_i| \geq 1$ and $a_i = |A_{X_i} \cap C|$. The sets $X_1, X_2 \subseteq R$ and $A_{X_1} \cap C, A_{X_2} \cap C \subseteq V(H)$ are pairwise disjoint (Lemma E.3(2),(3)), so $|C| \geq x_1 + x_2 + a_1 + a_2$. Let $\Delta_i$ be the cost change of moving $X_i$ out of $C$. Since $\Pi$ is an optimum, $\Delta_1 \geq 0$ and $\Delta_2 \geq 0$, i.e. by Lemma E.4 $|C| \leq x_i + 2a_i$ for $i = 1, 2$. With the lower bound on $|C|$: $x_2 + a_2 \leq a_1$ and $x_1 + a_1 \leq a_2$. Adding gives $x_1 + x_2 \leq 0$, absurd. So no cluster contains two classes.

An edit with both ends in $R$ joins two vertices $r, s$. If they lie in one class, they are adjacent and (by the above) in one cluster, so $rs$ is no edit. If they lie in different classes they are non-adjacent, and $rs$ is an edit iff they share a cluster, which was excluded. □

The strictness of the inequality is what makes the statement hold for every optimum (no ties, and no iteration or termination argument).

## E.4 Consequences

Let $(G, H, \ell)$ be a *YES instance* with $H$ pair-maximal, $\Pi$ any optimum, $S$ its edit set, so $|S| = \mathrm{OPT} \leq |H| + \ell$. Let $U$ be the hard edits. By Lemma 2.2, $|S| - |H| = \sum_i (|x_i| - 1) + |U|$, so $|U| \leq \ell$. Write $q = \max_u |Q(u)| \leq 2c$.

**Corollary E.6** (Normal form)**.** *For every class $X$ let $T_X = \Pi(X) \cap V(H)$.*

*(N1)* Types of hard edits. *Every hard edit is either an* HH-edit *(both ends in $V(H)$) or an* RH-edit *$\{x, a\}$, $x \in X$ a class, $a \in V(H)$; the latter is a deletion if $a \in A_X \setminus T_X$ and an insertion if $a \in T_X \setminus A_X$, and $\{x, a\}$ is edited iff $\{x', a\}$ is, for all $x, x' \in X$. With $h$ the number of HH-edits, $|U| = h + \sum_X |X| \cdot |A_X \triangle T_X| \leq \ell - \sum_i (|x_i| - 1) \leq \ell$.*

*(N2)* Frozen classes. *If $|X| > \ell$, then $\Pi(X) = X \cup A_X$.*

*(N3)* Attachment. *Every $a \in V(H)$ lies in $A_X$ for at most one class $X$. If $z, z' \in A_X$ are non-adjacent then $zz'$ is soft. If $a \in V(H) \setminus A_X$, $z \in A_X$ and $az \in E$ then $az$ is soft; hence $|N(a) \cap A_X| \leq |Q(a)| \leq q$.*

*(N4)* Shrink rule. *If $|X| \geq \ell + 2$ and $x \in X$, then $(G, H, \ell)$ is a YES instance iff $(G - x, H, \ell)$ is. This holds for $H$ pair-maximal without the YES hypothesis. If $A_X = \emptyset$ then $X$ is a component of $G$ and may be deleted without changing* OPT.

*(N5)* Size. *For a pair-maximal $H$ and after exhausting the two rules in (N4), every class has at most $\ell + 1$ vertices and at most $|V(H)|$ classes have $A_X \neq \emptyset$, so $|R| \leq (\ell + 1)|V(H)|$ and $n \leq (\ell+2)|V(H)| \leq 3(\ell+2)|H|$. ($|V(H)|$ is not bounded in terms of $\ell$, so this is not a kernel; see Remark E.8.)*

*Proof.* (N1) Pairs with an end in $R$ are hard; by Theorem E.5 none has both ends in $R$. The vertices of $X$ are twins towards $V(H)$ and lie in one cluster, which gives the all-or-none statement; $xa$ is an edit iff exactly one of $a \in T_X$, $a \in A_X$ holds. Counting gives the identity; the bound is the first paragraph of this subsection.

(N2) If $\Pi(X) \neq X \cup A_X$ then, since $\Pi(X) = X \cup T_X$ (Theorem E.5), $A_X \triangle T_X \neq \emptyset$ and (N1) gives $|U| \geq |X| > \ell$, contradicting $|U| \leq \ell$.

(N3) The first claim is Lemma E.3(3). Let $x \in X$. If $z, z' \in A_X$ are non-adjacent, $z\,x\,z'$ is an induced $P_3$ with pairs $zx, xz'$ hard, so $zz'$ must be soft (pair-maximality). If $a \in V(H) \setminus A_X$ and $z \in A_X$ with $az \in E$, then $xa \notin E$ and $x\,z\,a$ is an induced $P_3$ with $xz, xa$ hard, so $az$ is soft, i.e. $z \in Q(a)$.

(N4) $x \in R$, so $H$ is still a packing of $G - x$, still pair-maximal (an induced $P_3$ of $G - x$ is one of $G$), and $X \setminus \{x\}$ is a class of $G - x$ with the same $A_X$. ($\Rightarrow$) Restricting a partition of $G$ to $V \setminus x$ does not increase its cost, so $\mathrm{OPT}(G-x) \leq \mathrm{OPT}(G) \leq |H|+\ell$. ($\Leftarrow$) Let $\Pi$ be an optimum of $G-x$ with cost $\leq |H| + \ell$. Now $|X \setminus x| \geq \ell + 1 > \ell$, so (N2) in $G - x$ gives $\Pi(X \setminus x) = (X \setminus x) \cup A_X$. Add $x$ to this cluster: $N[x] = X \cup A_X$, so every pair at $x$ is satisfied, and the cost does not change. Hence $\mathrm{OPT}(G) \leq |H| + \ell$. If $A_X = \emptyset$, $X$ is a clique component of $G$, which costs nothing, and $H$ is disjoint from it.

(N5) After the rules each class has at most $\ell + 1$ vertices; the non-empty sets $A_X \subseteq V(H)$ are pairwise disjoint (Lemma E.3), and $|V(H)| \leq 3|H|$. Classes with $A_X = \emptyset$ were deleted. □

*Remark* E.7 (Tightness of the threshold in (N4)). For $\ell = 0$ the threshold $|X| \geq 2$ cannot be lowered to $|X| \geq 1$: take the 4-cycle of Proposition Q.3, $H = \{(0, 1, 3)\}$, $X = \{2\}$. Then $(G, H, 0)$ is a NO instance, but $G - 2$ is the induced $P_3$ 0 1 3 itself, so $(G - 2, H, 0)$ is a YES instance. For $\ell = 1$ the threshold $|X| \geq \ell + 2 = 3$ cannot be lowered to $|X| \geq \ell + 1 = 2$: replace vertex 2 of the 4-cycle by two adjacent twins $x, x'$, i.e. $G$ has vertices $0, 1, 3, x, x'$ and edges $01, 13, 0x, 0x', 3x, 3x', xx'$, and $H = \{(0, 1, 3)\}$ (pair-maximal, $R = X = \{x, x'\}$). Then $\mathrm{OPT}(G) = 3$, so $(G, H, 1)$ is a NO instance, whereas $\mathrm{OPT}(G - x) = 2 = |H| + 1$, so $(G - x, H, 1)$ is a YES instance (both by enumerating all partitions). For $\ell = 2$ the same family with $|X| = 3$ does not separate ($\mathrm{OPT} = 3$ in both graphs). So the threshold is tight for $\ell \in \{0, 1\}$; tightness for $\ell \geq 2$ is open.

*Remark* E.8 (The rules and the input packing). The input packing $H$ of CEaMP need not be pair-maximal. Extend it greedily (polynomial time) to a pair-maximal $H' \supseteq H$ with $t = |H'| - |H|$ new members, set $\ell' = \ell - t$ (answer NO if $\ell' < 0$, Lemma 2.4) and apply this section to $(G, H', \ell')$. Then (N4), (N5) hold with $\ell'$ and $H'$; in particular $n \leq 3(\ell + 2)(|H| + \ell)$ after reduction, since $|H'| \leq |H| + \ell$. But $\mathrm{tw}(I(H'))$ may exceed $\mathrm{tw}(I(H))$, so these rules are a preprocessing for the parameters $(\ell, |H'|)$ or $(\ell, \mathrm{tw}(I(H')))$, not for $(\ell, \mathrm{tw}(I(H)))$ with the given $H$; and $|V(H')|$ is not bounded by $\ell$, so (N5) is not a kernel.

**Proposition E.9** (Partners of edits touching $R$)**.** *In the setting of Corollary E.6, let $X$ be a class, $T = T_X$. Then:*

*(P1)* Deletion. *If $\{x, a\}$ is an RH-deletion, $a \in A_X$, then $X$ is determined by $a$ (the unique class with $a \in A_X$): it is a unary flag at $a$ of weight $|X|$. Moreover $a$ is then cut off from every $z \in A_X \cap T$ with $z \notin Q(a)$, each such edge $az$ being a hard HH-deletion; so at least $|A_X \cap T| - q$ HH-edits accompany it.*

*(P2)* No detachment. *If $T \cap A_X = \emptyset$ then $T = \emptyset$.*

*(P3)* Insertion. *If $a \in T \setminus A_X$ then $T \cap A_X \neq \emptyset$, and for every $z \in T \cap A_X$ either $az$ is soft ($a \in Q(z)$, at most $q$ choices of $a$ per $z$) or $az$ is a hard non-edge, hence an inserted HH-edit.*

*(P4)* Wide classes. *If $|A_X| \geq \ell + q$ then $T \subseteq A_X$.*

*Consequently, every candidate partner $a \in V(H)$ of a class in an RH-edit is unary (P1), local to some $z$ (P3, soft case) or tied to an HH-edit (P3, second case; P1). This classifies the partners; it does not assert that the HH-edits are independent of the RH-edits.*

*Proof.* (P1) Uniqueness is (N3). Let $a \in A_X \setminus T$ and $z \in A_X \cap T$, $z \notin Q(a)$. If $az \notin E$ then $az$ would be soft by (N3), contradicting $z \notin Q(a)$; so $az \in E$, hard, and $a \notin \Pi(X) \ni z$, so $az$ is deleted. At most $q$ vertices $z$ lie in $Q(a)$.

(P2) Suppose $T \neq \emptyset = T \cap A_X$. The cluster is $C = X \cup T$ (Theorem E.5), $a_X = |A_X \cap C| = 0$, and by Lemma E.4 moving $X$ out changes the cost by $|X|(|X| - |X| - |T|) = -|X||T| < 0$, contradicting optimality.

(P3) $T \neq \emptyset$, so $T \cap A_X \neq \emptyset$ by (P2). Let $z \in T \cap A_X$. If $az \in E$ then $az$ is soft by (N3) ($a \in V(H) \setminus A_X$), i.e. $a \in Q(z)$. Otherwise either $az$ is soft (again $a \in Q(z)$) or it is a hard non-edge inside the cluster $\Pi(X)$, an insertion.

(P4) Suppose $T \not\subseteq A_X$. Put $\alpha = |A_X \cap T|$, $\delta = |A_X \setminus T|$, $\beta = |T \setminus A_X| \geq 1$. By (N1) there are $|X|\beta$ RH-insertions and $|X|\delta$ RH-deletions. For $b \in T \setminus A_X$ and $z \in A_X \cap T$ with $z \notin Q(b)$, the pair $bz$ is hard, lies in a cluster, and is a non-edge by (N3); so it is a hard insertion, an HH-edit. There are at least $\beta \max(0, \alpha - q)$ of them. Hence $\ell \geq |U| \geq \beta(|X| + \max(0, \alpha - q)) + |X|\delta$. If $\alpha \geq q$ this gives $\ell \geq 1 + (\alpha - q) + \delta$, i.e. $|A_X| = \alpha + \delta \leq \ell + q - 1$. If $\alpha < q$ it gives $\delta \leq \ell - 1$ and $|A_X| \leq q - 1 + \ell - 1$. Either way $|A_X| < \ell + q$. □

# F Counting lemmas for arbitrary packings

These lemmas hold for every packing $H$ (no maximality unless stated) and describe what a solution of size at most $|H| + \ell$ can look like. All proofs use only pair-disjointness, a budget, and counting. They give neither a fixed-parameter algorithm in $\ell + w$ nor hardness; they are not used in the main text.

## F.1 Hubs

In this subsection $H$ is an arbitrary packing, $c = c(H)$, and $(G, H, \ell)$ is a YES instance where stated.

**Lemma F.1** (HUB$^+$)**.** *Let $H$ be any packing, $(G, H, \ell)$ a YES instance, $u$ a vertex and $N_h(u) = \{x : ux \in E,\ ux \text{ not soft}\}$. Let $\Lambda_u$ be the graph on $N_h(u)$ whose edges are the pairs $\{x, y\}$ with $xy \notin E$ and $xy$ not soft (hard non-edges). Then:*

*(a) $\nu(\Lambda_u) \leq \ell$ (matching number);*

*(b)* $\Lambda_u$ *has a vertex cover* $T$ *with* $|T| \le 2\ell$*, and every pair of* $N_h(u) \setminus T$ *is an edge of* $G$ *or is soft;*

*(c)* $\alpha(G[N_h(u)]) \le 2\ell + c + 1$ *where* $c = c(H)$*.*

*Proof.* (a) Let $x_1y_1, \ldots, x_ty_t$ be a matching of $\Lambda_u$ and $P_i = (x_i, u, y_i)$. Since $ux_i, uy_i \in E$ and $x_iy_i \notin E$, $P_i$ is an induced $P_3$. Its three pairs $ux_i, uy_i, x_iy_i$ are hard. The $2t$ vertices $x_i, y_i$ are distinct and different from $u$, so the pairs $ux_i, uy_i$ ($i \le t$) are pairwise distinct, and the pairs $x_iy_i$ are pairwise distinct and do not contain $u$. So $\{P_1, \ldots, P_t\}$ is hard over $H$ and $t \le \ell$ by Lemma 2.3.

(b) The vertex set of a maximum matching of $\Lambda_u$ is a vertex cover (by maximality) of size $2\nu \le 2\ell$. A pair of $N_h(u) \setminus T$ is not an edge of $\Lambda_u$, i.e. it is an edge of $G$ or soft.

(c) Let $J$ be independent in $G[N_h(u)]$. By (b), every pair of $J \setminus T$ is a non-edge of $G$ and not an edge of $\Lambda_u$, hence soft. Put $J' = J \setminus T$. A soft pair of $J'$ lies in a member $P$ of $H$ and is a non-edge, so it is the pair of the two ends of $P$ (the other two pairs of $P$ are edges); in particular $|V(P) \cap J'| \le 2$. If $J' \ne \emptyset$ fix $x \in J'$; every $y \in J' \setminus \{x\}$ gives a member $P_y \ni x, y$, and $P_y \ne P_{y'}$ for $y \ne y'$ because $|V(P_y) \cap J'| \le 2$. So $|J'| - 1 \le c$. Hence $|J| \le |T| + |J'| \le 2\ell + c + 1$. □

**Corollary F.2** (HUB, hard-independent sets)**.** *In the setting of Lemma F.1, if* $Y \subseteq N_h(u)$ *and every pair of* $Y$ *is a hard non-edge, then* $|Y| \le 2\ell + 1$*.*

*Proof.* $Y$ is a clique of $\Lambda_u$, so it contains a matching of size $\lfloor |Y|/2 \rfloor$, and $\lfloor |Y|/2 \rfloor \le \ell$ by Lemma F.1(a). □

*Remark* F.3 (HUB$^+$ versus an independence-number bound). (1) Bounding $\alpha(G[N_h(u)])$ through the corollary (greedy choice of a subset of an independent set that is also free of soft pairs) gives only $(c+1)(2\ell+1) = (D/2+1)(2\ell+1)$, which is weaker than (c). (2) The matching form (a) is the relevant statement for Vertex Cover on the non-adjacency graph of $N(u)$: let $N_h(u)$ be two disjoint cliques of $G$ of size $m$ with no soft pair between them. Then $\alpha = 2$, but $(a_i, u, b_i)$, $a_i, b_i$ from different cliques, are $m$ pair-disjoint induced $P_3$'s, and $m \le \ell$ in a YES instance. Large independent sets are not the obstruction; large matchings of $\Lambda_u$ are. (3) For a general $H$ the lemma is applied to $H'$ of Lemma 2.4 with $c', \ell'$.

## F.2 Pieces and small merges

Fix an arbitrary packing $H$ and a solution $S$ with $|S| \le |H| + \ell$ (neither optimality nor pair-maximality is used). Let $U = U^- \cup U^+$ be its hard part and $u = |U| \le |S| - |H| \le \ell$ (Lemma 2.2). The *pieces of* $S$ are the connected components of the graph of kept hard edges, as in Appendix L; every piece lies inside one cluster of $G \triangle S$.

**Lemma F.4** (SM: small merge)**.** *Let* $S$*,* $u$ *and the pieces be as above,* $D \ge \max_v |Q(v)|$ *(e.g.* $D = D(H)$*), and let* $C$ *be a cluster of* $G \triangle S$*.*

*(i) If* $A \ne B$ *are pieces contained in* $C$*, then every pair* $ab$ *(*$a \in A, b \in B$*) is soft or inserted.*

*(ii) For such* $A, B$ *with* $|A| \ge |B|$*:* $|B|\,(|A| - D) \le |U^+ \cap (A \times B)| \le u$*, hence* $|A| \le D + u \le D + \ell$*.*

*(iii) If* $C$ *is the union of at least two pieces, then* $|C| \le \max\{D + u + 1,\ 2D,\ 2u\}$*.*

*Proof.* (i) is Lemma L.1.

(ii) Each vertex of $B$ has at most $D$ soft partners, so at most $D|B|$ pairs of $A \times B$ are soft. By (i) all other pairs of $A \times B$, at least $|A||B| - D|B|$, are inserted, and $|U^+| \le |U| = u$. If $|A| \ge D + u + 1$ then $|B|(|A| - D) \ge u + 1$ because $|B| \ge 1$, a contradiction; so $|A| \le D + u$.

(iii) Let $A$ be a largest piece in $C$, $a = |A|$ and $r = |C| - a \geq 1$. By (i), every pair $\{x, y\}$ with $x, y \in C$ in different pieces is soft or inserted.

*Case $a > D$.* Each $x \in C \setminus A$ has at most $D$ soft partners, so at least $a - D \geq 1$ vertices $y \in A$ give an inserted pair $xy$. These pairs are distinct for distinct $(x, y)$, so $r(a - D) \leq |U^+| \leq u$. Then $r \leq u$ and $a \leq D + u/r$, so $|C| = a + r \leq D + u/r + r$. The function $r \mapsto r + u/r$ is convex on $[1, u]$ and takes the value $u + 1$ at both ends, so $|C| \leq D + u + 1$. (The case does not occur if $u = 0$.)

*Case $a \leq D$.* All pieces of $C$ have at most $D$ vertices. Each $x \in C$ has at least $|C| - D$ vertices of $C$ outside its piece, at most $D$ of which are soft partners; so $x$ lies in at least $|C| - 2D$ inserted pairs $\{x, y\}$ with $y \in C$. Summing over $x$ counts each pair twice: $2|U^+| \geq |C|(|C| - 2D)$. If $|C| > 2D$, then $|C| - 2D \geq 1$ and $|C| \leq 2|U^+| \leq 2u$. So $|C| \leq \max\{2D, 2u\}$. □

*Remark* F.5 (Qualifiers and relation to the original bound). (1) The qualifier "pieces sharing a cluster" is essential: a single piece may be an entire cluster of unbounded size, and a statement "pieces have size at most $D+\ell$" is false. (2) The bound (iii) implies the weaker bound $|C| \leq \max\{2D+\ell, 2\ell\}$: $2u \leq 2\ell$ and $2D \leq 2D + \ell$ are clear; $D + u + 1 \leq 2D + \ell$ if $D \geq 1$; if $D = 0$ and $u \geq 1$ then $u + 1 \leq 2u \leq 2\ell$; and if $D = u = 0$ no cluster with two pieces exists (the two cases of the proof are both impossible). (3) The bound (iii) is attained for instance by two singleton pieces and one inserted pair ($D = 0$, $u = 1$, $|C| = 2$); we do not claim it is tight in general. (4) Lemma F.4 says nothing about the *degrees* in a graph of pieces; it bounds the sizes of pieces that share a cluster and the number of cross pairs that are not soft.

## F.3 The soft graph has bounded treewidth

Recall the soft graph $G_{\text{soft}}$ (Section 2).

**Proposition F.6** (Treewidth of the soft graph)**.** $\operatorname{tw}(G_{\text{soft}}) \leq 3(w+1) - 1 = 3w + 2$ *if $H \neq \emptyset$; $G_{\text{soft}}$ is edgeless if $H = \emptyset$. The bound is attained for $w = 0$ ($H$ a single $P_3$, $G_{\text{soft}}$ a triangle).*

*Proof.* Let $(\mathcal{T}, \{\beta_t\})$ be a tree decomposition of $I(H)$ of width $w$ and $\mathcal{T}_P = \{t : P \in \beta_t\}$, a subtree of $\mathcal{T}$ for each $P \in H$. Put $\beta'_t = \bigcup_{P \in \beta_t} V(P)$, so $|\beta'_t| \leq 3(w+1)$. For every vertex $x \notin V(H)$ add a new leaf bag $\{x\}$ to an arbitrary node. *Edges:* a soft pair lies in $V(P)$ for some $P \in H$, and $P \in \beta_t$ for some $t$, so the pair lies in $\beta'_t$. *Connectivity:* for $x \in V(H)$ the nodes $t$ with $x \in \beta'_t$ form $\bigcup_{P \ni x} \mathcal{T}_P$. The members through $x$ pairwise share $x$, so they are pairwise adjacent in $I(H)$ and, the edge being covered by a bag, their subtrees $\mathcal{T}_P$ pairwise intersect. A union of connected subtrees that pairwise intersect is connected (two points of the union lie in subtrees $\mathcal{T}_P, \mathcal{T}_{P'}$ that meet; join them through a common node). The new leaf bags are single bags of size 1. So this is a tree decomposition of $G_{\text{soft}}$ of width at most $3(w+1) - 1$. □

## F.4 Completing a fixed soft assignment

**Proposition F.7.** *Let $x$ assign a subset $x_i \subseteq \operatorname{pairs}(P_i)$ to each $P_i$ and let $b = |H| + \ell - \sum |x_i|$ be the excess budget left. Whether there is a set $U$ of at most $b$ hard pairs with $G \triangle (x \cup U)$ a cluster graph is decidable in $3^b n^{O(1)}$ time.*

*Proof.* Search tree on $G_x = G \triangle x$: if the current graph has an induced $P_3$, some pair of it must be flipped; soft pairs are frozen by $x$, so branch on its at most three hard pairs, reducing $b$ by one (if all its pairs are soft, reject). Depth is at most $b$; flipping a pair twice is useless, so the search is exhaustive. □

The proposition is a reformulation, not progress towards fixed-parameter tractability in $(\ell, w)$: finding $x$ is the whole problem, and the property "$x$ is completable" is not local in $x$.

# G Optimal solutions of pair-maximal packings

**Setting.** Throughout this section $H$ is pair-maximal (Section 2; for an arbitrary packing read the primed instance), $D \geq \max_v |Q(v)|$ is a number, and $S$ is a solution with $|S| \leq |H| + \ell$, so that its set $U$ of hard edits has $u = |U| \leq \ell$ (Lemma 2.2); $U^-$ and $U^+$ are the deleted hard edges (hard edges across clusters) and the inserted hard non-edges (hard non-edges inside clusters), and $\mathrm{del}_x, \mathrm{ins}_x$ count the elements of $U^-$ and $U^+$ at the vertex $x$. Solutions and partitions correspond (Section 2). A *hard component* is a connected component of the graph $(V, \text{hard edges of } G)$; it is *clean* if all its non-adjacent pairs are soft.

**Lemma G.1** (Cost formula)**.** *Let $\pi$ be a partition of $V$ and $f(C) = \binom{|C|}{2} - 2e(C)$, where $e(C)$ is the number of edges of $G$ inside $C$. Then* $\mathrm{cost}(\pi) = |E| + \sum_{C \in \pi} f(C)$.

*Proof.* The cost counts non-edges inside blocks, $\sum_C [\binom{|C|}{2} - e(C)]$, plus edges across blocks, $|E| - \sum_C e(C)$. □

## G.1 Clusters of optimal solutions: half-degree and diameter

**Lemma G.2** (Half-degree and diameter)**.** *Let $S$ be an optimal solution, $C$ a cluster of $G \triangle S$ and $v \in C$. Put $d_C(v) = |N_G(v) \cap C|$.*

*(a) $d_C(v) \geq (|C| - 1)/2$.*

*(b) If $a, b \in C$ and $ab \notin E$, then some $z \in C \setminus \{a, b\}$ has $az, zb \in E$.*

This is the single-vertex-move argument behind the critical-clique property of Cluster Editing [10, 7]; we include the short proof.

*Proof.* (a) Let $S' = S \triangle \{vx : x \in C \setminus \{v\}\}$. Then $G \triangle S'$ is $G \triangle S$ with $v$ cut off from the rest of $C$, a cluster graph. For $x \in C \setminus \{v\}$ with $vx \in E$ the pair $vx \notin S$ and $vx \in S'$: this adds $d_C(v)$ pairs. For $x \in C \setminus \{v\}$ with $vx \notin E$ the pair is in $S$ and not in $S'$: this removes $|C| - 1 - d_C(v)$ pairs. All other pairs are unchanged. So $|S'| = |S| + 2d_C(v) - (|C| - 1)$, and $|S'| \geq |S|$ by optimality.

(b) $N_G(a) \cap C$ and $N_G(b) \cap C$ are subsets of $C \setminus \{a, b\}$ (as $a \notin N(a)$ and $ab \notin E$), a set with $|C| - 2$ elements, while $d_C(a) + d_C(b) \geq |C| - 1$ by (a). So the two sets intersect. □

Lemma G.2 uses neither $H$ nor $\ell$; only local optimality under single-vertex moves.

## G.2 Hard pairs

**Lemma G.3** (HE and H2)**.** *Let $H$ be pair-maximal. (a) If $ab, bc$ are hard edges ($a, b, c$ distinct), then $ac \in E$ or $ac$ is soft. (b) If $uv$ is a hard non-edge, then $N_h(u) \cap N_h(v) = \emptyset$.*

*Proof.* (a) If $ac \notin E$ and $ac$ were hard, then $a\,b\,c$ would be an induced $P_3$ with all three pairs hard, contradicting pair-maximality. (b) If $z \in N_h(u) \cap N_h(v)$, then $u\,z\,v$ is an induced $P_3$ ($uz, zv \in E$, $uv \notin E$) all of whose pairs are hard. □

No YES hypothesis is needed. Without pair-maximality the lemma fails ($H = \emptyset$ and any induced $P_3$).

**Lemma G.4** (EX: bounds at a hard edit)**.** *Let $H$ be pair-maximal, $S$ a solution with hard edits $U$, $u = |U|$, let $xy \in U$ and let $C$ be the cluster of $x$. Then:*

*(a) if $xy$ is deleted, $d_h(x) \le D + \mathrm{del}_x + \mathrm{del}_y - 1$; if $xy$ is inserted, $d_h(x) \le D + \mathrm{ins}_y - 1 + \mathrm{del}_x$. In both cases $d_h(x) \le D + u$;*

*(b) $|C| \le 1 + D + d_h(x) + \mathrm{ins}_x$, hence $|C| \le 1 + 2D + u$;*

*(c) every $z \in C$ has $d_h(z) \le |C| - 1 + \mathrm{del}_z \le 2D + 2u$.*

*Proof.* (a) *Deleted:* $x \in C$, $y \notin C$. A hard neighbour $z$ of $x$ with $xz$ not deleted ("kept") lies in $C$, and $z \neq y$. Apply Lemma G.3(a) with centre $x$ to the hard edges $xz, xy$: $zy \in E$ or $zy$ is soft. If $zy$ is a hard edge, it joins $z \in C$ to $y \notin C$, so it is a deleted hard edge at $y$ other than $xy$ (there are $\mathrm{del}_y - 1$ of these); otherwise $z \in Q(y)$, at most $D$ vertices. So at most $D + \mathrm{del}_y - 1$ hard neighbours are kept, and $d_h(x) \le D + \mathrm{del}_y - 1 + \mathrm{del}_x$. *Inserted:* $x, y \in C$. A kept hard neighbour $z$ of $x$ lies in $C$ and $z \neq y$ (as $xy \notin E$); $zy$ is a pair inside $C$, hence an edge of $G \triangle S$. If $zy$ were a hard edge, $z \in N_h(x) \cap N_h(y)$, contradicting Lemma G.3. So $zy$ is soft ($z \in Q(y)$, at most $D$ choices) or a hard non-edge, inserted at $y$ and different from $xy$ ($\mathrm{ins}_y - 1$ choices). So $d_h(x) \le D + \mathrm{ins}_y - 1 + \mathrm{del}_x$. In the deleted case the pairs counted by $\mathrm{del}_x + \mathrm{del}_y - 1$ are distinct pairs of $U$, so the sum is $\le u$; in the inserted case $\mathrm{ins}_y - 1 + \mathrm{del}_x \le u - 1$ (an inserted pair at $y$ other than $xy$ and a deleted pair at $x$ are distinct elements of $U \setminus \{xy\}$).

(b) For $z \in C \setminus \{x\}$ the pair $xz$ is an edge of $G \triangle S$. Either $xz$ is soft (at most $D$ such $z$), or a kept hard edge (at most $d_h(x)$), or a hard non-edge, inserted at $x$ (at most $\mathrm{ins}_x$). With (a): in the deleted case $|C| \le 1 + D + (D + \mathrm{del}_x + \mathrm{del}_y - 1) + \mathrm{ins}_x \le 1 + 2D + u$ because the deleted pairs at $x$ or $y$ and the inserted pairs at $x$ are distinct elements of $U$; in the inserted case $\mathrm{ins}_x + \mathrm{ins}_y - 1$ counts the inserted pairs at $x$ or $y$ and $\mathrm{del}_x$ is disjoint from them, so the total is again $\le u$.

(c) Hard neighbours of $z$ inside $C$ are at most $|C| - 1$; hard neighbours outside $C$ are joined by deleted edges, at most $\mathrm{del}_z \le u$. Use (b). □

**Lemma G.5** (Connected clusters)**.** *In an optimal solution every cluster $C$ induces a connected subgraph $G[C]$.*

*Proof.* If $C = C_1 \cup C_2$ with $C_1, C_2 \neq \emptyset$ and no edge of $G$ between them, splitting $C$ into $C_1, C_2$ turns the $|C_1||C_2| > 0$ inserted pairs between them into non-edges outside $S$ and creates no deletion. □

## G.3 Normal form and candidates

Fix $\ell$ and put

$$B = 2D + 3\ell, \qquad L_B = \{v : \ d_h(v) \le B\}, \qquad Z = B + D + 1 = 3D + 3\ell + 1.$$

In an optimal solution of size at most $|H| + \ell$ call a cluster *exceptional* if it contains an endpoint of a pair of $U$. Let BIG be the union of the hard components that contain a vertex $v$ with $d_h(v) > B$.

**Lemma G.6** (NF: normal form)**.** *Let $H$ be pair-maximal and $S$ an optimal solution with $|S| \le |H| + \ell$, so $u \le \ell$.*

*($\alpha$) There are at most $2u \le 2\ell$ exceptional clusters. Each exceptional cluster $C$ has $|C| \le 1 + 2D + u \le 2D + \ell + 1$, lies in $L_B$, and $G[C]$ is connected.*

*($\beta$) Let $C$ be a non-exceptional cluster. Then no hard edge leaves $C$ and no hard non-edge lies in $C$; so $C$ is a union of whole hard components, each clean, and all pairs between different components inside $C$ are soft. If $C$ contains at least two components, $|C| \le 2D$.*

*($\gamma$) Every hard component $K$ of* BIG *is a whole cluster of $S$, and this cluster is non-exceptional. Consequently every exceptional cluster is disjoint from* BIG*, and every component of* BIG *is clean.*

*Proof.* ($\alpha$) Each pair of $U$ has at most two clusters at its ends, which gives the count. The bound on $|C|$ is Lemma G.4(b) (for an endpoint $x \in C$ of a pair in $U$), the inclusion $C \subseteq L_B$ is Lemma G.4(c) together with $2D + 2u \le B$, and connectivity is Lemma G.5.

($\beta$) A hard edge leaving $C$ would be deleted, with an endpoint in $C$; a hard non-edge inside $C$ would be inserted. Both would make $C$ exceptional. If $K$ is a hard component meeting $C$, hard edges at a vertex of $K \cap C$ do not leave $C$, so $K \subseteq C$ by connectivity of $K$; $K$ is clean since $C$ contains no hard non-edge. Two vertices of different components inside $C$ are not joined by a hard edge, and are not a hard non-edge, so their pair is soft. If $C$ contains components $K_1 \neq K_2$, a vertex $x \in K_1$ has all vertices of $C \setminus K_1$ as soft partners, so $|C \setminus K_1| \le D$; likewise $|C \setminus K_2| \le D$; since $K_1 \subseteq C \setminus K_2$, $|C| = |K_1| + |C \setminus K_1| \le 2D$.

($\gamma$) Let $d_h(v) > B$ and let $K = K(v)$ be its hard component. By Lemma G.4(a) no endpoint of a pair of $U$ has hard degree above $D + u \le B$, and by ($\alpha$) no vertex of an exceptional cluster has hard degree above $B$. So the cluster $C$ of $v$ is non-exceptional, and by ($\beta$) $K \subseteq C$ and $C$ is a union of components. Now $|K| \ge d_h(v) + 1 > B + 1 > 2D$, so by ($\beta$) $C$ cannot contain two components; hence $C = K$. Every other vertex of $K$ lies in the same cluster. Cleanness is ($\beta$). The disjointness statement follows since a component of BIG is a non-exceptional cluster. □

**Corollary G.7** (CAND: bounded candidate partners)**.** *Let $H$ be pair-maximal, $S$ optimal with $|S| \le |H| + \ell$, and $xy \in U$. Then $x, y \in L_B$ and:*

*(i) if $xy$ is deleted, $y \in N_h(x)$, and $|N_h(x)| \le D + \ell$;*

*(ii) if $xy$ is inserted, $x$ and $y$ have a common neighbour in their cluster, so their distance in $G[L_B]$ is exactly 2;*

*(iii) in both cases $y$ lies in the ball* Ball$(x, 2)$ *of $G[L_B]$, which has at most $Z^2$ vertices.*

*The same holds with the roles of $x$ and $y$ exchanged.*

*Proof.* $x, y \in L_B$ by Lemma G.4(a) ($d_h \le D + u \le B$). (i) is the definition together with Lemma G.4(a). (ii) $x, y$ lie in the exceptional cluster $C$ and are non-adjacent, so Lemma G.2(b) gives $z \in C$ adjacent to both; $C \subseteq L_B$ by Lemma G.6($\alpha$), so the path $x\,z\,y$ lies in $G[L_B]$ and the distance is 2 (not 1 since $xy \notin E$). (iii) A vertex of $G[L_B]$ has at most $B$ hard neighbours and at most $D$ soft ones, so degree at most $B + D = Z - 1$, and a ball of radius 2 has at most $1 + (Z-1) + (Z-1)^2 \le Z^2$ vertices; deleted pairs have distance 1. □

Lemma G.2(b) is what makes the radius 2 independent of $D$ and $\ell$: the candidates for the other end of a hard edit lie within distance 2 of one end in $G[L_B]$. Corollary G.7 says that *one* endpoint of a hard edit determines a set of at most $Z^2$ candidates for the other, for every hard edit, including those of type T2 and deleted edges (Appendix L). What it does not bound is the number of candidates for the first endpoint, which is $n$.

# H Components of $I(H)$

Let $\mathcal{C}_1, \ldots, \mathcal{C}_r$ be the connected components of $I(H)$, $p_j = |\mathcal{C}_j|$ the number of members in $\mathcal{C}_j$, and $W_j = \bigcup_{P \in \mathcal{C}_j} V(P)$. Put $s = \max_j |W_j|$, the largest number of *vertices* spanned by a component.

If $s_P = \max_j p_j$ counts members instead, then $s \le 2s_P + 1$: order the members of a component so that each shares a vertex with an earlier one; the first spans 3 vertices and each later one adds at most 2.

**Theorem H.1** (Component theorem). *CEaMP can be solved in time $1.62^{(\binom{s}{2}+1)\ell}\, n^{O(1)} + n\, 3^s$ and admits a kernel with at most $(s^2 - s + 2)\,\ell$ vertices, where $s$ is the maximum number of vertices spanned by a connected component of $I(H)$. No maximality hypothesis is used. The kernel is an instance of CEaMP with empty packing. The proof uses the soundness of Reduction Rule 6.2 of [17] (their Lemma 6.3), the algorithm of Böcker [2] and the $2k$-vertex kernel of Chen and Meng [4].*

**Corollary H.2.** *If $I(H)$ is a perfect matching (so $w = 1$, $c = 2$), CEaMP is fixed-parameter tractable in $\ell$: here $s = 5$, the time is $1.62^{11\ell} n^{O(1)}$ and the kernel has at most $22\ell$ vertices. More generally components with at most $s_P$ members give Theorem H.1 with $s = 2s_P + 1$.*

*Proof.* Two adjacent nodes of $I(H)$ share exactly one vertex ($H$ is modification-disjoint), so $|W_j| = 3 + 3 - 1 = 5$; $s^2 - s + 2 = 22$ and $\binom{5}{2} + 1 = 11$. □

**What is cited and what is proved.** We use the problem of van Bevern, Froese and Komusiewicz [17]: an instance $(G, \mathcal{G}, k)$ consists of a graph, a set $\mathcal{G}$ of *vertex-disjoint induced subgraphs* $G_j$ with $1 \le \tau(G_j) \le t$, where $\tau$ is the cluster-editing optimum, and a budget $k$; its parameters are $t$ and $k - \sum_j \tau(G_j)$ (Definition 1.2 and Problem 1.3 of [17]; connectivity of the $G_j$ is not required, and their size enters only through $t$). We prove below that the components of $I(H)$ define such an instance with excess at most $\ell$ (Lemma H.3), that the reduction rule can be applied at component level without disturbing the other components (Lemma H.4), and the bound $r \le 2\bar{\ell}$ on the number of irreducible components (Lemma H.6); the last re-derives, for components, the counting argument of [17, Lemmas 3.2 and 6.5]. The following are *cited, not reproved*: (i) soundness of their Reduction Rule 6.2 for an arbitrary induced subgraph $G[W]$ (their Lemma 6.3, whose proof uses only $G$, $W$ and the optimal solution $S$ of $G[W]$); (ii) the time to test and apply the rule (their Lemma 6.4); (iii) the algorithm of Böcker [2] running in $O(1.62^k + n + m)$ and the kernel with at most $2k$ vertices for Cluster Editing [4], both as used in [17].

## H.1 Components as a packing in the sense of van Bevern et al.

**Lemma H.3** (Components form a packing in the sense of [17]). *Let $G_j = G[W_j]$, $\tau_j = \mathrm{OPT}(G_j)$, $k = |H| + \ell$ and $\bar{\ell} = k - \sum_j \tau_j$.*

1. *The sets $W_j$ are pairwise disjoint and each $G_j$ is an induced subgraph.*
2. *$p_j \le \tau_j \le \lfloor \binom{s}{2}/2 \rfloor =: t_s$ and $\tau_j \ge 1$.*
3. *$\bar{\ell} = \ell - \sum_j (\tau_j - p_j) \le \ell$.*
4. *For every solution $S$ of $G$ and every $j$, $S_j := S \cap \binom{W_j}{2}$ is a solution of $G_j$, so $|S_j| \ge \tau_j$; the sets $\binom{W_j}{2}$ are pairwise disjoint, so $|S| = \sum_j |S_j| + |S_{\mathrm{ext}}|$ where $S_{\mathrm{ext}}$ are the pairs of $S$ inside no $W_j$. In particular $\mathrm{OPT}(G) \ge \sum_j \tau_j$, and if $\bar{\ell} < 0$ then $(G, H, \ell)$ is a NO instance.*

*Proof.* (1) A vertex in $W_i \cap W_j$, $i \ne j$, lies in a member of $\mathcal{C}_i$ and in a member of $\mathcal{C}_j$; these are adjacent in $I(H)$, contradicting that $\mathcal{C}_i, \mathcal{C}_j$ are different components. Induced by definition.

(2) The members of $\mathcal{C}_j$ are induced $P_3$'s of $G_j$ with pairwise disjoint pair sets inside $\binom{W_j}{2}$. A solution of $G_j$ must edit a pair of each of them (Lemma 2.2); hence $\tau_j \ge p_j \ge 1$. For the upper

bound, let $m_j$ be the number of edges of $G_j$; deleting all edges, or inserting all non-edges, yields a cluster graph, so $\tau_j \le \min(m_j, \binom{|W_j|}{2} - m_j) \le \binom{|W_j|}{2}/2 \le \binom{s}{2}/2$.

(3) $|H| = \sum_j p_j$ because the components partition $H$.

(4) $(G \triangle S)[W_j] = G_j \triangle S_j$ is an induced subgraph of a cluster graph, hence a cluster graph. The rest is counting with disjoint pair sets. □

Thus $(G, H, \ell)$ is a YES instance iff $(G, \{G_j\}, k)$ is a YES instance of the problem of [17] with $t \le t_s$ and excess $\bar{\ell} \le \ell$ (answer NO if $\bar{\ell} < 0$). Components, not single members, are needed: a pair $uv$ with $u \in W_j$ and $v \notin W_j$ lies in no $W_i$ at all when $v$ is outside $V(H)$ or in another component, hence is not counted in $\sum_j \tau_j$ and is an excess edit (Lemma H.3(4)); for a packing whose members share vertices, the same pair may be a soft pair of another member and is then free, which is the obstruction of [15].

## H.2 Reducible sets and the counting lemma

A set $\mathcal{F}$ of members is *pair-closed* if no member outside $\mathcal{F}$ has two vertices in $W_\mathcal{F} := \bigcup_{P \in \mathcal{F}} V(P)$; every component of $I(H)$ is pair-closed, and then no outside member meets $W_\mathcal{F}$ at all. An optimal solution $S_\mathcal{F}$ of $G[W_\mathcal{F}]$ satisfies the *boundary condition* if for every $y \notin W_\mathcal{F}$ the set $N(y) \cap W_\mathcal{F}$ is empty or exactly one cluster of $G[W_\mathcal{F}] \triangle S_\mathcal{F}$. $\mathcal{F}$ is *reducible* if some optimal $S_\mathcal{F}$ satisfies it, and *irreducible* otherwise. This is the condition of Reduction Rule 6.2 of [17] for the induced subgraph $G[W_\mathcal{F}]$ (stated there as two clauses: vertices in one cluster have the same neighbours outside $W_\mathcal{F}$; vertices in different clusters have no common neighbour outside $W_\mathcal{F}$; the two forms are equivalent). A component $\mathcal{C}_j$ is *tight* if $\tau_j = p_j$.

**Lemma H.4** (Rule 6.2 at component level)**.** *If component $\mathcal{C}_j$ is reducible with $S_j$, then $(G, \{G_i\}, k)$ is a YES instance iff $(G \triangle S_j, \{G_i\}_{i \ne j}, k - \tau_j)$ is. In the new instance the graphs $G_i$ ($i \ne j$) are unchanged, still vertex-disjoint and induced, with the same $\tau_i$; the excess $\bar{\ell}$ is unchanged; and for every $i \ne j$ the boundary condition, hence reducibility, is unchanged. So one pass over the components, in any order, reaches an instance in which no component is reducible.*

*Proof.* The equivalence is Lemma 6.3 of [17] (cited; sound for every induced subgraph). Both $k$ and $\sum \tau$ drop by $\tau_j$. The set $S_j$ consists of pairs with both ends in $W_j$, so no pair incident to a vertex of $W_i$, $i \ne j$, changes: $G[W_i]$ and all neighbourhoods $N(u)$, $u \in W_i$, are the same before and after, and the boundary condition for $i$ refers only to these. □

**Lemma H.5** (Boundary violations force an edited boundary pair)**.** *Let $S$ be any solution, $\mathcal{F}$ pair-closed, $S_\mathcal{F} = S \cap \binom{W_\mathcal{F}}{2}$ and $y \notin W_\mathcal{F}$. If $N(y) \cap W_\mathcal{F}$ is neither empty nor a cluster of $G[W_\mathcal{F}] \triangle S_\mathcal{F}$, then $S$ contains a pair $\{y, u\}$ with $u \in W_\mathcal{F}$.*

*Proof.* Suppose not. Then $N_{G \triangle S}(y) \cap W_\mathcal{F} = N_G(y) \cap W_\mathcal{F}$. In the cluster graph $G \triangle S$ this set is $\mathrm{cluster}(y) \cap W_\mathcal{F}$ (as $y \notin W_\mathcal{F}$), which is empty or a cluster of the induced subgraph $(G \triangle S)[W_\mathcal{F}] = G[W_\mathcal{F}] \triangle S_\mathcal{F}$. □

**Lemma H.6** (Counting irreducible components)**.** *Let $S$ be a solution of $G$ and $J$ a set of irreducible components of $I(H)$. Then $|S| \ge \sum_j \tau_j + |J|/2$. In particular, if $|S| \le k = \sum_j \tau_j + \bar{\ell}$, then $|J| \le 2\bar{\ell}$.*

*Proof. Claim:* for every $j \in J$, (a) $|S_j| \ge \tau_j + 1$, or (b) $S$ contains a pair with exactly one end in $W_j$ (an *external* pair of $j$). Suppose (a) fails; then $|S_j| = \tau_j$ by Lemma H.3(4), so $S_j$ is an optimal solution of $G_j$, and since $\mathcal{C}_j$ is irreducible, $S_j$ violates the boundary condition at some $y \notin W_j$. A component is pair-closed, so Lemma H.5 gives a pair $\{y, u\} \in S$ with $u \in W_j$.

Let $J_a$ be the indices of $J$ satisfying (a) and $J_b$ the others, which satisfy (b). An external pair of $j$ has exactly one end in $W_j$; it lies in no $\binom{W_i}{2}$, so it belongs to $S_{\text{ext}}$, and it is external for at most two indices (the components containing its ends). Hence $|S_{\text{ext}}| \geq |J_b|/2$. Since $|S| = \sum_j |S_j| + |S_{\text{ext}}|$ and $|S_j| \geq \tau_j$ for all $j$, we get $|S| \geq \sum_j \tau_j + |J_a| + |J_b|/2 \geq \sum_j \tau_j + |J|/2$. □

The bound concerns irreducible components, not components with positive internal excess $\tau_j > p_j$: a reducible component may have large internal excess, and it is removed by Lemma H.4 at no cost in $\bar{\ell}$.

*Proof of Theorem H.1. Algorithm.* (1) Compute the components of $I(H)$, the optima $\tau_j$ by the subset dynamic programme over $W_j$ ($O(3^{|W_j|})$), and $\bar{\ell}$; if $\bar{\ell} < 0$ answer NO (Lemma H.3(4)). (2) For each component test reducibility and, if reducible, apply the rule (Lemma H.4); the test of [17, Lemma 6.4] needs a constant number of exact cluster-editing computations on subgraphs of $G_j$ plus polynomial time, which the subset programme provides in $O(3^{|W_j|})$ each. (3) Let $r^*$ be the number of remaining components. If $r^* > 2\bar{\ell}$, answer NO (Lemma H.6 with $J$ the set of all remaining components). Otherwise the current instance has budget $k^* = \sum_{\text{remaining}} \tau_j + \bar{\ell} \leq r^* t_s + \bar{\ell} \leq (2t_s+1)\bar{\ell}$. Forget the packing and solve plain Cluster Editing with budget $k^*$ by Böcker's algorithm [2] in $O(1.62^{k^*} + n + m)$.

*Correctness* follows from Lemma H.3 (the instance is of the required form and $\bar{\ell}$ is the excess), Lemma H.4 (steps preserve the answer), and Lemma H.6.

*Time.* The components have at least 3 vertices and are disjoint, so there are at most $n/3$ of them and steps (1), (2) take $O(n3^s) + n^{O(1)}$. Step (3) takes $1.62^{k^*} n^{O(1)}$ with $k^* \leq (2t_s+1)\bar{\ell} \leq (\binom{s}{2}+1)\ell$, since $2t_s \leq \binom{s}{2}$ and $\bar{\ell} \leq \ell$.

*Kernel.* After step (2) and the checks of step (3), apply the kernelization of Cluster Editing with at most $2k^*$ vertices [4] to $(G, k^*)$ and add the empty packing: an equivalent CEaMP instance with empty packing, budget $\leq k^*$ and at most $2k^* \leq (4t_s+2)\ell \leq (s^2 - s + 2)\ell$ vertices. □

*Remark* H.7 (Scope). Theorem H.1 is fixed-parameter tractable in $(\ell, s)$; it is not an FPT algorithm in $\ell + w$ (which does not exist unless FPT = W[1], Theorem 5.1), since $s$ is not bounded by a function of $w$ (a path or tree $I(H)$ has $w \leq 1$ and arbitrarily large components), and the exponent $\binom{s}{2}\ell$ is of no practical interest. The argument uses neither pair-maximality nor Appendix E. For a perfect matching it covers the smallest non-trivial instance of the case $c = 2$ left open in [15] for fixed-parameter tractability in $\ell$.

### H.3 Components of arbitrary size

For components too large for the subset programme, tightness and reducibility can still be decided when $w$ is small.

**Proposition H.8.** *Given a tree decomposition of $I(H)$ of width $w$ and a component $\mathcal{C}$ of $I(H)$, one decides in time $3^{w+1}n^{O(1)}$ whether $\mathcal{C}$ is tight; if it is tight, one decides in the same time whether it is reducible and finds a witnessing $S_{\mathcal{C}}$.*

*Proof.* A tight solution of $G[W_{\mathcal{C}}]$ consists of one edited pair per member and nothing else (each member needs at least one edited pair and the pair sets are disjoint), and every such solution is optimal since $\tau \geq p$; so for a tight $\mathcal{C}$ the optimal solutions are exactly the tight ones. Variables: the three choices of each member. Constraint: $G[W_{\mathcal{C}}] \triangle S$ has no induced $P_3$, i.e. every triple of $W_{\mathcal{C}}$ is fine. A triple with a soft pair involves at most three members (one per soft pair, pair sets being disjoint); each owns two of the three vertices of the triple, so any two of them share a vertex, they form a clique of $I(H)$ and lie in a common bag. A triple without soft pair is checked statically. So

$\mathcal{C}$ is tight iff this CSP is satisfiable, and Lemma 3.4 decides this in $3^{w+1}n^{O(1)}$ time. The boundary condition at $y \notin W_{\mathcal{C}}$ with $A_y = N(y) \cap W_{\mathcal{C}} \neq \emptyset$ says: (i) every pair inside $A_y$ is an edge of $G \triangle S$ and (ii) every pair between $A_y$ and $W_{\mathcal{C}} \setminus A_y$ is a non-edge of $G \triangle S$ (given that $G[W_{\mathcal{C}}] \triangle S$ is a cluster graph this says $A_y$ is a cluster). Each such pair is hard (static) or owned by one member (a unary constraint). Reducibility is therefore the same CSP with additional unary constraints. □

(For a non-tight component the optimal $S_{\mathcal{C}}$ has more than $p$ edits and is not a $3^p$ assignment; the proposition says nothing about it.)

**Lemma H.9** (Reduction preserves pair-maximality)**.** *Let $H$ be pair-maximal, $\mathcal{F} \subseteq H$ pair-closed and tight with $S_{\mathcal{F}}$ an optimal solution of $G[W_{\mathcal{F}}]$ satisfying the boundary condition, $G^\circ = G \triangle S_{\mathcal{F}}$ and $H^\circ = H \setminus \mathcal{F}$. Then $H^\circ$ is a modification-disjoint packing of $G^\circ$, $(G, H, \ell)$ and $(G^\circ, H^\circ, \ell)$ are equivalent, $H^\circ$ is pair-maximal in $G^\circ$, and $I(H^\circ)$ is an induced subgraph of $I(H)$, so* $\mathrm{tw}(I(H^\circ)) \leq \mathrm{tw}(I(H))$.

*Proof.* Members outside $\mathcal{F}$ have at most one vertex in $W_{\mathcal{F}}$, so none of their pairs lies in $W_{\mathcal{F}}$; $S_{\mathcal{F}}$ does not touch them, so they remain induced $P_3$'s with the same pairs, and $H^\circ$ is a modification-disjoint packing of $G^\circ$. Since $\mathcal{F}$ is tight, $|S_{\mathcal{F}}| = |\mathcal{F}|$, and Rule 6.2 (Lemma 6.3 of [17]) gives $\mathrm{OPT}(G) = \mathrm{OPT}(G^\circ) + |\mathcal{F}|$, hence the equivalence with unchanged $\ell$.

Pair-maximality. Let $Q$ be an induced $P_3$ of $G^\circ$; we show that it has a soft pair with respect to $H^\circ$. Let $q = |V(Q) \cap W_{\mathcal{F}}|$. If $q = 3$, $Q$ is an induced $P_3$ of $G^\circ[W_{\mathcal{F}}] = G[W_{\mathcal{F}}] \triangle S_{\mathcal{F}}$, a cluster graph: impossible. If $q = 2$, $V(Q) = \{p, p', y\}$ with $p, p' \in W_{\mathcal{F}}$, $y \notin W_{\mathcal{F}}$. The pairs $yp, yp'$ are unchanged, and by the boundary condition $N_{G^\circ}(y) \cap W_{\mathcal{F}}$ is empty or one cluster $K$ of the cluster graph $G^\circ[W_{\mathcal{F}}]$. If it is empty, $Q$ has at most one edge. If both $p, p' \in K$, they are adjacent and $Q$ is a triangle. If exactly one is in $K$, the other is a non-neighbour of $y$ and of the first (different clusters), so $Q$ has one edge. If neither is in $K$, $Q$ has at most one edge. In no case is $Q$ an induced $P_3$. If $q \leq 1$, every pair of $Q$ has at most one end in $W_{\mathcal{F}}$, so $Q$ is an induced $P_3$ of $G$ with the same pairs and, $H$ being pair-maximal, has a pair in pairs$(H)$; this pair is not in pairs$(\mathcal{F})$ (those lie inside $W_{\mathcal{F}}$), so it is soft with respect to $H^\circ$. □

**Theorem H.10** (Component reduction)**.** *Let $H$ be* any *packing, $\bar{\ell} = \ell - \sum_j (\tau_j - p_j)$, and apply Rule 6.2 to every reducible tight component (found by Proposition H.8, which needs a tree decomposition of $I(H)$ of width $w$ and costs $3^{w'+1}n^{O(1)}$ with $w' \leq w + \ell$ per test). Then, modulo the cited soundness of Rule 6.2 (Lemma 6.3 of [17]), the reduced instance is equivalent to $(G, H, \ell)$, and if it is a YES instance then $\bar{\ell} \geq 0$ and the number $r$ of components of $I(H^o)$, $H^o$ the remaining packing, satisfies $r \leq \ell + \bar{\ell} \leq 2\ell$. In particular* if more than $2\ell$ components remain, the answer is NO. *No $\tau_j$ of a non-tight component is needed by the algorithm.*

*Proof. The rule.* The set $S_j$ used for a reducible tight component $\mathcal{C}_j$ consists of pairs inside $W_j$. So $G[W_i]$ and the sets $N(y) \cap W_i$ $(i \neq j)$ do not change, $\tau_i$, tightness and (ir)reducibility of the other components are unchanged, $\ell$ and $\bar{\ell}$ are unchanged (Lemma H.4), and one pass reaches an instance with no reducible tight component.

*Counting.* Work in the reduced instance. Let $\mathcal{A}$ be the set of tight (irreducible) components and $\mathcal{B}$ the set of non-tight ones. Since $\sum_j \tau_j = |H| + \sum_j (\tau_j - p_j)$, we have $|H| + \ell = \sum_j \tau_j + \bar{\ell}$. If $\bar{\ell} < 0$ the instance is a NO instance (Lemma H.3). Otherwise let $S$ be an optimal solution with $|S| \leq \sum_j \tau_j + \bar{\ell}$. By Lemma H.6 with $J = \mathcal{A}$, $|\mathcal{A}| \leq 2\bar{\ell}$. Each $j \in \mathcal{B}$ has $\tau_j - p_j \geq 1$, so $|\mathcal{B}| \leq \sum_j (\tau_j - p_j) = \ell - \bar{\ell}$. Therefore $r \leq 2\bar{\ell} + \ell - \bar{\ell} = \ell + \bar{\ell} \leq 2\ell$ (as $\bar{\ell} \leq \ell$). □

The theorem is phrased through the number of remaining components so that it does not need the $\tau_j$ of non-tight components, which are not computable in polynomial time for a large component.

**Corollary H.11.** *(a) For $\ell = 0$ a YES instance is reduced to the empty packing; this agrees with Theorem 3.5. (b) For an arbitrary packing $H$, compute $M$ and $H' = H \cup M$ (Lemma 2.4) and then reduce tight reducible components repeatedly. This ends, in time $3^{w+\ell+1}n^{O(1)}$ given a tree decomposition of $I(H)$ of width $w$, with a pair-maximal packing $H^*$, a budget $\ell^* = \ell - |M| \le \ell$, $\mathrm{tw}(I(H^*)) \le w + |M| \le w + \ell$, and at most $2\ell^*$ components of the intersection graph, or with a correct NO answer.*

*Proof.* (a) $r \le 2\ell = 0$. (b) After the single maximalisation $H'$ is pair-maximal and $\mathrm{tw}(I(H')) \le w + |M|$ (Lemma 2.4). Each reduction step removes the members of a tight reducible component, keeps the budget (Lemma H.9), keeps the packing pair-maximal and does not increase the treewidth of the intersection graph (it is an induced subgraph of the previous one). The loop ends after at most $|H'|$ steps, when no tight component is reducible; Theorem H.10 bounds the number of components by $2\ell^*$. □

What remains after the reduction is the case of *few but arbitrarily large* components of $I(H)$ (for $w = 1$, a single tree of members), which is Open Problem 5.

# I An $n^\ell$ algorithm and the limits of the cluster-identity route

This section uses Corollary G.7 in two ways: guessing the positions of the hard edits (Theorem I.2), and following the clusters directly (Theorem I.3). The first halves the exponent of Theorem 3.6; the second is fixed-parameter tractable for a stronger parameter $\omega$ which, by Proposition I.4, is not bounded by any function of $(\ell, w)$. **Whether CEaMP is fixed-parameter tractable in $\ell + w$ remains open.**

## I.1 The $n^\ell$ algorithm

*Remark* I.1 (Fixed set of hard edits). The proof of Theorem 3.6 shows: for a *fixed* set $U'$ of hard pairs, $m_{U'} = \min \sum_i |x_i|$ over all $x \in \prod \mathcal{D}_i$ with $G \triangle (U' \cup \bigcup x_i)$ a cluster graph is computed in $5^{w+1}n^{O(1)}$ time from a tree decomposition of $I(H)$ of width $w$, and for *every* $U'$ of hard pairs the value $m_{U'} + |U'|$ is the size of an actual solution.

**Theorem I.2** ($n^\ell$ algorithm)**.** *Given $(G, H, \ell)$ for an arbitrary packing $H$ and a tree decomposition of $I(H)$ of width $w$, CEaMP is solvable in time*

$$(n+1)^\ell \, Z_*^{2\ell} \, 5^{\,w+\ell+1} \, n^{O(1)}, \qquad D_* = D(H) + 2\ell, \quad Z_* = 3D_* + 3\ell + 1 = 3D(H) + 9\ell + 1.$$

*Since $D(H) \le 2(w+1)$, $Z_* \le 6w + 9\ell + 7$, so the factor $Z_*^{2\ell}$ is a function of $(\ell, w)$ only. This improves the exponent $2\ell$ of Theorem 3.6 to $\ell$; it is still XP in $\ell$.*

*Proof.* Compute $M$ and $H' = H \cup M$ by Lemma 2.4; reject if $|M| > \ell$. Then $H'$ is pair-maximal, $\ell' = \ell - |M| \le \ell$, $D(H') \le D(H) + 2|M| \le D_*$, a tree decomposition of $I(H')$ of width $w' \le w + |M| \le w + \ell$ is available, and the instance is equivalent to $(G, H', \ell')$. All lemmas of Appendix G apply to $H'$ with $D(H')$ and $\ell'$; put $B, L_B, Z$ accordingly. Then $Z = 3D(H') + 3\ell' + 1 \le Z_*$.

*Algorithm.* Compute $L_B$ and, for $x \in L_B$, the set $P(x)$ of vertices $y \in \mathrm{Ball}_{G[L_B]}(x, 2) \setminus \{x\}$ with $xy$ hard; $|P(x)| \le Z^2$. Enumerate all sequences of at most $\ell'$ pairs $(x_1, y_1), \ldots, (x_j, y_j)$ with $x_i \in L_B$ and $y_i \in P(x_i)$; there are at most $(1 + nZ^2)^{\ell'} \le (n+1)^{\ell'} Z^{2\ell'}$ of them. For each, let $U' = \{x_i y_i\}$ (a set of at most $\ell'$ hard pairs), compute $m_{U'}$ by Remark I.1 for $H'$ in $5^{w'+1}n^{O(1)}$ time, and accept iff $m_{U'} + |U'| \le |H'| + \ell'$.

*Soundness.* By Remark I.1, $m_{U'} + |U'|$ is the size of an actual solution of $G$ for every $U'$ of hard pairs of $H'$; acceptance gives a solution of size $\leq |H'| + \ell' = |H| + \ell$.

*Completeness.* Suppose the instance is a YES instance and let $S$ be an optimal solution, $|S| \leq |H'| + \ell'$, with hard edits $U$, $|U| \leq \ell'$ (Lemma 2.2 for $H'$). For each pair of $U$ take one endpoint $x$; it lies in $L_B$ and the other endpoint lies in $P(x)$ by Corollary G.7. So $U$ is among the enumerated sets, and for it $m_U + |U| \leq |S| \leq |H'| + \ell'$ (the soft part of $S$ is a feasible choice of $x$ for $U$ by Lemma 2.2).

*Time.* $(n+1)^{\ell'} Z^{2\ell'} 5^{w'+1} n^{O(1)} \leq (n+1)^{\ell} Z_*^{2\ell} 5^{w+\ell+1} n^{O(1)}$. □

The exponent is halved because the candidate list is attached to *one* endpoint (Corollary G.7); Theorem 3.6 guesses both. Note that $\ell$ bounds the *number* of hard edits but not the cost of locating them: the $n^{\ell}$ comes from the position of the first endpoint of each pair. A fixed-parameter algorithm along this route needs a bound on these positions as well.

## I.2 A fixed-parameter algorithm for a stronger parameter

Let $H$ be pair-maximal and $\ell, D, B, Z, L_B, \mathrm{BIG}$ as in Appendix G. Let $V' = V \setminus \mathrm{BIG}$ and let $\Gamma'$ be the graph on $V'$ whose edges are all edges of $G$ and all soft pairs with both ends in $V'$.

**Theorem I.3** (Fixed-parameter tractability in $(\ell, D, \omega)$)**.** *Let $\omega = \mathrm{tw}(\Gamma')$ and assume a tree decomposition of $\Gamma'$ of width $\omega$ is given. Then one can compute in time $(Z(\omega+1))^{O(\omega)} n + n^{O(1)}$ a number $m^*$ such that*

$$\mathrm{OPT}(G) \leq |H| + \ell \iff |E| + \sum_K f(K) + m^* \leq |H| + \ell,$$

*where $K$ ranges over the hard components of* BIG *and $f$ is as in Lemma G.1. The proof, with the dynamic program, is in Appendix J.*

All quantities in Theorem I.3 refer to the pair-maximal instance $(G, H', \ell')$; they are not functions of the input $(G, H, \ell)$ (Section 2).

## I.3 Why $\omega$ is not bounded by $(\ell, w)$

**Proposition I.4** ($\omega$ is unbounded at $\ell = 0$, $w = 0$)**.** *For every cubic graph $F$ there is an instance $(G_F, H_F, 0)$ with $H_F$ a vertex-disjoint pair-maximal packing ($w = 0$, $c = 1$, $D = 2$), $\mathrm{OPT}(G_F) = |H_F|$ (a YES instance with $\ell = 0$), $\mathrm{BIG} = \emptyset$, every hard component of size at most 6 and every hard degree at most 4, and $\mathrm{tw}(\Gamma') \geq \mathrm{tw}(F)$. By [9] (cited, not re-proved here) there are cubic graphs $F$ (expanders) with $\mathrm{tw}(F) = \Omega(|V(F)|)$. Hence $\omega$ is not bounded by any function of $(\ell, w)$.*

*Proof.* Orient each edge of $F$ arbitrarily; for each oriented edge $e = (p \to q)$ introduce three vertices $a_e, b_e, c_e$. Let $\mathcal{K}_v = \{a_e : e \text{ leaves } v\} \cup \{b_e, c_e : e \text{ enters } v\}$ and make each $\mathcal{K}_v$ a clique; add the cross edges $a_e b_e$ and nothing else. Then $|\mathcal{K}_v| = \mathrm{out}(v) + 2\,\mathrm{in}(v) = 3 + \mathrm{in}(v) \leq 6$. $H_F = \{P_e = (a_e, b_e, c_e)\}$: $a_e b_e, b_e c_e \in E$ and $a_e c_e \notin E$ since $a_e \in \mathcal{K}_p$, $c_e \in \mathcal{K}_q$, $p \neq q$ and cross edges are of the form $a_{e'} b_{e'}$; so $P_e$ is an induced $P_3$. The members are vertex-disjoint, so modification-disjoint, $c = 1$, $w = 0$ and $D = 2$.

*Pair-maximality.* In an induced $P_3$ $x\,y\,z$ not both edges can lie inside cliques, since then $x, z$ would be in the clique of $y$ (each vertex is in exactly one clique) and adjacent. So one of $xy, yz$ is a cross edge $a_e b_e$, which is soft.

$\mathrm{OPT} = |H_F|$. The partition $\{\mathcal{K}_v\}$ has no inserted pair and deletes exactly the cross edges, $|H_F|$ of them; and $\mathrm{OPT} \geq |H_F|$ by Lemma 2.2.

*Hard structure.* The soft pairs inside $\mathcal{K}_v$ are $b_e c_e$ ($e$ entering $v$); all other pairs inside $\mathcal{K}_v$ are hard edges. So each $\mathcal{K}_v$ is one hard component and it is clean. The hard degree of $a_e \in \mathcal{K}_p$ is $|\mathcal{K}_p| - 1 \le 4$ (as $a_e$ is an out-edge, $\mathrm{in}(p) \le 2$, $|\mathcal{K}_p| \le 5$); that of $b_e$ or $c_e \in \mathcal{K}_q$ is $|\mathcal{K}_q| - 2 \le 4$. So $d_h \le 4 = 2D + 3\ell$ at $\ell = 0$, hence no vertex has $d_h > B$ and $\mathrm{BIG} = \emptyset$.

*Minor.* $V' = V$, each $\mathcal{K}_v$ is connected in $\Gamma'$ (a clique), and an edge or soft pair of $\Gamma'$ between $\mathcal{K}_p \neq \mathcal{K}_q$ is one of $a_e b_e$, $a_e c_e$, which exist only for the oriented edge $e = (p \to q)$ of $F$. Contracting the $\mathcal{K}_v$ gives exactly $F$, so $F$ is a minor of $\Gamma'$ and $\mathrm{tw}(\Gamma') \ge \mathrm{tw}(F)$. □

*Remark* I.5 (Scope of Proposition I.4). The proposition refutes only the claim "$\omega \le f(\ell, w)$"; it does *not* refute fixed-parameter tractability. The family has $w = 0$ and $\ell = 0$; it is solved in polynomial time by Theorem 3.6 and has $s = 3$ in Theorem H.1. The obstruction is to the representation by $\Gamma'$ (and by any single decomposition used both for the soft pairs and for the clusters): hard edges between far vertices are free (not in the budget), arbitrary in number, and make footprints of clusters disconnected in any decomposition of $I(H)$ or of $G_{\mathrm{soft}}$. At $\ell = 0$ they are harmless because the CSP of Theorem 3.5 needs no identities; for $\ell > 0$ an exceptional cluster can be assembled from vertices that are far apart (a $c = 1$ instance with two vertex-disjoint packed $P_3$'s joined by one hard edge that is deleted). So a dynamic program on one decomposition cannot be used for both purposes without a preprocessing step; a reduction lemma that would supply it is Open Problem 7.

# J The dynamic program for $\omega$ and the structure of $\Gamma'$

This appendix proves Theorem I.3, using the notation $f(C) = \binom{|C|}{2} - 2e(C)$ of Lemma G.1, and records two further statements on the graph $\Gamma'$. Throughout, $H$ is pair-maximal.

## J.1 Proof of Theorem I.3

**Theorem J.1** (Theorem I.3 with bounds on $m^*$)**.** *Let $H$ be pair-maximal and let $\ell$, $D$, $B$, $Z$, $L_B$, BIG, $V'$, $\Gamma'$, $\omega$ be as in Appendix I, and assume a tree decomposition of $\Gamma'$ of width $\omega$ is given. Then one can compute in time $(Z(\omega+1))^{O(\omega)}n + n^{O(1)}$ a number $m^*$ such that*

$$\mathrm{OPT}(G) \le |H| + \ell \iff |E| + \sum_K f(K) + m^* \le |H| + \ell,$$

*where $K$ ranges over the hard components of* BIG *and $f$ is as in Lemma G.1. Moreover $m_{\mathrm{all}} \le m^* \le m_{\mathrm{conn}}$, where $m_{\mathrm{all}}$ ($m_{\mathrm{conn}}$) is the minimum of $\sum_{C\in\pi'} f(C)$ over all partitions $\pi'$ of $V'$ with clusters of at most $Z$ vertices (and, for $m_{\mathrm{conn}}$, connected in $\Gamma'$).*

*Proof.* *($\Leftarrow$)* Let $\pi'$ be the partition of $V'$ realising $m^*$ (the dynamic program below produces a real partition) and $\pi = \pi' \cup \{K : K \text{ a component of BIG}\}$. By Lemma G.1, $\mathrm{cost}(\pi) = |E| + \sum_K f(K) + m^* \ge \mathrm{OPT}$, so the right-hand side gives $\mathrm{OPT} \le |H| + \ell$. No structure of $G$ is needed for this direction.

*($\Rightarrow$)* Let $S$ be optimal with $|S| \le |H| + \ell$. By Lemma G.6($\gamma$) every component of BIG is a whole cluster; let $\pi'$ be the restriction of $S$'s partition to $V'$. Its clusters are $\Gamma'$-connected and have at most $Z$ vertices:

- *Exceptional $C$.* $C \cap \mathrm{BIG} = \emptyset$ by ($\gamma$), so $C \subseteq V'$; $G[C]$ is connected by ($\alpha$) and its edges are edges of $\Gamma'$. $|C| \le 2D + \ell + 1 \le Z$.
- *Non-exceptional $C$ not in* BIG. By ($\beta$), $C$ is a union of whole hard components, none of which is in BIG (a component of BIG is a cluster by itself), so $C \subseteq V'$. Each component is

connected by hard edges, which are edges of $\Gamma'$. If there are at least two components, all pairs between them are soft, hence edges of $\Gamma'$, so $C$ is $\Gamma'$-connected, and $|C| \le 2D$. If there is exactly one component $K = C$, it is clean, so every vertex $y$ of $K$ has all other vertices of $K$ as hard neighbours or soft partners, $|K| \le d_h(y) + D + 1 \le B + D + 1 = Z$ because $y \notin \mathrm{BIG}$ has $d_h(y) \le B$.

So $\pi'$ is feasible for $m_{\mathrm{conn}}$ and $\mathrm{cost}(S) = |E| + \sum_K f(K) + \sum_{C \in \pi'} f(C) \ge |E| + \sum_K f(K) + m_{\mathrm{conn}} \ge |E| + \sum_K f(K) + m^*$.

*The dynamic program.* Take a nice tree decomposition of $\Gamma'$ of width $\omega$ with empty root bag. A state at node $t$ is a partition $\mathcal{P}$ of the bag $X_t$ into blocks together with a size $s(\mathsf{b}) \in \{|\mathsf{b}|, \dots, Z\}$ for each block $\mathsf{b}$ (the number of vertices introduced so far, in the subtree of $t$, of the cluster containing $\mathsf{b}$). Its value is the minimum, over partitions of the introduced vertices into clusters that realise $(\mathcal{P}, s)$ (every cluster either contains a block of $\mathcal{P}$ or has been *closed*, i.e. has no vertex left in the bag), of $\sum_C g(C)$, with $g(C) = \binom{|C|}{2} - 2e(C)$ evaluated on the introduced vertices of $C$. *Introduce* $v$ into the block $\mathsf{b}$ of size $s$ (or into a new block of size 1): the value increases by $s - 2k$, where $k$ is the number of $G$-neighbours of $v$ in $\mathsf{b}$; here we use that an introduced vertex $y$ that has been forgotten cannot be a $G$-neighbour of $v$, because $yv$ would be an edge of $\Gamma'$ and $y, v$ would share a bag. *Forget* $v$: if its block has other bag vertices the block shrinks; otherwise the cluster is closed and is never extended. *Join* of two children with the same partition $\mathcal{P}$: for a block with bag part $\mathsf{b}$ and sizes $s_1, s_2$ in the two children, put $x_1 = s_1 - |\mathsf{b}|$, $x_2 = s_2 - |\mathsf{b}|$; the new size is $s_1 + s_2 - |\mathsf{b}|$ and the value is $g_1 + g_2 - g(\mathsf{b}) + x_1 x_2$ summed over blocks, where $g(\mathsf{b}) = \binom{|\mathsf{b}|}{2} - 2e(\mathsf{b})$ is read off the bag. This is correct because pairs inside $\mathsf{b}$ and edges inside $\mathsf{b}$ are counted in both children, pairs between the two exclusive parts number $x_1 x_2$, and there is no $G$-edge between the exclusive parts (such an edge would be an edge of $\Gamma'$ between vertices forgotten in different subtrees, which share no bag). *The join term $x_1 x_2$ and the subtraction of $g(\mathsf{b})$ are essential*: omitting them gives a different value on some inputs. At the root the value is $\sum_C f(C)$ over a partition of $V'$ with clusters of size $\le Z$. Call it $m^*$ (the minimum over states; we treat the closed clusters as final).

Every state sequence yields a real partition of $V'$ with cluster sizes $\le Z$ and correct cost, so $m^* \ge m_{\mathrm{all}}$. A $\Gamma'$-connected partition is represented: if the last bag vertex of a cluster $C$ is forgotten at $t$, the vertex subtrees of $C$ have a connected union (as $C$ is $\Gamma'$-connected), which cannot contain the parent of $t$, so it lies below $t$ and all of $C$ has been introduced; hence no later vertex needs to join a closed cluster. So $m^* \le m_{\mathrm{conn}}$. There are at most $(\omega+1)^{\omega+1} Z^{\omega+1}$ states per node, a join combines at most $Z^{2(\omega+1)}$ pairs of states with equal partition, and the tree has $O(\omega n)$ nodes; the total time is $(Z(\omega+1))^{O(\omega)} n$. Computing $d_h$, BIG and $\Gamma'$ takes $n^{O(1)}$.

Combining: $\mathrm{OPT} \le |H| + \ell$ implies $|E| + \sum_K f(K) + m^* \le \mathrm{cost}(S) \le |H| + \ell$ by ($\Rightarrow$), and the converse is ($\Leftarrow$). □

*Remark* J.2. (1) Cleanliness of the components of BIG need not be tested: it is implied when the instance is a YES instance (Lemma G.6($\gamma$)), and the inequality itself accounts for the true cost $f(K)$; rejecting an unclean component is a harmless optimisation. (2) The proof uses that an exceptional cluster is disjoint from BIG and hence lies in $V'$. It is *not* true that $L_B \subseteq V'$: a component of BIG can contain vertices of small hard degree (pendant vertices of a hub component). (3) Here no pointer is guessed: far merges are carried by $\Gamma'$-connectivity. As for Theorem I.2, $D, B, Z, \ell$ and $\omega$ refer to the pair-maximal instance $(G, H', \ell')$ of Lemma 2.4. Without a tree decomposition of $\Gamma'$, one is computed in $2^{O(\omega)} n$ time with width $2\omega + 1$ [14].

### J.2 Further statements

**Lemma J.3** (Cost decomposition along a decomposition of the soft graph)**.** *Let $\pi$ be a partition of $V$, let $(\mathcal{T}, \beta)$ be a tree decomposition of $G_{\text{soft}}$ (Proposition F.6) and call a hard pair* local *if it lies in a common bag. Then* $\text{cost}(\pi) = \sum_t \text{loc}_t + \text{far}$*, where each edited soft pair and each edited local hard pair is charged to one bag containing it,* $\text{loc}_t$ *depends only on* $\pi|_{\beta(t)}$*, and* far *is the number of edited non-local hard pairs, so* $\text{far} \le |U|$*.*

*Proof.* Bookkeeping: all edited pairs are soft, or hard and local, or hard and non-local; soft pairs always lie in a bag. □

(The $\ell = 0$ CSP of Theorem 3.5 is the local part; all non-locality is in far.)

**Proposition J.4** (Structure of $\Gamma'$)**.** *Let $(G, H, \ell)$ be a YES instance with $H$ pair-maximal.*

*(a)* $\Gamma' = G_{\text{soft}}[V'] \cup G_{\text{hard}}[V']$*, where $G_{\text{hard}}$ is the graph of hard edges.*

*(b) Put* $\widehat{Z} = \max\{Z, (\ell+1)(2D+\ell+1)\}$*. Every hard component $K$ not in* BIG *has* $|K| \le \widehat{Z}$*. So "reject if a non-*BIG *hard component has more than $\widehat{Z}$ vertices" is a safe polynomial rule.*

*(c) Let* $\Gamma'_{/K}$ *be $\Gamma'$ with every hard component contracted. Then* $\text{tw}(\Gamma'_{/K}) \le \text{tw}(\Gamma') \le \widehat{Z}(\text{tw}(\Gamma'_{/K}) + 1) - 1$*.*

*Proof.* (a) Every edge of $G$ is hard or soft, and soft pairs are edges of $G_{\text{soft}}$. (b) Take an optimal $S$ with $|S| \le |H| + \ell$ and let $U$ be its edited hard pairs. If no edge of $K$ is deleted, $K$ lies in one cluster $C$. If $C$ is exceptional, $|C| \le 2D+\ell+1 \le Z$ (Lemma G.6($\alpha$)). If not, $C$ is a union of whole hard components (Lemma G.6($\beta$)); with several components $|C| \le 2D$, and a single component $K = C$ is clean, so each $y \in K$ has all other vertices of $K$ as hard neighbours or soft partners and $|K| \le d_h(y) + D + 1 \le B + D + 1 = Z$ ($K \not\subseteq$ BIG). If edges of $K$ are deleted, there are at most $\ell+1$ pieces of $K$ (components of the kept hard edges); $K$ is connected, so each piece contains an endpoint of a deleted edge and lies in an exceptional cluster of size $\le 2D+\ell+1$; thus $|K| \le (\ell+1)(2D+\ell+1)$. (c) Hard components are connected in $\Gamma'$ by hard edges, so $\Gamma'_{/K}$ is a minor of $\Gamma'$. Conversely replace every node of a tree decomposition of $\Gamma'_{/K}$ by the vertices of its component (of size $\le \widehat{Z}$ by (b)). □

By Proposition F.6 the soft graph $G_{\text{soft}}$ has treewidth $\le 3w + 2$, so the unbounded treewidth of $\Gamma'$ in Proposition I.4 comes from hard edges between far vertices, which are free.

## K Conflict weights, pruning and far pointers

This appendix concerns the two kinds of hard edit that the locality results of Appendix L leave open: inserted pairs of type T2 and deleted hard edges. It proves a *conflict lemma* that gives every soft pair a static weight, a defect bound and a polynomial NO-test that follow from it, bounded candidate lists, a pruned version of Theorem I.3, and the statement that putting "locally bad" members into every bag does not repair the far pointers of Proposition L.8 for distances 2 and 3. Nothing here proves or refutes fixed-parameter tractability in $\ell + w$.

**Setting.** $H$ is pair-maximal (otherwise apply Lemma 2.4 and read the primed instance), $Q(v)$ is the set of soft partners, $N_h(v)$ the set of hard neighbours, $D \ge \max_v |Q(v)|$. A solution is identified

with a partition $\pi$ of $V$; write $x(p,q) = 1$ if $p, q$ lie in one block and 0 otherwise; $U$ is the set of edited hard pairs (so $|U| \le |S| - |H|$ by Lemma 2.2). For a soft pair $\sigma = (z, b)$ put

$$\Delta_{\neq}(z,b) = \{y \notin \{z,b\} : \ zy, by \text{ hard, exactly one an edge of } G\},$$

$$\Delta_{=}(z,b) = N_h(z) \cap N_h(b).$$

Both sets are computed from $(G, H)$ alone. For a solution, the *weight* of the soft pair $\sigma = (z, b)$ is $\mathrm{wt}(\sigma) = |\Delta_{\neq}(z,b)|$ if $z, b$ lie in one cluster (*merged*) and $\mathrm{wt}(\sigma) = |\Delta_{=}(z,b)|$ otherwise. A *conflict* is a pair $(\sigma, y)$ with $y$ in the relevant set.

## K.1 The conflict lemma and its consequences

**Lemma K.1** (W1: conflicts)**.** *Let $\pi$ be* any *solution and $\sigma = (z, b)$ a soft pair.*

*(a) If $z, b$ are in one cluster, every $y \in \Delta_{\neq}(z,b)$ has $zy \in U$ or $by \in U$.*

*(b) If $z, b$ are in different clusters, every $y \in \Delta_{=}(z,b)$ has $zy \in U$ or $by \in U$.*

*Distinct $y$ give distinct edited pairs, hence $|U| \ge \mathrm{wt}(\sigma)$ for every soft pair $\sigma$.*

*Proof.* (a) In a cluster graph, $z, b$ in one cluster implies $x(z,y) = x(b,y)$ for all $y$. For $y \in \Delta_{\neq}(z,b)$ the $G$-values of the two hard pairs $zy, by$ differ, so at least one of them is changed. (b) If neither $zy$ nor $by$ is edited, then $x(z,y) = x(b,y) = 1$ (both are edges of $G$), so $z, b, y$ share a cluster, a contradiction. The edited pair chosen for $y$ contains $y$ and one of $z, b$, so different $y$ give different pairs. □

Neither maximality nor optimality is used. Lemma L.6 is the merged case of (a): for $z, b$ in one cluster $\Theta(z,b) \subseteq \Delta_{\neq}(z,b)$, so $|\Theta(z,b)| \le |U|$. (The hypothesis that $z, b$ lie in one cluster is essential: for unmerged $z, b$ the set $\Theta(z,b)$ can be large.)

**Lemma K.2** (W2: defect bound)**.** *For every solution, $\sum_{\sigma \text{ soft}} \mathrm{wt}(\sigma) \le 2D\,|U|$.*

*Proof.* Map each conflict $((z,b), y)$ to an edited pair of $\{zy, by\}$ (Lemma K.1). An edited pair $e = \{p, y\}$ is chosen by conflicts $(\sigma, y)$ with $\sigma \ni p$, at most $|Q(p)| \le D$ of them, and by conflicts $(\sigma', p)$ with $\sigma' \ni y$, at most $D$ of them. □

The constant $2D$ is attained for $D = 2$ ($c = 1$) by instances with $n = 6, 7$ found by computer search; for $D = 4$ and $D = 6$ the same search reached only $1/2$ and $1/3$ of the bound, so attainment for $D \ge 4$ is open.

**Corollary K.3** (LB: a lower bound from the weights)**.** *Let $\kappa_i \ge 1$ be the number of edited pairs among* pairs$(P_i)$ *and let* $\mathrm{Wt} = \sum_\sigma \mathrm{wt}(\sigma)$ *be the total weight of a solution $\pi$. Then* $\mathrm{cost}(\pi) \ge \sum_i \kappa_i + \lceil \mathrm{Wt}/2D \rceil$.

*Proof.* $\mathrm{cost}(\pi) = \sum_i \kappa_i + |U|$ (Lemma 2.2) and $|U| \ge \mathrm{Wt}/2D$ by Lemma K.2. □

The right-hand side is not exact: minimised over assignments of the soft pairs that are transitive on all-soft triples, it can equal $|H|$ although $\mathrm{OPT} > |H|$ (found by computer search on small random instances; the triples with two soft pairs and one hard pair are not checked). We only use it as a lower bound.

**Lemma K.4** (Static pruning)**.** *Let $|S| \le |H| + \ell$. If $|\Delta_{\neq}(u,v)| > \ell$, the soft pair $(u,v)$ is split in $S$; if $|\Delta_{=}(u,v)| > \ell$ it is merged in $S$.*

*Proof.* Lemma K.1 and $|U| \le |S| - |H| \le \ell$. □

**Lemma K.5** (DEL: rigid hard edges)**.** *If $zy$ is a deleted hard edge of a solution, then $|U| \ge 1 + |N_h(z) \cap N_h(y)|$. Hence, if $|S| \le |H| + \ell$, a hard edge with at least $\ell$ common hard neighbours is never deleted (it is* rigid*), and the hard components of rigid edges lie inside single clusters.*

*Proof.* $z$ and $y$ are in different clusters. A common hard neighbour $w$ cannot be kept with both, so $wz$ or $wy$ is a deleted hard edge; distinct $w$ give distinct pairs (each contains $w$), all different from $zy$. □

A member $P = (x, y, z)$ with centre $y$ has three states of member-cost one: delete $xy$, delete $yz$, insert $xz$. A state fixes whether each of the three pairs of $P$ is merged, hence a weight for each. Call $P$ *locally bad* if in each of its three states some pair of $P$ has positive weight. This is a static property.

**Lemma K.6** (BAD: few locally bad members)**.** *For every solution of size $|H| + \varepsilon$ the number of locally bad members is at most $2c\varepsilon$. Hence more than $2c\ell$ locally bad members is a NO-instance of CEaMP with budget $\ell$ (a polynomial test).*

*Proof.* By Lemma 2.2, $\varepsilon = \sum_i (\kappa_i - 1) + |U|$. A bad member with $\kappa_i \ge 2$ contributes at least 1 to the first sum. A bad member with $\kappa_i = 1$ is in one of its three cost-one states, which has a conflict $(\sigma, y)$ with $\sigma$ a pair of $P_i$; by Lemma K.1 an edit of $U$ is incident to $V(P_i)$. An edit has two ends and each vertex lies in at most $c$ members, so there are at most $2c|U|$ such members. In total at most $\sum_i (\kappa_i - 1) + 2c|U| \le 2c\varepsilon$. □

**Proposition K.7** (Candidate lists for (T2) pairs and deleted edges)**.** *Let $H$ be pair-maximal, $S$ an* optimal *solution with $|S| \le |H| + \ell$, and $ab \in U$. Define, from $(G, H)$ alone,*

$$L_1(a) = \begin{cases} N_h(a) \cup \bigcup_{z \in N_h(a)} Q(z) & \text{if } d_h(a) \le D + \ell, \\ \emptyset & \text{otherwise,} \end{cases} \qquad L_2(b) = \bigcup_{z \in Q(b),\ |\Delta_{\neq}(z,b)| \le \ell} \Delta_{\neq}(z, b).$$

*Then $|L_1(a)| \le (D + \ell)(D + 1)$ and $|L_2(b)| \le D\ell$, and:*

*(i) if $ab$ is a deleted hard edge, then $b \in L_1(a)$ and $a \in L_1(b)$;*

*(ii) if $ab$ is an inserted pair of type (T2) with middle vertex $z$, $az$ hard (kept) and $zb$ soft (Theorem L.2), then $b \in L_1(a)$ and $a \in L_2(b)$.*

*Pairs of type (T1) are not covered; they are bag-local by Proposition L.5.*

*Proof.* Every endpoint $x$ of a pair of $U$ has $d_h(x) \le D + |U| \le D + \ell$ (Lemma G.4(a)), so the guard in $L_1$ holds for the ends. (i) The deleted pair is a hard edge, so $b \in N_h(a)$, and symmetrically. (ii) $z \in N_h(a)$ and $b \in Q(z)$, so $b \in L_1(a)$, and $|L_1(a)| \le (D + \ell) + (D + \ell)D$. For the soft end: $z \in Q(b)$ ($\le D$ choices); since $az$ is kept and $ab$ is inserted, $a, z, b$ lie in one cluster, so $zb$ is merged, and $az$ is a hard edge while $ab$ is a hard non-edge, so $a \in \Delta_{\neq}(z, b)$, and $|\Delta_{\neq}(z, b)| \le |U| \le \ell$ by Lemma K.1(a). Hence $a \in L_2(b)$ and $|L_2(b)| \le D\ell$. □

Optimality is used only through Theorem L.2. The lists say where the partner of a given end can be; they do not say where the first end is (Corollary G.7).

## K.2 A pruned form of Theorem J.1

Let BIG, $V'$, $Z$ be as in Theorem J.1 and let $\Gamma^\ell$ be the graph on $V'$ whose edges are all hard edges inside $V'$ and all soft pairs $(u,v)$ inside $V'$ with $|\Delta_{\neq}(u,v)| \le \ell$. It is a spanning subgraph of $\Gamma'$, so $\mathrm{tw}(\Gamma^\ell) \le \mathrm{tw}(\Gamma')$, and it is computable from $(G,H)$ alone.

**Theorem K.8** (P: pruned $\omega$-theorem)**.** *Let $H$ be pair-maximal and let $\omega_\ell = \mathrm{tw}(\Gamma^\ell)$, with a tree decomposition of that width given. Then one can compute in time $(Z(\omega_\ell+1))^{O(\omega_\ell)}n+n^{O(1)}$ a number $m^*$ with*

$$\mathrm{OPT}(G) \le |H| + \ell \iff |E| + \sum_K f(K) + m^* \le |H| + \ell,$$

*$K$ ranging over the hard components of* BIG*. So CEaMP is FPT in $(\ell, D, \mathrm{tw}(\Gamma^\ell))$.*

*Proof.* Let $e^+(C)$ be the number of edges of $G$ inside $C$ that are edges of $\Gamma^\ell$, and $\tilde f(C) = \binom{|C|}{2} - 2e^+(C)$. Run the dynamic program of the proof of Theorem J.1 on $\Gamma^\ell$, with $e^+$ in place of $e$ everywhere (in the introduce step $k$ counts the $\Gamma^\ell$-neighbours of $v$ in the block, and in the join step there is no $\Gamma^\ell$-edge between the exclusive parts). Exactly as there, it computes $m^* =$ the minimum of $\sum_C \tilde f(C)$ over $\Gamma^\ell$-connected partitions of $V'$ with clusters of at most $Z$ vertices, in the stated time.

($\Leftarrow$) $e^+ \le e$, so $f(C) \le \tilde f(C)$ and the partition realising $m^*$, completed by the components of BIG, has true cost at most $|E| + \sum_K f(K) + m^*$ (Lemma G.1); no structure is needed.

($\Rightarrow$) Let $S$ be optimal, $|S| \le |H| + \ell$. By the proof of Theorem J.1, the clusters of $S$ inside $V'$ have at most $Z$ vertices and are connected through edges of $G$ inside the cluster and (for a cluster with several hard components) soft pairs inside the cluster. A soft pair inside a cluster is merged, so $|\Delta_{\neq}| \le \ell$ by Lemma K.4 and it is an edge of $\Gamma^\ell$; every hard edge is an edge of $\Gamma^\ell$. Hence these clusters are $\Gamma^\ell$-connected and $e^+(C) = e(C)$, $\tilde f(C) = f(C)$. So $\mathrm{cost}(S) = |E| + \sum_K f(K) + \sum_C f(C) \ge |E| + \sum_K f(K) + m^*$. □

Theorem K.8 does not use optimality of the merges, only their cost, and it therefore cannot decide the open question.

**Proposition K.9** (Theorem K.8 is strictly stronger than Theorem I.3)**.** *Let $F$ be a cubic graph with an orientation in which every in-degree is 1 or 2 (so every vertex has in-degree $\ge 1$; for a 2-edge-connected $F$ such an orientation exists by Robbins' theorem [16]), and let $(G_F, H_F)$ be the instance of Proposition I.4. For every budget $\ell \le 4$, every component of $\Gamma^\ell$ has at most 5 vertices (so $\mathrm{tw}(\Gamma^\ell) \le 4$), whereas $\mathrm{tw}(\Gamma') \ge \mathrm{tw}(F)$.*

*Proof.* For the oriented edge $e = (p \to q)$ the soft pairs $a_eb_e$ and $a_ec_e$ join $\mathcal{K}_p$ and $\mathcal{K}_q$. The soft partners of $a_e$ are $b_e, c_e$ and those of $b_e$ are $a_e, c_e$ (the three pairs of $P_e$). Hence $\Delta_{\neq}(a_e, b_e)$ consists of the vertices $y \in \mathcal{K}_p \setminus \{a_e\}$ (the pair $a_ey$ is a hard edge and $b_ey$ a hard non-edge), of which there are $2 + \mathrm{in}(p)$, and of the vertices $y \in \mathcal{K}_q \setminus \{b_e, c_e\}$ (here $b_ey$ is a hard edge and $a_ey$ a hard non-edge), of which there are $1 + \mathrm{in}(q)$. So $|\Delta_{\neq}(a_e, b_e)| = 3 + \mathrm{in}(p) + \mathrm{in}(q) \ge 5$, and the same count holds for $a_ec_e$. For $\ell \le 4$ both cross pairs are pruned, the only edges of $\Gamma^\ell$ are those inside the cliques $\mathcal{K}_v$, and $|\mathcal{K}_v| = 3 + \mathrm{in}(v) \le 5$. The bound on $\mathrm{tw}(\Gamma')$ is Proposition I.4. □

The in-degree condition matters: with a source $p$ (in-degree 0) the weight can be 4.

### K.3 Far pointers and locally bad members

An edited pair $e = (a, b)$ of a solution $S$ is *two-sided* if there are vertices $t, t'$ with $(t, a)$ soft, $(t', b)$ soft, and $\{a, b, t\}$, $\{a, b, t'\}$ induced $P_3$'s of $G \triangle (S \setminus \{e\})$. It is *far* (at distance $d$) if the members containing $a$ and $b$ are at distance $\geq d$ in $I(H)$.

**Proposition K.10** (NF-bad fails for distance 2 and 3)**.** *The following statement is false for "far" meaning distance $\geq 2$ and for "far" meaning distance $\geq 3$ in $I(H)$:* there is an optimum in which every far two-sided edit touches a locally bad member.

*Witnesses.* Both instances have $n = 10$, a pair-maximal modification-disjoint packing, a unique optimum, $\mathrm{OPT} = |H| + 1 = 6$, no locally bad member, and exactly one edit, a hard deleted edge, which is two-sided (found by enumerating all set partitions). *Distance 2:* $E = \{01, 06, 12, 23, 29, 34, 45, 56, 67, 78, 79\}$, $H = \{(0, 1, 2), (2, 3, 4), (4, 5, 6), (6, 7, 8), (2, 9, 7)\}$; the optimum $\{0, 1\}, \{2, 9\}, \{3, 4\}, \{5, 6\}, \{7, 8\}$ deletes $\{0, 6\}$ (witnesses 1 and 5). *Distance 3:* $E = \{01, 08, 12, 14, 23, 34, 45, 49, 56, 67, 78\}$, $H = \{(0, 1, 2), (2, 3, 4), (4, 5, 6), (6, 7, 8), (1, 4, 9)\}$; the optimum $\{0, 1\}, \{2, 3\}, \{4, 9\}, \{5, 6\}, \{7, 8\}$ deletes $\{0, 8\}$ (witnesses 1 and 7). In both, $I(H)$ has $c = 3$. □

*Remark* K.11 (Scope). Pointers at distance $\leq K$ are bag-local in a decomposition of $I(H)^K$, whose width is a function of $(w, c, K)$, so the version of the statement that matters is the one with "far" larger than any function of $(\ell, w)$. Proposition K.10 does not refute it, and Proposition L.8 does not either: the far ends of its edit are exactly the two locally bad members (for $m = 3, 4, 5$ these are the only locally bad members, by enumeration). So the idea "put the locally bad members into every bag" is excluded for distances 2 and 3 only; for unbounded distance it is open.

## L Locality of inserted pairs and a non-local obstruction

Appendix G bounds the *number* of hard edits and the *candidates for the second endpoint* of a hard edit, but not the number of candidate *positions* of the first. This section studies the inserted hard pairs of an *optimal* solution. The main tool is the diameter bound of Lemma G.2, which, combined with pair-maximality, classifies every inserted pair into one of two types (Theorem L.2). Pairs of the first type are local in a decomposition of $I(H)$ (Corollary L.3, Proposition L.5); the second type, which includes the generic merge of two pieces, is *not* local in the same sense (Proposition L.7). Finally, deleted hard edges can join packed $P_3$'s at arbitrary distance in $I(H)$ (Proposition L.8). **Nothing in this section gives a fixed-parameter algorithm in $\ell + w$ or a hardness result.**

**Setting.** $H$ is pair-maximal (for an arbitrary packing read the primed instance), $D(H) = 2c$, $S$ is an *optimal* solution with $|S| \leq |H| + \ell$, $C$ is a cluster of $G \triangle S$, and $U = U^- \cup U^+$, $u = |U| \leq \ell$ are as in Appendix G. An *inserted pair* is a pair of $U^+$: a hard non-edge of $G$ inside one cluster. A hard edge of $G$ inside a cluster is an edge of $G \triangle S$ and not in $S$, hence *kept*. The *pieces* of $S$ are the connected components of the graph $(V, E_{\mathrm{keep}})$ where $E_{\mathrm{keep}}$ is the set of kept hard edges (isolated vertices are singleton pieces); every piece lies inside one cluster. Pieces depend on $S$, not only on $(G, H)$.

**Lemma L.1** (Cross pairs)**.** *If $A \neq B$ are pieces contained in a cluster $C$, then every pair $ab$ ($a \in A, b \in B$) is soft or inserted. This holds for every solution $S$; neither optimality nor pair-maximality is used.*

*Proof.* Since $a, b \in C$, $ab$ is an edge of $G \triangle S$, i.e. $ab \in E$ xor $ab \in S$. Suppose $ab$ is hard. If $ab \in E$ then $ab \notin S$, so $ab$ is a kept hard edge and $a, b$ lie in one piece, a contradiction. So $ab \notin E$, hence $ab \in S$, and as $ab$ is hard, $ab \in U^+$. □

(Appendix F bounds the sizes of pieces that share a cluster.)

## L.1 The T1/T2 dichotomy

For an inserted pair $ab$ with $a, b \in C$, a *witness* is a vertex $z \in C \setminus \{a, b\}$ with $az, zb \in E$.

**Theorem L.2** (T1/T2)**.** *Let $H$ be pair-maximal, $S$ optimal and $ab$ an inserted pair, $a, b \in C$. Then $ab$ has a witness, and for every witness $z$ at least one of $az, zb$ is soft. Moreover:*

*(T1) If $az$ and $zb$ are both soft, they are owned by two* distinct *members $P_a \neq P_b$ of $H$, and $z \in V(P_a) \cap V(P_b)$.*

*(T2) If $az$ is hard (the case of $zb$ hard is symmetric), then $az$ is a kept hard edge, so $a$ and $z$ lie in one piece, and $zb$ is soft.*

*We say that $ab$ is of* type T1 *if some witness has both steps soft, and of* type T2 *otherwise; then every witness has exactly one hard step.*

The type is a property of the pair $ab$ (not of a chosen witness): it is T1 iff a witness with two soft steps exists.

*Proof.* A witness exists by Lemma G.2(b). If $az$ and $zb$ were both hard for a witness $z$, then $a\,z\,b$ would be an induced $P_3$ with all three pairs hard ($ab$ is hard by assumption), contradicting pair-maximality. If both are soft, let $P_a, P_b$ own $az, zb$. If $P_a = P_b$ then $a, b \in V(P_a)$ and $ab$ is soft, contradicting that $ab$ is hard; so $P_a \neq P_b$, and both contain $z$. If $az$ is hard, it is a hard edge inside $C$, hence kept. □

**Corollary L.3** (Locality of soft-free merges)**.** *Let $H$ be pair-maximal, $S$ optimal, and let $A \neq B$ be pieces inside one cluster $C$ such that* no soft pair joins $A$ and $B$. *Then for every $(a, b) \in A \times B$:*

*(i) $ab$ is an inserted pair, of type T1;*

*(ii) there is $z \in C \setminus (A \cup B)$ with $az, zb$ soft, owned by distinct members $P_a \neq P_b$ with $z \in V(P_a) \cap V(P_b)$;*

*(iii) given $a$, the number of possible partners $b$ in any such pair is at most $D(H)(D(H) - 2) = 4c(c - 1)$.*

*In particular $a$ and $b$ are at distance exactly 2 in the soft graph $G_{\text{soft}}$, and $A \cup B$ lies in the ball of radius 4 of $G_{\text{soft}}$ around any of its vertices.*

*Proof.* (i) The cross pair $ab$ is not soft by hypothesis, so it is hard, hence inserted by Lemma L.1; its type is T1 by (ii). (ii) Apply Theorem L.2 to $ab$ with a witness $z$. Suppose $az$ is hard (the other case is symmetric); then $z$ lies in the piece $A$ of $a$. The pair $zb$ is an edge of $G$ and joins $A$ to $B$: if it were hard it would be kept and $z, b$ would share a piece, contradicting $A \neq B$; if soft it contradicts the hypothesis. So $az$ and $zb$ are both soft, which is (T1). If $z \in A$, then $zb$ (soft) is a soft pair between $A$ and $B$; similarly for $z \in B$ and $az$. So $z \notin A \cup B$. (iii) Fix $a$. By (ii) $z \in Q(a)$: at most $D(H)$ choices. For such $z$, let $P_a$ own $az$. The vertex $b$ satisfies $bz$ soft owned by $P_b \neq P_a$, and $b \notin V(P_a)$ (otherwise $ab \in \text{pairs}(P_a)$ is soft). Now $Q(z) \cap V(P_a) = V(P_a) \setminus \{z\}$ has two elements, so $b \in Q(z) \setminus V(P_a)$, a set of at most $|Q(z)| - 2 \leq D(H) - 2$ vertices. Altogether at most $D(H)(D(H) - 2) = 2c(2c - 2) = 4c(c - 1)$ partners. (For $c = 1$ the count is 0: (T1) needs two members through $z$.)

The distance statement: $ab$ is not soft and $a, z, b$ is a soft path; so $d_{G_{\text{soft}}}(a, b) = 2$. Every $a' \in A$ has $d_{G_{\text{soft}}}(a', b) = 2$ for a fixed $b \in B$, and every $b' \in B$ has $d_{G_{\text{soft}}}(b', a') = 2$; hence all of $A \cup B$ is within distance 4 of $b$ (and of any other vertex of $A \cup B$, by symmetry). □

*Remark* L.4 (Scope). If no soft pair joins $A$ and $B$ then all $|A||B|$ cross pairs are inserted (Lemma L.1), so $|A||B| \le u \le \ell$. The corollary covers *only* merges without a soft pair between the pieces. Merges in which some cross pairs are soft (the generic case when the pieces are neighbours in the soft graph) are not covered, and Proposition L.7 shows they can genuinely be of type T2.

**Proposition L.5** (Bag locality of type T1 pairs)**.** *Let $(\mathcal{T}, \{\beta_t\})$ be a tree decomposition of $I(H)$ of width $w$, let $c = c(H)$, and for a set $Y$ of nodes of $I(H)$ let $N[Y]$ be the set of nodes at distance at most 1 from $Y$ (a set of* members of $H$*).*

*(a) If $ab$ is an inserted pair of type T1 with a witness $z$ and members $P_a, P_b$ as in Theorem L.2, then $P_a, P_b$ are adjacent in $I(H)$, so some bag $\beta_t$ contains both and $a, b \in \bigcup_{P \in \beta_t} V(P)$, a set of at most $3(w+1)$ vertices. Moreover the member through $z$ and every member through $a$ or $b$ lies in $N[\beta_t]$.*

*(b) $(\mathcal{T}, \{N[\beta_t]\})$ is a tree decomposition of the square $I(H)^2$ of width at most $(w+1)(3c-2)-1 \le (w+1)(3w+1)-1$.*

*Proof.* (a) $P_a, P_b$ share $z$, so they are adjacent in $I(H)$ and, the edge being covered, lie in a common bag. Every member through $a$ contains $a \in V(P_a)$, hence meets $P_a \in \beta_t$ and lies in $N[\beta_t]$; similarly for $b$ and $z$.

(b) *Size.* Each member has 3 vertices, each lying in at most $c$ members, so $I(H)$ has maximum degree at most $3(c-1)$, $|N[\beta_t]| \le (w+1)(1+3(c-1)) = (w+1)(3c-2)$, and $c \le w+1$. *Connectivity.* For a node $v$ let $\mathcal{T}_v = \{t : v \in \beta_t\}$ (a subtree). The nodes $t$ with $v \in N[\beta_t]$ form $\bigcup_{y \in N[v]} \mathcal{T}_y$. Each $\mathcal{T}_y$, $y \in N(v)$, meets $\mathcal{T}_v$ because the edge $yv$ lies in some bag; a union of subtrees that all meet the connected set $\mathcal{T}_v$ is connected. *Edges.* Let $yv$ be an edge of $I(H)^2$. If $y, v$ are adjacent in $I(H)$, any $t \in \mathcal{T}_y$ has $y \in \beta_t$ and $v \in N[y]$. If they have a common neighbour $m$, any $t \in \mathcal{T}_m$ has $y, v \in N[m] \subseteq N[\beta_t]$. □

A type T1 pair is therefore a pair of vertices inside one bag of a decomposition of bounded width, and, given one end, it has at most $4c(c-1)$ candidate partners in the soft-free case. In particular each *cross pair* $(a, b)$ of a soft-free merge lies in a bag; this does not put the pieces $A, B$ themselves in one bag.

**Lemma L.6** (Near-twin agreement)**.** *Let $S$ be any solution (optimal or not), $u = |U|$, and let $z \ne b$ lie in one cluster of $G \triangle S$. Then*

$$A(z,b) = \big\{a \notin \{z,b\} : \ az, ab \text{ both hard}, \ [az \in E] \ne [ab \in E]\big\}$$

*has at most $u$ elements. In particular, if $z, b$ lie in one cluster of $G \triangle S$ and $zb$ is soft, the set $\Theta(z,b) = N_h(z) \setminus (N[b] \cup Q(b))$ satisfies $|\Theta(z,b)| \le u \le |S| - |H|$ and is computable from $G$ and $H$ alone.*

*Proof.* Let $a \in A(z,b)$. Exactly one of $az, ab$ is a hard edge and the other a hard non-edge. If $a$ lies in the cluster of $z, b$, the hard non-edge is a hard non-edge inside a cluster, so it is in $U$. If $a$ lies in another cluster, the hard edge joins two different clusters, so it is deleted and in $U$. Assign to $a$ this pair. Its ends are $a$ and one of $z, b$; since $a \notin \{z, b\}$, the map $a \mapsto$ pair is injective. So $|A(z,b)| \le u$. For the second statement, $a \in \Theta(z,b)$ means $a \ne z$, $az$ a hard edge, $a \notin N[b]$ (so $a \ne b$ and $ab \notin E$) and $ab$ not soft. Thus $a \in A(z,b)$. □

In a type T2 pair with $az$ hard and $zb$ soft, $a \in \Theta(z,b)$: the far end $a$ is one of at most $\ell$ vertices determined by the soft pair $(z,b)$. The lemma does not say where $a$ is relative to $z$ in $I(H)$: $a$ is a hard neighbour of $z$.

## L.2 Merge pointers need not be bag-local

**Proposition L.7** (Examples of type T2 merges)**.** *Each of the following instances (graph $G$ on vertices $0, \dots, n-1$ given by its edge list, $H$ given by triples $(x, y, z)$ with centre $y$) has a pair-maximal modification-disjoint packing $H$ and a unique optimal solution, found by enumerating all set partitions of the vertex set.*

*(a)* $n = 7$, $E = \{03, 04, 12, 13, 14, 15, 23, 35, 36\}$, $H = \{(4, 1, 5), (0, 3, 2), (1, 3, 6)\}$. $\mathrm{OPT} = 4 = |H| + 1$*; the clusters are* $\{0, 4\}, \{1, 2, 3, 5\}, \{6\}$*. The pieces of* $\{1, 2, 3, 5\}$ *are* $\{1, 2\}$ *and* $\{3, 5\}$ *(kept hard edges* 12 *and* 35*); the cross pairs* $13, 15, 23$ *are soft and* 25 *is inserted.* $Q(2) = \{0, 3\}$ *and* $Q(5) = \{4, 1\}$ *are disjoint, so the pair* 25 *has no witness with two soft steps: it is of type T2.*

*(b)* $n = 8$, $E = \{02, 06, 14, 16, 17, 23, 25, 37, 45, 46, 47, 57\}$, $H = \{(0, 6, 4), (3, 7, 4), (2, 5, 4), (0, 2, 3), (6, 1, 7)\}$. $\mathrm{OPT} = 6 = |H| + 1$*; the clusters are* $\{0, 6\}, \{1, 4, 5, 7\}, \{2, 3\}$*; the only hard edit of the solution is the inserted pair* 15*; the pieces are* $\{1, 4\}$ *and* $\{5, 7\}$*, the cross pairs* $17, 45, 47$ *are soft,* $Q(1) = \{6, 7\}$*,* $Q(5) = \{2, 4\}$*, so* 15 *has no witness with two soft steps and is of type T2.*

*(c)* $n = 6$, $E = \{02, 03, 04, 05, 12, 13, 15, 25, 35, 45\}$, $H = \{(3, 5, 4), (0, 2, 1)\}$*. The unique optimum has clusters* $\{0, 1, 2, 3, 5\}, \{4\}$ *and* $\mathrm{OPT} = 4$*; it contains the inserted pair* 23 *(type T2:* 30 *hard and kept,* 02 *soft), but* 2 *and* 3 *lie in the* same *piece (the kept hard edges* $03, 05, 13, 15, 25$ *connect* $\{0, 1, 2, 3, 5\}$*).*

*Consequently "every inserted pair with both ends in $V(H)$ has a common soft neighbour" is false, and Corollary L.3 cannot be extended to merges of two pieces that are joined by a soft pair: (a) and (b) are such merges, with inserted pairs of type T2. Example (c) is T2 but intra-piece.*

*Proof.* Direct enumeration (all 877, 4140 and 203 set partitions of 7, 8 and 6 vertices respectively) of the cost $|\{ab : (ab \in E) \neq (a, b \text{ in one block})\}|$, together with a check that each triple of $H$ is an induced $P_3$, that the triples have pairwise disjoint pair sets and that no induced $P_3$ of $G$ has all pairs hard. The structural claims (pieces, soft pairs, $Q$) are read off from the lists above. In (a) and (b), every witness $z$ of the inserted pair has $az$ or $zb$ hard because $Q(a) \cap Q(b) = \emptyset$, so the pair is not of type T1. □

## L.3 An obstruction: deleted hard edges between far members

An optimum may be forced to delete a hard edge whose ends lie in members that are far apart in $I(H)$.

**Proposition L.8** (FAR)**.** *For $m \geq 3$ let $G_m$ be the path $0, 1, \dots, 2m$ with the extra edge $e^* = \{1, 2m-1\}$ and $H_m = \{P_i = (2i, 2i+1, 2i+2) : 0 \leq i < m\}$ (centres $2i+1$). Then:*

*(i) $H_m$ is a pair-maximal modification-disjoint packing, $I(H_m)$ is a path on $m$ nodes ($c = 2$, $w = 1$);*

*(ii)* $\mathrm{OPT}(G_m) = m + 1 = |H_m| + 1$*;*

*(iii) in* every *optimum the hard edge $e^*$ is deleted. Its ends lie in $P_0$ and $P_{m-1}$, at distance $m-1$ in $I(H_m)$.*

*Proof.* (i) The chord joins 1 and $2m-1$, which are $2m-2 \geq 4$ apart, so no triple $P_i$ contains both of them and each $P_i$ is an induced $P_3$; the pair sets are disjoint. Every path edge lies in some $P_i$ and is soft, so $e^*$ is the only hard edge. An induced $P_3$ has two edges, at most one of which is $e^*$, so it contains a soft pair: $H_m$ is pair-maximal.

(ii) $|E| = 2m+1$ and $G_m$ is connected with $2m+1$ vertices, hence unicyclic (the unique cycle is on $1, \ldots, 2m-1$, of length $\geq 5$). An induced subgraph on $s$ vertices has $e \leq s-1$ edges unless it contains the cycle, in which case $s \geq 5$ and $e \leq s$. With $f(C) = \binom{|C|}{2} - 2e(C)$ this gives $f(C) \geq (s-1)(s-4)/2 \geq -1$ in the first case, with equality iff $s \in \{2,3\}$ and $G[C]$ is connected, and $f(C) \geq s(s-5)/2 \geq 0$ in the second. Clusters with $f = -1$ have $\geq 2$ vertices, so there are at most $m$ of them, and by Lemma G.1 cost $\geq (2m+1) - m = m+1$. The partition $\{0,1\}, \{2,3\}, \ldots, \{2m-2, 2m-1\}, \{2m\}$ has cost $m+1$.

(iii) In an optimum $\sum f = -m$, so exactly $m$ clusters have $f = -1$ (each connected, of size 2 or 3) and all others have $f = 0$; at most one vertex lies outside the $f = -1$ clusters. A cluster with $f = -1$ that contains the pendant vertex 0 contains its only neighbour 1; likewise for $2m$ and $2m-1$. Suppose $1, 2m-1$ lie in one cluster $C^*$. If $f(C^*) = -1$ and $|C^*| = 2$, then $C^* = \{1, 2m-1\}$ contains no pendant, and both pendants are outside the $f = -1$ clusters: two vertices outside, a contradiction. If $f(C^*) = -1$ and $|C^*| = 3$, it contains at most one pendant; any pendant not in $C^*$ has its only neighbour in $C^*$ and lies in no other $f = -1$ cluster, so exactly one pendant is outside and the remaining $2m-3$ vertices must form $m-1$ clusters of size $\geq 2$, impossible. If $f(C^*) \geq 0$, at most $2m-1$ vertices remain for $m$ clusters of size $\geq 2$. So 1 and $2m-1$ are in different clusters, i.e. $e^*$ is deleted. □

(After one exchange of a member this obstruction disappears: Proposition O.10 shows that it is an artifact of the packing $H_m$, while the excess 1 is intrinsic.) So the pointer realised by the deleted edge $e^*$ joins members at distance $m-1$ in $I(H_m)$: its two ends are not adjacent in any $I(H)^r$ with $r < m-1$, and a decomposition of $I(H)^r$ has no reason to put them in one bag. Corollary G.7 bounds the candidates for the *other* end of $e^*$ (at most $D+\ell$ hard neighbours), not the position of the first. For $m = 3, 4, 5$ a computer enumeration gives OPT $= m+1$ with $7, 9, 11$ optima, each separating the ends of $e^*$. Appendix K shows that the idea of putting "locally bad" members into every bag does not repair this for distances 2 and 3 (Proposition K.10); for unbounded distance this is open.

# M Reduction on unions of members fails: the spine family

Theorem H.10 applies Rule 6.2 of [17] to components of $I(H)$. This appendix shows that the natural generalisation to pair-closed unions of *members* (Appendix H) cannot prove the reduction lemma (Open Problem 7), already for $w = 1$ and $\ell = 1$. Soundness of the rule is cited from [17]; the proposition itself is proved here, and does not use the rule. Throughout, $\Gamma'$, BIG, $B$, $Z$ are as in Appendix I; a set $\mathcal{F}$ of members is pair-closed, reducible and irreducible as in Appendix H.

**Proposition M.1** (The spine family $\mathrm{Sp}(F)$)**.** *Let $F$ be a cubic graph with an orientation of in-degree at most 2 at every vertex. Let $F'$ be $F$ plus a new vertex $*$ and an edge $e_0 = (t \to *)$, $t \in V(F)$. Order the edges of $F'$ as $e_0, e_1, \ldots, e_N$ so that $e_0$ comes first,* consecutive edges have different heads and $\mathrm{head}(e_1) \neq \mathrm{tail}(e_0) = t$ *(possible, since every head occurs at most twice and $t$ has in-degree at most 2). Write $h(i)$ and $\mathrm{t}(i)$ for the head and tail of $e_i$. The instance $\mathrm{Sp}(F)$ has vertices $a_i, b_i, c_i, u_i$ $(0 \leq i \leq N)$, $z$ and $y$; for $v \in V(F')$ the sets $\mathcal{K}_v^+ = \{a_i : \mathrm{t}(i) = v\} \cup \{b_i, c_i, u_i : h(i) = v\}$ (so $\mathcal{K}_*^+ = \{b_0, c_0, u_0\}$). Edges: all pairs inside each $\mathcal{K}_v^+$; the cross edges $a_i b_i$; the spine edges*

$b_iu_{i+1}$ *($i < N$) and* $b_Nz$*; and* $yu_0$*,* $yc_0$ *(and no other edge at* $y$*). Packing:* $P_i = (a_i, b_i, c_i)$ *and* $X_i = (u_i, b_i, u_{i+1})$*, both with centre* $b_i$ *(*$u_{N+1} := z$*);* $\ell = 1$*. Then:*

*(S1) the members are induced* $P_3$*'s with pairwise disjoint pair sets,* $H$ *is pair-maximal,* $I(H)$ *is a caterpillar (spine* $X_0, \ldots, X_N$*, leaf* $P_i$ *at* $X_i$*), so* $w = 1$*,* $c = 2$*,* $D = 4$*;*

*(S2)* $(G, H, 1)$ *is a YES instance and* $\mathrm{OPT} = |H| + 1$*;*

*(S3) all hard degrees are at most* $7 \le B = 11$*, so* $\mathrm{BIG} = \emptyset$*;*

*(S4)* $F$ *is a minor of* $\Gamma'$*, so* $\mathrm{tw}(\Gamma') \ge \mathrm{tw}(F)$*, which is* $\Omega(|V(F)|)$ *for cubic expanders [9] (cited);*

*(S5)* $I(H)$ *is connected and tight:* $\tau = |H|$*;*

*(S6) every nonempty pair-closed set* $\mathcal{F}$ *of members (connected or not) is irreducible.*

*Proof.* (S1) Every induced $P_3$ of $G$ uses a cross edge or a spine edge or a $y$-edge, because two edges inside cliques at a common vertex lie in one clique. $P_i$ is induced: $a_ib_i$ is a cross edge, $b_ic_i$ lies in $\mathcal{K}^+_{h(i)}$, $a_ic_i \notin E$ because $\mathrm{t}(i) \neq h(i)$. $X_i$ is induced: $u_ib_i$ lies in $\mathcal{K}^+_{h(i)}$, $b_iu_{i+1}$ is a spine edge, and $u_iu_{i+1} \notin E$ since consecutive heads differ ($z$ is adjacent to $b_N$ only). The pair sets $\{a_ib_i, b_ic_i, a_ic_i\}$, $\{u_ib_i, b_iu_{i+1}, u_iu_{i+1}\}$ are pairwise disjoint. Pair-maximality: no two hard edges form an induced $P_3$. The soft edges are $a_ib_i$, $b_ic_i$, $u_ib_i$, $b_iu_{i+1}$, so the hard edges are the other clique edges and $yu_0$, $yc_0$. Two hard clique edges at a vertex lie in one clique, so their other ends are adjacent; the hard edges at $y$ go to $u_0, c_0$, which are adjacent; and a hard edge at $u_0$ or $c_0$ inside $\mathcal{K}^+_* = \{b_0, c_0, u_0\}$ can only be $u_0c_0$, whose ends are both adjacent to $y$.

(S2) The partition $\{\mathcal{K}^+_v\}$ together with $\{z\}$, with $y$ put into $\mathcal{K}^+_*$, deletes exactly the $2(N + 1) = |H|$ cross and spine edges (all soft) and inserts the single pair $yb_0$; its cost is $|H| + 1$. For $\mathrm{OPT} \ge |H| + 1$ see (S6) below.

(S3) A vertex of $\mathcal{K}^+_v$ has at most $|\mathcal{K}^+_v| - 1 = 2 + 2\,\mathrm{in}(v) \le 6$ hard neighbours inside its clique, and $u_0, c_0$ have one more hard neighbour, $y$ (which has hard degree 2).

(S4) Contract every $\mathcal{K}^+_v$ (a clique). The cross edges $a_ib_i$ join $\mathcal{K}^+_{\mathrm{t}(i)}$ and $\mathcal{K}^+_{h(i)}$, so the quotient contains $F$.

(S5) In the partition of (S2), the pair $a_ib_i$ is deleted, $b_ic_i$ kept, $a_ic_i$ a non-edge in different clusters; $u_ib_i$ kept, $b_iu_{i+1}$ deleted, $u_iu_{i+1}$ a non-edge in different clusters. So its restriction to $V^- = V \setminus \{y\}$ edits exactly one pair per member: $\tau \le |H|$, hence $\tau = |H|$.

(S6) For pair-closed $\mathcal{F}$ the restriction of the solution of (S2) to $W_\mathcal{F}$ has at most $|\mathcal{F}|$ edits (no member outside $\mathcal{F}$ has a pair inside $W_\mathcal{F}$, and $yb_0$ leaves $W_\mathcal{F}$), so $\tau(G[W_\mathcal{F}]) = |\mathcal{F}|$ and the optimal $S_\mathcal{F}$ are the *tight* ones: exactly one edited pair per member of $\mathcal{F}$ and no hard pair is edited. Suppose such an $S_\mathcal{F}$ satisfied the boundary condition. We use repeatedly: two vertices $p, q \in W_\mathcal{F}$ joined by a hard non-edge are non-adjacent in $G[W_\mathcal{F}] \triangle S_\mathcal{F}$, and a hard edge is kept.

*Case 1: $\mathcal{F}$ contains a spine member; let $X_i$ be the one of smallest index.*

$i \ge 1$. Then $X_{i-1} \notin \mathcal{F}$, and $P_{i-1} \notin \mathcal{F}$: otherwise $b_{i-1}, u_i \in W_\mathcal{F}$ and the pair $b_{i-1}u_i$ of the outside member $X_{i-1}$ would lie inside $W_\mathcal{F}$. So $a_{i-1}, b_{i-1}, u_{i-1} \notin W_\mathcal{F}$ and $N(b_{i-1}) \cap W_\mathcal{F} = \{u_i\} \cup Y$ with $Y = \mathcal{K}^+_{h(i-1)} \cap W_\mathcal{F}$. For $m \in Y$ the pair $u_im$ is a non-edge ($h(i) \neq h(i-1)$) and is soft only if $m = u_{i+1}$. If $Y$ contains some $m \neq u_{i+1}$, then $u_i, m$ are two non-adjacent vertices of $N(b_{i-1}) \cap W_\mathcal{F}$: the boundary condition fails. If $Y = \{u_{i+1}\}$, the set $\{u_i, u_{i+1}\}$ must be a cluster, so $X_i$ inserts $u_iu_{i+1}$ and keeps $u_ib_i$, $b_iu_{i+1}$; then $b_i$ lies in this cluster, but $b_i \notin N(b_{i-1})$: contradiction. So $Y = \emptyset$ and $\{u_i\}$ is a cluster, i.e. $X_i$ deletes $u_ib_i$. If $P_i \notin \mathcal{F}$, the vertex $c_i \notin W_\mathcal{F}$ is adjacent to $u_i$ and $b_i$, which lie in different clusters: the boundary condition fails at $c_i$. If $P_i \in \mathcal{F}$, the hard edge $u_ic_i$ is kept, contradicting that $u_i$ is a singleton.

$i = 0$. The vertex $y \notin W_{\mathcal{F}}$ has $N(y) \cap W_{\mathcal{F}} = \{u_0\}$ if $P_0 \notin \mathcal{F}$ and $\{u_0, c_0\}$ if $P_0 \in \mathcal{F}$. First case: $\{u_0\}$ is a cluster, so $X_0$ deletes $u_0b_0$, and $c_0 \notin W_{\mathcal{F}}$, adjacent to $u_0, b_0 \in W_{\mathcal{F}}$ in different clusters, violates the condition. Second case: $\{u_0, c_0\}$ must be a cluster, so $b_0$ is separated from both: $X_0$ deletes $u_0b_0$ and $P_0$ deletes $b_0c_0$. Then $a_0b_0$ and $b_0u_1$ are kept, so $a_0, u_1$ are adjacent to $b_0$. Now $a_0 \in \mathcal{K}^+_t$ and $u_1 \in \mathcal{K}^+_{h(1)}$ with $h(1) \neq t$ *by the ordering condition*, so $a_0u_1$ is a hard non-edge (the soft partners of $a_0$ are $b_0, c_0$), and $a_0b_0u_1$ is an induced $P_3$ of $G[W_{\mathcal{F}}] \triangle S_{\mathcal{F}}$: contradiction.

*Case 2: $\mathcal{F}$ contains only leaves.* Take $P_i \in \mathcal{F}$; $u_i, u_{i+1} \notin W_{\mathcal{F}}$ (for $i = N$ read $z$ for $u_{i+1}$). If $i = N$ then $N(z) \cap W_{\mathcal{F}} = \{b_N\}$ must be a cluster, i.e. $b_N$ is a singleton; but each of the three single edits of $P_N$ (delete $a_Nb_N$, delete $b_Nc_N$, insert $a_Nc_N$) keeps $b_N$ with $a_N$ or $c_N$. So let $i < N$. Then $N(u_{i+1}) \cap W_{\mathcal{F}} = \{b_i\} \cup Y$ with $Y = \mathcal{K}^+_{h(i+1)} \cap W_{\mathcal{F}}$. For $m \in Y$ the pair $b_im$ is a non-edge ($h(i+1) \neq h(i)$) unless $m = a_i$ (possible when $\mathrm{t}(i) = h(i+1)$), and it is soft only if $m = a_i$ (the soft partners of $b_i$ are $a_i, c_i, u_i, u_{i+1}$). *(2a)* If $Y$ contains some $m \neq a_i$, then $b_i, m$ are non-adjacent in $N(u_{i+1}) \cap W_{\mathcal{F}}$: failure. *(2b)* If $Y = \emptyset$, then $\{b_i\}$ is a cluster, impossible as above. *(2c)* If $Y = \{a_i\}$, then $P_{i+1} \notin \mathcal{F}$ and $b_{i+1} \in \mathcal{K}^+_{h(i+1)}$ is outside $W_{\mathcal{F}}$ with $N(b_{i+1}) \cap W_{\mathcal{F}} = \{a_i\}$; this must be a cluster, so $a_i$ is a singleton and $a_ib_i$ is deleted. But then $\{a_i, b_i\} = N(u_{i+1}) \cap W_{\mathcal{F}}$ is not a cluster ($a_ib_i$ is not an edge of $G \triangle S_{\mathcal{F}}$): contradiction.

So $\mathcal{F}$ is irreducible. For the lower bound of (S2): if $\mathrm{OPT} = |H|$, an optimum $S$ has $|S| = \tau$ for the whole component $\mathcal{C} = I(H)$, so $S = S_{\mathcal{C}}$ is tight and optimal on $V^- = V \setminus \{y\}$; by (S6) it violates the boundary condition at $y$, and Lemma H.5 gives an edited pair $\{y, u\}$, which contradicts $S = S_{\mathcal{C}}$. So $\mathrm{OPT} \geq |H| + 1$. □

*Remark* M.2 (Finite checks). Claims (S1)–(S5) and $\mathrm{OPT} = |H| + 1$ were verified by computer for $F$ the triangle ($n = 18$, OPT by an exact subset dynamic program) and for $K_4$ ($n = 30$). Claim (S6) was also confirmed by computer on all small pair-closed subfamilies, in agreement with the proof.

**Consequence: an obstruction, not a refutation of the reduction lemma.** No reduction obtained by iterating Rule 6.2 on pair-closed unions of members (the form used for components in Appendix H) can prove Open Problem 7 by itself: on $\mathrm{Sp}(F)$ it removes nothing, although $\ell = 1$, $w = 1$, $\mathrm{BIG} = \emptyset$ and $\mathrm{tw}(\Gamma') = \Omega(|V(F)|)$. In particular the counting argument of Lemma H.6 does not extend from components to members: all $2(N+1)$ members are irreducible although $\ell = 1$. The mechanism is Lemma H.5: along the spine each member is made irreducible by an edit at a shared vertex of its neighbour, down to the trigger $y$. Other generalisations of the rule (to vertex sets that are not unions of members) are not excluded. Repacking, by contrast, removes the obstruction: Proposition O.11 gives a packing $H'$ of size $|H| + 1 = \mathrm{OPT}$ with $\mathrm{tw}(I(H')) \leq 2$, and also a two-line proof of $\mathrm{OPT} \geq |H| + 1$.

# N Tunnels: locks, memory and a restricted model

The natural route to W[1]-hardness in $\ell + w$ is a reduction from Grid Tiling in which hard edges between far-apart vertices (*tunnels*) carry equalities between distant parts of $I(H)$ of bounded width. This appendix proves what a tunnel can and cannot do: the lock lemmas used by the reduction of Section 5 (Lemmas N.1, N.2, N.3), the cores and the splice lemmas, and the *restricted* tunnel-release model, in which, without soft triangles, tunnels cannot carry the reduction (Theorem 6.10). The reduction of Section 5 shows that with soft triangles they can; the results of this appendix about the restricted model are not used in its proof, which does not assume the restricted model.

## N.1 Zero-excess solutions and tunnels

Soft and hard pairs and $Q(v)$ are as in Section 2. A solution $S$ is *zero-excess* (ZE) if $|S| = |H|$. A member $(a, b, c)$ (centre $b$) is in state 1, 2 or 3 if its edited pair is $ab$, $bc$ or $ac$. By Lemma 2.2 a ZE solution has exactly one edit in each member and no other edit, so it is determined by its vector of states; let $\Sigma = \Sigma(G, H) \subseteq \{1, 2, 3\}^H$ be the set of these vectors. None of the lemmas of this section except Lemma N.7 needs pair-maximality.

**Lemma N.1** (Zero-excess structure)**.** *Let $S$ be a ZE solution. (a) Every hard edge lies inside a cluster and every hard non-edge joins two clusters. (b) Every member is in one of the states $1, 2, 3$. (c) Every hard component lies in one cluster, and two vertices of one cluster in different hard components form a soft pair.*

*Proof.* (a),(b) are Lemma 2.2 with $|S| = |H|$: no hard pair is edited and each member has exactly one edit. A member $(a, b, c)$ with togetherness pattern $(t_{ab}, t_{bc}, t_{ac})$ has the cluster patterns $000, 100, 010, 111$ of costs $2, 1, 1, 1$, so the one-edit patterns are the states $2, 1, 3$. (c) A hard edge is unedited, so its ends are together; two vertices of one cluster in different hard components are not joined by a hard edge, and a hard non-edge inside a cluster would be edited, so the pair is soft. □

A *tunnel* is a hard edge $xx'$ with $N_h(x) = \{x'\}$, $N_h(x') = \{x\}$ and $Q(x) \cap Q(x') = \emptyset$; tunnels are always taken pairwise vertex-disjoint. A *stub* at $x$ is a new vertex $x'$ adjacent to $x$ only (then $Q(x') = \emptyset$). A vertex is an *anchor* if it lies in a member and has no hard edge. *Cutting* a tunnel means editing $xx'$ (one hard edit).

**Lemma N.2** (A tunnel is a lock)**.** *Let $xx'$ be a tunnel and $S$ a solution in which all hard pairs at $x$ and at $x'$ are unedited. Then the cluster of $x$ is $\{x, x'\}$. If $S$ is ZE, then no member has $x$ as centre, every member through $x$ has $x$ as an end with the pair at $x$ that is an edge deleted, and the other two vertices of that member are together.*

*Proof.* Let $t \neq x, x'$ be in the cluster of $x$. If $t \notin Q(x)$ then $xt$ is hard, hence a non-edge as $N_h(x) = \{x'\}$, hence edited, a contradiction; so $t \in Q(x)$, and likewise $t \in Q(x')$, contradicting $Q(x) \cap Q(x') = \emptyset$. In a ZE solution a member $(y, x, z)$ with centre $x$ would have both pairs at $x$ apart, hence two edits. If $x$ is an end, the pair $x$–centre is apart (an edit), the non-edge at $x$ is apart (no edit), so the remaining pair is together. □

**Lemma N.3** (Anchors: singleton = forced states; inactive anchors cost nothing)**.** *Let $x$ be an anchor and $\sigma$ a ZE solution of $(G, H)$. Then $x$ is a singleton of $\sigma$ iff every member through $x$ has $x$ as an end and is in the state that deletes the edge at $x$. Moreover: (a) if $G'$ is $G$ with a stub at $x$, then $G'$ has a ZE solution iff $G$ has a ZE solution in which $x$ is a singleton; (b) if $x_1, x_2$ are anchors that lie in no common member, are non-adjacent and have $Q(x_1) \cap Q(x_2) = \emptyset$, and $G'$ is $G$ plus the edge $x_1x_2$, then $G'$ has a ZE solution iff $G$ has a ZE solution in which $x_1$ and $x_2$ are both singletons.*

*Proof.* The first statement: ($\Rightarrow$) as in Lemma N.2; ($\Leftarrow$) the soft partners of $x$ are its mates in members, and these are apart from $x$; $x$ has no hard edge, and its hard non-edges are unedited. (a),(b) ($\Rightarrow$): by Lemma N.2 the cluster is $\{x, x'\}$ (resp. $\{x_1, x_2\}$); split it into singletons. No other pair changes status or togetherness. ($\Leftarrow$) join the singleton to its stub (resp. join the two singletons). □

Thus an anchor costs nothing when it is a singleton in some ZE solution, and the point of a tunnel is precisely that this singleton requirement is a constraint on the member states through $x$: a *lock*. (A lock fixes the states of the members through $x$ to unit values; it never forces more.)

**Lemma N.4** (Rigidity at a lock)**.** *Let $S$ be ZE. (a) If $y, z$ are together then for every $q \notin \{y, z\}$, $q$ is together with $y$ iff with $z$. Hence if $qz$ is hard and $qy$ soft, the soft pair $qy$ is forced to be together iff $qz \in E$; if both are soft, the owners of $yz$, $qy$, $qz$ form a clique of $I(H)$ (a triangle if distinct). (b) If $u, w$ and $w, v$ are together, then $uv$ is an edge or a soft pair.*

*Proof.* Transitivity of "together"; for (b) a hard pair inside a cluster is an unedited edge (Lemma N.1(a)). □

Consequently the two vertices that a lock bridges (the mates $y, z$ of a locked member) are *occupied*: $y \sim z$ is a cluster that cannot also be bridged to a third vertex unless the pair is an edge or soft.

## N.2 Memory across a tree decomposition

Let $(\mathcal{T}, \beta)$ be a tree decomposition of $I(H)$, $e = tt'$ an edge of $\mathcal{T}$, $\mathcal{B}_e$ the members in both bags of $e$, and $\mathcal{M}_1$, $\mathcal{M}_2$ the members that occur only on the side of $t$, resp. $t'$. A set $S$ of anchors is a *face* if some ZE solution of $(G, H)$ makes every $x \in S$ a singleton; equivalently (Lemma N.3) the partial vector of forced states is the restriction of an element of $\Sigma$. A *minimal non-face* (a *core*) is a non-face all of whose proper subsets are faces.

**Lemma N.5** (Splice)**.** *(a) If $\sigma, \sigma' \in \Sigma$ agree on $\mathcal{B}_e$, the vector equal to $\sigma$ on $\mathcal{M}_1 \cup \mathcal{B}_e$ and to $\sigma'$ on $\mathcal{M}_2$ is in $\Sigma$. (b) Let $S = S_1 \cup S_2$ be a core with all members through anchors of $S_i$ in $\mathcal{M}_i$ ($i = 1, 2$, both parts non-empty). For $b \in S$ let $\sigma^b \in \Sigma$ make all anchors of $S \setminus \{b\}$ singletons. Then $\{\sigma^b|_{\mathcal{B}_e} : b \in S_1\}$ and $\{\sigma^a|_{\mathcal{B}_e} : a \in S_2\}$ are disjoint.*

*Proof.* (a) $\sigma \in \Sigma$ iff no vertex triple is an induced $P_3$ after editing; the status of a triple's pairs depends on the owners of its soft pairs, a clique of $I(H)$ (Observation 3.2), so contained in one bag, hence in $\mathcal{M}_1 \cup \mathcal{B}_e$ or in $\mathcal{M}_2 \cup \mathcal{B}_e$; all constraints are satisfied by the combined vector. (b) If $\sigma^b|_{\mathcal{B}_e} = \sigma^a|_{\mathcal{B}_e}$ with $b \in S_1$, $a \in S_2$, splice $\sigma^a$ on $\mathcal{M}_1 \cup \mathcal{B}_e$ with $\sigma^b$ on $\mathcal{M}_2$: the anchors of $S_1$ are singletons (they are in $S \setminus \{a\}$, and their members are in $M_1$), those of $S_2$ likewise, so $S$ is a face, a contradiction. □

**Lemma N.6** (Majority closure bounds cores)**.** *Suppose $\Sigma$ is closed under* some *majority operation $m : \{1, 2, 3\}^3 \to \{1, 2, 3\}$ applied coordinatewise ($m(a, a, b) = m(a, b, a) = m(b, a, a) = a$). Then every core has at most two anchors.*

*Proof.* Let $S$ be a core with $|S| \geq 3$; we derive a contradiction.

*(1) Each member contains at most one anchor of $S$, and as an end.* If an anchor $x \in S$ is the centre of a member, $x$ is never a singleton of a ZE solution (first statement of Lemma N.3: $x$ a singleton and the centre of $(y, x, z)$ would mean both edges at $x$ are edited, two edits in one member), so $\{x\}$ is a non-face, contradicting minimality of $S$. If both ends $a, c$ of a member $(a, b, c)$ are anchors of $S$, then making $a$ a singleton forces state 1 and making $c$ a singleton forces state 2 (Lemma N.3), so $\{a, c\}$ is a non-face, again contradicting minimality, since $|S| \geq 3$. (For cores of size at most 2 a member can carry two anchors.)

*(2) The forced vector.* Let $T$ be the set of members through anchors of $S$. By Lemma N.3 and (1), the requirement that every anchor of $S$ be a singleton fixes the state of each member of $T$ to a single value, giving a vector $t \in \{1, 2, 3\}^T$ without conflicts, and $S$ is a face iff $t \in \Sigma_T := \mathrm{proj}_T(\Sigma)$, which is closed under $m$. The same holds for every $S' \subseteq S$ with its forced vector $t'$ on its members $T'$.

*(3) Binary projections determine $t$.* Claim: if for every two members $P, P' \in T$ the pair $(t_P, t_{P'})$ lies in $\mathrm{proj}_{P,P'}(\Sigma)$, then $t \in \Sigma_T$. (This is the Baker–Pixley argument [1].) Induction on $|T|$; the base

case $|T| = 2$ is the hypothesis. For $|T| \geq 3$ choose three members $i_1, i_2, i_3$ and, by induction applied to $T \setminus \{i_j\}$ and the $m$-closed relation $\mathrm{proj}_{T\setminus\{i_j\}}(\Sigma)$, choose $u_j \in \Sigma_T$ that agrees with $t$ on $T \setminus \{i_j\}$. Then $m(u_1, u_2, u_3) = t$: at $i_1$ the vectors $u_2, u_3$ agree with $t$, similarly at $i_2, i_3$, and elsewhere all three agree.

*(4)* Since $S$ is a non-face, $t \notin \Sigma_T$, so by (3) some pair of members $P, P' \in T$ has $(t_P, t_{P'}) \notin \mathrm{proj}_{P,P'}(\Sigma)$. Let $S'$ be the set of anchors of $S$ lying in $P$ or $P'$; by (1), $|S'| \leq 2 < |S|$. The vector forced by $S'$ agrees with $t$ on $P$ and $P'$. If $S'$ were a face, some $\sigma \in \Sigma$ would extend it and $(\sigma_P, \sigma_{P'}) = (t_P, t_{P'})$, a contradiction. So $S'$ is a non-face and a proper subset of $S$, contradicting that $S$ is a core. □

## N.3 Fan-out of released tunnels, and the tunnel-release model

**Lemma N.7** (Released ends have small hard degree)**.** *Let $H$ be pair-maximal, $D \geq \max_v |Q(v)|$, and $S$ an optimal solution with $|S| \leq |H| + \ell$. (a) Every endpoint of an edited hard pair (in particular of a cut hard edge) has hard degree at most $D + \ell$. (b) Every hard component containing a vertex of hard degree larger than $2D + 3\ell$ is a whole cluster of $S$ in which no* hard *pair is edited.*

*Proof.* (a) Lemma G.4(a) gives $d_h(x) \leq D + u$ and $u \leq |S| - |H| \leq \ell$. (b) Lemma G.6($\gamma$) with $B = 2D + 3\ell$; hard pairs inside the cluster are unedited by Lemma G.6($\beta$). Soft pairs inside the component may be edited (take $K_9$ minus one edge $ac$ with $H = \{(a, b, c)\}$). □

**Proposition N.8** (Bounded-degree Grid Tiling is easy)**.** *Grid Tiling [6] (find $a_1, \ldots, a_k, b_1, \ldots, b_k \in [n]$ with $(a_i, b_j) \in S_{ij}$ for all $i, j$) is solvable in time $n\, d^{2k}$* poly *on instances where every value has at most $d$ partners in every $S_{ij}$.*

*Proof.* Guess $a_1$ ($n$ ways); each $b_j$ has at most $d$ candidates, being a partner of $a_1$ in $S_{1j}$; then each $a_i$ has at most $d$ candidates, being a partner of $b_1$ in $S_{i1}$. □

## N.4 The tunnel-release model

Fix a packing $H$ and a set $\mathcal{J}$ of tunnels (stubs allowed). A *TR-solution* is a solution with exactly one edit in every member whose hard edits are cuts of tunnels; its excess is the number of cuts. For $Y \subseteq \mathcal{J}$ let $f(Y) = 1$ iff $G$ minus the tunnel edges of $Y$ has a ZE solution; call $S \subseteq \mathcal{J}$ *lockable* if $f(\mathcal{J} \setminus S) = 1$, and a minimal non-lockable set a *core*.

**Proposition N.9** (Bounded cores give FPT in the model)**.** *(i) $f$ is monotone. (ii) $f(Y) = 1$ iff $Y$ meets every core. (iii) The minimum number of cuts of a TR-solution is the minimum size of a hitting set of the cores. (iv) If every core has at most $d$ tunnels, this minimum is computable in time $d^\ell 3^{w+1} n^{O(1)}$ (or decides that it exceeds $\ell$), given a tree decomposition of $I(H)$ of width $w$.*

*Proof.* (i) Given a ZE solution of $G$ minus $Y'$ and $Y' \subseteq Y$, each tunnel of $Y \setminus Y'$ is a cluster $\{x, x'\}$ (Lemma N.2); splitting it into singletons changes no edit. (ii),(iii): $f(Y) = 1$ iff $\mathcal{J} \setminus Y$ is lockable iff it contains no core, as non-lockability is upward closed; a TR-solution with cut set $Y$ exists iff $f(Y) = 1$. (iv) Test $f$ with the $\ell = 0$ dynamic program of Theorem 3.5; find a core inside $\mathcal{J} \setminus Y$ by deleting tunnels one at a time while the rest stays non-lockable; branch on which of its at most $d$ tunnels is cut, to depth $\ell$. □

This is a statement about a restricted model, not about CEaMP: solutions in which a member receives two edits, or a hard pair other than a tunnel edge is edited, are excluded, and they can be strictly cheaper. Inside the model, a W[1]-hardness proof therefore needs cores of unbounded size; such cores exist with soft triangles (Remark 5.15), and the hardness of Section 5 is proved for the unrestricted problem.

### N.5 Examples

**Example N.10** (A local flip gadget)**.** Let $G^*$ have vertices $0, 1, 3, 4, 5, 6, 7$ and edges $01, 04, 05$, $14, 15, 45, 03, 13, 36, 37, 06, 56$, with members $(6, 3, 7), (4, 1, 3), (5, 0, 3), (4, 5, 6), (6, 0, 1)$ and anchor $x = 7$ ($c = 3$; the only hard edges are 04 and 15, and 7 has none). It is valid, pair-maximal and tight ($\mathrm{OPT} = |H| = 5$), and has exactly five ZE solutions (found by enumerating all partitions): with $x$ a singleton, $\{0, 1, 3, 4, 5, 6\}$ and $\{0, 1, 4, 5\}, \{3, 6\}$; with $x$ not a singleton, $\{0, 1, 4, 5, 6\}, \{3, 7\}$, $\{0, 1, 4, 5\}, \{3, 6, 7\}$ and $\{0, 1, 4, 5\}, \{3, 7\}, \{6\}$. On the pair of members consisting of the fourth member $(4, 5, 6)$ and the second member $(4, 1, 3)$ in the order listed the locked relation (anchor a singleton) is the equality on $\{2, 3\}$, while releasing the anchor admits $(3, 2)$: the stored value flips only when the tunnel at $x$ is cut. The relation of $G^*$ is not determined by its binary projections (22233 is pairwise consistent but not a solution).

**Example N.11** (The uniform majority operation need not close $\Sigma$)**.** Let $n = 6$, $E = \{03, 05$, $12, 14, 23, 24, 25, 34, 35, 45\}$ and $H = \{(0, 5, 4), (1, 2, 5), (0, 3, 2), (1, 4, 3)\}$ ($c = 2$, tight). Then $\Sigma = \{1111, 1313, 3131, 2222\}$ (by enumeration of all partitions). The uniform majority $m_3$ ($m_3(a, b, c) = 3$ for pairwise distinct $a, b, c$) maps $(1313, 3131, 2222)$ to $3333 \notin \Sigma$, so $\Sigma$ is not closed under $m_3$. It is, however, closed under other majority operations, for instance the one with value 1 on all pairwise distinct triples, so Lemma N.6 still applies to this instance. In particular the example refutes closure under the specific operation $m_3$ for $c \leq 2$, not closure under every majority operation. (The packing of this example has four soft triangles; without soft triangles $\Sigma$ is closed under all majority operations, Theorem 6.4.)

The object that a hardness proof through tunnels needs is a family of tight instances with bounded $c$ and $\mathrm{tw}(I(H))$ and a core of unbounded size. Lemma N.4(b) explains why the obvious compositions of copies of $G^*$ fail: the two vertices bridged by a lock are occupied and cannot be shared with the next gadget. By Lemma N.6, such a family needs a set $\Sigma$ that is closed under no majority operation, and by Theorem 6.4 it must contain a soft triangle (Section 6). The thick selection chain of Section 5 is such a family (Remark 5.15); it needs $c = 4p + 2 \geq 26$.

## O Repacking: the parameters belong to the packing

The pair $(\ell, w)$ is a property of the packing $H$, not of the graph $G$. If $H^*$ is another modification-disjoint packing with $|H^*| \geq |H|$, then $(G, H^*, \ell^*)$ with $\ell^* = |H| + \ell - |H^*| \leq \ell$ is an equivalent instance (the case $H^* = H \cup M$ is Lemma 2.4), with a new excess and a new width $\mathrm{tw}(I(H^*))$. The *intrinsic excess* $\mathrm{OPT}(G) - \nu_{\mathrm{mod}}(G)$, where $\nu_{\mathrm{mod}}(G)$ is the maximum size of a modification-disjoint packing, is a lower bound for $\ell^*$ whatever $H^*$ is. This section asks which of the obstructions of Appendix L and 6 survive when $H$ may be replaced. The answer is a precise map of the remaining difficulty, not an algorithm: both known obstruction families to locality are artifacts of the choice of $H$ (Propositions O.10 and O.11), a one-sided fixed-parameter algorithm exists for a class of instances that contains them after repacking (Theorem O.13), no polynomial repacking procedure can have bounds in $\ell$ alone (Theorem O.15), and what remains is to *locate* the repacking (Open Problem 8). Fixed-parameter tractability in $\ell + w$ stays open. All proofs are in Appendix P.

**Conventions.** A vertex is *covered* if it lies in $V(H)$ and *uncovered* otherwise (the set $R$ of Appendix E). $H$ is pair-maximal unless said otherwise, $S$ is an optimal solution with $|S| \leq |H| + \ell$, and $U$ is its set of hard edits, so $|U| \leq \ell$ (Lemma 2.2). $\mathrm{Ball}(x)$ is the ball of radius 2 around $x$ in $G[L_B]$ and $D, B, Z, L_B$ are as in Appendix G. For a tree decomposition $(\mathcal{T}, \{\beta_t\})$ of $I(H)$ put $X_t = \bigcup_{P \in \beta_t} V(P)$ and $\mathcal{T}_v = \{t : v \in X_t\}$ for $v \in V(H)$. The members through $v$ pairwise intersect,

so their subtrees pairwise meet and $\mathcal{T}_v$ is a subtree (Helly). Two covered vertices $a, b$ are *co-bagged* if $\mathcal{T}_a \cap \mathcal{T}_b \neq \emptyset$. Statements labelled "computer" are finite checks that support, and do not prove, the claim; they are summarised in Appendix S.

## O.1 An exchange calculus

**Lemma O.1** (Exchange). *Let $H$ be a packing, $Q \in H$, and $T$ an induced $P_3$ of $G$ whose three pairs are not soft pairs of $H - Q$. Then $H' = (H - Q) + T$ is a packing with $|H'| = |H|$, and $I(H') - T = I(H) - Q$; hence $\mathrm{tw}(I(H')) \leq \mathrm{tw}(I(H)) + 1$. If $Q_1, \ldots, Q_j \in H$ are distinct and $T_1, \ldots, T_j$ are induced $P_3$'s that are pairwise pair-disjoint and whose pairs are not soft pairs of $H - \{Q_1, \ldots, Q_j\}$, the exchanged packing has size $|H|$ and $\mathrm{tw} \leq \mathrm{tw}(I(H)) + j$. The packing $H'$ need not be pair-maximal. If $(G, H, \ell)$ is a YES instance, a maximal family $M$ hard over $H'$ has $|M| \leq \ell$, and $H'' = H' \cup M$ is a pair-maximal packing with $\mathrm{tw}(I(H'')) \leq \mathrm{tw}(I(H)) + 1 + |M| \leq w + 1 + \ell$ and $(G, H'', \ell - |M|)$ equivalent to $(G, H, \ell)$.*

Simultaneous exchanges need the $T_i$ to be pairwise pair-disjoint; sequential exchanges, each applied to the current packing, need no such condition. The packing $H'$ need not be pair-maximal: the two other pairs of $Q$ become hard and can complete an all-hard induced $P_3$.

**Lemma O.2** (Closed twins). *Let $S$ be an optimal solution and $N[a] = N[b]$ in $G$. Then $a$ and $b$ lie in one cluster of $G \triangle S$.*

**Corollary O.3** (Hard edits lie in induced $P_3$'s). *Every edit of an optimal solution $S$ lies in an induced $P_3$ of $G$. For an inserted pair $ab$ one is $(a, z, b)$ with $z$ as in Lemma G.2(b); for a deleted pair $ab$ one is $(x, a, b)$ or $(x, b, a)$ with $x$ adjacent to exactly one of $a, b$.*

This is the standard fact that optimal solutions of Cluster Editing edit only pairs lying in an induced $P_3$; we need the form with a witness.

**Definition O.4** (Bridged pairs). A hard pair $\{a, b\}$ with $a, b \in V(H)$ is *bridged* if $a$ and $b$ have a common neighbour $z$ in the soft graph $G_{\mathrm{soft}}$. Equivalently, there are members $P \ni a$, $P' \ni b$ with $P \neq P'$ and $V(P) \cap V(P') \neq \emptyset$. Then $P, P'$ are adjacent in $I(H)$, so $a, b$ are co-bagged in *every* tree decomposition of $I(H)$. Inserted pairs of type T1 (Theorem L.2) are bridged.

**Lemma O.5** (Witness case split). *Let $H$ be pair-maximal, $S$ optimal and $e = \{a, b\} \in U$. Then $e$ lies in an induced $P_3$ $T$ of $G$ (a* witness*), and every witness has exactly one or two soft pairs. Let $z$ be the vertex of $T$ opposite to $e$; it is the centre of $T$ if $e$ is inserted and an end of $T$ if $e$ is deleted.*

*(a) If some witness has two soft pairs, they are $\{a, z\}$ and $\{b, z\}$, owned by two distinct members $P \ni a$ and $P' \ni b$ with $z \in V(P) \cap V(P')$; so $e$ is bridged.*

*(b) Otherwise every witness has exactly one soft pair $s$, owned by some $Q$, and $(H - Q) + T$ is a packing of the same size in which $e$ is soft (Lemma O.1); we call $e$* absorbable.

The two cases are a case split, not a dichotomy: an edit can have witnesses of both kinds . The shared vertex $z$ is the centre of $T$ only for inserted edits; for a deleted edit $e = ab$ with witness $(x, a, b)$ the two soft pairs $xa, xb$ share the end $x$. The conclusion of (a) is the same in both cases.

**Corollary O.6** (Non-bridged edits are absorbable). *Let $H$ be pair-maximal, $S$ optimal and $e \in U$ not bridged (this includes every edit with an uncovered end). Then there is $Q \in H$ and an induced $P_3$ $T \supset e$ such that $H' = (H - Q) + T$ is a packing of size $|H|$ in which $e$ is soft, with $\mathrm{tw}(I(H')) \leq \mathrm{tw}(I(H)) + 1$. The price is that the pairs of $Q$ other than $s$ become hard and may carry edits of $S$ (Lemma O.8).*

**Lemma O.7** (Uncovered ends). *Let $H$ be pair-maximal and $S$ optimal. No hard edit of $S$ has both ends uncovered.*

So every hard edit of an optimum has at least one covered end; with Corollary G.7, its other end is covered or is an uncovered vertex within Ball-distance 2 of it.

**Lemma O.8** (Displacement, single-edit case). *Let $H$ be pair-maximal, $S$ optimal, $e = ab \in U$ inserted, $z$ a witness vertex in the cluster of $a, b$ (Lemma G.2(b)) such that $T = (a, z, b)$ has exactly one soft pair $az$, owned by $Q = (a, z, r)$ (as a vertex set $\{a, z, r\}$). Then $S$ contains exactly one pair $f$ of* pairs$(Q)$*, namely one of $ar, zr$. After the exchange of Lemma O.1, the set of hard edits relative to $(H - Q) + T$ is $(U - e) + f$. The pair $f$ has the ends $r$ and one vertex of $T$. If $r$ lies in no member of $H - Q$, then $r$ is uncovered after the exchange; otherwise $f$ may again be non-bridged ("the defect moves").*

Iteration of Lemma O.8 is the typical case (computer experiments suggest that this is the typical case). Iterating gives an alternating chain of exchanges; whether it terminates at an uncovered end or a bridged pair is **open** (Conjecture O.18).

**Lemma O.9** (Hard components of uncovered vertices). *Let $H$ be pair-maximal and $r$ uncovered. The hard component of $r$ is $N[r]$, with $N_h(r) = N(r)$, and every pair inside $N[r]$ is a hard edge or a soft pair. In particular two vertices of $N[r]$ that are not co-bagged are joined by a hard edge.*

## O.2 The two known obstructions are artifacts of the packing

**Proposition O.10** (FAR after one exchange). *Let $G_m$, $H_m$ be as in Proposition L.8, $m \geq 3$.*

*(a) $\nu_{\mathrm{mod}}(G_m) = m$ and $\mathrm{OPT}(G_m) = m+1$. So the excess 1 is intrinsic: $\ell^* \geq 1$ for every packing.*

*(b) $H^* = \{P_0, \ldots, P_{m-2}, T\}$ with $T = (1, 2m-1, 2m-2)$ (centre $2m-1$) is a pair-maximal packing of size $m$, equal to $H_m - P_{m-1} + T$, and $I(H^*)$ is a cycle on $m$ nodes, so $\mathrm{tw}(I(H^*)) = 2$.*

*(c) The optimum $\{0,1\}, \{2,3\}, \ldots, \{2m-2, 2m-1\}, \{2m\}$ has cost $m+1$, and relative to $H^*$ its only hard edit is the deleted pair $\{2m-1, 2m\}$, whose end $2m$ is uncovered. The chord $e^*$ is a soft pair of $T$. The vertex $2m$ is a local hub in every decomposition, since $\mathrm{Ball}(2m) \cap V(H^*) \subseteq V(T)$.*

So the far pointer of Proposition L.8 becomes a hub edit at one uncovered vertex. By Proposition L.8(iii) the original packing $H_m$ has an optimum-forced deletion at distance $m-1$ in $I(H_m)$; the point of (a) is that the repacking does not remove the excess, only the non-locality.

**Proposition O.11** (The spine family after one exchange and one augmentation). *Let* $\mathrm{Sp}(F)$*, $H$, $\ell = 1$ be as in Proposition M.1 and put $Y = (y, u_0, b_0)$, $Z = (b_0, u_1, c_1)$, $H' = (H - X_0) + Y + Z$. Then $H'$ is a packing with $|H'| = |H| + 1$, hence $\mathrm{OPT} = |H| + 1 = |H'|$: the excess is 0 after repacking. Moreover $\mathrm{tw}(I(H')) \leq 2$. Also, the original packing $H$ lies in the class $\mathcal{S}$ of Definition O.12 (with $w = 1$, $\ell = 1$): the unique hard edit $\{y, b_0\}$ of the optimum of Proposition M.1(S2) is a hub edit at the local hub $y$.*

This gives a two-line proof of $\mathrm{OPT} \geq |H| + 1$ for $\mathrm{Sp}(F)$: $H'$ is a packing of size $|H| + 1$, and Lemma 2.2 applies; it replaces the case analysis (S6) for this purpose (S6 itself, irreducibility of pair-closed unions, is a different statement and is not affected).

## O.3 A one-sided algorithm for local instances

**Definition O.12** (Local hubs and the class $\mathcal{S}$)**.** Let $H$ be pair-maximal, $\ell \geq 0$, $(\mathcal{T}, \{\beta_t\})$ a tree decomposition of $I(H)$. An uncovered vertex $r \in L_B$ is a *local hub* if the vertices of $V(H) \cap \mathrm{Ball}(r)$ are pairwise co-bagged. A hard pair is *eligible* if it is (i) $\{a, b\}$ with $a, b \in V(H) \cap L_B$, $b \in \mathrm{Ball}(a)$, co-bagged, or (ii) $\{r, v\}$ with $r$ a local hub and $v \in V(H) \cap \mathrm{Ball}(r)$. Let $\mathcal{S}(H, \mathcal{T}, \beta)$ be the set of solutions $S$ all of whose hard edits are eligible and such that every covered vertex lies in at most $\ell$ of them.

By Corollary G.7 and Lemma O.7, every hard edit of an optimal solution of size at most $|H| + \ell$ is a pair $\{x, y\}$ with $x, y \in L_B$, $y \in \mathrm{Ball}(x)$, at least one end covered; the class asks that such pairs are co-bagged (covered ends) or hub-local (one uncovered end). Bridged pairs are co-bagged in every decomposition (Definition O.4).

**Theorem O.13** (One-sided local algorithm)**.** *Let $H$ be pair-maximal and $(\mathcal{T}, \{\beta_t\})$ a tree decomposition of $I(H)$ of width $w$. There is an algorithm of running time $5^{w+1}(Z^2+1)^{3\ell(w+1)}n^{O(1)}$, where $Z = 3D + 3\ell + 1 \leq 6w + 3\ell + 7$, that outputs a number $m^* = \min\{|S| : S \in \mathcal{S}(H, \mathcal{T}, \beta)\} \geq \mathrm{OPT}(G)$. If $\mathrm{OPT} \leq |H| + \ell$ and some optimal solution lies in $\mathcal{S}$, then $m^* = \mathrm{OPT}$. Consequently, accepting iff $m^* \leq |H| + \ell$ never accepts a NO instance and accepts every YES instance with an optimum in $\mathcal{S}$.*

*This is a one-sided algorithm.* Its promise (some optimum lies in $\mathcal{S}$) cannot be tested: $m^* > |H| + \ell$ does not certify NO. For $\mathrm{FAR}_m$ with the path tiling the YES instance $(G_m, H_m, 1)$ is rejected, with $m^* = m + 2 = \mathrm{OPT} + 1$; with $H^*$ of Proposition O.10 it is accepted. The algorithm is therefore not comparable with the unconditional Theorem I.3: for the family $\mathrm{Sp}(F)$ of Proposition M.1 the instance $(G, H, 1)$ is in the class with $w = 1$ whereas $\omega = \mathrm{tw}(\Gamma') \geq \mathrm{tw}(F)$ is unbounded; for $\mathrm{FAR}_m$ with the path tiling $\Gamma'$ has treewidth at most 3 (it is $G_m$ plus the soft pairs $\{2i, 2i+2\}$), but the instance is outside the class.

Where the proof needs care (Appendix P): the label of a covered vertex must be restricted to *co-bagged covered* partners and to *local hubs*, otherwise the symmetry check between the two ends of a pair cannot be enforced across a separator and the algorithm is unsound (computer: with that restriction lifted, the algorithm returned a value below OPT on some small instances); there is no edit between two uncovered vertices (Lemma O.7); and every triple must be visible in one bag. The running time uses the five states of Lemma 3.3, $|X_t| \leq 3(w+1)$ and $|\mathrm{Ball}| \leq Z^2$.

**Corollary O.14** (Static form)**.** *Let $\Lambda(H)$ be the set of pairs $\{a, b\} \subseteq V(H) \cap L_B$ with $a, b \in \mathrm{Ball}(r)$ for some $r \in L_B$. Let $w_\Lambda$ be the minimum width of a tree decomposition of $I(H)$ that co-bags every pair of $\Lambda(H)$; equivalently, the minimum over the choices, for every $\{a, b\} \in \Lambda(H)$, of a member through $a$ and a member through $b$, of the treewidth of $I(H)$ plus these links. Then CEaMP is solvable in time $5^{w_\Lambda+1}(Z^2+1)^{3\ell(w_\Lambda+1)}n^{O(1)}$, and the promise holds for every instance.*

Corollary O.14 is only as strong as $w_\Lambda$, which is large on tunnel-rich instances; by Theorem 5.1 it cannot be bounded by a function of $(\ell, w)$ unless $\mathrm{FPT} = \mathrm{W}[1]$.

## O.4 Limits of repacking

**Theorem O.15** (No repacking with bounds in $\ell$ alone)**.** *(a) If there is a polynomial-time algorithm that maps every YES instance $(G, H, 0)$ to a packing $H^*$ with excess $\ell^* \leq s(0)$ and $\mathrm{tw}(I(H^*)) \leq f(0)$, for constants $s(0), f(0)$, then $\mathrm{P} = \mathrm{NP}$. The statement is stated for functions $s, f$ of $\ell$ but only $\ell = 0$ is used.*

*(b) Under ETH there is no polynomial-time algorithm that maps every YES instance $(G, H, 0)$ to a packing $H^*$ with $|H^*| = |H|$ and $\mathrm{tw}(I(H^*)) \leq |H|^{1-\varepsilon}$, $\varepsilon > 0$.*

The guarantees hold only on YES inputs, so the proof validates the output: it tests that $H^*$ is a modification-disjoint packing of induced $P_3$'s of $G$, computes a tree decomposition of the stated width or rejects, and tests the excess before running the algorithm of the CSP theorem (Theorem 3.6). The theorem used is the *CSP algorithm* of Section 3, valid for every packing; it is not the majority-closure theorem (Theorem 6.4), which concerns soft-triangle-free packings. The reductions of [15] contain soft triangles, which is consistent: the NP-hardness at $\ell = 0$ is not contradicted by Theorem 6.4, and Theorem O.15 does not use it. Hence any usable repacking must have $\mathrm{tw}(I(H^*)) \le w + g(\ell)$ *and* a locality property; with $H^* = H$ the width bound alone is trivial. The hope "if $\mathrm{tw}(I(H))$ is large relative to $\ell$ then a better packing exists, hence reject" is false for the same reason: every tight instance ($\ell = 0$, $\nu_{\mathrm{mod}} = \mathrm{OPT} = |H|$) has no better packing, and tight instances have unbounded treewidth under ETH.

**Proposition O.16** (The naive normal form is false)**.** *Let $G = K_2 \vee 3K_1$ (hubs $3, 4$ adjacent, leaves $0, 1, 2$ each adjacent to both hubs; $n = 5$). Then $\nu_{\mathrm{mod}} = 2$, $\mathrm{OPT} = 3$ with four optima, every maximum packing covers all five vertices, and for every maximum packing every optimum has a hard edit with both ends covered. Hence "every hard edit has an uncovered end for some maximum packing and some optimum" is false. The number of connected graphs with at least one induced $P_3$ on which it fails is 3 of 20 ($n = 5$), 2 of 111 ($n = 6$), 3 of 852 ($n = 7$) and 305 of 11,116 ($n = 8$) (computer, exhaustive). The surviving hard edit in the example is bridged.*

**Proposition O.17** (Bounded-radius augmentation is insufficient)**.** *Let $P$ be the path $0, \ldots, 2m+2$ and $H_0 = \{(2i+1, 2i+2, 2i+3) : 0 \le i < m\}$. Then $H_0$ is a pair-maximal packing, $\nu_{\mathrm{mod}} = \mathrm{OPT} = m + 1$, $\ell = 1$, $w = 1$, and the unique packing of size $m + 1$ shares no member with $H_0$. So every improving exchange (remove $r$ members, add $r + 1$ pair-disjoint ones) has $r = m$ (computer: $m = 2, \ldots, 6$).*

The scope is exactly bounded-$r$ exchange search; the instance is a path. Moreover, in $\mathrm{FAR}_m$ the original tiling is already a maximum packing, so no improving exchange exists, and the helpful move of Proposition O.10 is *neutral*: its benefit is invisible to any objective computed from $(G, H)$, because it is chosen for the chord that the optimum cuts.

## O.5 What remains: locating the repacking

**Conjecture O.18** (C1, normal form NF$^*$)**.** *For every graph $G$ there are a maximum modification-disjoint packing $H^*$ and an optimal solution each of whose hard edits (relative to $H^*$) has an uncovered end or is bridged.*

**Status: open.** Experimental evidence: no counterexample among all connected graphs with $n \le 8$ and among random connected graphs with $n = 9$ and $n = 10$. By Proposition O.16 the clause "bridged" is necessary. C1 alone is not enough for Theorem O.13, which also needs the hub clause of Definition O.12 (the experiments above concern the covered-covered clause only), and it is an existence statement; it does not say how to find $H^*$.

**Open Problem 8** (LOC: locating the repacking; **no, unless** $\mathrm{FPT} = \mathrm{W}[1]$)**.** Is there an algorithm that, given $(G, H, \ell)$ and a decomposition of $I(H)$ of width $w$, runs in time $f(\ell, w)n^{O(1)}$ and outputs a pair-maximal packing $H^*$ with $|H^*| \ge |H|$ and $\mathrm{tw}(I(H^*)) \le g(\ell, w)$, together with a decomposition of that width, such that on every YES instance some optimal solution lies in $\mathcal{S}(H^*, \mathcal{T}^*, \beta^*)$ for $\ell^* = |H| + \ell - |H^*|$?

**Proposition O.19.** *A positive answer to Open Problem 8 implies that CEaMP is fixed-parameter tractable in $\ell + w$.*

**Corollary O.20.** *Unless* FPT = W[1]*, the answer to Open Problem 8 is negative: no algorithm locates a repacking of bounded width and excess in time $f(\ell, w)n^{O(1)}$ on every YES instance.*

*Proof.* Proposition O.19 and Theorem 5.1. □

By Theorem O.15 the width bound $g$ cannot depend on $\ell$ alone, so $g$ must involve $w$.

*Why this is a different problem (heuristic).* If $S$ is an optimum with far hard edits $e_1, \ldots, e_k$, $k \leq \ell$, and $H^*$ absorbs them (Corollary O.6), then $H^* - H$ consists of witnesses $T_i$ of the $e_i$, and conversely the $e_i$ are read off from $H^*$ as the hard pairs inside the new members. So computing $H^*$ is computing the far edits: repacking moves the information problem (which of about $n^2$ candidate pairs is cut) into the packing, and does not remove it. A dynamic program that must carry for each pending far edit the identity of its partner across a separator would need more than $f(\ell, w)$ states; colour coding repairs completeness but not soundness. This paragraph is not a proof.

**Precise status.** (1) The obstructions of Proposition L.8 (deleted hard edges between far members) and of Proposition M.1 (spine family, unbounded $\mathrm{tw}(\Gamma')$) are artifacts of the chosen packing: after one exchange, respectively one exchange and one augmentation, they have $\mathrm{tw}(I(H^*)) \leq 2$ and a single hard edit at an uncovered vertex, respectively excess 0. (2) The intrinsic excess and, for tight instances, the unbounded width are not artifacts (Proposition O.10(a), Theorem O.15). (3) Locating the repacking in fixed-parameter time is impossible unless FPT = W[1] (Corollary O.20); the existence of a good repacking (Conjecture O.18) is not affected by this.

# P Proofs for Appendix O

Notation and conventions are those of Appendix O.

## P.1 The exchange calculus

*Proof of Lemma O.1.* The pairs of $T$ are not soft pairs of $H - Q$, and the members of $H - Q$ have pairwise disjoint pair sets; hence $H' = (H - Q) + T$ is modification-disjoint, and $T \notin H - Q$ (a member of $H - Q$ has soft pairs), so $|H'| = |H|$ (if $T = Q$ there is nothing to prove). As graphs on the node set $H - Q$, $I(H') - T = I(H) - Q$, an induced subgraph of $I(H)$. Adding one node to a graph raises the treewidth by at most one (put it in every bag), so $\mathrm{tw}(I(H')) \leq \mathrm{tw}(I(H) - Q) + 1 \leq \mathrm{tw}(I(H)) + 1$. For $j$ exchanges the same argument applies with $H - \{Q_1, \ldots, Q_j\}$, the $T_i$ being pairwise pair-disjoint and disjoint from its soft pairs, and $j$ added nodes.

Now let $(G, H, \ell)$ be a YES instance. Since $|H'| = |H|$, $\mathrm{OPT} \leq |H'| + \ell$, so $(G, H', \ell)$ is a YES instance. Let $M$ be a maximal family hard over $H'$. By Lemma 2.3, $|M| \leq \ell$, by Lemma 2.4 $H'' = H' \cup M$ is pair-maximal, $\mathrm{tw}(I(H'')) \leq \mathrm{tw}(I(H')) + |M| \leq \mathrm{tw}(I(H)) + 1 + |M|$, and $(G, H'', \ell - |M|)$ is equivalent to $(G, H', \ell)$, hence to $(G, H, \ell)$. That $H'$ need not be pair-maximal is shown by the computer test of Appendix O. □

*Proof of Lemma O.2.* Suppose $a \in C_1$, $b \in C_2$ with $C_1 \neq C_2$. Since $N[a] = N[b]$, $ab \in E$, and $ab$ is an edge across clusters, so it is edited. Let $c_a$ be the number of pairs $\{a, x\}$, $x \notin \{a, b\}$, that disagree with $S$ (that is, are an edge across clusters or a non-edge inside a cluster), and $c_b$ likewise. These pairs together with $ab$ contribute $c_a + c_b + 1$ to the cost; all other pairs do not involve $a$ or $b$. Move $a$ into $C_2$. For $x \notin \{a, b\}$, $a$ and $x$ have the same adjacency as $b$ and $x$, and $a$ is now in the cluster of $b$, so $\{a, x\}$ disagrees exactly when $\{b, x\}$ did before: $c_b$ pairs; the pairs at $b$ are unchanged, $c_b$, and $ab$ is now an edge inside a cluster and agrees. The new cost of these pairs is $2c_b$, all other pairs are

unchanged. Symmetrically, moving $b$ into $C_1$ gives $2c_a$. Since $\min\{2c_a, 2c_b\} \le c_a + c_b < c_a + c_b + 1$, $S$ is not optimal. □

*Proof of Corollary O.3.* Let $ab \in S$ be inserted: $a, b$ lie in one cluster $C$ and $ab \notin E$. By Lemma G.2(b) there is $z \in C$ with $az, zb \in E$, so $(a, z, b)$ is an induced $P_3$ containing $ab$. Let $ab \in S$ be deleted: $ab \in E$ and $a, b$ lie in different clusters. By Lemma O.2, $a, b$ are not closed twins, so some $x \notin \{a, b\}$ is adjacent to exactly one of them, say to $a$; then $x\,a\,b$ is an induced $P_3$ containing $ab$ (adjacent to $b$ instead: $x\,b\,a$). □

*Proof of Lemma O.5.* By Corollary O.3 witnesses exist. Let $T$ be a witness; the pair $e$ is hard, so $T$ has at most two soft pairs. As $H$ is pair-maximal, $T$ does not have all three pairs hard, so it has at least one soft pair. If $e$ is inserted, a non-edge of $G$, then in the induced $P_3$ $T$ the pair $e$ joins the two ends and the opposite vertex $z$ is the centre. If $e$ is deleted, an edge of $G$, then $e$ is an edge of $T$ and $z$ is an end.

(a) Let $T$ have two soft pairs. They are the two pairs other than $e$, namely $\{a, z\}$ and $\{b, z\}$; let $P, P'$ own them, so $a, z \in V(P)$ and $b, z \in V(P')$. If $P = P'$ then $a, b \in V(P)$ and $e \in \text{pairs}(P)$ would be soft, a contradiction. So $P \ne P'$, $z \in V(P) \cap V(P')$, and $z$ is a common soft neighbour of $a$ and $b$ ($z \ne a, b$): $e$ is bridged.

(b) Otherwise every witness has exactly one soft pair $s$, owned by $Q$, and its other two pairs (one of them $e$) are hard. So the pairs of $T$ are not soft pairs of $H - Q$, and Lemma O.1 shows that $(H - Q) + T$ is a packing of size $|H|$; $e \in \text{pairs}(T)$ is soft in it. □

*Proof of Corollary O.6.* If $e$ is not bridged, no witness has two soft pairs by Lemma O.5(a) (whose conclusion needs both ends covered), so case (b) applies to any witness, and one exists. The width bound is Lemma O.1. □

*Proof of Lemma O.7.* Let $e = \{r, r'\} \in U$ with $r, r'$ uncovered. By Corollary O.3 it has a witness $T = \{r, r', x\}$. The pairs $\{r, x\}$ and $\{r', x\}$ are hard, since $r$ and $r'$ lie in no member, and $e$ is hard. So $T$ is an induced $P_3$ with all pairs hard, contradicting pair-maximality. □

*Proof of Lemma O.8.* The pair $az$ joins two vertices of the cluster $C$ of $a, b$ and is an edge of $G$ (a step of the witness), so it is not in $S$. $Q$ is an induced $P_3$ of $G$ on $\{a, z, r\}$ with the edge $az$, so exactly one of $ar, zr$ is an edge of $G$ and the other a non-edge. If $r \in C$, all three pairs of $Q$ are edges of $G \triangle S$: the edge among $ar, zr$ is kept, and the non-edge is inserted. If $r \notin C$, then $ar, zr$ join different clusters: the edge is deleted and the non-edge is untouched. In both cases $S \cap \text{pairs}(Q) = \{f\}$ with $f \in \{ar, zr\}$. Also $r \ne b$: otherwise $\text{pairs}(Q) = \text{pairs}(T) \ni e$ would be soft. The pairs of $T$ in $S$ are $e$ only ($az \notin S$, and $zb$ is an edge inside $C$). So relative to $H' = (H - Q) + T$ the hard edits are $(U \setminus \{e\}) \cup \{f\}$: $e$ is soft, $f \notin \text{pairs}(T)$ is hard. The ends of $f$ are $r$ and $a$ or $z$. As $r \notin V(T)$, $r$ is covered after the exchange iff it lies in a member of $H - Q$. □

*Proof of Lemma O.9.* All pairs at $r$ are hard, so $N_h(r) = N(r)$. Let $z \in N(r)$ and $w \in N_h(z)$, $w \ne r$. Then $rz$ and $zw$ are hard edges, so by Lemma G.3(a) (centre $z$) $rw \in E$ or $rw$ is soft; $rw$ is not soft as $r$ is uncovered, so $w \in N(r)$. Hence the hard component of $r$ is $N[r]$. For $z, z' \in N(r)$, Lemma G.3(a) with centre $r$ gives: $zz' \in E$ (a hard edge or a soft pair) or $zz'$ is soft. A soft pair lies in one member, so its ends are co-bagged; a pair of $N[r]$ that is not co-bagged is therefore not soft, and is an edge, hence a hard edge. □

## P.2 Proofs of Propositions O.10 and O.11

*Proof of Proposition O.10.* (a) An induced $P_3$ has exactly two edges. The members of a packing have disjoint pair sets, hence are edge-disjoint. As $|E(G_m)| = 2m+1$, a packing has at most $m$ members, and $H_m$ has $m$. OPT $= m+1$ is Proposition L.8(ii). By Lemma 2.2 every packing $H^*$ has $\ell^* = |H| + \ell - |H^*| \geq \text{OPT} - \nu_{\text{mod}} = 1$ in a YES instance of excess 1.

(b) Pairs of $P_i$: $\{2i, 2i+1\}, \{2i+1, 2i+2\}, \{2i, 2i+2\}$. $T = (1, 2m-1, 2m-2)$: $1 \sim 2m-1$ (chord), $2m-1 \sim 2m-2$ (path edge), and $1 \not\sim 2m-2$ because the neighbours of 1 are $0, 2, 2m-1$ and $2m-2 \geq 4$ for $m \geq 3$. So $T$ is an induced $P_3$ with centre $2m-1$ and pairs $\{1, 2m-1\}$, $\{2m-2, 2m-1\}$, $\{1, 2m-2\}$. The first is in no $P_i$ ($2m-2 \geq 4$ apart, both odd); the second lies only in $P_{m-1}$; the third is neither a consecutive pair nor an even-even pair ($2m-3 \geq 3$). So $T$ avoids the pairs of $P_0, \ldots, P_{m-2}$ and Lemma O.1 applies with $Q = P_{m-1}$. Soft edges of $H^*$: the path edges $\{j, j+1\}$, $j \leq 2m-3$, and $\{2m-2, 2m-1\}$, and the chord. The only hard edge is $\{2m-1, 2m\}$; an induced $P_3$ has two edges, at most one hard, so it contains a soft pair: $H^*$ is pair-maximal. $V(T) = \{1, 2m-2, 2m-1\}$ meets $V(P_i) = \{2i, 2i+1, 2i+2\}$, $i \leq m-2$, exactly for $i = 0$ (vertex 1) and $i = m-2$ (vertex $2m-2$); and $V(P_i) \cap V(P_{i+1}) = \{2i+2\}$. So $I(H^*)$ is the cycle $P_0, P_1, \ldots, P_{m-2}, T, P_0$ on $m \geq 3$ nodes, of treewidth 2.

(c) The partition has the edits: the path edges $\{2i+1, 2i+2\}$ for $0 \leq i \leq m-1$ and the chord, $m+1$ pairs, and no inserted pair; so its cost is $m+1 = \text{OPT}$. Relative to $H^*$ the edits $\{2i+1, 2i+2\}$, $i \leq m-2$, lie in $P_i$, the chord lies in $T$; the only hard edit is $\{2m-1, 2m\}$ and $2m \notin V(H^*) = \{0, \ldots, 2m-1\}$. The vertex $2m$ has the single neighbour $2m-1$, whose neighbours are $2m-2, 2m, 1$. So $\text{Ball}(2m) \cap V(H^*) \subseteq \{2m-1, 2m-2, 1\} = V(T)$, pairwise co-bagged in every decomposition (one member), and $2m \in L_B$ (hard degree 1). So $2m$ is a local hub and $\{2m-1, 2m\}$ is eligible; this optimum lies in $\mathcal{S}$ for $H^*$, $\ell^* = 1$. □

*Proof of Proposition O.11.* We use the description of $\text{Sp}(F)$ in Proposition M.1. *$Y$ is an induced $P_3$:* $yu_0, u_0b_0 \in E$ ($b_0, u_0 \in \mathcal{K}^+_*$) and $yb_0 \notin E$ ($y$ is adjacent only to $u_0, c_0$). Its pairs $\{y, u_0\}$, $\{y, b_0\}$ are hard ($y$ is uncovered) and $\{u_0, b_0\} \in \text{pairs}(X_0)$ only. *$Z$ is an induced $P_3$:* $b_0u_1$ is a spine edge, $u_1c_1$ is an edge inside $\mathcal{K}^+_{h(1)}$ (both belong to it), and $b_0c_1 \notin E$, because $c_1$ has no edge outside $\mathcal{K}^+_{h(1)}$ and $b_0 \in \mathcal{K}^+_*$ with $* \neq h(1)$ (only $e_0$ has head $*$). The soft edges are $a_ib_i, b_ic_i, u_ib_i, b_iu_{i+1}$ and the soft non-edges are $a_ic_i$, $u_iu_{i+1}$; so $\{u_1, c_1\}$ and $\{b_0, c_1\}$ are hard, and $\{b_0, u_1\} \in \text{pairs}(X_0)$ only. The pairs of $Y$ and $Z$ are disjoint, and none is a soft pair of $H - X_0$. So $H' = (H - X_0) + Y + Z$ is a packing and $|H'| = |H| + 1$. By Lemma 2.2 $\text{OPT} \geq |H'| = |H| + 1$, and $\text{OPT} \leq |H| + 1$ by the partition of Proposition M.1(S2).

*Width.* $I(H)$ is a caterpillar with spine $X_0, \ldots, X_N$ and leaf $P_i$ at $X_i$. Removing $X_0$ isolates $P_0$. $Y$ meets only $P_0$ and $Z$ (all through $b_0$; $u_0$ lies in $X_0$ and in no other member, and $y$ in no member of $H$). $Z$ meets $P_0, Y$ (via $b_0$), $X_1$ (via $u_1$) and $P_1$ (via $c_1$). So $I(H')$ is obtained from the caterpillar $X_1, \ldots, X_N$ with leaves $P_1, \ldots, P_N$ by adding the triangle $\{P_0, Y, Z\}$ and joining $Z$ to $X_1$ and $P_1$ (the triangle $\{Z, X_1, P_1\}$, since $X_1P_1$ is an edge). Its blocks are triangles and edges, so $\text{tw}(I(H')) \leq 2$.

*Class.* The optimum of (S2) has the single hard edit $\{y, b_0\}$ with $y$ uncovered. All hard degrees are at most $7 \leq B = 11$ (S3), so $y, b_0 \in L_B$. $\text{Ball}(y) = \{y, u_0, c_0, b_0\}$ (the neighbours of $u_0$ and $c_0$ are $\mathcal{K}^+_*$ and $y$). The covered vertices $u_0, b_0 \in V(X_0)$ and $c_0, b_0 \in V(P_0)$, and $X_0, P_0$ are adjacent through $b_0$, so they are co-bagged in every decomposition: $y$ is a local hub and the edit is eligible. □

## P.3 Proof of Theorem O.13 and Corollary O.14

Fix $H$, $\ell$ and a decomposition; *eligibility* (Definition O.12) is then computable from $G, H$ and the decomposition. For a covered $v \in L_B$ let $\text{Elig}(v)$ be the set of partners $p$ with $\{v, p\}$ eligible;

$|\mathrm{Elig}(v)| \le |\mathrm{Ball}(v)| \le Z^2$. Eligibility of a covered pair is symmetric. Every eligible pair has a covered end in $L_B$.

*Labels and status.* A *labelling* assigns to every member $P$ a state $x_P$ among the five of Lemma 3.3, and to every covered $v \in L_B$ a set $E_v \subseteq \mathrm{Elig}(v)$ with $|E_v| \le \ell$. It defines $S = \bigcup_P x_P \cup U$ with $U = \{\{v,p\} : p \in E_v\}$ and is *symmetric* if for covered $v, v'$ with $\{v,v'\}$ eligible, $v' \in E_v \iff v \in E_{v'}$. The status of a pair in $G \triangle S$ is $G$-status xor membership in $S$. At a bag $t$, the status of a pair is *determined by the state at $t$* if it is soft with owner in $\beta_t$ (status from $x_P$), or hard and not eligible (status of $G$), or hard and eligible with an end $p \in X_t \cap L_B$ (status from $[q \in E_p]$).

**Lemma P.1** (Visibility)**.** *Let $H$ be pair-maximal. For every vertex triple $\Delta$ of $G$ there is a bag $t$ at which the statuses of all three pairs of $\Delta$ are determined by the state, or $\Delta$ has no soft pair and no eligible pair (then all its statuses are static and $\Delta$ is not an induced $P_3$ of $G$).*

*Proof.* *$\Delta$ has a soft pair.* The owners of the soft pairs of $\Delta$ form a clique of $I(H)$ (Observation 3.2), so some bag $t$ contains all of them (Helly). Every vertex of $\Delta$ incident to a soft pair lies in $X_t$. A hard pair of $\Delta$ has an end among these vertices: if $\Delta$ has one soft pair $\{a,b\}$, the hard pairs are $\{a,c\},\{b,c\}$; if it has two or three, all vertices of $\Delta$ lie in $X_t$. So each hard pair is static or has an end in $X_t$ with a label, and is determined at $t$ (a vertex outside $L_B$ is in no eligible pair).

*$\Delta$ has no soft pair and no eligible pair.* Then $\Delta$ has three hard pairs, all static, and it is not an induced $P_3$ of $G$ by pair-maximality.

*$\Delta = \{p,q,c\}$ has no soft pair and the eligible pair $\{p,q\}$.* (i) $p,q$ covered, co-bagged: take $t \in \mathcal{T}_p \cap \mathcal{T}_q$; the pairs $\{p,c\},\{q,c\}$ are static or determined by the labels of $p$ and $q$. (ii) $p = r$ uncovered, $q = v$ covered, $r$ a local hub. The pair $\{r,v\}$ is determined by the label of $v$. If $c$ is uncovered then $\{r,c\}$ is not eligible, hence static, and $\{v,c\}$ is static or determined by the label of $v$: any bag of $\mathcal{T}_v$ works. If $c$ is covered and $c \in \mathrm{Ball}(r)$, then $v, c \in V(H) \cap \mathrm{Ball}(r)$ are co-bagged; take $t \in \mathcal{T}_v \cap \mathcal{T}_c$: $\{r,c\}$ is determined by the label of $c$, $\{v,c\}$ by that of $v$. If $c$ is covered and $c \notin \mathrm{Ball}(r)$, then $\{r,c\}$ is static and a bag of $\mathcal{T}_v$ works. By Lemma O.7 no eligible pair has two uncovered ends (by definition of eligibility an eligible pair has a covered end), so the cases are exhaustive. □

*Proof of Theorem O.13. Algorithm.* Take a nice tree decomposition of $I(H)$ of width $w$. A state at a bag $t$ is $(x_P)_{P \in \beta_t} \in [5]^{\beta_t}$ together with $E_v \subseteq \mathrm{Elig}(v)$, $|E_v| \le \ell$, for every $v \in X_t \cap V(H) \cap L_B$. A vertex $v$ is introduced when the first member through $v$ enters, its label is chosen there, it is carried unchanged through the subtree $\mathcal{T}_v$, and forgotten with the last member through $v$; a join node requires equal states of the two children. At each bag the table keeps, for each state, the minimum cost, where the cost is $\sum_P |x_P| + \sum_v |E_v \setminus V(H)| + \frac{1}{2}\sum_v |E_v \cap V(H)|$, each term charged at the introduction of its member or vertex. A state is *consistent at $t$* if (S) for $v, v' \in X_t \cap L_B$ covered: $v' \in E_v \iff v \in E_{v'}$; and (V) for every triple $\Delta$ whose three statuses are determined at $t$, $\Delta$ is not an induced $P_3$ of $G \triangle S$, where the statuses are those given by the state. Inconsistent states are discarded. The output $m^*$ is the minimum cost at the root, over all states in the root table (and $\infty$ if none).

*Number of states.* $|\beta_t| \le w+1$ gives $5^{w+1}$ choices of $(x_P)$; $|X_t| \le 3(w+1)$ vertices have a label, each a set of at most $\ell$ elements of a set of at most $Z^2$ elements, at most $(Z^2+1)^\ell$ choices. Together $5^{w+1}(Z^2+1)^{3\ell(w+1)}$ states. Each transition and each test (enumerating triples) takes $n^{O(1)}$ time, and $Z = 3D + 3\ell + 1 \le 6(w+1) + 3\ell + 1$ since $D = 2c \le 2(w+1)$ (Lemma 2.1).

*Soundness.* A consistent labelling of the whole decomposition gives labels $E_v$ constant along $\mathcal{T}_v$ (carried through, equal at joins) and states $x_P$; define $S = \bigcup x_P \cup U$ with $U = \{\{v,p\} : p \in E_v\}$. Two covered vertices $v, v'$ with $\{v,v'\}$ eligible are co-bagged, so some bag contains both, and (S) there gives $v' \in E_v \iff v \in E_{v'}$ globally: a covered pair of $U$ is seen from both ends, so the cost term $\frac{1}{2}\sum_v |E_v \cap V(H)|$ counts it once, and $|S|$ equals the cost. By Lemma P.1, every triple is either static

and not an induced $P_3$ of $G$ (so unaffected by $S$) or has all three statuses determined at some bag, where it is verified. Statuses at different bags agree, since they come from the same global labelling (the symmetry of labels makes the status of an eligible covered pair independent of the end that is used). Hence no triple is an induced $P_3$ of $G \triangle S$: $S$ is a solution, all of whose hard edits are eligible, each covered vertex in at most $\ell$ of them: $S \in \mathcal{S}$, and $m^*$ is at least $\min\{|S| : S \in \mathcal{S}\} \geq \mathrm{OPT}$.

*Completeness.* Let $S \in \mathcal{S}$. Put $x_P = S \cap \mathrm{pairs}(P)$, which is one of the five states because $V(P)$ is a cluster graph in $G \triangle S$ (Lemma 2.2, Lemma 3.3), and $E_v = \{p : \{v, p\} \in S \text{ hard}\} \subseteq \mathrm{Elig}(v)$, $|E_v| \leq \ell$. The labelling is symmetric by construction, its cost is $|S|$, and every triple is not an induced $P_3$ of $G \triangle S$; so all states are consistent and $m^* \leq |S|$. Hence $m^* = \min\{|S| : S \in \mathcal{S}\}$. If some optimal solution of size at most $|H| + \ell$ is in $\mathcal{S}$ then $m^* = \mathrm{OPT}$. The last sentence of the theorem follows. □

The restriction of the labels to $\mathrm{Elig}(v)$ is used twice: it makes (S) checkable in a bag common to both ends, and it gives the local-hub condition that makes the triples at an uncovered vertex visible; an uncovered $r$ lies in no bag, so two edits at $r$ with partners that are not co-bagged could not be compared.

*Proof of Corollary O.14.* Let a decomposition co-bag every pair of $\Lambda(H)$. Let $S$ be an optimal solution of size at most $|H| + \ell$ and $\{x, y\} \in U$ with $x$ covered. By Corollary G.7, $x, y \in L_B$ and $y \in \mathrm{Ball}(x)$; so if $y$ is covered, $\{x, y\} \in \Lambda(H)$ (take $r = x$), hence co-bagged and eligible. If $y$ is uncovered, $y \in L_B$ is a hub: any two covered $a, b \in \mathrm{Ball}(y)$ lie in $\Lambda(H)$ (take $r = y$), so they are co-bagged, and $\{y, x\}$ is eligible. Lemma O.7 excludes two uncovered ends, and $|U| \leq \ell$. So $S \in \mathcal{S}$ and Theorem O.13 is exact. The equivalence with the treewidth of $I(H)$ plus links: a decomposition of $I(H)$ that puts a member $P \ni a$ and a member $P' \ni b$ in a common bag is a decomposition of $I(H)$ plus the link $PP'$, and conversely a decomposition of $I(H)$ plus links is one of $I(H)$ in which the linked members share a bag. □

## P.4 Proofs for the limits of repacking

*Proof of Theorem O.15.* (a) At $\ell = 0$ CEaMP is NP-hard [15, Thm. 1], even with every vertex in at most 23 members (Theorem 4.1). Suppose the algorithm $\mathcal{R}$ exists with constants $s = s(0)$, $f = f(0)$. Given $(G, H, 0)$, run $\mathcal{R}$ and let $H^*$ be its output. The guarantee holds only on YES instances, so validate: (i) test that $H^*$ is a modification-disjoint packing of induced $P_3$'s of $G$, else reject; (ii) compute, in time $f^{O(f^3)}|H^*|$ [3], a tree decomposition of $I(H^*)$ of width at most $f$, else reject; (iii) put $\ell^* = |H| - |H^*|$ and reject unless $0 \leq \ell^* \leq s$. Finally run the CSP algorithm (Theorem 3.6) on $(G, H^*, \ell^*)$ with this decomposition, in time $5^{f+1}n^{2s+O(1)}$, polynomial, and answer accordingly. Correctness: for a valid packing $H^*$, $(G, H^*, \ell^*)$ is a YES instance iff $\mathrm{OPT} \leq |H^*| + \ell^* = |H|$, i.e. iff $(G, H, 0)$ is. So whenever a decision is made it is correct; the rejections in (i)–(iii) are unreachable on YES instances (where the guarantees hold, and $\ell^* \geq 0$ because $|H^*| \leq \mathrm{OPT} \leq |H|$), and are correct on NO instances. Hence NP is solved in polynomial time. The theorem used is the algorithm of Section 3, valid for every packing; the majority-closure theorem (Theorem 6.4) is not used, and the soft triangles in the gadgets of [15] play no role.

(b) Suppose $\mathcal{R}$ maps every YES instance $(G, H, 0)$ to $H^*$ with $|H^*| = |H|$ and $\mathrm{tw}(I(H^*)) \leq |H|^{1-\varepsilon}$. Take the instances of the proof of Theorem 4.1 from a sparsified 3-SAT formula on $N$ variables, with $|H| = O(N)$. Run $\mathcal{R}$, validate that $H^*$ is a packing of induced $P_3$'s with $|H^*| = |H|$ (else reject), compute a tree decomposition of $I(H^*)$ of width at most $2\mathrm{tw} + 1$ in time $2^{O(|H|^{1-\varepsilon})}|H|$ [14], rejecting if its width exceeds $2|H|^{1-\varepsilon} + 1$, and apply Theorem 3.5 with it, in time $3^{2|H|^{1-\varepsilon}+3}n^{O(1)}$. Since $(G, H^*, 0)$ is equivalent to $(G, H, 0)$ for a valid packing of the same size, this decides 3-SAT in

time $2^{O(N^{1-\varepsilon})}$ (plus the $2^{o(N)}$ of sparsification and the polynomial running time of $\mathcal{R}$), contradicting ETH.

The two remarks follow: a tight instance ($\ell = 0$, $\nu_{\mathrm{mod}} = \mathrm{OPT} = |H|$) has no packing larger than $H$, so "large treewidth implies a better packing" fails on it, and tight instances of unbounded treewidth exist under ETH by Theorem 4.1 (they are the instances of its proof). □

*Proof of Proposition O.16.* The induced $P_3$'s of $G$ are $(x, h, y)$ with $x, y$ leaves and $h$ a hub (a leaf and both hubs form a triangle). Every member has a hub as its centre. Two members with the same centre $h$ and leaf pairs $\{x, y\}, \{x', y'\}$ have disjoint pair sets only if $\{x, y\} \cap \{x', y'\} = \emptyset$, impossible among three leaves. So each hub is the centre of at most one member, and $\nu_{\mathrm{mod}} \leq 2$. A packing of size 2 has one member with centre 3 and leaf pair $\{x, y\}$ and one with centre 4 and a different leaf pair $\{x', y'\}$; two different 2-subsets of three leaves cover all leaves, so every maximum packing covers all five vertices ($\nu_{\mathrm{mod}} = 2$, six maximum packings). Enumerating all 52 partitions gives $\mathrm{OPT} = 3$ and the four optima $\{0, 1, 2, 3, 4\}$, $\{0\}, \{1, 2, 3, 4\}$, $\{1\}, \{0, 2, 3, 4\}$, $\{2\}, \{0, 1, 3, 4\}$; for each of the six maximum packings each optimum has a hard edit (all vertices are covered, so it is covered-covered). For example, for $H^* = \{(0, 3, 1), (1, 4, 2)\}$ and the optimum $\{0\}, \{1, 2, 3, 4\}$ the hard edit is $\{0, 4\}$ and $0, 4$ have the common soft neighbour 1: it is bridged. The counts quoted in Section O come from an exhaustive enumeration of all connected graphs with at least one induced $P_3$, over all maximum packings and all optima. □

*Proof of Proposition O.17.* The hard edges of $H_0$ are $\{0, 1\}$ and $\{2m+1, 2m+2\}$, which do not share a vertex; an induced $P_3$ of the path is three consecutive vertices, with two edges, at most one hard, so it contains a soft pair: $H_0$ is pair-maximal. A packing has at most $\lfloor (2m+2)/2 \rfloor = m+1$ members (two edges each, edge-disjoint), and $\{(2i, 2i+1, 2i+2) : 0 \leq i \leq m\}$ attains it; $\mathrm{OPT} \leq m+1$ by the partition $\{0, 1\}, \ldots, \{2m, 2m+1\}, \{2m+2\}$, so $\nu_{\mathrm{mod}} = \mathrm{OPT} = m+1$ and $\ell = 1$ for $H_0$ ($|H_0| = m$), $w = 1$. A packing of size $m+1$ must use all $2m+2$ edges. The edge $\{0, 1\}$ lies only in $(0, 1, 2)$, so this is a member; then $\{2, 3\}$ lies only in $(2, 3, 4)$ among the remaining ones, and so on: the packing is unique, $\{(2i, 2i+1, 2i+2)\}$, and it shares no member with $H_0$ (whose members start at odd vertices). An improving exchange removes $r$ members of $H_0$ and adds $r+1$ members pair-disjoint from the rest; the result is a packing of size $m+1$, hence the unique one, so all $m$ members of $H_0$ are removed: $r = m$. □

*Proof of Proposition O.19.* Run the algorithm of Open Problem 8, validate that $H^*$ is a pair-maximal packing of induced $P_3$'s with $\ell^* \geq 0$ (else reject, correct because then $\mathrm{OPT} \geq |H^*| > |H| + \ell$ or the output is invalid on a NO instance), and run Theorem O.13 on $(G, H^*, \ell^*)$ with the decomposition of width at most $g(\ell, w)$; the time is $5^{g+1}(Z^2+1)^{3\ell^*(g+1)} n^{O(1)}$, a function of $(\ell, w)$ times a polynomial. Accept iff $m^* \leq |H^*| + \ell^*$. Soundness holds for every output by Theorem O.13, and on a YES instance some optimum lies in the class, so $m^* = \mathrm{OPT} \leq |H^*| + \ell^* = |H| + \ell$. □

# Q Small examples: weaker hypotheses and the star

This appendix collects explicit instances referred to in Sections 3 and 7 and in the appendices.

## Q.1 Treewidth of the graph versus of the intersection graph

**Proposition Q.1.** *(a) For the star $G = K_{1,2m}$ ($m \geq 1$) every packing $H \neq \emptyset$ has $I(H) = K_{|H|}$ and every YES instance $(G, H, \ell)$ satisfies $\ell + \mathrm{tw}(I(H)) \geq 2m - 2$, while $\mathrm{tw}(G) = 1$. (b) For the $k \times k$ grid $G$ and a family $H$ of vertex-disjoint induced $P_3$'s along rows, $\mathrm{tw}(I(H)) = 0$ and $\mathrm{tw}(G) = k$.*

*Proof.* (a) Leaves have degree one, so the centre of any induced $P_3$ is the hub $v$; all packed $P_3$'s are $(a, v, b)$ with distinct leaves (the pairs $va$ are disjoint), pairwise sharing $v$, so $I(H) = K_p$, $p = |H| \le m$, $\mathrm{tw} = p - 1$. If the hub's cluster contains $j$ leaves, the cost is $(2m - j) + \binom{j}{2}$ (the other leaves are best left as singletons), which is minimized, with value $2m - 1$, at $j \in \{1, 2\}$; so $\mathrm{OPT} = 2m - 1$ and a YES instance has $\ell \ge 2m - 1 - p$, i.e. $\ell + \mathrm{tw}(I(H)) \ge 2m - 2$. (b) Rows of a grid are induced $P_3$'s (no triangles) and $\mathrm{tw}(k \times k \text{ grid}) = k$ is standard. □

Proposition Q.1 concerns the numbers $\mathrm{tw}(G)$ and $\mathrm{tw}(I(H))$ only, which are incomparable. Part (a) is the only part that bears on the pair $(\ell, w)$: bounded $\mathrm{tw}(G)$ does not make $\ell + w$ small, because on a star small $\mathrm{tw}(I(H))$ forces large $\ell$. Part (b) says nothing about $\ell$.

### Q.2 Weaker maximality hypotheses fail

The following two instances show that the structure theory of Appendix E and Appendix G needs pair-maximality and that edits touching vertices outside $V(H)$ cannot be excluded.

**Proposition Q.2** (Pair-maximality cannot be replaced by vertex-maximality)**.** *Let $G$ have vertices $0, \dots, 4$ and edges $03, 13, 14, 24, 34$, and $H = \{(0, 3, 1)\}$ (pairs $03, 13, 01$). Then $H$ is maximal among vertex-disjoint packings (only the two vertices $2, 4$ remain),* $\mathrm{OPT} = 2$ *and the unique optimum is $\{03, 24\}$; it contains the edit $24$ with both endpoints outside $V(H)$. Here $\{3, 4, 2\}$ is an induced $P_3$ all of whose pairs are hard, so $H$ is not pair-maximal.*

*Proof.* The partition $\{\{0\}, \{1, 3, 4\}, \{2\}\}$ has the two edits $03, 24$. Conversely, $0\,3\,1$ and $3\,4\,2$ are induced $P_3$'s with disjoint pair sets $\{03, 13, 01\}$ and $\{34, 24, 23\}$, so $\mathrm{OPT} \ge 2$. That this is the only optimum is checked by enumerating all 52 partitions of the five vertices. □

A second example: for the path $0\,1\,2\,3\,4$ and $H = \{(0, 1, 2)\}$ the remaining vertices $3, 4$ cannot form a $P_3$, so $H$ is maximal among vertex-disjoint packings, but $(2, 3, 4)$ is an induced $P_3$ whose pairs $23, 34, 24$ are disjoint from $\mathrm{pairs}(H)$, so $H$ is not pair-maximal; the pair-maximal extension is no longer vertex-disjoint.

**Proposition Q.3** (Strong locality fails)**.** *Let $G = C_4$ with edges $01, 02, 13, 23$ and $H = \{(0, 1, 3)\}$ (pairs $01, 13, 03$; vertex $2 \notin V(H)$). Then $H$ is pair-maximal,* $\mathrm{OPT} = 2 = |H| + 1$*, so $(G, H, 0)$ is a NO instance, and there are exactly three optima $\{03, 12\}$, $\{01, 23\}$, $\{02, 13\}$, each containing a hard pair at vertex $2$.*

*Proof.* Every induced $P_3$ of $C_4$ contains one of the soft pairs $01, 13, 03$, so $H$ is pair-maximal. $\mathrm{OPT} \ge |H| = 1$, and a single edit leaves an induced $P_3$ (checked on the six pairs), so $\mathrm{OPT} = 2$; the three optima are found by enumerating all 15 partitions. □

This refutes the claim that all hard edits of some optimum have both endpoints in $V(H)$; what remains true is that no edit has both endpoints outside $V(H)$ (Theorem E.5 in Appendix E).

## R Scope and status of the results

The following table summarises the scope and status of the results. "Input $H$" means that the statement is in terms of $w$, $\ell$ and $n$ of the input packing; "pair-maximal $H'$" means that it is stated for a pair-maximal packing, and for an arbitrary input applies to a pair-maximal extension $H'$ with the primed parameters of Section 2. "YES" and "optimal" mean that the statement needs a YES instance, respectively an optimal solution.

| Result | Applies to | Needs | Status |
|---|---|---|---|
| Thm 3.5, 3.6 (CSP, $3^{w+1}$, $5^{w+1}n^{2\ell}$) | input $H$ | – | proved; Lean: Obs. 3.1, 3.2 (owner cliques) and the five-state lemma, not the algorithms |
| Thm 6.4, Cor. 6.5, 6.7, B.1 (no soft triangle: majority closure, $\ell = 0$ polynomial, cores $\leq 2$) | input $H$, soft-triangle-free | – | proved; 40-relation classification by computer; cites [12] |
| Lemmas B.3–B.5, Thm B.7 (far-lift model, $6^{\ell}$), Lemma B.8 | input $H$ (soft-triangle-free for Thm B.7) | – | proved; restricted model |
| Prop. B.9, B.10 (RM2 FPT; not faithful) | model; explicit instance | – | proved (hub: enumeration) |
| Lemma B.12, Thm B.13, B.14 (certificate branching, leaf test) | input $H$ (adjacency-local) | – | proved |
| Thm B.15 (certificates $\leq 12$ at $\ell = 0$) | soft-triangle-free | $\ell = 0$ | proved |
| Thm B.16 (unbounded minimal certificates, $c = 2$) | explicit family | – | proved; block structure by computer; the value $\lvert H\rvert + 1$ and the connected $\ell = 1$ family are supported by experiments only |
| Lemma B.18 (no conservative NU) | $c \geq 3$ language | – | proved |
| Thm 6.9, 6.10 (abstract W[1]-hardness; tunnels cannot realise it without soft triangles) | abstract; soft-triangle-free tunnel model | – | proved; remarks (iv), (v) are heuristic |
| **Thm 5.1** (W[1]-hardness in $\ell + w$, $\ell + \mathrm{pw}$, $\ell + c$; no $f(\ell, w)n^{o(\sqrt{\ell})}$ under ETH, Cor. 5.2) | pair-maximal $H$, $c \leq 4p + 2$, $\ell = k^2$ | – | **proved, computer-assisted**: Lemma 5.3 (transfer matrix, finite checks for $m \leq 60$), Lemma 5.4, counts of Lemma 5.8; the rest by hand; Lean 4 checks the counting core of Thm 5.11, the Grid Tiling encoding, the chain lemma for $m \leq 9$ and the matrix algebra, not the graph-level statements (Section 9) |
| Remark 5.15 (chain with unbounded core, bounded $c$ and width) | single chain, $p \geq 8$ | – | follows from the proof of Prop. 5.7 |
| Thm B.11 (forests, conditional) | forest $I(H)$ | BC, oracle (O) | implication proved; BC, (O) open |
| Thm 4.1, Prop. 4.3 (ETH; $w$ alone) | input $H$ | – | proved from [15] |
| Thm H.1 (components, kernel) | input $H$ | – | proved; cites [17] Lemma 6.3, [2], [4] |
| Thm H.10 (components of any size) | input $H$ | – | proved; cites [17] Lemma 6.3 |
| Thm I.2 ($n^{\ell}$ algorithm) | input $H$ (via $H'$) | – | proved |
| Lemmas G.2–G.6, Cor. G.7 | pair-maximal $H'$ | optimal, YES | proved; Lean: twin, half-degree, diameter, MAX, HUB, SM (abstract form) |
| Thm L.2, Cor. L.3, Prop. L.5 | pair-maximal $H'$ | optimal, YES | proved |

| Result | Applies to | Needs | Status |
|---|---|---|---|
| Prop. L.7, L.8 | explicit instances | – | proved (cex: enumeration) |
| Thm I.3, Prop. I.4 | pair-maximal $H'$ | YES | proved; expander treewidth cited [9] |
| Appendices E–K | pair-maximal $H'$ (App. F: any $H$) | mostly YES | proved |
| Appendix M, N | explicit families; tunnel model | – | proved |
| Lemma O.1, O.2, Cor. O.3, Lemma O.5, Cor. O.6, Lemma O.7, O.9 (exchange, twins, witness case split, uncovered ends) | pair-maximal $H$ (exchange: any $H$) | optimal (not Lemma O.1, O.9) | proved |
| Lemma O.8 (displacement, single edit) | pair-maximal $H$ | optimal | proved; iteration and termination open |
| Prop. O.10, O.11 (FAR and Sp($F$) repack) | explicit instances | – | proved; verified by computer for $m \leq 9$ and $F \in \{\text{triangle}, K_4, K_{3,3}\}$ |
| Thm O.13, Cor. O.14 (one-sided local algorithm) | pair-maximal $H$ | promise: optimum in $\mathcal{S}$ | proved; one-sided, promise untestable |
| Thm O.15 (no repacking bounded in $\ell$) | any $H$, $\ell = 0$ | P $\neq$ NP resp. ETH | proved from [15] |
| Prop. O.16, O.17 (naive normal form false; augmentation depth $m$) | explicit instances | – | proved; small cases verified by computer |
| **FPT in** $\ell + w$ (the main question) | | | **no unless** FPT = W[1] (Thm 5.1) |
| Hardness at bounded $c$, in particular $c = 2$ (Open Problem 1) | | | **open** |
| Exact exponent under ETH (Open Problem 3): $N^{o(\sqrt{\ell})}$ excluded, $N^{\ell}$ achieved | | | **open** |
| Reduction lemma (Open Problem 7) | | | **no unless** FPT = W[1] |
| Positions of T2 / deleted pointers (Open Problem 7) | | | **open on restricted classes**; no in general unless FPT = W[1] |
| LOC (Open Problem 8); implies FPT (Prop. O.19) | | | **no unless** FPT = W[1] (Cor. O.20) |
| Conjecture C1 / NF* (Conj. O.18) | | | **open**; no counterexample among all connected graphs with $n \leq 8$ and random graphs with $n = 9, 10$ |
| Thm D.4, 6.11 (certificates at $\ell = 0$: reduction to the unit-lock Helly number; at most 4 remote pairs for $c \leq 2$, soft triangles allowed), Lemmas D.5, D.6 | input $H$ ($c \leq 2$ for Thm 6.11) | $\ell = 0$ (Lemma D.6: any $\ell$) | proved; finite classification by computer (Lemma D.5, Thm 6.11); tightness of 4 verified by computer |
| BC for $\ell \geq 1$, BC for $c \geq 3$ with soft triangles (Open Problem 2); soft triangles at $c \leq 2$ (Open Problem 6) | | | **open** (experimental evidence only) |

# S Computer verifications and formal verification

The proofs in this paper rely on computation in three places only: (i) statements about explicit small instances were verified by exhaustive enumeration of set partitions, as indicated where they occur; (ii) the finite classifications marked *by computer* (Theorem 6.4, Theorem B.16, Lemma D.5 and Theorem 6.11); and (iii) the four computer-assisted steps (CA1)–(CA4) in the proof of Theorem 5.1. All other computations reported below are supporting experiments: they test statements that are proved by hand, and no proof uses them. Where two implementations are mentioned, the second was written directly from the definitions (all set partitions, an exact subset dynamic programme, SAT or CP-SAT) without using the structure lemmas.

**Reproducibility.** The code for all computations and the Lean 4 development are available in the repository `vinhqdang/correlation_clustering` on GitHub, directory `paper2_p3_interaction` (subdirectories `code` and `lean/P3Proofs`).

## S.1 Table of computations

| Statement | Computation | Role |
|---|---|---|
| Chain lemma, (a) | every $P_a$, $7 \le m \le 60$ (cost $\lvert H\rvert$, one edit per member, no hard pair edited) | used in proof (CA1) |
| Chain lemma, (b) | transfer matrix ($6 \times 6$, $\chi = x^4(x-1)^2$), $N_m = m$ for $7 \le m \le 40$; equality of the matrix at the cells $7, \dots, 19$ of $m = 26$ | used in proof (CA2) |
| Chain lemma | independent SAT enumeration of all ZE partitions: exactly $m$ partitions for $m = 7, \dots, 14$, a sample up to $m = 100$, and $m = 14, 18, 26, 42$ | supporting |
| Host profiles | all even $m \in [8, 60]$ and all groups (1,836 cases), including both caps | used in proof (CA3) |
| Parameters $u_{\max} \le 3$, $c \le 4p + 2$ | count for $1 \le k \le 5$, $2 \le n \le 4$: $u_{\max} = 2$ for $k \le 2$ and 3 for $k \ge 3$; member counts through vertices for $k \le 4$, $n \le 3$ | used in proof (CA4) |
| Tunnel-release semantics | random Grid Tiling instances, every choice of one tile per cell (a sample when large): ZE feasibility with those tunnels cut agrees with "tiling" ($k = 2, n = 2, 3$; $k = 3, n = 2$; $k = 1, n = 4, 5$; thousands of cases, two implementations, no mismatch); $k = n = 2$: all 386 cut sets of size $\le 4$, exactly the 15 tilings feasible | supporting |
| Comparator | all 100 pairs of positions of two chains with $m = 10$, thin model and thick instance $p = 8$ (515 vertices): feasible exactly when ($a \le 3$ or $b \ge 6$) and neither is a permanently forbidden position; thick $k = 2, n = 2, p = 8$ (2,916 vertices, 23,176 members) agrees with the thin model on four choices | supporting |

| Statement | Computation | Role |
|---|---|---|
| Exact optimum, small thick instances | all pairs editable, SAT with cardinality bound: $k = 1$, $p = 8$ (428 vertices, $\|H\| = 3258$): $\|H\| < \mathrm{OPT} \leq \|H\| + 1$; same without the tile: $\mathrm{OPT} \geq \|H\| + 2$; comparator instance ($p = 6$, 401 vertices, $\ell = 2$): three YES choices satisfiable, one NO choice unsatisfiable at $\|H\| + 2$ | supporting |
| Exact optimum, full thick instances | after contraction of each $K_v$ (justified by Lemma 5.9; validated against the unrestricted solver for $k = 1$), CP-SAT; table in Appendix S.2 below: no mismatch with Grid Tiling for $k \leq 3$ | supporting |
| Validity, pair-maximality | $(k, n, p) = (1, 2, 8)$, $(1, 2, 6), \ldots, (1, 5, 6)$, $(2, 2, 10)$: packing of induced $P_3$'s, no induced $P_3$ with three hard pairs | supporting |
| Pathwidth | measured bounds 497 ($k = 1$, $p = 6$) and 5,049 ($k = 2$, $p = 10$); interval decomposition verified exhaustively, largest bag independent of $n$ | supporting |
| Soft triangles: $R_{AB}$ | classification of 144 configurations into 40 relations; independent reconstruction from 576 five-vertex graphs; all 729 majorities against all relations | used in proof (Thm 6.4) |
| Soft triangles: Cor. 6.5 | path-consistency solver vs. enumeration on 20,920 random soft-triangle-free instances and 3,400 tree-like ones ($\Sigma$ closed under all 729 majorities in all); with soft triangles 451 failures of closure | supporting |
| Remark 6.6 | 100 soft triangles in the variable gadget of [15] | supporting |
| Lemma B.4, Thm B.7 | 803 strict cases, no violation; far-lift algorithm vs. exhaustive search on 1,420 and (independent) 1,050 instances | supporting |
| Prop. B.9, B.10 | RM2 search tree vs. exhaustive search on $5{,}000 + 2{,}100$ instances; hub instance by enumeration and subset dynamic programme | supporting / enumeration |
| Thm B.13, B.14, B.15 | certificate branching vs. exact OPT on 449 and 237 pairs; 786 random soft-triangle-free instances: largest minimal inconsistent family of guards had two members; radius 3 suffices on random tree-like instances | supporting |
| Thm B.16 | block structure of the ZE set of $G_N$ by computer; $F'(T_N) = \|H\| + 1$ for $N \leq 10$; $\mathrm{OPT}(G_N) = \|H\| + N$ for $N \leq 5$ | used in proof (blocks) / supporting |
| Remark B.2 | tight triangle blocks (3,712 configurations, 163 distinct $\Sigma$), tight $K_4$ gadgets (521), 3,000 random prism instances: all 2-decomposable, no core with three anchors | supporting |
| Thm 6.9 | exhaustive search and two independent solvers on 1,710 instances | supporting |
| Thm 6.11 | about 25,000 random $c \leq 2$ instances; exact minimum certificates $\leq 4$; tightness instance ($n = 13$) | classification used in proof; rest supporting |

| Statement | Computation | Role |
|---|---|---|
| Normal-form lemmas, Cor. G.7, locality (Sections of Appendices G, L) | random small graphs ($n \leq 10$; random, clustered, hub, sparse; random and maximal packings) against all partitions of cost $\leq \lvert H\rvert + 3$, independent code; weakening a constant by one produces violations (so the tests are not vacuous) | supporting |
| Thm I.3 | dynamic programme vs. exhaustive search on 2,778 pairs (instance, budget) with $n \leq 10$ | supporting |
| Explicit instances | all set partitions enumerated (Remark 3.7, Props. Q.2, Q.3, L.7, K.10, FAR for $m = 3, 4, 5$, the shrink-rule example, $G^*$ and the majority example of Appendix N); packing validity and pair-maximality checked | enumeration (proves the instance statements) |
| Repacking: exchange calculus | 5,867 random instances ($6 \leq n \leq 10$), 26,387 hard edits: all in induced $P_3$'s, none with two uncovered ends, all 27,177 exchanges gave packings of size $\lvert H\rvert$ with tw $\leq$ tw$(I(H)) + 1$ (59% pair-maximal); displacement: 5,027 inserted edits, $r$ uncovered in 1,650 | supporting |
| Repacking: FAR, spine | $m = 3, \ldots, 9$ (MILP, enumeration for $m \leq 5$); spine family for triangle ($n = 18$), $K_4$ ($n = 30$), $K_{3,3}$ ($n = 42$) | supporting |
| Thm O.13 | implementation on 984 instances ($n \leq 9$): value $\geq$ OPT always, $=$ OPT in all 948 with an optimum in $\mathcal{S}$; negative control without the co-bagging restriction: 17 of 177 instances below OPT | supporting |
| Conj. O.18, Prop. O.16 | all connected graphs $n \leq 8$, 1,500 random $n = 9$, 300 random $n = 10$: no counterexample to C1; naive form fails on 3/20, 2/111, 3/852, 305/11,116 graphs for $n = 5, \ldots, 8$ | supporting (exhaustive for $n \leq 8$) |
| Prop. O.17 | $m = 2, \ldots, 6$: smallest improving exchange has $r = m$ | enumeration |

## S.2 Exact optima of full thick instances

This experiment supports the reduction and is not part of the proof. By Lemma 5.9, every solution with ex $\leq k^2$ leaves each $K_v$ inside one cluster, so OPT $\leq \lvert H\rvert + k^2$ iff the minimum over solutions with every $K_v$ unsplit is at most $\lvert H\rvert + k^2$. That minimum is the optimum of a weighted correlation clustering instance on the *contraction* (classes: the super-vertices and the single guest vertices; for two classes $A, B$ the weights are the numbers of edges and of non-edges of $G$ between them, computed from the real thick graph, so that every pad pair, host pair, thin member pair, hard non-edge and tunnel is counted; plus the internal non-edges of each class). All pairs of classes are editable and nothing is assumed about the states of the members; the problem is solved exactly by CP-SAT with lazily added transitivity constraints. The contraction was validated against the unrestricted exact solver (all pairs of $V(G)$ editable, no contraction) for $k = 1$: with one tile $p = 8$ both give OPT $= \lvert H\rvert + 1$ ($\lvert H\rvert = 3258$), and without a tile ($p = 4$) both give OPT $= \lvert H\rvert + 8$ ($\ell = 0, \ldots, 7$ unsatisfiable, $\ell = 8$ satisfiable, no split class). Results, with ex $=$ OPT $- \lvert H\rvert$ (expected ex $= k^2$ if Grid Tiling is solvable and ex $\geq k^2 + 1$ otherwise):

| $k$ | $n$ | $p$ | instances | YES: ex | NO: ex | mismatches |
|---|---|---|---|---|---|---|
| 1 | 3 | 8 | 3 (random) | 3 YES, ex $= 1$ | – | 0 |
| 2 | 2 | 10 | 11 (3 fixed, 8 random) | 6 YES, ex $= 4$ | 5 NO, ex $= 5$ | 0 |
| 2 | 3 | 10 | 2 (random) | 2 YES, ex $= 4$ | – | 0 |
| 3 | 2 | 14 | 2 (fixed) | 1 YES, ex $= 9$ | 1 NO, ex $= 10$ | 0 |

The largest instance has $|V(G)| = 11{,}262$ and $|H| = 153{,}498$ ($k = 3$). On every instance solved, YES instances have ex $= k^2$ exactly and NO instances have ex $= k^2 + 1$ exactly (the cheapest deviation found by the solver is one comparator guest left alone, one member with two edits). That NO instances sit at exactly $k^2+1$ is an observation on these instances, not a theorem; the proof only uses ex $\geq k^2+1$. The computation is an end-to-end exact test for $k \geq 2$ that the unrestricted solver could not perform; it relies on Lemma 5.9, which is proved by hand and was confirmed against the unrestricted solver for $k = 1$.

## S.3 Status of the reduction proof

| Statement | Status |
|---|---|
| Lemma 5.3(a) ($P_a$ is ZE, all $m$) | proved modulo (CA1): finite check for $m \leq 60$ plus the window argument; Lean: $m = 7, 8, 9$ only |
| Lemma 5.3(b) (no other ZE partition) | proved modulo (CA2): transfer matrix with $\chi = x^4(x-1)^2$, $N_m = m$ for $7 \leq m \leq 40$, translation invariance as in Appendix A; Lean: the classification for $m = 7, 8, 9$ and the algebra of the matrix, not the link between graph and matrix and not $m \geq 10$ |
| Lemma 5.4, Cor. 5.5 (profiles) | read off the formula; (CA3) verified for even $m \leq 60$ and all groups |
| Lemma 5.6 (blow-up) | proved by hand |
| Prop. 5.7 (tunnel release, comparators) | proved by hand from the above; Lean: the block encoding and the equivalence of the comparator conditions with the Grid Tiling constraints, not the graph-level statement |
| Lemma 5.8 ($u_{\max} \leq 3$, $c \leq 4p + 2$) | hand count (host 2 or 3, non-host $\leq 2$); (CA4) confirmed by computer |
| Lemma 5.9 (no split) | proved by hand (needs $p \geq k^2 + 5$); Lean: the arithmetic inequality only |
| Lemma 5.10, Thm. 5.11 (soundness) | proved by hand; Lean: the counting core as an abstract inequality system, not at graph level; exact optima of full thick instances for $k \leq 3$ agree with Grid Tiling (supporting, Appendix S.2) |
| Thm. 5.12 (completeness) | proved by hand |
| Lemma 5.13 (pathwidth $< 24kp^2$) | proved by hand (level sweep); confirmed on small instances |
| Lemma 5.14 (pair-maximal) | proved by hand; confirmed on small instances |

## S.4 Formal verification in Lean 4

Part of the argument was formalised in Lean 4 with Mathlib (see the repository in the paragraph on reproducibility above). All files are free of `sorry` and of `native_decide`; every theorem depends at most on the axioms `propext`, `Classical.choice` and `Quot.sound`, as reported by `#print axioms`. The finite computations use `decide +kernel`, which is evaluated by the Lean kernel and does not add `Lean.ofReduceBool`.

*Machine-checked.*

- *CSP representation (Section 3).* Clusterings versus cluster graphs and the cost/OPT equivalence; OPT $\geq |H|$ for a modification-disjoint packing; Observations 3.1 and 3.2 (a solution with $|S| \leq |H|$ has one pair per member and nothing else, $(G, H, 0)$ is YES iff the CSP is satisfiable, the constraint of a triple depends only on its owners, owners of a triple pairwise share

a vertex; the bound "at most three owners" and the treewidth statements are not included); the five-state lemma, in its finite form and in its graph form.

- *Structural lemmas.* The twin lemma (twins lie in one cluster of every optimum); half-degree and diameter at most 2 of an optimal cluster, and separation of vertices at distance at least 3; the lemmas on $M$ (MAX) and on the hub (HUB, in the bounds $\nu(\Lambda_u) \le \ell$, vertex cover of size at most $2\ell$ and $|Y| \le 2\ell + 1$); the small-merge lemma (SM) in abstract form, with its hypotheses (cross-pair condition, size and soft-degree bounds) taken as assumptions.

- *Block encoding of Grid Tiling.* The encoding of tiles by slot and separator groups, the comparator clauses, and the equivalence of the comparator conditions on the A-chains and B-chains with the Grid Tiling constraints (the last step of Proposition 5.7, $\Rightarrow$).

- *Counting core of Theorem 5.11.* As an abstract system in the data $h_c$, $E_c$, $q$, $h_{\mathrm{in}}$, $|S_{\mathrm{rest}}| - |H_{\mathrm{cmp}}|$, ex, with the hypotheses $\sum_c h_c = h_{\mathrm{in}} + 2q$, $r_c \ge \max\{1, h_c\}$ and $\mathrm{ex} + q = \sum_c r_c + (|S_{\mathrm{rest}}| - |H_{\mathrm{cmp}}|)$: the bound $\mathrm{ex} \ge \max\{q, 2k^2 - q\} \ge k^2$, the inequality $\mathrm{ex} \ge |Z| + h_{\mathrm{in}} + q + \sum_{c \notin Z} E_c$, the equality case ($q = k^2$, $Z = \emptyset$, $h_{\mathrm{in}} = 0$, all $E_c = 0$, all $h_c = 1$, $|S_{\mathrm{rest}}| = |H_{\mathrm{cmp}}|$), and that the bound is attained by a consistent system. The summation must run over $c \notin Z$ only: with $\sum_c E_c$ in its place the inequality fails for $k = 1$, $h = (0, 0)$, $E = (1, 1)$, $\mathrm{ex} = 2$, where the right-hand side would be 4.

- *Chain lemma, finite version.* Lemma 5.3 for $m = 7, 8, 9$ only: every $P_a$ is ZE, and every ZE partition of $\mathrm{OH}_m$ is one of the $P_a$ with spine word $R^a M L^{m-1-a}$ (kernel computation).

- *Transfer-matrix algebra.* The characteristic polynomial $x^4(x-1)^2$ of the $6 \times 6$ matrix, $M^4(M-1)^2 = 0$, $M^t = M^4 + (t-4)(M^5 - M^4)$ for $t \ge 4$, and the affine-growth lemma: if $N_m = m$ at two consecutive arguments from the stable range on, then $N_m = m$ for all $m$ in that range.

- *Parameter arithmetic.* $p \le k^2 + 6$, $c \le 4p + 2 \le 4k^2 + 26$, and the no-split inequality of Lemma 5.9 ($x(p - x - u) > k^2$ under the stated bounds).

*Not machine-checked.* Theorem 5.11 at graph level; the blow-up lemma; host profiles (Lemma 5.4); tunnel release at graph level; the region bound (Lemma 5.10); pathwidth and pair-maximality of the reduction instance; the chain lemma for $m \ge 10$; the link between the graph $\mathrm{OH}_m$ and the transfer matrix (so the steps (CA1)–(CA4) remain computer-assisted, with Lean covering only $m \le 9$ and the matrix algebra); the dynamic programmes and running-time theorems; the ETH statements; and the reduction as a whole (Theorem 5.1). The hypotheses of the counting core are taken as given, so the Lean result certifies the arithmetic of Theorem 5.11, not the claim that the instance $I_{\mathrm{tile}}$ satisfies them.